PROOF OF THE ORBIFOLD ABELIANITY CONJECTURE WITH FINITE MULTIPLICITIES

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Abstract. We prove Campana's orbifold abelianity conjecture with finite multiplicities. The conjecture predicts that if a compact Kähler manifold carries a simple normal crossing divisor whose components have integral multiplicities at least two, and the orbifold so defined is special in the sense of Campana, then the orbifold fundamental group, namely the quotient of the fundamental group of the complement of the divisor by the normal subgroup generated by the meridians of the components raised to their multiplicities, contains an abelian subgroup of finite index.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let X be a connected compact Kähler manifold, let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossing divisor on X with smooth irreducible components D_i, let $m_1, \dots, m_r \geq 2$ be integers, and let

$$\Delta = \sum_{i=1}^{r} \Big(1 - \frac{1}{m_i}\Big) D_i.$$

Campana's theory of orbifolds [Cam04b, Cam11a] studies such pairs (X, Δ) (Definition 2.2), regarding D_i as a divisor of multiplicity m_i. The orbifold fundamental group of (X, Δ) is the group

$$\pi_1(X \backslash D) / \langle\langle \gamma_1^{m_1}, \dots, \gamma_r^{m_r} \rangle\rangle,$$

where γ_i is a positively oriented meridian of D_i joined to the base point by a path, and $\langle\langle \cdot \rangle\rangle$ denotes the normal closure [Cam11a, Définition 12.2]. When $D = 0$, the orbifold fundamental group is $\pi_1(X)$.

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Roughly speaking, the pair (X, Δ) is *special* (Definition 2.11) if there exists no meromorphic fibration $f : X \dashrightarrow Z$ with $\dim Z > 0$ such that Z is of general type once the multiplicities m_i and the multiplicities of the fibers of f over the prime divisors of Z are counted on Z [Cam11a, Définition 8.1]. Definition 2.11 is formulated with differential forms instead of fibrations, and it is equivalent to [Cam11a, Définition 8.1] by [Cam11a, Théorème 9.9]. A group is *virtually abelian* if it contains an abelian subgroup of finite index.

Campana conjectured that the fundamental group of a special compact Kähler manifold is virtually abelian [Cam04b, Conjecture 7.1] (see also [CH13, Conjecture A.1]), and he conjectured that the orbifold fundamental group of a special orbifold pair is virtually abelian [Cam04b, Conjecture 7.3]. The orbifold statement appears again in [Cam11a, Conjecture 13.10(1)] and [Cam11b, Conjecture 6.2], and, for smooth projective orbifolds, in [Cam10, Conjecture 11.2]. In [Cam04b, Conjecture 7.3] and [Cam11a, Conjecture 13.10], D is only assumed to have normal crossings: its irreducible components need not be smooth, and its crossings need not be simple. This case reduces to the case of a simple normal crossing divisor with smooth irreducible components, see Section 2.1 after Definition 2.2. For finite multiplicities and a simple normal crossing divisor D with smooth irreducible components, the conjecture reads as follows.

Conjecture 1.1 (Campana)**.** *Let X be a connected compact Kähler manifold, let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossing divisor on X with smooth irreducible components D_i, let $m_1, \dots, m_r \geq 2$ be integers, and let $\Delta := \sum_{i=1}^{r}(1 - \frac{1}{m_i})D_i$. Assume that (X, Δ) is special. Then the orbifold fundamental group of (X, Δ) is virtually abelian.*

In [Cam11a, Conjecture 13.10] the multiplicities may take the value $+\infty$ [Cam11a, Définition 2.1]. With infinite multiplicities the group need not be virtually abelian: [CDY25, Example 4.27] constructs a special smooth quasi-projective surface whose fundamental group is not virtually abelian (see also [AAC25, Example 23 and Remark 24] and [GLM23, Example 10.2]). Thus Conjecture 1.1 only concerns finite multiplicities. The orbifold statement is still listed as a conjecture in [KR24a, Conjecture 6.16.1], where, as in Conjecture 1.1, all multiplicities are assumed to be finite. For special smooth quasi-projective varieties, [CDY25, Conjecture 2.6] proposes a nilpotency conjecture instead, and [CDY25, Theorem A] shows that the image of any representation of the fundamental group in $\mathrm{GL}_N(\mathbb{C})$ is virtually nilpotent. Linear images of the fundamental group of a smooth special quasi-projective variety are virtually nilpotent of class at most two [OAI26d, Theorem 1.1].

Before this paper, the following cases and partial forms of Conjecture 1.1 were known.

(1) $\dim X = 1$. In this case (X, Δ) is special if and only if $\deg(K_X + \Delta) \leq 0$ [Cam11a, Exemple 8.2(2)]. The orbifold fundamental group is finite if $\deg(K_X + \Delta) < 0$, and it contains a subgroup of finite index isomorphic to $\mathbb{Z}^2$ if $\deg(K_X + \Delta) = 0$ [CC11, Example 4.1].

(2) Projective orbifold surfaces. For projective orbifold surfaces that are special in the sense of [CC11, Definition 3.7], that is, that admit no fibration onto a curve whose orbifold base, with multiplicities defined by greatest common divisors, is of general type, and satisfy $\kappa(X, K_X + \Delta) < 2$, [CC11, Theorem 2.2] shows that the orbifold fundamental group is virtually abelian of even rank at most 4. Here the rank of a finitely generated virtually abelian group is the maximal rank of an abelian subgroup of finite index [CC11, Theorem 4.2, footnote].

(3) $D = 0$ and $\dim X = 3$ [CC11, Theorem 1.1].

(4) $D = 0$, in any dimension [OAI26c, Theorem 1.1]. Earlier, for projective X, the case $D = 0$ was reduced to showing that $\pi_1(X)$ is finite when every finite unramified cover of X has irregularity zero [CC16, Proposition 3.2], using that the general fibers of the Albanese map of a special smooth projective variety are special [CC16, Théorème 2.4]. For compact Kähler manifolds, the linear Shafarevich reduction of representations of the fundamental group was constructed in [CCE15, Theorems 1 and 4], and the Albanese

map of a special compact Kähler manifold was shown to be surjective with connected fibers [Cam23, Theorem 1.2].

(5) $c_1(K_X+\Delta)=0$, or $-c_1(K_X+\Delta)$ positive. Compact Kähler orbifolds with $c_1=0$ whose local groups act freely in codimension one, that is, without ramification along divisors, have orbifold fundamental group containing a free abelian subgroup of finite index and of even rank at most twice the dimension [Cam04a, Corollaire 6.3]. Such orbifolds have no ramification along divisors, so for the pairs of Conjecture 1.1 they only give the case $D=0$. For X, D and $m_1,\dots,m_r\geq 2$ as in Conjecture 1.1, [CC11, Theorem 4.2] shows that the orbifold fundamental group is virtually abelian of even rank at most $2\dim X$ if $c_1(K_X+\Delta)=0$, and finite if $-c_1(K_X+\Delta)$ is positive, that is, represented by a closed real $(1,1)$-form on $X\backslash D$ whose pullbacks to the local uniformizations of (X,Δ) extend to Kähler forms [CC11, §4.3].

(6) X projective and $-(K_X+\Delta)$ nef and big. In this case the orbifold fundamental group is finite [Bra21, Theorem 2]. Another proof is in [LT26, Theorem 1.7].

(7) Linear and virtually solvable quotients. If (X,Δ) is special, then any quotient of the orbifold fundamental group that is virtually solvable, or isomorphic to a subgroup of $\mathrm{GL}_N(\mathbb{C})$, is virtually abelian [Cam11b, Corollaire 6.5].

(8) X projective, $\dim X=4$, $r=1$, and $m_1=2$. Assuming the case $D=0$ [OAI26b, Assumption 1.2], [OAI26b, Theorem 1.3] shows the following: if X is a smooth projective fourfold, D is a non-empty smooth connected reduced divisor, and $(X,\frac{1}{2}D)$ is special, then $\pi_1(X\backslash D)/\langle\langle\gamma_1^2\rangle\rangle$ is virtually abelian. The assumption is now [OAI26c, Theorem 1.1].

Two related results concern weaker properties of the orbifold fundamental group. For effective compact Kähler orbifolds with nef anti-canonical bundle, the orbifold fundamental group has polynomial growth [Liu23]. For special pairs, [KRT26, Theorem 1.3] proves an upper bound for the irregularity.

We now explain why the case $D=0$ does not imply Conjecture 1.1 by passing to a finite cover. For $D=0$, any finite unramified cover of X is again a compact Kähler manifold. For the pair (X,Δ), the analogous object is a finite cover $Y\to X$ by a connected compact complex manifold Y whose ramification index over each D_i is exactly m_i and which is unramified elsewhere. Such a cover exists only if the orbifold fundamental group of (X,Δ) has a finite quotient in which the image of each γ_i has order m_i. For example, let p and q be two distinct points of $\mathbb{P}^1$. The pair $(\mathbb{P}^1,\frac{1}{2}p+\frac{2}{3}q)$ is special, since

$$\deg\left(K_{\mathbb{P}^1}+\frac{1}{2}p+\frac{2}{3}q\right)=-2+\frac{1}{2}+\frac{2}{3}=-\frac{5}{6}<0.$$

The group $\pi_1(\mathbb{P}^1\backslash\{p,q\})$ is infinite cyclic, generated by γ_p, with $\gamma_q=\gamma_p^{-1}$. Since $\gamma_p^2=1$ and $\gamma_p^3=1$ in the quotient, the group

$$\pi_1(\mathbb{P}^1\backslash\{p,q\})/\langle\langle\gamma_p^2,\gamma_q^3\rangle\rangle$$

is trivial, and the image of γ_p in it has order 1, not 2. Indeed, if $C\to\mathbb{P}^1$ were a finite cover of degree n by a compact Riemann surface C of genus g, with ramification index 2 over p, ramification index 3 over q, and no other ramification, then $6\mid n$, and the Riemann–Hurwitz formula would imply

$$2g-2=-2n+\frac{n}{2}\cdot(2-1)+\frac{n}{3}\cdot(3-1)=-\frac{5n}{6}\leq -5,$$

which contradicts $g\geq 0$. A finite cover of X only sees a finite quotient of the orbifold fundamental group of (X,Δ), so the intersection of all normal subgroups of finite index in the orbifold fundamental group is not detected by any finite cover. Accordingly, [OAI26b, Theorem 1.3] does not assume that X has a finite cover by a manifold with ramification index 2 over D. It uses a smooth projective variety Y over X whose fundamental group maps onto the orbifold fundamental group [OAI26b, Propositions 3.1 and 4.1].

Moreover, the compact Kähler case does not have the covering constructions of the projective case. The adapted Kawamata covers [CP19, Lemma 5.2] are constructed for a polarized X [CP19, §5.1], the Kawamata coverings of [Laz04, Proposition 4.1.12] are constructed on quasi-projective varieties from Bloch–Gieseker covers and cyclic covers, and the auxiliary branched covers [OAI26b, Lemma A.1] are built from general members of very ample linear systems by Bertini's theorem. A compact Kähler manifold carrying a big line bundle is projective [Moi66, Part I, Chapter I, Theorem 11]. If the algebraic dimension $a(X)$ [Uen75, Chapter I, §3] is zero, then

$$\dim H^0(X, \mathcal{L}) \leq 1$$

for any line bundle $\mathcal{L}$ on X, since the ratio of two linearly independent sections of $\mathcal{L}$ is a non-constant meromorphic function on X. Thus X carries no linear system $|\mathcal{L}|$ with $\dim |\mathcal{L}| \geq 1$ from which auxiliary divisors can be taken. Nor can the compact Kähler case be reduced to the projective case by deformation: in any dimension at least 4 there exist compact Kähler manifolds that are not homotopy equivalent to any projective manifold [Voi04, Theorem 2], while the group of Conjecture 1.1 is determined by the topology of the pair (X, D) and the integers m_i.

When $D \neq 0$, the case $\dim X \geq 3$ and the case of non-projective compact Kähler surfaces remained open before this paper, apart from the cases of [CC11, Theorem 4.2], [Bra21, Theorem 2], and [OAI26b, Theorem 1.3], and the linear and virtually solvable quotients of [Cam11b, Corollaire 6.5].

In this paper we prove Conjecture 1.1 in the following form.

Theorem 1.2. *Let X be a connected compact Kähler manifold, let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossing divisor on X with smooth irreducible components D_i, where $r \geq 0$, and let $m_1, \ldots, m_r \geq 2$ be integers. Let $\Delta := \sum_{i=1}^{r}(1 - \frac{1}{m_i})D_i$, fix a base point $x_0 \in X \backslash D$, and for each i let $\gamma_i \in \pi_1(X \backslash D, x_0)$ be a positively oriented meridian of D_i joined to x_0 by a path. Assume that (X, Δ) is special (see Definition 2.11). Then the orbifold fundamental group*

$$\pi_1^{\mathrm{orb}}(X, \Delta) = \pi_1(X \backslash D, x_0) / \langle\langle \gamma_1^{m_1}, \ldots, \gamma_r^{m_r} \rangle\rangle$$

is virtually abelian, that is, it contains an abelian subgroup of finite index. Here $\langle\langle \cdot \rangle\rangle$ denotes the normal closure.

Let X be a compact Kähler manifold of algebraic dimension zero, let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossing divisor on X with smooth irreducible components D_i, and let $m_1, \ldots, m_r \geq 2$ be integers. Then the pair (X, Δ), where $\Delta := \sum_{i=1}^{r}(1 - \frac{1}{m_i})D_i$, is special (Corollary 3.6(2)). Thus Theorem 1.2 implies the following corollary.

Corollary 1.3. *Let X be a connected compact Kähler manifold of algebraic dimension zero, let $D = \sum_{i=1}^{r} D_i$ be a reduced simple normal crossing divisor on X with smooth irreducible components D_i, where $r \geq 0$, and let $m_1, \ldots, m_r \geq 2$ be integers. Then $\pi_1^{\mathrm{orb}}\big(X, \sum_{i=1}^{r}(1 - \frac{1}{m_i})D_i\big)$ is virtually abelian.*

Sketch of the proof. Proposition 7.8 replaces a counterexample to Theorem 1.2 of least dimension by a special pair (Y, Δ_Y) with $a(Y) = 0$, $\pi_1(Y) = \{1\}$, and $G := \pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ infinite without proper subgroups of finite index. By Proposition 9.1, Y dominates a pair (Z, Δ_Z) with $\dim Z \geq 1$ and $K_Z + \Delta_Z$ big, so $a(Y) \geq a(Z) = \dim Z \geq 1$ by Lemma 2.15(1) and (2), a contradiction.

(1) Proposition 7.8 is an induction on $\dim X$, along the algebraic reduction if $a(X) > 0$ (Proposition 7.5) and along Albanese maps of finite covers if $a(X) = 0$ (Propositions 6.5 and 7.7).
(2) Theorem 5.1, the case of Theorem 1.2 with X projective, is used in (1) and follows from Theorem 2.13 for a special smooth projective variety whose fundamental group maps onto $\pi_1^{\mathrm{orb}}(X, \Delta)$ (Construction-Proposition 4.5 and Propositions 5.2 and 5.5).

(3) Proposition 9.1 follows from Theorem 9.2 for the base of the Γ-reduction (Definition-Theorem 8.2) of an infinite quotient of G of minimal Γ-dimension, with multiplicities the orders of the images of meridians (Construction-Proposition 9.4).

The key step is (3): [OAI26c, §4] constructs a line bundle $\mathcal{A}$ on Z whose curvature is, over a dense open subset of Z, the pullback of the Kähler form of a Kähler manifold W of dimension s by a submersion, and a quotient I of G acting on W by isometries. Lemma 8.4 bounds the Γ-dimension of I by s, so minimality forces $s = \dim Z$, and $\mathcal{A}$ is big.

Items (2), (3), and Lemma 7.6 are new, and the rest of (1) extends known results to orbifold pairs. The proof does not bound the index of an abelian subgroup.

(a) The projective case. As noted before Theorem 1.2, Theorem 5.1 is new for $\dim X \geq 3$ and $D \neq 0$, apart from the cases listed there. A Kummer cover $S \to X$ with ramification index m_α over D_α (Construction-Proposition 4.3) is also branched over auxiliary divisors, so S need not be special. For one divisor with $m_1 = 2$, [OAI26b, Lemma A.2 and Proposition A.3] removes the auxiliary branching by commuting involutions of elliptic curves and a blow-up of their fixed loci. The involutions have no analogue of order m_α for general m_α, since the automorphisms of an elliptic curve fixing a point form a group of order 2, 4, or 6 [Har77, Chapter IV, Corollary 4.7].

Instead, the Galois group A of $S \to X$ acts by cyclic permutations and translations on a product F of copies of $E_\alpha^{m_\alpha}$, where E_α is an elliptic curve, one copy for each α and each pair of auxiliary divisors attached to D_α (Construction-Proposition 4.2). A point $x \in D_\alpha$ is contained in at most $\dim X - 1$ of the $\dim X + 1$ auxiliary divisors of D_α (Construction-Proposition 4.3(2)). Thus an element fixing a point over x that moves the m_α-th root of a local equation of D_α has equal non-zero coefficients on the basis elements of A attached to two auxiliary divisors not containing x, and acts freely on F (Construction-Proposition 4.2(2)). Hence the m_α-th roots descend to $(S \times F)/A$ (Construction-Proposition 4.5(2)). For $m_\alpha \geq 3$, blowing up fixed loci once leaves multiplicity m_α over the auxiliary divisors (Remark 5.3), so Proposition 5.2 extracts over them prime divisors of multiplicity one by monomial valuations divided by m_α.

(b) Algebraic dimension zero. If $a(X) = 0$, the base of the algebraic reduction is a point, so the induction runs along the Albanese maps of finite covers and needs very general Albanese fibers to be special. For smooth projective special varieties, this is [CC16, Théorème 2.4], an answer to [Cam04b, Question 5.4] in the projective case, but here the Albanese torus need not be algebraic. Lemma 7.6 uses $a(X) = 0$ instead: a non-special very general fiber would yield, by the relative core of [Lu02, Proposition 8.7] and the subadditivity over a complex torus of [Wan21, Main Theorem A(II)], a compact Kähler manifold V dominated by X and a $\mathbb{Q}$-divisor B on V with $\kappa(V, K_V + B) > 0$, whereas $a(V) \leq a(X) = 0$. On the group side, Theorem 2.13 only concerns the fundamental groups of the compact Kähler manifolds underlying the finite-index models of subgroups of finite index in $\pi_1^{\mathrm{orb}}(X, \Delta)$ (Construction-Proposition 6.1), so the unknown part of $\pi_1^{\mathrm{orb}}(X, \Delta)$ is its finite residual (Definition 2.16), which no finite cover detects (Proposition 6.5).

(c) Comparison with the case $D = 0$. The proof of [OAI26c, Theorem 1.1], the case $D = 0$, has five parts: the exact sequence of fundamental groups of the Albanese map [OAI26c, Lemma 6.1], the FAb property of the fiber group, that is, any subgroup of finite index has finite abelianization [OAI26c, Proposition 6.2], a minimal base over the Albanese torus [OAI26c, Proposition 6.3], positivity on that base, separately for groups with and without infinite linear images [OAI26c, §6.5], and the descent over a torus [OAI26c, Theorem 5.8], which contradicts specialness.

For (Y, Δ_Y), $\mathrm{Alb}(Y)$ is a point, G has no proper subgroup of finite index, and any linear representation of G is trivial (Lemma 2.17(6)). Thus the first two parts are trivial, infinite linear images do not occur, the minimal base is the base of a Γ-reduction on Y itself, and the inequality $a(Y) \geq \dim Z$ replaces the descent. In particular, Proposition 9.1 does not assume specialness, which enters only through (1) and (2). Only the positivity theorem [OAI26c, Theorem 4.1] for groups without infinite linear images remains. By contrast, a projective pair (X, Δ) has $a(X) = \dim X$, and $\pi_1^{\mathrm{orb}}(X, \Delta)$ may have proper subgroups of finite index and infinite linear images, so (1) treats projective pairs by (2).

(d) Orbifold pairs instead of manifolds. [OAI26c, Theorem 4.1] needs a compact Kähler manifold M dominating the base and a homomorphism from $\pi_1(M)$ onto a subgroup of finite index of the quotient, and states the minimality condition for finite étale covers of M. The transverse volume bound [OAI26c, Lemma 4.7] also starts from the fundamental group of a compact Kähler manifold. Here any compact Kähler manifold bimeromorphic to Y is simply connected, and G has no non-trivial finite quotient. We minimize the Γ-dimension over the infinite quotients of G itself, which needs Γ-reductions of arbitrary quotients (Definition-Theorem 8.2 and Remark 8.3). We prove the transverse volume bound with compactness on a compact Kähler manifold bimeromorphic to Y and monodromy on $Y\setminus \operatorname{Supp}\Delta_Y$ (Lemma 8.4). The images in G/C of the meridians of the base, where C is the image of the fundamental group of a general fiber, have finite orders, which the base carries as multiplicities (Construction-Proposition 9.4).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce root orbifolds, orbifold symmetric differentials and specialness in the sense of Campana [Cam11a, Définition 8.1], divisible orbifold morphisms [Cam11a, Définition 2.4], and finite residuals of groups.

In Section 3 we rewrite the specialness of Campana [Cam11a, §2.5 and Théorème 9.9] as a condition on line bundles on root orbifolds: a compact Kähler orbifold pair (X,Δ) is special if and only if, for every $1\leq p\leq \dim X$, no line bundle of Iitaka dimension p on its root orbifold admits a non-zero map to the sheaf of p-forms.

In Section 4 we introduce the first new construction of this paper: the quotient of the product of a Kummer cover of a smooth projective variety with an abelian variety by the diagonal action of the Galois group of the Kummer cover.

In Section 5 we prove Theorem 1.2 when X is projective, and the new inputs are the extraction of auxiliary divisors and the proof, extending [OAI26b, Proposition 3.1], that the specialness of (X,Δ) passes to a resolution of the quotient constructed in Section 4.

In Section 6 we realize any subgroup of finite index in $\pi_1^{\mathrm{orb}}(X,\Delta)$ as the orbifold fundamental group of a compact Kähler orbifold pair which admits a generically finite holomorphic map onto X, and, for special (X,Δ), we extend the exact sequence of fundamental groups for the Albanese map [OAI26c, Lemma 6.1] to orbifold pairs and deduce Proposition 6.5.

In Section 7 we argue by induction on the dimension and reduce to a counterexample of least dimension, which has algebraic dimension zero, combining results of [Lu02], [Wan21], and [CC11] with a new argument showing that if $a(X)=0$, then the restriction of (X,Δ) to a very general fiber of a holomorphic map with connected fibers onto a complex torus is special.

In Section 8 we extend the Γ-reduction of [Cam94, Théorème 3.5], [Cla08, Theorem 0.2], and [OAI26c, Theorem 2.3] (see also [Cam11a, Théorème 12.23]) to arbitrary quotients of orbifold fundamental groups, and extend the transverse volume bound [OAI26c, Lemma 4.7], which constructs compact subvarieties from positive currents, to orbifold fundamental groups.

In Section 9, which contains the new input for the non-projective case, we prove a version of the positivity theorem [OAI26c, Theorem 4.1] in which the base is dominated by a compact Kähler orbifold pair instead of a compact Kähler manifold, and deduce Proposition 9.1.

In Section 10 we prove Theorem 1.2 and Corollary 1.3.

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2. Preliminaries

In this section, we define orbifold pairs, their orbifold fundamental groups, root orbifolds and specialness, and we collect the facts on compact Kähler spaces and on finite residuals of groups that are used in the later sections.

2.1. **Orbifold pairs and orbifold fundamental groups.**

Definition 2.1. Let X be a connected complex manifold, let $R \subset X$ be a closed analytic subset of pure codimension one, let R_1 be an irreducible component of R, and let $x_0 \in X \backslash R$. A *meridian* of R_1 is an element of $\pi_1(X\backslash R, x_0)$ represented by a path in $X\backslash R$ from x_0 to a point near a point $y \in R_1$ at which R is smooth, followed by a small loop in $X\backslash R$ turning once positively around R_1 in a disc transverse to R_1 at y, followed by the inverse path.

Definition 2.2. An *orbifold pair* (X, Δ) consists of a connected compact complex manifold X and a $\mathbb{Q}$-divisor

$$\Delta = \sum_{i=1}^{r} \left(1 - \frac{1}{m_i}\right) D_i$$

on X, where $r \geq 0$, the D_i are distinct smooth prime divisors such that $D := \sum_{i=1}^{r} D_i$ is a simple normal crossing divisor, and $m_1, \ldots, m_r \geq 2$ are integers. If X is Kähler, then we say that (X, Δ) is a *compact Kähler orbifold pair*.

Fix $x_0 \in X\backslash D$, and for each i let $\gamma_i \in \pi_1(X\backslash D, x_0)$ be a meridian of D_i (Definition 2.1). The *orbifold fundamental group* of (X, Δ) is

$$\pi_1^{\mathrm{orb}}(X, \Delta) := \pi_1(X\backslash D, x_0)/\langle\langle \gamma_1^{m_1}, \ldots, \gamma_r^{m_r} \rangle\rangle,$$

where $\langle\langle \gamma_1^{m_1}, \ldots, \gamma_r^{m_r} \rangle\rangle$ is the normal subgroup generated by $\gamma_1^{m_1}, \ldots, \gamma_r^{m_r}$.

We require each D_i to be smooth and D to be a simple normal crossing divisor so that near any point of X each D_i is a single coordinate hyperplane. This is the form used by the charts of the root orbifold (Definition 2.8) and by the local computation of Lemma 2.3(3). It loses no generality. Indeed, suppose that X is Kähler and that D only has normal crossings, as for the lisse orbifolds of [Cam11a, §2.1]. By [BM97, Theorem 1.10 and (1.2)(1)], a sequence of blow-ups along smooth centers contained in D gives $\mu : X' \to X$ such that $\mu^{-1}(D)$ is a simple normal crossing divisor, and X' is compact Kähler by Lemma 2.14(1). Give the strict transform of D_i the multiplicity m_i, and give each μ-exceptional prime divisor the multiplicity $\mathrm{lcm}(m_1, \ldots, m_r)$. The complement of the boundary does not change. By Lemma 2.3(3), a meridian of a μ-exceptional prime divisor maps to a product of pairwise commuting powers of meridians of the local branches of D, and each of these is conjugate to some γ_i by Lemma 2.3(5). Thus the $\mathrm{lcm}(m_1, \ldots, m_r)$-th power of such a meridian is contained in $\langle\langle \gamma_1^{m_1}, \ldots, \gamma_r^{m_r} \rangle\rangle$, and the orbifold fundamental group does not change.

Moreover, μ is an elementary bimeromorphic orbifold morphism in the sense of [Cam11a, Définition 2.8]. By [Cam11a, Théorème 9.9], each of the pairs on X and on X' is special in the sense of [Cam11a, Définition 8.1] if and only if it carries no Δ-Bogomolov sheaf in the sense of [Cam11a, Définition 3.14], since X and X' are compact Kähler and D and $\mu^{-1}(D)$ have normal crossings. Let $p > 0$ and let $\mathcal{F}' \subset \Omega^p_{X'}$ be a coherent subsheaf of rank one. Then $\mathcal{F} := \mu_* \mathcal{F}'$ is a subsheaf of $\mu_* \Omega^p_{X'} = \Omega^p_X$ of rank one, and $\mu^* \mathcal{F} = \mathcal{F}'$ outside the exceptional locus of μ, so the orbifold Iitaka dimensions of $\mathcal{F}'$ and $\mathcal{F}$ in the sense of [Cam11a, §3.1] are equal by [Cam11a, Théorème 3.5]. Thus, if $\mathcal{F}'$ is a Δ-Bogomolov sheaf of the pair on X', then $\mathcal{F}$ is a Δ-Bogomolov sheaf of the pair on X. Hence, if the pair on X is special in the sense of Campana, then so is the pair on X'.

Lemma 2.3. *Let X be a connected complex manifold. Then the following hold.*

1. *If $V \subsetneq X$ is a closed analytic subset, then the map $\pi_1(X\backslash V) \to \pi_1(X)$ is surjective, and it is an isomorphism if $\mathrm{codim}_X V \geq 2$.*
2. *If $R \subset X$ is a closed analytic subset of pure codimension one, then the kernel of $\pi_1(X\backslash R) \to \pi_1(X)$ is the normal subgroup generated by meridians of the irreducible components of R.*
3. *Let $W \subset \mathbb{C}^d$ be a polydisc with coordinates $z_1, \ldots, z_d$ and let $1 \leq k \leq d$. Let $W^\circ := W\backslash\{z_1 \cdots z_k = 0\}$, and let $\lambda_j \in \pi_1(W^\circ) \cong \mathbb{Z}^k$ be the meridian of $\{z_j = 0\}$. Let*

$f : X \to W$ be a holomorphic map and $E \subset X$ a smooth prime divisor with $f(X\backslash E) \subset W^\circ$. Then f_* maps a meridian of E to $\prod_{j=1}^k \lambda_j^{a_j}$, where $a_j := \operatorname{mult}_E f^*\{z_j = 0\}$.

(4) *If X is compact and $D \subset X$ is a simple normal crossing divisor, then $\pi_1(X\backslash D)$ is finitely generated.*

(5) *If $R \subset X$ is a closed analytic subset of pure codimension one, R_1 is an irreducible component of R, and $x_0, x_1 \in X\backslash R$, then any two meridians of R_1 in $\pi_1(X\backslash R, x_0)$ are conjugate, and any path from x_0 to x_1 in $X\backslash R$ induces an isomorphism $\pi_1(X\backslash R, x_0) \to \pi_1(X\backslash R, x_1)$ which maps meridians of R_1 to meridians of R_1.*

Proof. (1) This is [Kol95, Proposition 2.10].

(2) We refer to [Kol95, Proposition 2.10 and the discussion after it], and we explain the reason briefly. Since $\operatorname{codim}_X \operatorname{Sing} R \geq 2$, the map $\pi_1(X \setminus \operatorname{Sing} R) \to \pi_1(X)$ is an isomorphism by (1), and meridians are defined at points where R is smooth. Thus, possibly replacing X with $X \setminus \operatorname{Sing} R$, we may assume that R is a closed complex submanifold of codimension one. The connected components of R then correspond to the irreducible components of R. Let N be a tubular neighborhood of R. Then $N\backslash R$ retracts onto a circle bundle over R whose fibers are represented by meridians, and N retracts onto R. By the homotopy exact sequence of the circle bundle [Hat02, Theorem 4.41 and Proposition 4.48] and the van Kampen theorem [Hat02, Theorem 1.20], applied to $X = (X\backslash R) \cup N$ for one connected component of R at a time so that $N\backslash R$ is path-connected, the kernel of $\pi_1(X\backslash R) \to \pi_1(X)$ is normally generated by the classes of the fibers, that is, by meridians of the irreducible components of R.

(5) We prove (5) before (3), since the proof of (3) uses it. The set of points of R_1 at which R is smooth is the complement in R_1 of the proper closed analytic subset $R_1 \cap \operatorname{Sing} R$. Since R_1 is irreducible, the regular locus of R_1 is connected, and it remains connected after removing a proper closed analytic subset. Moving the transverse disc of Definition 2.1 along a path in the set of points of R_1 at which R is smooth, we see that the small loops at any two such points are freely homotopic in $X\backslash R$. Hence any two meridians of R_1 in $\pi_1(X\backslash R, x_0)$ are conjugate. A path β from x_0 to x_1 in $X\backslash R$ induces the isomorphism $[\alpha] \to [\bar\beta \cdot \alpha \cdot \beta]$ by [Hat02, Proposition 1.5], where $\bar\beta$ is the inverse path. Composing the path in Definition 2.1 with β, we see that $[\alpha] \to [\bar\beta \cdot \alpha \cdot \beta]$ maps meridians of R_1 to meridians of R_1.

(3) By (5), any two meridians of E are conjugate. Since $\pi_1(W^\circ) \cong \mathbb{Z}^k$ is abelian, it suffices to compute the image of one meridian, and the base point plays no role. Since $f(X\backslash E) \subset W^\circ$, the divisor $f^*\{z_j = 0\}$ is supported on E, so $f^*\{z_j = 0\} = a_j E$ for $1 \leq j \leq k$. Choose a point $y \in E$, and let t be a coordinate on a small disc in X transverse to E at y, with $t = 0$ at y. On the disc

$$z_j \circ f = t^{a_j} u_j$$

for $1 \leq j \leq k$, where u_j is a nowhere vanishing holomorphic function. Hence, along a small circle around $t = 0$, the function $z_j \circ f$ has winding number a_j around 0. Thus f_* maps the meridian of E at y to $\prod_{j=1}^k \lambda_j^{a_j}$.

(4) The real oriented blow-up of X along D is a compact manifold with corners whose interior is diffeomorphic to $X\backslash D$. Smoothing the corners, we obtain a compact smooth manifold with boundary, which is homotopy equivalent to the interior $X\backslash D$ by the collar theorem. By [Mun66, Theorem 10.6], a compact smooth manifold with boundary admits a smooth triangulation. Thus $X\backslash D$ has the homotopy type of a finite CW complex, and $\pi_1(X\backslash D)$ is finitely generated. □

By Lemma 2.3(5), the normal subgroup $\langle\langle \gamma_1^{m_1}, \dots, \gamma_r^{m_r} \rangle\rangle$ does not depend on the choice of the γ_i, and a path between base points induces an isomorphism of orbifold fundamental groups. We therefore omit the base point from the notation.

Lemma 2.4. *Let X be a connected complex manifold, let $h : X \to T$ be a proper surjective holomorphic map onto a reduced irreducible complex space T, and let $D = \sum_{j=1}^s D_j$ be a simple normal crossing divisor on X with prime components D_j. For a non-empty subset $I \subset \{1, \dots, s\}$ let $D_I := \bigcap_{j \in I} D_j$. Then there exists a proper closed analytic subset $T' \subsetneq T$ such that $T\backslash T'$ is*

smooth, h is a submersion over $T\backslash T'$, the restriction of h to each D_I is a submersion over $T\backslash T'$, and $h^{-1}(T\backslash T')\backslash D \to T\backslash T'$ is a locally trivial fiber bundle. In particular, if the fibers of h are connected, then for $t \in T\backslash T'$ the sequence

$$\pi_1(h^{-1}(t)\backslash D) \to \pi_1(h^{-1}(T\backslash T')\backslash D) \to \pi_1(T\backslash T') \to 1$$

is exact.

Proof. The union of $h^{-1}(\operatorname{Sing} T)$ and the set of points of $h^{-1}(T\setminus \operatorname{Sing} T)$ at which h is not a submersion is a closed analytic subset of X: after embedding a neighborhood of a point of T into some $\mathbb{C}^N$, it is the union of $h^{-1}(\operatorname{Sing} T)$ and the locus where the Jacobian matrix of h has rank less than $\dim T$. For each I, the union of $D_I \cap h^{-1}(\operatorname{Sing} T)$ and the set of points of $D_I \cap h^{-1}(T\setminus \operatorname{Sing} T)$ at which $h|_{D_I}$ is not a submersion is a closed analytic subset of D_I, by the Jacobian rank description applied to $h|_{D_I}$. Let T' be the union of $\operatorname{Sing} T$, the image under h of the closed analytic subset of X just described, the images under h of the corresponding closed analytic subsets of the D_I, and the images of the connected components of the D_I whose image is not T. By Remmert's proper mapping theorem [Rem57] (see also [Dem12, Chapter II, (8.8)]), each of the images under h in the definition of T' is a closed analytic subset of T. Each of the sets in the definition of T' is different from T. The set $\operatorname{Sing} T$ is a proper closed analytic subset of T since T is reduced. The image of the closed analytic subset of X is contained in the union of $\operatorname{Sing} T$ and the set of critical values of h in $T\setminus \operatorname{Sing} T$, and the image of the corresponding subset of D_I is contained in the union of $\operatorname{Sing} T$ and the set of critical values of $h|_{D_I}$ in $T\setminus \operatorname{Sing} T$. The sets of critical values of h and of $h|_{D_I}$ in $T\setminus \operatorname{Sing} T$ have measure zero by Sard's theorem [Sar42]. The image of a connected component of D_I whose image is not T is different from T by definition. Since h is proper, the sets in the definition of T' form a locally finite family, so T' is a closed analytic subset of T. Since T is irreducible, each proper closed analytic subset of T is nowhere dense in T, and hence so is the locally finite union T'. Thus $T' \subsetneq T$.

By construction, $T\backslash T'$ is smooth, h is a submersion over $T\backslash T'$, and the restriction of h to each D_I is a submersion over $T\backslash T'$. The subsets $X\backslash D$ and $D_I\setminus \bigcup_{j\notin I} D_j$ form a Whitney stratification of X, since D is locally a union of coordinate hyperplanes. The restriction of h to $h^{-1}(T\backslash T')$ is proper and a submersion on each stratum. By Thom's first isotopy lemma [Mat12, Proposition 11.1], $h^{-1}(T\backslash T') \to T\backslash T'$ is locally trivial as a stratified map, and in particular $h^{-1}(T\backslash T')\backslash D \to T\backslash T'$ is a locally trivial fiber bundle. If $D = 0$, then the local triviality is Ehresmann's theorem [Voi02, Proposition 9.3].

Now assume that the fibers of h are connected. Since T is irreducible, $T\setminus \operatorname{Sing} T$ is connected, and $T\backslash T'$ is connected because T' is a proper closed analytic subset of T. Let $t \in T\backslash T'$ and $F := h^{-1}(t)$. Then F is a connected complex manifold. Since the restriction of h to each D_I is a submersion over $T\backslash T'$, $D \cap F$ is a simple normal crossing divisor on F, and hence $F\backslash D$ is connected. Thus $h^{-1}(T\backslash T')\backslash D$ is connected. The homotopy exact sequence of the fiber bundle [Hat02, Theorem 4.41 and Proposition 4.48] yields the exact sequence

$$\pi_1(F\backslash D) \to \pi_1(h^{-1}(T\backslash T')\backslash D) \to \pi_1(T\backslash T') \to \pi_0(F\backslash D),$$

and $\pi_0(F\backslash D)$ is a point. Since $F\backslash D$, $h^{-1}(T\backslash T')\backslash D$, and $T\backslash T'$ are path-connected, we may omit the base points. □

Divisible orbifold morphisms were introduced in [Cam11a, Définition 2.4].

Definition 2.5. Let (X, Δ) and (X', Δ') be orbifold pairs with $\Delta = \sum_i (1 - \frac{1}{m_i})D_i$, and let $h : X' \to X$ be a surjective holomorphic map. For a prime divisor E on X', let m'_E be the integer with $\operatorname{mult}_E \Delta' = 1 - \frac{1}{m'_E}$, so that $m'_E = 1$ if $E \not\subset \operatorname{Supp} \Delta'$. We say that $h : (X', \Delta') \to (X, \Delta)$ is a *divisible orbifold morphism* if m_i divides $m'_E \operatorname{mult}_E h^* D_i$ for any i and any prime divisor E on X'. A divisible orbifold morphism h is called a *divisible elementary modification* if h is bimeromorphic and $h_*\Delta' = \Delta$.

Let E be a prime divisor on X' which h maps into the support of Δ. Let U be a coordinate polydisc of X around the image of a general point of E, and connect U to the base point by a path. By Lemma 2.3(5), the path sends the meridians of the coordinate hyperplanes of U to conjugates of the γ_i. Since $U\setminus\operatorname{Supp}\Delta$ is a product of discs and punctured discs, $\pi_1(U\setminus\operatorname{Supp}\Delta)$ is abelian, so the images of the meridians of the coordinate hyperplanes of U commute pairwise. By Lemma 2.3(3), applied to h near a general point of E, the image of a meridian of E in $\pi_1(X\setminus\operatorname{Supp}\Delta)$ is a product of pairwise commuting powers of meridians of the D_i, with exponents the multiplicities of the pullbacks of the D_i along E. The divisibility condition guarantees that the m'_E-th power of the image of a meridian of E is contained in the normal subgroup generated by the $\gamma_i^{m_i}$, so that h induces a homomorphism of orbifold fundamental groups (Theorem 2.7). The weaker condition of [Cam11a, Définition 2.3] only asks for an inequality between the multiplicities m'_E, m_i, and $\operatorname{mult}_E h^*D_i$, and it does not guarantee that h induces such a homomorphism, as Example 2.6 shows. Since the divisibility condition implies the inequality of [Cam11a, Définition 2.3], the divisible elementary modifications are exactly the elementary bimeromorphic orbifold morphisms in the sense of [Cam11a, Définition 2.8] between orbifold pairs which are divisible orbifold morphisms.

Example 2.6. Let $h:\mathbb{P}^1\to\mathbb{P}^1$ be the identity map, and consider

$$h:\left(\mathbb{P}^1,\tfrac{2}{3}[0]+\tfrac{2}{3}[\infty]\right)\to\left(\mathbb{P}^1,\tfrac{1}{2}[0]+\tfrac{1}{2}[\infty]\right).$$

For $E=[0]$ and $D_1=[0]$, we have $m'_E\operatorname{mult}_E h^*D_1=3\geq 2=m_1$, and similarly for $[\infty]$, but 2 does not divide 3. Let $\gamma_1\in\pi_1(\mathbb{C}^*)\cong\mathbb{Z}$ be a meridian of 0. Then γ_1 generates $\pi_1(\mathbb{C}^*)$, and γ_1^{-1} is a meridian of ∞. Hence

$$\pi_1^{\mathrm{orb}}\left(\mathbb{P}^1,\tfrac{2}{3}[0]+\tfrac{2}{3}[\infty]\right)\cong\mathbb{Z}/3\quad\text{and}\quad\pi_1^{\mathrm{orb}}\left(\mathbb{P}^1,\tfrac{1}{2}[0]+\tfrac{1}{2}[\infty]\right)\cong\mathbb{Z}/2.$$

A homomorphism $\mathbb{Z}/3\to\mathbb{Z}/2$ compatible with the identity of $\pi_1(\mathbb{C}^*)$ maps the class of γ_1 to the class of γ_1. Then it maps the class of γ_1^3, which is trivial in $\mathbb{Z}/3$, to the class of γ_1^3, which is the non-trivial element of $\mathbb{Z}/2$. Thus there exists no such homomorphism.

Theorem 2.7 ([Cam11a, Propositions 12.5 and 12.7])**.** *Let $h:(X',\Delta')\to(X,\Delta)$ be a divisible orbifold morphism between orbifold pairs, and let $X'^{\circ}:=X'\setminus(\operatorname{Supp}\Delta'\cup h^{-1}(\operatorname{Supp}\Delta))$. Then h induces a homomorphism $h_*:\pi_1^{\mathrm{orb}}(X',\Delta')\to\pi_1^{\mathrm{orb}}(X,\Delta)$ compatible with the homomorphism $\pi_1(X'^{\circ})\to\pi_1(X\setminus\operatorname{Supp}\Delta)$ induced by h. If the fibers of h are connected, then h_* is surjective. If h is a divisible elementary modification, then h_* is an isomorphism.*

2.2. **Root orbifolds and specialness.** The root orbifold of an orbifold pair is the complex orbifold obtained by adjoining an m_i-th root of a local equation of D_i for each i.

Definition 2.8. Let (X,Δ) be an orbifold pair with $\Delta=\sum_{i=1}^r(1-\frac{1}{m_i})D_i$ and $d:=\dim X$, and let $\mu_m\subset\mathbb{C}^*$ denote the group of m-th roots of unity. The *root orbifold* $\mathcal{X}$ of (X,Δ) is the complex orbifold with coarse space X whose charts are described as follows. Let $U\subset X$ be a coordinate polydisc with coordinates $z_1,\dots,z_d$ such that $D_{i_j}\cap U=\{z_j=0\}$ for $1\leq j\leq k$ and $D_i\cap U=\emptyset$ for $i\notin\{i_1,\dots,i_k\}$. Then $\mathcal{X}$ has the chart

$$\widetilde{U}\to U,\qquad(w_1,\dots,w_k,z_{k+1},\dots,z_d)\to(w_1^{m_{i_1}},\dots,w_k^{m_{i_k}},z_{k+1},\dots,z_d),$$

with the group $\prod_{j=1}^k\mu_{m_{i_j}}$ acting on the coordinates $w_1,\dots,w_k$. We write $\pi:\mathcal{X}\to X$ for the coarse map.

Vector bundles, coherent sheaves, and maps between them on $\mathcal{X}$ are equivariant holomorphic objects on the charts $\widetilde{U}$ together with compatible isomorphisms on overlaps, and a global section is a compatible family of invariant sections. We write $\Omega^p_{\mathcal{X}}$ for the orbifold bundle of holomorphic p-forms. A line bundle on $\mathcal{X}$ is also called an *orbifold line bundle*.

For a coherent sheaf $\mathcal{G}$ on $\mathcal{X}$, the *invariant direct image* $\pi_*\mathcal{G}$ is the sheaf on X whose sections over U are the invariant sections of $\mathcal{G}$ over $\widetilde{U}$. For a coherent sheaf $\mathcal{E}$ on X, the sheaf $\pi^*\mathcal{E}$ on $\mathcal{X}$ is the pullback of $\mathcal{E}$ along $\widetilde{U}\to U$ on each chart, and pulling back forms defines a map $\pi^*\Omega^p_X\to\Omega^p_{\mathcal{X}}$.

For a holomorphic vector bundle $\mathcal{E}$ on a complex manifold and a coherent subsheaf $\mathcal{F}_0 \subset \mathcal{E}$, the *saturation* of $\mathcal{F}_0$ in $\mathcal{E}$ is the sheaf of local sections s of $\mathcal{E}$ such that hs is a section of $\mathcal{F}_0$ for some holomorphic function $h \not\equiv 0$ on a connected open set, and $\mathcal{F}_0$ is *saturated* in $\mathcal{E}$ if $\mathcal{F}_0$ is equal to the saturation of $\mathcal{F}_0$ in $\mathcal{E}$. A coherent subsheaf $\mathcal{F}$ of an orbifold vector bundle $\mathcal{V}$ on $\mathcal{X}$ is *saturated* if $\mathcal{F}|_{\widetilde{U}}$ is saturated in $\mathcal{V}|_{\widetilde{U}}$ for any chart $\widetilde{U}$.

For $1 \le i \le r$, the *root divisor* R_i over D_i and the line bundle $\mathcal{O}_{\mathcal{X}}(R_i)$ are given on a chart with $i = i_j$ by $\{w_j = 0\}$ and $w_j^{-1}\mathcal{O}_{\widetilde{U}}$, and on a chart with $D_i \cap U = \emptyset$ by the empty divisor and $\mathcal{O}_{\widetilde{U}}$. Thus $\pi^*\mathcal{O}_X(D_i) \cong \mathcal{O}_{\mathcal{X}}(R_i)^{m_i}$, and we write $\pi^*D_i = m_iR_i$.

Definition 2.9. Let (X, Δ) be an orbifold pair with root orbifold $\mathcal{X}$, and let $\mathcal{L}$ be a line bundle on $\mathcal{X}$. The *Iitaka dimension* $\kappa(\mathcal{X}, \mathcal{L})$ is $-\infty$ if $H^0(\mathcal{X}, \mathcal{L}^q) = 0$ for any $q \ge 1$, and otherwise it is the maximum, over the integers $q \ge 1$ with $H^0(\mathcal{X}, \mathcal{L}^q) \ne 0$, of the dimensions of the images of the meromorphic maps $X \dashrightarrow \mathbb{P}(H^0(\mathcal{X}, \mathcal{L}^q)^*)$ defined by $H^0(\mathcal{X}, \mathcal{L}^q)$. Here the ratio of two sections of $\mathcal{L}^q$ is a meromorphic function on each chart of $\mathcal{X}$ which is invariant under the group of the chart, hence a meromorphic function on X. If $\Delta = 0$, then $\mathcal{X} = X$ and we write $\kappa(X, \mathcal{L})$.

A line bundle $\mathcal{L}$ on X is *big* if $\kappa(X, \mathcal{L}) = \dim X$. A $\mathbb{Q}$-divisor D on X is *big* if ND is an integral divisor and $\mathcal{O}_X(ND)$ is big for some integer $N \ge 1$.

The sheaves of orbifold symmetric differentials, the Iitaka dimension $\kappa(X, \Delta; \mathcal{F})$, and the Δ-Bogomolov sheaves in the next definition are those of [Cam11a, §2.5], [Cam11a, §3.1], and [Cam11a, Définition 3.14], respectively.

Definition 2.10. Let (X, Δ) be an orbifold pair with $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i})D_i$ and $D := \operatorname{Supp} \Delta$, and let $1 \le p \le \dim X$ and $q \ge 1$. In a coordinate polydisc as in Definition 2.8, for p-element subsets $I_1, \dots, I_q \subset \{1, \dots, \dim X\}$ let dz_{I_a} be the wedge product of the dz_l, $l \in I_a$, in increasing order, and for $1 \le j \le k$ let t_j be the number of indices a with $j \in I_a$. The *sheaf of orbifold symmetric differentials* $S^q\Omega^p(X, \Delta) \subset \operatorname{Sym}^q \Omega^p_X(\log D)$ is the locally free subsheaf generated in such charts by the tensors

$$\Big(\prod_{j=1}^k z_j^{-\lfloor t_j(1-1/m_{i_j})\rfloor}\Big) dz_{I_1} \odot \cdots \odot dz_{I_q},$$

and $S^0\Omega^p(X, \Delta) := \mathcal{O}_X$.

For a coherent subsheaf $\mathcal{F} \subset \Omega^p_X$ of rank one, let $\mathcal{F}_{0,\Delta} := \mathcal{O}_X$, for $q \ge 1$ let $\mathcal{F}_{q,\Delta}$ be the saturation in $S^q\Omega^p(X, \Delta)$ of the image of $\mathcal{F}^{\otimes q}$, and let $\kappa(X, \Delta; \mathcal{F}) := -\infty$ if $H^0(X, \mathcal{F}_{q,\Delta}) = 0$ for any $q \ge 1$, and otherwise

$$\kappa(X, \Delta; \mathcal{F}) := \operatorname{trdeg}_{\mathbb{C}} \operatorname{Frac}\Big(\bigoplus_{q \ge 0} H^0(X, \mathcal{F}_{q,\Delta})\Big) - 1.$$

We call $\mathcal{F}$ a *Δ-Bogomolov sheaf* if $\kappa(X, \Delta; \mathcal{F}) = p$.

The rounding down in Definition 2.10 determines exactly which logarithmic symmetric differentials become holomorphic after pulling back to the charts of the root orbifold (Proposition 3.4). Because the rounding down is not additive in q, the sheaves $\mathcal{F}_{q,\Delta}$ are in general not the powers of $\mathcal{F}_{1,\Delta}$, and hence the algebra of sections in Definition 2.10 is not the section ring of $\mathcal{F}_{1,\Delta}$. On the root orbifold, the algebra of sections in Definition 2.10 is the section ring of a single line bundle (Definition-Theorem 3.5).

Definition 2.11. Let (X, Δ) be a compact Kähler orbifold pair. We say that (X, Δ) is *special* if no Δ-Bogomolov sheaf $\mathcal{F} \subset \Omega^p_X$ exists for any $1 \le p \le \dim X$. We say that X is *special* if $(X, 0)$ is special.

Remark 2.12. By [Cam11a, Théorème 9.9], a compact Kähler orbifold pair (X, Δ) is special in the sense of Definition 2.11 if and only if it is special in the sense of [Cam11a, Définition 8.1], that is, no fibration of (X, Δ) is of general type in the sense of [Cam11a, §5], where the orbifold base of a fibration is defined with infimum multiplicities [Cam11a, Définition 4.2].

The hypothesis of Theorem 2.13 is the definition of special in [OAI26c, §1, p. 2]. By the second sentence of Theorem 3.1, a connected compact Kähler manifold X satisfies the hypothesis of Theorem 2.13 if and only if X is special in the sense of Definition 2.11.

Theorem 2.13 ([OAI26c, Theorem 1.1])**.** *Let X be a connected compact Kähler manifold. Assume that for every $1 \leq p \leq \dim X$ no line bundle $\mathcal{L}$ on X admits a non-zero map $\mathcal{L} \to \Omega_X^p$ with $\kappa(X, \mathcal{L}) = p$. Then $\pi_1(X)$ is virtually abelian.*

2.3. Compact Kähler spaces and algebraic dimension.

Lemma 2.14. *The following hold.*

1. *Let X be a compact Kähler space and let $Y \subset X$ be a closed complex subspace. Then the blow-up of X along Y is a compact Kähler space.*
2. *Let X be a reduced compact Kähler space. Then there exists a resolution $X' \to X$ by a compact Kähler manifold X' which is an isomorphism over the smooth locus of X.*
3. *Let X be a compact Kähler manifold, let Z be an irreducible compact complex space, and let $X \dashrightarrow Z$ be a dominant meromorphic map. Then Z is bimeromorphic to a compact Kähler manifold.*

Proof. (1) This is [Xia19, Corollary 2.17]. For the definition of compact Kähler spaces, see [Xia19, Definition 2.13].

(2) By [Kol05, Paragraph 44 and Theorem 45], X has a resolution $X' \to X$ such that X' is smooth, $X' \to X$ is an isomorphism over the smooth locus of X, and, since X is compact, $X' \to X$ is a composite of finitely many blow-ups. Applying (1) to each blow-up in the composite, X' is a compact Kähler space. Since X' is smooth, X' is a compact Kähler manifold [Xia19, Example 2.14].

(3) Fujiki's class $\mathcal{C}$ [Fuj78] consists of the compact reduced complex spaces that are meromorphic images of compact Kähler manifolds, equivalently, that are bimeromorphic to compact Kähler manifolds [Var89, Chapter IV, Theorem 5]. Since Z is irreducible and $X \dashrightarrow Z$ is dominant, Z is the image of the compact Kähler manifold X under $X \dashrightarrow Z$. Thus $Z \in \mathcal{C}$, and therefore Z is bimeromorphic to a compact Kähler manifold, see [Cam04b, §2.1]. □

Lemma 2.15. *Let X be a connected compact complex manifold. Then the following hold.*

1. *If $f : X \dashrightarrow Z$ is a dominant meromorphic map onto an irreducible compact complex space Z, then $a(Z) \leq a(X)$, with equality if f is generically finite.*
2. *For every line bundle $\mathcal{L}$ on X, $\kappa(X, \mathcal{L}) \leq a(X)$. In particular, if $a(X) = 0$, then every meromorphic function on X is constant and $\kappa(X, \mathcal{L}) \leq 0$ for every line bundle $\mathcal{L}$ on X.*
3. *If X is Kähler and $a(X) = \dim X$, then X is projective.*

Proof. (1) Since f is dominant, pulling back meromorphic functions along f defines an injection f^* from the field of meromorphic functions on Z into the field of meromorphic functions on X. Thus $a(Z) \leq a(X)$. If f is generically finite, then the field of meromorphic functions on X is algebraic over the image of f^*, and hence $a(Z) = a(X)$. See [Uen75, Chapter I, §3].

(2) The inequality $\kappa(X, \mathcal{L}) \leq a(X)$ is [Uen75, Chapter II, §5]. Assume that $a(X) = 0$. Then any meromorphic function ϕ on X is algebraic over $\mathbb{C}$, hence takes only finitely many values on the connected manifold X, and therefore ϕ is constant. If $\kappa(X, \mathcal{L}) \geq 1$, then there exist an integer $q \geq 1$ and two linearly independent sections $s_0, s_1 \in H^0(X, \mathcal{L}^q)$, and s_1/s_0 is a non-constant meromorphic function on X, a contradiction. Thus $\kappa(X, \mathcal{L}) \leq 0$.

(3) This is Moishezon's theorem [Moi66, Part I, Chapter I, Theorem 11]. □

2.4. Finite residuals of groups.

Definition 2.16. For a group P, the *finite residual* $R_{\mathrm{fin}}(P)$ is the intersection of all normal subgroups of finite index in P.

Thus $R_{\mathrm{fin}}(P)$ consists of the elements of P that are mapped to the identity by any homomorphism from P to a finite group. By Lemma 2.17(2), passing to a subgroup of finite index does not change the finite residual. By Proposition 6.5, for a special compact Kähler orbifold pair (X, Δ), some subgroup of finite index in $\pi_1^{\mathrm{orb}}(X, \Delta)$ is an extension of the fundamental group of a compact complex torus by its finite residual. If Theorem 1.2 fails, then Proposition 7.8 provides a special compact Kähler orbifold pair (X, Δ) whose dimension is the least dimension of a counterexample and such that $R_{\mathrm{fin}}(\pi_1^{\mathrm{orb}}(X, \Delta)) = \pi_1^{\mathrm{orb}}(X, \Delta)$.

Lemma 2.17. *Let P be a group. Then the following hold.*

1. *If $\phi : P \to P'$ is a surjective homomorphism and $P_0 \leq P$ is an abelian subgroup of finite index, then $\phi(P_0)$ is an abelian subgroup of P' of index at most $[P : P_0]$.*
2. *If $P_0 \leq P$ is a subgroup of finite index, then $R_{\mathrm{fin}}(P_0) = R_{\mathrm{fin}}(P)$, and every subgroup of finite index in P contains $R_{\mathrm{fin}}(P)$.*
3. *If P is finitely generated, then P has only finitely many subgroups of any given finite index.*
4. *If P is finitely generated and virtually abelian, then P is residually finite, that is, $R_{\mathrm{fin}}(P) = \{1\}$. If moreover every subgroup of finite index in P equals P, then $P = \{1\}$.*
5. *If $1 \to F \to P \to \mathbb{Z}^s \to 1$ is an exact sequence with F finite, then P has a free abelian subgroup P_1 of finite index with $P_1 \cap F = \{1\}$.*
6. *If P is finitely generated and every subgroup of finite index in P equals P, then every homomorphism $P \to \mathrm{GL}_k(\mathbb{C})$, $k \geq 1$, is trivial.*

Proof. (1) The group $\phi(P_0)$ is abelian as a quotient of P_0. Since ϕ is surjective, the map of coset sets $P/P_0 \to P'/\phi(P_0)$, $gP_0 \to \phi(g)\phi(P_0)$, is surjective. Thus $[P' : \phi(P_0)] \leq [P : P_0]$.

(2) Let $H \leq P$ be a subgroup of finite index. The normal core $\bigcap_{g \in P} gHg^{-1}$ of H is the kernel of the action of P on the finite set P/H, hence it is a normal subgroup of finite index in P contained in H. Thus $H \supset R_{\mathrm{fin}}(P)$, which is the second statement. If N is a normal subgroup of finite index in P, then $N \cap P_0$ is a normal subgroup of finite index in P_0, hence $R_{\mathrm{fin}}(P_0) \subset N$. Thus $R_{\mathrm{fin}}(P_0) \subset R_{\mathrm{fin}}(P)$. Conversely, if N_0 is a normal subgroup of finite index in P_0, then $[P : N_0] = [P : P_0][P_0 : N_0] < +\infty$, so the second statement implies that $R_{\mathrm{fin}}(P) \subset N_0$. Thus $R_{\mathrm{fin}}(P) \subset R_{\mathrm{fin}}(P_0)$. Here P_0 need not be normal in P.

(3) Let $n \geq 1$. Numbering the cosets of a subgroup $H \leq P$ of index n by $1, \ldots, n$ with H numbered 1, the action of P on P/H defines a homomorphism $P \to S_n$ in which H is the stabilizer of 1. Since P is finitely generated, a homomorphism $P \to S_n$ is determined by the images of finitely many generators, so there are only finitely many such homomorphisms. Thus P has only finitely many subgroups of index n.

(4) Let $P_0 \leq P$ be an abelian subgroup of finite index, and let C be the normal core of P_0 in P. The subgroup C is the kernel of the action of P on the finite set P/P_0, so C is a normal abelian subgroup of finite index in P. By the Reidemeister–Schreier theorem, C is finitely generated. For $k \geq 1$, the subgroup $C^k := \{c^k \mid c \in C\}$ is characteristic in C, hence normal in P, and C/C^k is a finitely generated abelian group of exponent dividing k, hence finite. Thus any C^k is a normal subgroup of finite index in P. By the structure theorem for finitely generated abelian groups, $\bigcap_{k \geq 1} C^k = \{1\}$. Thus $R_{\mathrm{fin}}(P) \subset \bigcap_{k \geq 1} C^k = \{1\}$. If moreover every subgroup of finite index in P equals P, then the only normal subgroup of finite index in P is P, and hence $P = R_{\mathrm{fin}}(P) = \{1\}$.

(5) The action of P on F by conjugation defines a homomorphism $P \to \mathrm{Aut}(F)$ to a finite group, so the centralizer $P_2 \leq P$ of F has finite index in P. The image Λ of P_2 in $\mathbb{Z}^s$ has finite index in $\mathbb{Z}^s$, hence $\Lambda \cong \mathbb{Z}^s$. Let $e_1, \ldots, e_s$ be a basis of Λ, let $x_1, \ldots, x_s \in P_2$ be lifts of $e_1, \ldots, e_s$, and let $n := |F|$. For $1 \leq i, j \leq s$, the commutator $c_{ij} := x_i x_j x_i^{-1} x_j^{-1}$ maps to 0 in $\mathbb{Z}^s$, hence $c_{ij} \in F$, and c_{ij} commutes with x_i and x_j since $x_i, x_j \in P_2$. A direct calculation shows that $x_i^a x_j^b = c_{ij}^{ab} x_j^b x_i^a$ for any integers $a, b \geq 0$, and $c_{ij}^n = 1$. Thus

$$x_i^n x_j^n = c_{ij}^{n^2} x_j^n x_i^n = x_j^n x_i^n.$$

Let P_1 be the subgroup generated by $x_1^n, \dots, x_s^n$. Then P_1 is abelian, and the composite of the surjection $\mathbb{Z}^s \to P_1$ sending the l-th standard basis vector to x_l^n with the map $P_1 \to n\Lambda$ sends a basis of $\mathbb{Z}^s$ to the basis $ne_1, \dots, ne_s$ of $n\Lambda$. Thus $\mathbb{Z}^s \to P_1$ and $P_1 \to n\Lambda$ are isomorphisms. In particular, P_1 is free abelian and $P_1 \cap F = \{1\}$. Since $[\mathbb{Z}^s : n\Lambda] < +\infty$ and $|F| < +\infty$, P_1 has finite index in P.

(6) Let $\rho : P \to \mathrm{GL}_k(\mathbb{C})$ be a homomorphism. The group $\rho(P)$ is a finitely generated linear group, hence residually finite by Mal'cev's theorem [Mal40] (see [Weh73, Theorem 4.2]). The preimage in P of a subgroup of finite index in $\rho(P)$ is a subgroup of finite index in P, hence equals P. Thus every subgroup of finite index in $\rho(P)$ equals $\rho(P)$, and therefore $\rho(P) = R_{\mathrm{fin}}(\rho(P)) = \{1\}$. □

3. A differential criterion for specialness

In this section, we prove the following criterion, which expresses the specialness of a compact Kähler orbifold pair through line bundles on the root orbifold.

Theorem 3.1. *Let (X, Δ) be a compact Kähler orbifold pair with root orbifold $\mathcal{X}$. Then (X, Δ) is special if and only if, for every $1 \le p \le \dim X$, no line bundle $\mathcal{L}$ on $\mathcal{X}$ admits a non-zero map $\mathcal{L} \to \Omega^p_{\mathcal{X}}$ with $\kappa(\mathcal{X}, \mathcal{L}) = p$. In particular, X is special if and only if, for every $1 \le p \le \dim X$, no line bundle $\mathcal{L}$ on X admits a non-zero map $\mathcal{L} \to \Omega^p_X$ with $\kappa(X, \mathcal{L}) = p$.*

Definition 2.11 is formulated with saturated powers of orbifold differentials on X, while Theorem 3.1 only involves powers of line bundles on the root orbifold. In Section 5, Theorem 3.1 allows us to pass specialness from an orbifold base to the total space of a fibration (Proposition 5.5 and Theorem 5.1). By the second sentence of Theorem 3.1, a connected compact Kähler manifold is special in the sense of Definition 2.11 if and only if it satisfies the hypothesis of Theorem 2.13, and Section 6 uses the second sentence together with Corollary 3.6(1).

We begin with saturated subsheaves of vector bundles. Lemma 3.2 is also used in Sections 5 and 6.

Lemma 3.2. *Let X be a complex manifold, $\mathcal{E}$ a holomorphic vector bundle on X, and $\mathcal{F}_0 \subset \mathcal{E}$ a coherent subsheaf of rank one. Let $\mathcal{F} \subset \mathcal{E}$ be the saturation of $\mathcal{F}_0$ (Definition 2.8). Then $\mathcal{F}$ is a line bundle. Moreover, for every $q \ge 1$ the subsheaf $\mathcal{F}^q \subset \mathrm{Sym}^q \mathcal{E}$ is saturated: if t is a local section of $\mathrm{Sym}^q \mathcal{E}$ and ht is a section of $\mathcal{F}^q$ for some holomorphic function $h \not\equiv 0$ on a connected open set, then t is a section of $\mathcal{F}^q$.*

Proof. Let $x \in X$ and $\mathcal{O}_x := \mathcal{O}_{X,x}$. The ring $\mathcal{O}_x$ is isomorphic to a convergent power series ring, hence factorial [GR65, Chapter II, §B]. See also [Mat86, Theorem 20.3]. We first show the following: if $u = (u_1, \dots, u_n) \in \mathcal{O}_x^n$ is a vector whose entries have no common irreducible factor, then

$$a \in \mathrm{Frac}\, \mathcal{O}_x \quad \text{and} \quad au \in \mathcal{O}_x^n \implies a \in \mathcal{O}_x. \tag{3.1}$$

Write $a = a_1/a_2$ with $a_1, a_2 \in \mathcal{O}_x$ without common irreducible factor. If an irreducible element $c \in \mathcal{O}_x$ divides a_2, then c divides $a_1 u_l$ for any l, hence c divides u_l for any l, which is impossible. Thus a_2 is a unit, and $a \in \mathcal{O}_x$.

Let $b := \mathrm{rank}\, \mathcal{E}$ and fix an isomorphism $\mathcal{E}_x \cong \mathcal{O}_x^b$. Choose a non-zero germ $f \in \mathcal{F}_{0,x}$ and write $f = gv$, where $g \in \mathcal{O}_x$ is a greatest common divisor of the entries of f and $v \in \mathcal{O}_x^b$. Then the entries of v have no common irreducible factor. We show that

$$\mathcal{F}_x = \mathcal{O}_x v. \tag{3.2}$$

Since $gv = f \in \mathcal{F}_{0,x}$ and $g \neq 0$, we have $v \in \mathcal{F}_x$. Conversely, let $s \in \mathcal{F}_x$ and let $h \in \mathcal{O}_x$ be non-zero with $hs \in \mathcal{F}_{0,x}$. Since $\mathcal{F}_0$ has rank one, $\mathcal{F}_{0,x}$ spans a one-dimensional subspace of $(\mathrm{Frac}\, \mathcal{O}_x)^b$, and the subspace contains f, hence v. Thus $s = av$ for some $a \in \mathrm{Frac}\, \mathcal{O}_x$, and (3.1) implies $a \in \mathcal{O}_x$. This proves (3.2).

The sheaf $\mathcal{F}$ is the kernel of the map from $\mathcal{E}$ to the quotient of $\mathcal{E}/\mathcal{F}_0$ by the torsion subsheaf of $\mathcal{E}/\mathcal{F}_0$. Since the torsion subsheaf of a coherent analytic sheaf is coherent, $\mathcal{F}$ is coherent. By (3.2), the germ v extends to a section of $\mathcal{F}$ over an open neighborhood V of x, and the kernel and the cokernel of the induced map $\mathcal{O}_V \to \mathcal{F}|_V$ are coherent sheaves with zero stalk at x. Possibly shrinking V, since the support of a coherent analytic sheaf is a closed analytic subset [Dem12, Chapter II, Theorem 5.10], we obtain $\mathcal{F}|_V = \mathcal{O}_V v$. Thus $\mathcal{F}$ is a line bundle.

Now let $y \in V$. Applying (3.2) at y, we have $\mathcal{F}_y = \mathcal{O}_{X,y} v'$ for a vector v' whose entries have no common irreducible factor in $\mathcal{O}_{X,y}$. Since v also generates $\mathcal{F}_y$, the germ of v at y is a unit multiple of v', so the entries of v have no common irreducible factor in $\mathcal{O}_{X,y}$. Let $q \geq 1$. The coordinates of v^q with respect to the monomial basis of $\operatorname{Sym}^q \mathcal{O}_{X,y}^b$ contain the powers $v_1^q, \dots, v_b^q$, which have no common irreducible factor. Let t and h be as in the statement, defined near y. Since h is not identically zero on a connected open set containing y, the germ of h at y is non-zero. Since ht is a section of $\mathcal{F}^q = \mathcal{O}_V v^q$, we have $t = a v^q$ for some $a \in \operatorname{Frac} \mathcal{O}_{X,y}$, and (3.1) implies $a \in \mathcal{O}_{X,y}$. Thus the germ of t at y belongs to $\mathcal{F}_y^q$. Since x and y are arbitrary, $\mathcal{F}^q$ is saturated in $\operatorname{Sym}^q \mathcal{E}$. □

The following lemma is used three times in this section, in the proof of Proposition 3.4 and in **Steps 3 and 5** of the proof of Definition-Theorem 3.5.

Lemma 3.3. *Let (X, Δ) be an orbifold pair with $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$ and root orbifold $\mathcal{X}$, let U, $z_1, \dots, z_d$, and $i_1, \dots, i_k$ be as in Definition 2.8, and let $\pi_U : \widetilde{U} \to U$ be the chart of $\mathcal{X}$ over U, with coordinates $w_1, \dots, w_k, z_{k+1}, \dots, z_d$ on $\widetilde{U}$. Let $G_U := \prod_{j=1}^k \mu_{m_{i_j}}$, and for $\zeta = (\zeta_1, \dots, \zeta_k) \in G_U$ let g_ζ be the automorphism of $\widetilde{U}$ which multiplies w_j by ζ_j for $1 \leq j \leq k$ and fixes $z_{k+1}, \dots, z_d$. Let $c_1, \dots, c_k$ be integers, and for $1 \leq j \leq k$ let $a_j \in \{0, \dots, m_{i_j} - 1\}$ be the residue of c_j modulo m_{i_j}. Then a holomorphic function u on $\widetilde{U}$ satisfies $u \circ g_\zeta = (\prod_{j=1}^k \zeta_j^{c_j}) u$ for any $\zeta \in G_U$ if and only if $u = (\prod_{j=1}^k w_j^{a_j}) \cdot (h \circ \pi_U)$ for a holomorphic function h on U.*

Proof. Expand u in the Taylor series at the origin, which converges on the polydisc $\widetilde{U}$. A monomial $w_1^{\alpha_1} \cdots w_k^{\alpha_k} z_{k+1}^{\beta_{k+1}} \cdots z_d^{\beta_d}$ is multiplied by $\prod_{j=1}^k \zeta_j^{\alpha_j}$ under g_ζ. Thus $u \circ g_\zeta = (\prod_{j=1}^k \zeta_j^{c_j}) u$ holds for any ζ if and only if only monomials with $\alpha_j \equiv c_j \pmod{m_{i_j}}$ for $1 \leq j \leq k$ occur in u. Grouping the monomials, this holds if and only if $u = (\prod_{j=1}^k w_j^{a_j}) \cdot (h \circ \pi_U)$ for a holomorphic function h on U. Here the Taylor series of h converges on U, because $|z_j| = |w_j|^{m_{i_j}}$ and $\widetilde{U} = \pi_U^{-1}(U)$. □

The next result compares symmetric differentials on the root orbifold with the sheaves of Definition 2.10.

Proposition 3.4. *Let (X, Δ) be an orbifold pair with root orbifold $\pi : \mathcal{X} \to X$. Then for $1 \leq p \leq \dim X$ and $q \geq 0$, pullback of functions and of symmetric tensors along the charts of $\mathcal{X}$ induces isomorphisms, compatible with products of symmetric tensors,*

$$\mathcal{O}_X \to \pi_* \mathcal{O}_{\mathcal{X}}, \qquad \Omega_X^p \to \pi_* \Omega_{\mathcal{X}}^p, \qquad S^q \Omega^p(X, \Delta) \to \pi_* \operatorname{Sym}^q \Omega_{\mathcal{X}}^p.$$

Proof. Write $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$ and $d := \dim X$. Let U, $z_1, \dots, z_d$, $i_1, \dots, i_k$, π_U, G_U, and g_ζ be as in Lemma 3.3, and let $y = (y_1, \dots, y_d) := (w_1, \dots, w_k, z_{k+1}, \dots, z_d)$ be the coordinates on $\widetilde{U}$. Pullback commutes with the chart embeddings of $\mathcal{X}$ and with products of symmetric tensors. Since the polydiscs U of Definition 2.8 form a basis of the topology of X, it suffices to show that pullback by π_U maps $\mathcal{O}_X(U)$, $\Omega_X^p(U)$, and $S^q \Omega^p(X, \Delta)(U)$ bijectively onto the G_U-invariant sections of $\mathcal{O}_{\widetilde{U}}$, $\Omega_{\widetilde{U}}^p$, and $\operatorname{Sym}^q \Omega_{\widetilde{U}}^p$ over $\widetilde{U}$. Pullback by π_U is injective, because π_U is surjective and is a local biholomorphism over $U \backslash D$.

Fix $q \geq 0$. For a q-tuple $\mathbf{I} = (I_1, \dots, I_q)$ of p-element subsets of $\{1, \dots, d\}$, let

$$dz_{\mathbf{I}} := dz_{I_1} \odot \cdots \odot dz_{I_q}, \qquad dy_{\mathbf{I}} := dy_{I_1} \odot \cdots \odot dy_{I_q},$$

and let $t_j(\mathbf{I})$ be the number of indices a with $j \in I_a$, as in Definition 2.10. Let $\mathcal{B}$ be a set of such tuples which contains exactly one ordering of each multiset $\{I_1, \dots, I_q\}$. Then $(dy_{\mathbf{I}})_{\mathbf{I}\in\mathcal{B}}$ is a basis of $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ and $(dz_{\mathbf{I}})_{\mathbf{I}\in\mathcal{B}}$ is a basis of $\operatorname{Sym}^q \Omega^p_U$. For $q = 0$ the only tuple is empty, $dy_{\mathbf{I}} = 1$, and $t_j(\mathbf{I}) = 0$.

Since $g_\zeta^* dy_{\mathbf{I}} = \big(\prod_{j=1}^k \zeta_j^{t_j(\mathbf{I})}\big) dy_{\mathbf{I}}$, a section $\tau = \sum_{\mathbf{I}\in\mathcal{B}} b_{\mathbf{I}} dy_{\mathbf{I}}$ of $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ is invariant if and only if

$$b_{\mathbf{I}} \circ g_\zeta = \Big(\prod_{j=1}^k \zeta_j^{-t_j(\mathbf{I})}\Big) b_{\mathbf{I}} \quad \text{for any } \zeta \in G_U \text{ and } \mathbf{I} \in \mathcal{B}. \tag{3.3}$$

Let $a_j(\mathbf{I}) \in \{0, \dots, m_{i_j} - 1\}$ be the residue of $-t_j(\mathbf{I})$ modulo m_{i_j}. By Lemma 3.3 with $c_j = -t_j(\mathbf{I})$, (3.3) holds if and only if

$$b_{\mathbf{I}} = \Big(\prod_{j=1}^k w_j^{a_j(\mathbf{I})}\Big) \cdot (h_{\mathbf{I}} \circ \pi_U) \quad \text{for some } h_{\mathbf{I}} \in \mathcal{O}_X(U) \tag{3.4}$$

for any $\mathbf{I} \in \mathcal{B}$.

For $1 \le j \le k$ we have $\pi_U^* dz_j = m_{i_j} w_j^{m_{i_j}-1} dw_j$, hence

$$\pi_U^* dz_{\mathbf{I}} = \Big(\prod_{j=1}^k m_{i_j}^{t_j(\mathbf{I})} w_j^{t_j(\mathbf{I})(m_{i_j}-1)}\Big) dy_{\mathbf{I}}.$$

For integers $t \ge 0$ and $m \ge 2$, let $a \in \{0, \dots, m-1\}$ be the residue of $-t$ modulo m. Then $t + a = m\lceil t/m \rceil$, so

$$t(m-1) - a = m\big(t - \lceil t/m \rceil\big) = m\big\lfloor t(1 - \tfrac{1}{m})\big\rfloor. \tag{3.5}$$

Let $e_j(\mathbf{I}) := \lfloor t_j(\mathbf{I})(1 - 1/m_{i_j})\rfloor$. By (3.5) and $\pi_U^* z_j = w_j^{m_{i_j}}$,

$$\pi_U^*\Big(\Big(\prod_{j=1}^k z_j^{-e_j(\mathbf{I})}\Big) dz_{\mathbf{I}}\Big) = \Big(\prod_{j=1}^k m_{i_j}^{t_j(\mathbf{I})}\Big)\Big(\prod_{j=1}^k w_j^{a_j(\mathbf{I})}\Big) dy_{\mathbf{I}}. \tag{3.6}$$

The tensors on the left-hand side of (3.6), for $\mathbf{I} \in \mathcal{B}$, generate $S^q\Omega^p(X, \Delta)$ over U. By (3.4) and (3.6), pullback by π_U maps $S^q\Omega^p(X, \Delta)(U)$ onto the invariant sections of $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ over $\widetilde{U}$. This proves the third isomorphism. For $q = 0$, (3.4) reads $b = h \circ \pi_U$, which proves $\pi_* \mathcal{O}_{\mathcal{X}} = \mathcal{O}_X$. For $q = 1$, we have $t_j(\mathbf{I}) \in \{0, 1\}$ and $e_j(\mathbf{I}) = 0$, so $S^1\Omega^p(X, \Delta) = \Omega^p_X$, which proves $\pi_* \Omega^p_{\mathcal{X}} = \Omega^p_X$. □

The sheaves of Definition 2.10 are defined through round-downs and saturations. Definition-Theorem 3.5 replaces them by the powers of one line bundle on the root orbifold, and shows that any non-zero map from a line bundle on the root orbifold to a sheaf of holomorphic p-forms factors through such a line bundle. Thus it suffices to check the condition in Theorem 3.1 for the root lines of Definition-Theorem 3.5(1).

Definition-Theorem 3.5. Let (X, Δ) be a compact Kähler orbifold pair with root orbifold $\pi : \mathcal{X} \to X$, where $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$, let R_i be the root divisor over D_i, let $N := \operatorname{lcm}(m_1, \dots, m_r)$, with $N := 1$ if $r = 0$, and let $1 \le p \le \dim X$. Then the following hold.

(1) Let $\mathcal{F}_0 \subset \Omega^p_X$ be a coherent subsheaf of rank one with saturation $\mathcal{F} \subset \Omega^p_X$. There exists a unique saturated line bundle $\mathcal{M}_{\mathcal{F}} \subset \Omega^p_{\mathcal{X}}$ on $\mathcal{X}$ whose restriction over $X \setminus \operatorname{Supp} \Delta$ equals $\pi^*\mathcal{F}$. We call $\mathcal{M}_{\mathcal{F}}$ the *root line* of $\mathcal{F}$. For every $q \ge 0$ we have $\pi_* \mathcal{M}^q_{\mathcal{F}} = \mathcal{F}_{q,\Delta}$, and

$$\kappa(\mathcal{X}, \mathcal{M}_{\mathcal{F}}) = \kappa(X, \Delta; \mathcal{F}_0) = \kappa(X, \Delta; \mathcal{F}).$$

(2) Let $\mathcal{F}$ be as in (1). For each i, let $\epsilon_i := 1$ if the composite $\mathcal{F}|_{D_i} \to \Omega^p_X|_{D_i} \to \Omega^p_{D_i}$ vanishes at a general point of D_i, and $\epsilon_i := 0$ otherwise. Then

$$\mathcal{M}_{\mathcal{F}} = \pi^*\mathcal{F} \otimes \mathcal{O}_{\mathcal{X}}\Big(\sum_{i=1}^{r} \epsilon_i(m_i-1)R_i\Big) \quad \text{and} \quad \mathcal{F}_{q,\Delta} = \mathcal{F}^q \otimes \mathcal{O}_X\Big(\sum_{i=1}^{r} \epsilon_i \big\lfloor q(1-\tfrac{1}{m_i})\big\rfloor D_i\Big) \text{ for } q \geq 0.$$

(3) For every line bundle $\mathcal{L}$ on $\mathcal{X}$, the sheaf $\mathcal{A}_{\mathcal{L}} := \pi_*\mathcal{L}^N$ is a line bundle on X, $\mathcal{L}^N \cong \pi^*\mathcal{A}_{\mathcal{L}}$, and $\kappa(\mathcal{X},\mathcal{L}) = \kappa(X,\mathcal{A}_{\mathcal{L}})$.

(4) Every non-zero map $\mathcal{L} \to \Omega^p_{\mathcal{X}}$ from a line bundle $\mathcal{L}$ on $\mathcal{X}$ factors as $\mathcal{L} \to \mathcal{M}_{\mathcal{F}'} \to \Omega^p_{\mathcal{X}}$ for a unique saturated subsheaf $\mathcal{F}' \subset \Omega^p_X$ of rank one, and $\kappa(\mathcal{X},\mathcal{L}) \leq \kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}'}) \leq p$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct $\mathcal{M}_{\mathcal{F}}$ and prove that it is unique.

Let $D := \operatorname{Supp}\Delta$. For a polydisc U as in Definition 2.8, let π_U, G_U, and g_ζ be as in Lemma 3.3, and let $H_U := \{w_1 \cdots w_k = 0\} = \pi_U^{-1}(D \cap U)$. The group G_U acts freely on $\widetilde{U}\backslash H_U$, and π_U restricts to the quotient map $\widetilde{U}\backslash H_U \to U\backslash D$, which is a local biholomorphism.

We first show the following inclusion property. Let W be a connected complex manifold, $\mathcal{V}$ a vector bundle on W, $\mathcal{S} \subset \mathcal{V}$ a saturated subsheaf which is a line bundle, and $\mathcal{S}' \subset \mathcal{V}$ a subsheaf which is a line bundle. If $\mathcal{S}' \subset \mathcal{S}$ over a dense open subset $W^\circ \subset W$, then $\mathcal{S}' \subset \mathcal{S}$. In particular, two saturated subsheaves of $\mathcal{V}$ which are line bundles and coincide on W° are equal. To prove this, let $x \in W$, and let s and s' be generators of $\mathcal{S}$ and $\mathcal{S}'$ on a connected open neighborhood W_x of x over which $\mathcal{V} \cong \mathcal{O}_{W_x}^{\oplus n}$. Write $s = (s_1,\dots,s_n)$ and $s' = (s'_1,\dots,s'_n)$. Since s' is a multiple of s on $W_x \cap W^\circ$, we have $s \wedge s' = 0$ on $W_x \cap W^\circ$, hence on W_x. Choose l with $s_l \not\equiv 0$ on W_x. The vanishing of the coordinates $s_l s'_{l'} - s_{l'} s'_l$ of $s \wedge s'$ implies $s_l s' = s'_l s$, which is a section of $\mathcal{S}$. Since $\mathcal{S}$ is saturated, s' is a section of $\mathcal{S}$. Thus $\mathcal{S}' \subset \mathcal{S}$. If $\mathcal{S}'$ is also saturated and $\mathcal{S}' = \mathcal{S}$ over W°, then $\mathcal{S} \subset \mathcal{S}'$ by symmetry.

By Lemma 3.2, $\mathcal{F}$ is a line bundle. The composite $\pi_U^*\mathcal{F} \to \pi_U^*\Omega^p_U \to \Omega^p_{\widetilde{U}}$ is injective over $\widetilde{U}\backslash H_U$, hence injective, since $\widetilde{U}$ is connected. Let $\mathcal{M}_U \subset \Omega^p_{\widetilde{U}}$ be the saturation of the image of $\pi_U^*\mathcal{F}$. By Lemma 3.2, $\mathcal{M}_U$ is a line bundle and $\mathcal{M}_U^q$ is saturated in $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ for any $q \geq 1$. For $\zeta \in G_U$ we have $\pi_U \circ g_\zeta = \pi_U$, so g_ζ^* preserves the image of $\pi_U^*\mathcal{F}$, hence preserves $\mathcal{M}_U$. If $\lambda : \widetilde{V} \to \widetilde{U}$ is a chart embedding of $\mathcal{X}$ over an inclusion $V \subset U$, then $\pi_U \circ \lambda = \pi_V$, so λ^* carries the image of $\pi_U^*\mathcal{F}$ onto the image of $\pi_V^*\mathcal{F}$, hence carries $\mathcal{M}_U$ onto $\mathcal{M}_V$. Thus the sheaves $\mathcal{M}_U$ glue to a saturated line bundle $\mathcal{M}_{\mathcal{F}} \subset \Omega^p_{\mathcal{X}}$. Since π_U is a local biholomorphism on $\widetilde{U}\backslash H_U$ and $\mathcal{F}$ is saturated in Ω^p_X, the image of $\pi_U^*\mathcal{F}$ is saturated over $\widetilde{U}\backslash H_U$. Thus $\mathcal{M}_{\mathcal{F}}$ restricts to $\pi^*\mathcal{F}$ over $X\backslash D$.

If $\mathcal{M}' \subset \Omega^p_{\mathcal{X}}$ is another saturated line bundle which restricts to $\pi^*\mathcal{F}$ over $X\backslash D$, then $\mathcal{M}'|_{\widetilde{U}}$ and $\mathcal{M}_U$ coincide on the dense open subset $\widetilde{U}\backslash H_U$ of $\widetilde{U}$. By the inclusion property above, $\mathcal{M}' = \mathcal{M}_{\mathcal{F}}$.

Step 2. In this step we prove $\pi_*\mathcal{M}_{\mathcal{F}}^q = \mathcal{F}_{q,\Delta}$ and the equalities of Iitaka dimensions in (1).

For $q = 0$ both sides equal $\mathcal{O}_X$ by Proposition 3.4. Let $q \geq 1$. By Proposition 3.4, pullback by π maps $S^q\Omega^p(X,\Delta)$ isomorphically onto $\pi_* \operatorname{Sym}^q \Omega^p_{\mathcal{X}}$. Since $\pi_*\mathcal{M}_{\mathcal{F}}^q \subset \pi_* \operatorname{Sym}^q \Omega^p_{\mathcal{X}}$ and $\mathcal{F}_{q,\Delta} \subset S^q\Omega^p(X,\Delta)$, it suffices to show the following: if U is a polydisc as in Definition 2.8 with $\mathcal{F}|_U = \mathcal{O}_U v$, $V \subset U$ is a connected open subset, and σ is a section of $S^q\Omega^p(X,\Delta)$ over V, then σ is a section of $\mathcal{F}_{q,\Delta}$ if and only if $\pi^*\sigma$ is a section of $\mathcal{M}_{\mathcal{F}}^q$. Such polydiscs U form a basis of the topology of X. Note that the exponents in Definition 2.10 are non-negative, so $\operatorname{Sym}^q \Omega^p_X \subset S^q\Omega^p(X,\Delta)$ and $\mathcal{F}^q = \mathcal{O}_U v^q$ is contained in $S^q\Omega^p(X,\Delta)$ over U.

Assume that σ is a section of $\mathcal{F}_{q,\Delta}$. Then near any point of V, $h\sigma$ is a section of $\mathcal{F}^q$ for some holomorphic $h \not\equiv 0$. Thus $(h \circ \pi_U)\pi^*\sigma$ is a section of $\pi_U^*\mathcal{F}^q \subset \mathcal{M}_{\mathcal{F}}^q$, and $\pi^*\sigma$ is a holomorphic section of $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ by Proposition 3.4. Since $\mathcal{M}_{\mathcal{F}}^q$ is saturated in $\operatorname{Sym}^q \Omega^p_{\widetilde{U}}$ (**Step 1**), $\pi^*\sigma$ is a section of $\mathcal{M}_{\mathcal{F}}^q$.

Conversely, assume that $\pi^*\sigma$ is a section of $\mathcal{M}_{\mathcal{F}}^q$. Since $\mathcal{M}_{\mathcal{F}}$ restricts to $\pi^*\mathcal{F}$ over $X\backslash D$ (**Step 1**) and π_U is a local biholomorphism over $U\backslash D$, σ is a section of $\mathcal{F}^q$ over $V\backslash D$. Write $\sigma = \sum_{\mathbf{I}} \sigma_{\mathbf{I}} dz_{\mathbf{I}}$ and $v^q = \sum_{\mathbf{I}} c_{\mathbf{I}} dz_{\mathbf{I}}$ with respect to the basis of $\mathrm{Sym}^q \Omega_U^p$ formed by the tensors $dz_{\mathbf{I}} = dz_{I_1} \odot \cdots \odot dz_{I_q}$. The coefficients $\sigma_{\mathbf{I}}$ are meromorphic on V and holomorphic on $V\backslash D$, and the $c_{\mathbf{I}}$ are holomorphic. Choose $\mathbf{I}_0$ with $c_{\mathbf{I}_0} \not\equiv 0$ on V. Since σ is a holomorphic multiple of v^q on $V\backslash D$, we have $c_{\mathbf{I}_0}\sigma = \sigma_{\mathbf{I}_0} v^q$ on $V\backslash D$, hence on V. Let $n \geq 0$ be an integer such that $z^n\sigma_{\mathbf{I}}$ is holomorphic for any $\mathbf{I}$, where $z := z_1 \cdots z_k$. Then $h := z^n c_{\mathbf{I}_0} \not\equiv 0$ and

$$h\sigma = (z^n \sigma_{\mathbf{I}_0}) v^q$$

is a section of $\mathcal{F}^q$ over V. Thus σ is a section of $\mathcal{F}_{q,\Delta}$.

Hence pullback by π induces an isomorphism $\mathcal{F}_{q,\Delta} \to \pi_* \mathcal{M}_{\mathcal{F}}^q$ for any $q \geq 0$, compatible with products by Proposition 3.4.

Next we compare $\mathcal{F}_0$ with $\mathcal{F}$. Near any point, $\mathcal{F} = \mathcal{O}v$ for a local generator v, and gv is a section of $\mathcal{F}_0$ for some holomorphic $g \not\equiv 0$. Thus $g^q v^q$ is contained in the image of $\mathcal{F}_0^{\otimes q}$, which is contained in $\mathcal{F}^q = \mathcal{O}v^q$. Hence the saturation of the image of $\mathcal{F}_0^{\otimes q}$ in $S^q\Omega^p(X,\Delta)$ equals the saturation of the image of $\mathcal{F}^{\otimes q}$, and $\kappa(X,\Delta;\mathcal{F}_0) = \kappa(X,\Delta;\mathcal{F})$.

For $q \geq 1$, $\mathcal{F}_{q,\Delta}$ is a saturated subsheaf of rank one of the vector bundle $S^q\Omega^p(X,\Delta)$, hence a line bundle by Lemma 3.2. Since pullback by π induces an isomorphism $\mathcal{F}_{q,\Delta} \to \pi_*\mathcal{M}_{\mathcal{F}}^q$, we have $H^0(X,\mathcal{F}_{q,\Delta}) \cong H^0(\mathcal{X},\mathcal{M}_{\mathcal{F}}^q)$ for any $q \geq 0$. If $s, s' \in H^0(X,\mathcal{F}_{q,\Delta})$ and $s \neq 0$, then s'/s is a meromorphic function on X which equals π^*s'/π^*s over $X\backslash D$. Thus the meromorphic maps defined by $H^0(X,\mathcal{F}_{q,\Delta})$ and by $H^0(\mathcal{X},\mathcal{M}_{\mathcal{F}}^q)$ coincide. By [Cam11a, Lemme 3.1], $\kappa(X,\Delta;\mathcal{F})$ is $-\infty$ if $H^0(X,\mathcal{F}_{q,\Delta}) = 0$ for any $q \geq 1$, and otherwise it is the maximum of the dimensions of the images of the meromorphic maps defined by $H^0(X,\mathcal{F}_{q,\Delta})$. By Definition 2.9, $\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}}) = \kappa(X,\Delta;\mathcal{F})$.

Step 3. In this step we prove (2).

By **Step 1**, the map $\pi^*\mathcal{F} \to \Omega_{\mathcal{X}}^p$ factors through a non-zero map $\pi^*\mathcal{F} \to \mathcal{M}_{\mathcal{F}}$ of line bundles. On any chart, the zero divisor of the map $\pi^*\mathcal{F} \to \mathcal{M}_{\mathcal{F}}$ is a G_U-invariant divisor $Z_U \geq 0$, which is supported on H_U because $\mathcal{M}_{\mathcal{F}}$ restricts to $\pi^*\mathcal{F}$ over $X\backslash D$. Let $c_i \geq 0$ be the multiplicity of the zero divisor along R_i. Then

$$\mathcal{M}_{\mathcal{F}} = \pi^*\mathcal{F} \otimes \mathcal{O}_{\mathcal{X}}\Big(\sum_{i=1}^r c_i R_i\Big).$$

Fix i and a point $x \in D_i$ which does not belong to $D_{i'}$ for $i' \neq i$. Let U be a polydisc centered at x as in Definition 2.8, so that $k = 1$, $i_1 = i$, and $D_i \cap U = \{z_1 = 0\}$. Write $w := w_1$, so that $\pi_U^* z_1 = w^{m_i}$. Possibly shrinking U, let e be a nowhere vanishing section of $\mathcal{F}$ over U, and write

$$e = \sum_{|I|=p} f_I dz_I.$$

If z_1 divided all f_I, then $z_1^{-1}e$ would be a section of Ω_X^p with $z_1 \cdot z_1^{-1}e \in \mathcal{F}$, hence a section of $\mathcal{F}$ because $\mathcal{F}$ is saturated, which is impossible because e generates $\mathcal{F}$. Thus

$$\min_I \mathrm{ord}_{D_i}(f_I) = 0,$$

where ord_{D_i} denotes the order of vanishing along $D_i \cap U = \{z_1 = 0\}$. Moreover, $\mathrm{ord}_{\{w=0\}}(f_I \circ \pi_U) = m_i \,\mathrm{ord}_{D_i}(f_I)$. Let e'' be a local generator of $\mathcal{M}_{\mathcal{F}}$ on $\widetilde{U}$ near a point of $\{w = 0\}$, and write $\pi_U^* e = \phi e''$ with ϕ holomorphic. Then $c_i = \mathrm{ord}_{\{w=0\}}(\phi)$. If w divided all coefficients of e'' with respect to the basis $(dy_I)_{|I|=p}$, then $w^{-1}e''$ would be a section of $\Omega_{\widetilde{U}}^p$ with $w \cdot w^{-1}e'' \in \mathcal{M}_{\mathcal{F}}$, hence a section of $\mathcal{M}_{\mathcal{F}}$, which is impossible. Thus c_i is the minimum of the orders along $\{w = 0\}$ of the coefficients of $\pi_U^* e$. Since $\pi_U^* dz_1 = m_i w^{m_i - 1} dw$, the coefficient of dy_I in $\pi_U^* e$ has order

$$\mathrm{ord}_{\{w=0\}}\left(\pi_U^* e\right)_I = \begin{cases} m_i \,\mathrm{ord}_{D_i}(f_I) + m_i - 1 & \text{if } 1 \in I, \\ m_i \,\mathrm{ord}_{D_i}(f_I) & \text{if } 1 \notin I. \end{cases}$$

The composite $\mathcal{F}|_{D_i} \to \Omega^p_{D_i}$ maps e to $\sum_{1\notin I}(f_I|_{D_i})dz_I|_{D_i}$, and the forms $dz_I|_{D_i}$ with $1 \notin I$ form a basis of $\Omega^p_{D_i}$ over $D_i \cap U$. Since D_i is connected, the composite vanishes at a general point of D_i if and only if it vanishes on $D_i \cap U$. Thus $\epsilon_i = 1$ if and only if $\mathrm{ord}_{D_i}(f_I) \geq 1$ for any I with $1 \notin I$. If $\epsilon_i = 0$, then some I with $1 \notin I$ has $\mathrm{ord}_{D_i}(f_I) = 0$, and $c_i = 0$. If $\epsilon_i = 1$, then the coefficients with $1 \notin I$ have order at least m_i, the coefficients with $1 \in I$ have order at least $m_i - 1$, and some I with $1 \in I$ has $\mathrm{ord}_{D_i}(f_I) = 0$. Thus $c_i = m_i - 1$. In both cases $c_i = \epsilon_i(m_i - 1)$, which proves the formula for $\mathcal{M}_{\mathcal{F}}$.

Now let $q \geq 0$, let U be a polydisc as in Definition 2.8 with $\mathcal{F}|_U = \mathcal{O}_U v$, and let $b_i := q\epsilon_i(m_i-1)$. By the formula for $\mathcal{M}_{\mathcal{F}}$, the sections of $\mathcal{M}^q_{\mathcal{F}}$ over $\widetilde{U}$ are the tensors $\pi_U^*(v^q) \cdot u \prod_{j=1}^k w_j^{-b_{i_j}}$ with u holomorphic on $\widetilde{U}$. Since $\pi_U^*(v^q)$ is invariant, such a tensor is invariant if and only if $u \circ g_\zeta = \big(\prod_j \zeta_j^{b_{i_j}}\big)u$ for any $\zeta \in G_U$. By Lemma 3.3 applied to the integers $b_{i_1}, \dots, b_{i_k}$, this holds if and only if $u = \big(\prod_j w_j^{a_j}\big)(h \circ \pi_U)$ for a holomorphic function h on U, where $a_j \in \{0, \dots, m_{i_j} - 1\}$ is the residue of b_{i_j} modulo m_{i_j}. In this case

$$u \prod_{j=1}^k w_j^{-b_{i_j}} = \pi_U^*\Big(h \prod_{j=1}^k z_j^{-\lfloor b_{i_j}/m_{i_j} \rfloor}\Big).$$

Since $\lfloor q\epsilon_i(m_i - 1)/m_i \rfloor = \epsilon_i \lfloor q(1 - \frac{1}{m_i}) \rfloor$, we obtain

$$\pi_* \mathcal{M}^q_{\mathcal{F}} = \mathcal{F}^q \otimes \mathcal{O}_X\Big(\sum_{i=1}^r \epsilon_i \big\lfloor q(1 - \tfrac{1}{m_i})\big\rfloor D_i\Big).$$

Together with **Step 2**, this proves the formula for $\mathcal{F}_{q,\Delta}$.

Step 4. In this step we prove (3).

Let $x \in X$ and let U be a polydisc centered at x as in Definition 2.8. The group G_U fixes the origin $0 \in \widetilde{U}$, so G_U acts linearly on the fiber of $\mathcal{L}$ at 0 by a character $\chi : G_U \to \mathbb{C}^*$. For $\zeta \in G_U$ and a section s of $\mathcal{L}$ over $\widetilde{U}$, let $\zeta \cdot s$ be the image of s under the action of ζ, so that $\zeta \cdot (us) = (u \circ g_\zeta^{-1})(\zeta \cdot s)$ for holomorphic functions u on $\widetilde{U}$. Possibly shrinking U, let e be a nowhere vanishing section of $\mathcal{L}$ over $\widetilde{U}$, and let

$$e_\chi := \frac{1}{|G_U|} \sum_{\zeta \in G_U} \chi(\zeta)^{-1} \zeta \cdot e.$$

Then $\zeta \cdot e_\chi = \chi(\zeta) e_\chi$ for any $\zeta \in G_U$ [Ser77, §2.6, Theorem 8]. Since $(\zeta \cdot e)(0) = \chi(\zeta)e(0)$, we have $e_\chi(0) = e(0) \neq 0$. Possibly shrinking U, e_χ is a local generator of $\mathcal{L}$ on $\widetilde{U}$ on which G_U acts by the character χ. Since the order of any element of G_U divides N, we have $\chi^N = 1$, so e_χ^N is a G_U-invariant generator of $\mathcal{L}^N$ on $\widetilde{U}$. By Proposition 3.4, the invariant sections of $\mathcal{L}^N$ over $\widetilde{U}$ are the sections $(h \circ \pi_U)e_\chi^N$ with $h \in \mathcal{O}_X(U)$. Thus $\pi_* \mathcal{L}^N|_U$ is free of rank one, and the natural map $\pi^* \pi_* \mathcal{L}^N \to \mathcal{L}^N$ is an isomorphism over $\widetilde{U}$. Hence $\mathcal{A}_{\mathcal{L}}$ is a line bundle and $\mathcal{L}^N \cong \pi^* \mathcal{A}_{\mathcal{L}}$.

For a line bundle $\mathcal{L}'$ on $\mathcal{X}$ and $q \geq 1$ with $H^0(\mathcal{X}, \mathcal{L}'^q) \neq 0$, let $\Phi_{\mathcal{L}',q} : X \dashrightarrow \mathbb{P}(H^0(\mathcal{X}, \mathcal{L}'^q)^*)$ be the meromorphic map defined by $H^0(\mathcal{X}, \mathcal{L}'^q)$. If $W \subset H^0(\mathcal{X}, \mathcal{L}'^q)$ is a non-zero subspace, then the meromorphic map defined by W is the composite of $\Phi_{\mathcal{L}',q}$ with the linear projection $\mathbb{P}(H^0(\mathcal{X}, \mathcal{L}'^q)^*) \dashrightarrow \mathbb{P}(W^*)$. Thus

$$\dim(\text{image of the map defined by } W) \leq \dim(\text{image of } \Phi_{\mathcal{L}',q}). \tag{3.7}$$

Let $q \geq 1$ with $H^0(\mathcal{X}, \mathcal{L}^q) \neq 0$, and let $s_0 \in H^0(\mathcal{X}, \mathcal{L}^q)$ be non-zero. Multiplication by s_0^{N-1} is an injective map $H^0(\mathcal{X}, \mathcal{L}^q) \to H^0(\mathcal{X}, \mathcal{L}^{qN})$, and $s s_0^{N-1}/s_0^N = s/s_0$ for any $s \in H^0(\mathcal{X}, \mathcal{L}^q)$. Thus $\Phi_{\mathcal{L},q}$ is the map defined by the subspace $s_0^{N-1} H^0(\mathcal{X}, \mathcal{L}^q)$ of $H^0(\mathcal{X}, \mathcal{L}^{qN})$, and (3.7) implies

$$\dim(\text{image of } \Phi_{\mathcal{L},q}) \leq \dim(\text{image of } \Phi_{\mathcal{L},qN}).$$

Moreover, $H^0(\mathcal{X},\mathcal{L}^q)\neq 0$ for some $q\geq 1$ if and only if $H^0(\mathcal{X},\mathcal{L}^{qN})\neq 0$ for some $q\geq 1$. Hence $\kappa(\mathcal{X},\mathcal{L})$ is $-\infty$ if $H^0(\mathcal{X},\mathcal{L}^{kN})=0$ for any $k\geq 1$, and otherwise it is the maximum, over the integers $k\geq 1$ with $H^0(\mathcal{X},\mathcal{L}^{kN})\neq 0$, of the dimensions of the images of $\Phi_{\mathcal{L},kN}$.

Since $\mathcal{L}^{kN}\cong\pi^*\mathcal{A}_{\mathcal{L}}^k$ and $\pi_*\mathcal{O}_{\mathcal{X}}=\mathcal{O}_X$ (Proposition 3.4), pullback by π induces an isomorphism $H^0(X,\mathcal{A}_{\mathcal{L}}^k)\to H^0(\mathcal{X},\mathcal{L}^{kN})$ for $k\geq 0$, and the meromorphic maps defined by $H^0(X,\mathcal{A}_{\mathcal{L}}^k)$ and by $H^0(\mathcal{X},\mathcal{L}^{kN})$ coincide, since $\pi^*s'/\pi^*s=\pi^*(s'/s)$ for $s,s'\in H^0(X,\mathcal{A}_{\mathcal{L}}^k)$ with $s\neq 0$. By Definition 2.9, $\kappa(\mathcal{X},\mathcal{L})=\kappa(X,\mathcal{A}_{\mathcal{L}})$.

Step 5. In this step we prove (4).

Let $j:\mathcal{L}\to\Omega_{\mathcal{X}}^p$ be non-zero. Since $\mathcal{X}$ is connected, j is non-zero on any chart by the identity theorem, hence injective. We first show that $\pi_*\mathcal{L}$ is a line bundle. Let x, U, χ, and e_χ be as in **Step 4**, and write $\chi(\zeta)=\prod_{l=1}^k\zeta_l^{c_l}$ for $\zeta\in G_U$ with integers c_l. Since $\zeta\cdot(ue_\chi)=\chi(\zeta)(u\circ g_\zeta^{-1})e_\chi$, a section ue_χ of $\mathcal{L}$ over $\widetilde{U}$ is invariant if and only if $u\circ g_\zeta=\chi(\zeta)u$ for any $\zeta\in G_U$. By Lemma 3.3 applied to $c_1,\dots,c_k$, this holds if and only if $u=\big(\prod_l w_l^{a_l}\big)(h\circ\pi_U)$ for a holomorphic function h on U, where $a_l\in\{0,\dots,m_{i_l}-1\}$ is the residue of c_l modulo m_{i_l}. Hence $\pi_*\mathcal{L}|_U$ is free of rank one.

Let $\iota:\pi_*\mathcal{L}\to\Omega_X^p$ be the composite of π_*j with the isomorphism $\pi_*\Omega_{\mathcal{X}}^p=\Omega_X^p$ of Proposition 3.4. Since taking invariant sections is left exact, ι is injective. Let $\mathcal{F}_0:=\iota(\pi_*\mathcal{L})$, a coherent subsheaf of Ω_X^p of rank one, and let $\mathcal{F}$ be the saturation of $\mathcal{F}_0$. Since G_U acts freely on $\widetilde{U}\backslash H_U$ with quotient $U\backslash D$, the natural map $\pi_U^*\pi_*\mathcal{L}\to\mathcal{L}$ is an isomorphism over $\widetilde{U}\backslash H_U$, so $j(\mathcal{L})=\pi_U^*\mathcal{F}_0\subset\pi_U^*\mathcal{F}\subset\mathcal{M}_{\mathcal{F}}$ over $\widetilde{U}\backslash H_U$. Since j is injective, $j(\mathcal{L})\subset\Omega_{\widetilde{U}}^p$ is a subsheaf which is a line bundle, and it is contained in the saturated line bundle $\mathcal{M}_{\mathcal{F}}$ over the dense open subset $\widetilde{U}\backslash H_U$. By the inclusion property in **Step 1**, $j(\mathcal{L})\subset\mathcal{M}_{\mathcal{F}}$ on $\widetilde{U}$. Thus j factors as $\mathcal{L}\to\mathcal{M}_{\mathcal{F}}\to\Omega_{\mathcal{X}}^p$.

If j also factors through $\mathcal{M}_{\mathcal{F}'}$ for a saturated subsheaf $\mathcal{F}'\subset\Omega_X^p$ of rank one, then over $X\backslash D$ both $\mathcal{F}$ and $\mathcal{F}'$ are saturated subsheaves of Ω_X^p containing $j(\mathcal{L})$, hence both contain the saturation $\mathcal{S}$ of $j(\mathcal{L})$ over $X\backslash D$. By Lemma 3.2, $\mathcal{F}$, $\mathcal{F}'$, and $\mathcal{S}$ are line bundles. If s' is a local generator of $\mathcal{F}'$ over $X\backslash D$, then as' is a section of $\mathcal{S}$ for some holomorphic $a\not\equiv 0$, so s' is a section of $\mathcal{S}$. Thus $\mathcal{F}'=\mathcal{S}=\mathcal{F}$ over $X\backslash D$, and the inclusion property in **Step 1** implies $\mathcal{F}'=\mathcal{F}$.

The map $\mathcal{L}\to\mathcal{M}_{\mathcal{F}}$ is a non-zero map of line bundles, that is, a non-zero section θ of $\mathcal{M}_{\mathcal{F}}\otimes\mathcal{L}^{-1}$. For $q\geq 1$, multiplication by θ^q is an injective map $H^0(\mathcal{X},\mathcal{L}^q)\to H^0(\mathcal{X},\mathcal{M}_{\mathcal{F}}^q)$ which preserves ratios of sections. By (3.7), $\kappa(\mathcal{X},\mathcal{L})\leq\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}})$. Finally, $\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}})=\kappa(X,\Delta;\mathcal{F})\leq p$ by **Step 2** and [Cam11a, Corollaire 3.13(1)]. □

Proof of Theorem 3.1. Assume that (X,Δ) is special, and suppose that there exist an integer p with $1\leq p\leq\dim X$, a line bundle $\mathcal{L}$ on $\mathcal{X}$, and a non-zero map $\mathcal{L}\to\Omega_{\mathcal{X}}^p$ with $\kappa(\mathcal{X},\mathcal{L})=p$. By Definition-Theorem 3.5(4), the map factors through $\mathcal{M}_{\mathcal{F}}$ for a saturated subsheaf $\mathcal{F}\subset\Omega_X^p$ of rank one, and

$$p=\kappa(\mathcal{X},\mathcal{L})\leq\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}})\leq p.$$

By Definition-Theorem 3.5(1), $\kappa(X,\Delta;\mathcal{F})=\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}})=p$, so $\mathcal{F}$ is a Δ-Bogomolov sheaf. This contradicts Definition 2.11.

Conversely, assume that (X,Δ) is not special. By Definition 2.11, there exist an integer p with $1\leq p\leq\dim X$ and a coherent subsheaf $\mathcal{F}_0\subset\Omega_X^p$ of rank one with $\kappa(X,\Delta;\mathcal{F}_0)=p$. Let $\mathcal{F}$ be the saturation of $\mathcal{F}_0$. By Definition-Theorem 3.5(1), the inclusion $\mathcal{M}_{\mathcal{F}}\to\Omega_{\mathcal{X}}^p$ is a non-zero map with $\kappa(\mathcal{X},\mathcal{M}_{\mathcal{F}})=\kappa(X,\Delta;\mathcal{F}_0)=p$.

If $\Delta=0$, then $\mathcal{X}=X$, and the second sentence is the first one for the pair $(X,0)$. □

Corollary 3.6. *Let (X,Δ) be a compact Kähler orbifold pair. Then the following hold.*

(1) If (X,Δ) is special, then X is special.
(2) If $a(X)=0$, then (X,Δ) is special.

Proof. Let $\pi:\mathcal{X}\to X$ be the root orbifold of (X,Δ) and $D:=\operatorname{Supp}\Delta$.

(1) Suppose that X is not special. By Theorem 3.1, there exist an integer p with $1 \leq p \leq \dim X$, a line bundle $\mathcal{L}$ on X, and a non-zero map $\mathcal{L} \to \Omega_X^p$ with $\kappa(X, \mathcal{L}) = p$. The composite $\pi^*\mathcal{L} \to \pi^*\Omega_X^p \to \Omega_{\mathcal{X}}^p$ is non-zero over $X \backslash D$, where π is an isomorphism. By the projection formula and $\pi_*\mathcal{O}_{\mathcal{X}} = \mathcal{O}_X$ (Proposition 3.4), $\pi_*\pi^*\mathcal{L}^q = \mathcal{L}^q$, so pullback by π is an isomorphism $H^0(X, \mathcal{L}^q) \to H^0(\mathcal{X}, \pi^*\mathcal{L}^q)$ for any $q \geq 1$ which preserves ratios of sections. Thus $\kappa(\mathcal{X}, \pi^*\mathcal{L}) = \kappa(X, \mathcal{L}) = p$, and Theorem 3.1 implies that (X, Δ) is not special.

(2) Let $1 \leq p \leq \dim X$ and let $\mathcal{L}$ be a line bundle on $\mathcal{X}$. By Definition-Theorem 3.5(3) and Lemma 2.15(2),

$$\kappa(\mathcal{X}, \mathcal{L}) = \kappa(X, \mathcal{A}_{\mathcal{L}}) \leq a(X) = 0 < p.$$

Thus no line bundle on $\mathcal{X}$ admits a non-zero map to $\Omega_{\mathcal{X}}^p$ with Iitaka dimension p, and (X, Δ) is special by Theorem 3.1. □

4. Kummer covers twisted by abelian varieties

In this section, we construct the auxiliary variety $h : Q \to X$ that Section 5 uses when X is projective, namely the quotient Q of the product of a Kummer cover $\pi : S \to X$ and an abelian variety F by a finite abelian group A (Construction-Proposition 4.5). Near any point of Q, the pullback of a local equation of any D_α is the m_α-th power of a holomorphic function. The cover π is also branched along auxiliary divisors $B_{\alpha,j}$, over which Q has only quotient singularities, from which Section 5 extracts divisors of multiplicity one.

Set-up 4.1. Let X, D, $m_1, \dots, m_r$, and N be as follows.

(1) X is a connected smooth projective variety of dimension $d \geq 1$.
(2) $D = \sum_{\alpha=1}^r D_\alpha$, where $r \geq 1$, is a reduced simple normal crossing divisor on X whose irreducible components D_α are smooth.
(3) $m_1, \dots, m_r \geq 2$ are integers.
(4) $N := d + 1$.

In Construction-Proposition 4.5, the group A acts diagonally on $S \times F$, and the action of A on F in Construction-Proposition 4.2 has two properties. First, each basis element $e_{\alpha,j}$ of A has fixed points on F (part (1)), so that Q has quotient singularities over $B_{\alpha,j}$, which over the general points of $B_{\alpha,j}$ are cyclic, with local group a subgroup of $\langle e_{\alpha,j} \rangle \cong \mathbb{Z}/m_\alpha$, instead of a smooth divisor of multiplicity m_α. Proposition 5.2 uses the quotient singularities of Q over $B_{\alpha,j}$ for condition (3) of Proposition 5.5. Second, an element whose coefficients of $e_{\alpha,i}$ and of $e_{\alpha,j}$ are equal and non-zero for some $i \neq j$ acts on F without fixed points (part (2)). By Construction-Proposition 4.3(2) and (4), every element of the stabilizer of a point of S that multiplies the m_α-th root of a local equation of D_α by a non-trivial root of unity is such an element, which is why F has one factor $E_\alpha^{m_\alpha}$, where E_α is an elliptic curve, for each pair $i < j$. For $m_\alpha = 2$, [OAI26b, Lemma A.2] uses $v \to -v$ and $v \to p - v$ on an elliptic curve, where p is a point of order two. Since the automorphisms of a complex elliptic curve fixing a point form a group of order 2, 4, or 6 [Har77, Chapter IV, Corollary 4.7], for general m_α we use cyclic permutations of the coordinates of $E_\alpha^{m_\alpha}$.

Construction-Proposition 4.2. Let $r \geq 1$ and $N \geq 2$ be integers. For $1 \leq \alpha \leq r$ let $m_\alpha \geq 2$ be an integer, E_α a complex elliptic curve, and $p_\alpha \in E_\alpha$ a point of order exactly m_α. Let $F := \prod_{\alpha=1}^r \prod_{1 \leq i < j \leq N} E_\alpha^{m_\alpha}$, where the coordinates x_k of each factor $E_\alpha^{m_\alpha}$ are indexed by $k \in \mathbb{Z}/m_\alpha\mathbb{Z}$, and let $A := \prod_{\alpha=1}^r (\mathbb{Z}/m_\alpha\mathbb{Z})^N$, where $e_{\alpha,1}, \dots, e_{\alpha,N}$ is the standard basis of the α-th factor. Let A act on F as follows: for $i < j$, the basis element $e_{\alpha,i}$ acts on the factor indexed by (α, i, j) by $(x_k)_k \to (x_{k-1})_k$, the basis element $e_{\alpha,j}$ acts on it by $(x_k)_k \to (x_{k+1} + p_\alpha)_k$, and all other basis elements act trivially on it. Then A acts on the abelian variety F by automorphisms of varieties, and the following hold.

(1) For every (α, j), the fixed locus of $e_{\alpha,j}$ on F is non-empty and smooth.

(2) An element $g \in A$ acts on F without fixed points if there exist α, indices $i \neq j$, and $a \in \mathbb{Z}/m_\alpha\mathbb{Z}$ with $a \neq 0$ such that the coefficients of $e_{\alpha,i}$ and of $e_{\alpha,j}$ in g are both equal to a.

Proof. For $1 \leq \alpha \leq r$, let σ_α and σ'_α be the automorphisms of the variety $E_\alpha^{m_\alpha}$ given by

$$\sigma_\alpha : (x_k)_k \to (x_{k-1})_k, \qquad \sigma'_\alpha : (x_k)_k \to (x_{k+1} + p_\alpha)_k.$$

Then $\sigma'_\alpha(x) = \sigma_\alpha^{-1}(x) + (p_\alpha, \dots, p_\alpha)$, and σ_α is a group automorphism fixing $(p_\alpha, \dots, p_\alpha)$. Thus for any integers u and v,

$$\sigma_\alpha^u \sigma_\alpha'^v(x) = \sigma_\alpha^{u-v}(x) + v(p_\alpha, \dots, p_\alpha). \tag{4.1}$$

In particular, $\sigma_\alpha\sigma'_\alpha = \sigma'_\alpha\sigma_\alpha$, and $\sigma_\alpha^{m_\alpha} = \sigma_\alpha'^{m_\alpha} = \mathrm{id}_{E_\alpha^{m_\alpha}}$ because $m_\alpha p_\alpha = 0$. Hence on the factor indexed by (α, i, j) the basis elements act by commuting automorphisms whose m_α-th powers are the identity, and the basis elements $e_{\beta,k}$ with $\beta \neq \alpha$ act trivially. Therefore the actions of the basis elements define a homomorphism from A to the group of automorphisms of the variety F. For $g \in A$, let $g_{\alpha,k} \in \mathbb{Z}/m_\alpha\mathbb{Z}$ be the coefficient of $e_{\alpha,k}$ in g. By (4.1), g acts on the factor indexed by (α, i, j), $i < j$, by

$$x \to \sigma_\alpha^{g_{\alpha,i} - g_{\alpha,j}}(x) + g_{\alpha,j}(p_\alpha, \dots, p_\alpha), \tag{4.2}$$

where we use any integer representatives of $g_{\alpha,i}$ and $g_{\alpha,j}$.

(1) The element $e_{\alpha,j}$ acts by σ'_α on the factors indexed by (α, i, j) with $i < j$, by σ_α on the factors indexed by (α, j, k) with $j < k$, and trivially on the remaining factors. The fixed locus of σ_α is defined by the equations $x_k = x_{k-1}$, which express $x_1, \dots, x_{m_\alpha - 1}$ in terms of x_0. Thus the fixed locus of σ_α is the diagonal $\Theta_\alpha = \{(y, \dots, y) \mid y \in E_\alpha\}$, which is smooth. Let $\theta_\alpha := (0, -p_\alpha, -2p_\alpha, \dots, -(m_\alpha - 1)p_\alpha)$. Since $m_\alpha p_\alpha = 0$, we have $\sigma'_\alpha(x + \theta_\alpha) = \sigma_\alpha^{-1}(x) + \theta_\alpha$, so the fixed locus of σ'_α is the translate $\theta_\alpha + \Theta_\alpha$ of the fixed locus of σ_α^{-1}, which is Θ_α. Hence the fixed locus of $e_{\alpha,j}$ on F is the product of copies of Θ_α, copies of $\theta_\alpha + \Theta_\alpha$, and the remaining factors. In particular, it is a translate of an abelian subvariety of F, so it is non-empty and smooth.

(2) Possibly exchanging i and j, we may assume that $i < j$. By (4.2), g acts on the factor indexed by (α, i, j) by the translation by $a(p_\alpha, \dots, p_\alpha)$. Since p_α has order exactly m_α and $a \neq 0$ in $\mathbb{Z}/m_\alpha\mathbb{Z}$, we have $ap_\alpha \neq 0$. Here a need not be invertible in $\mathbb{Z}/m_\alpha\mathbb{Z}$. A translation by a non-zero element has no fixed points, and a fixed point of g on F projects to a fixed point of g on the factor indexed by (α, i, j). Thus g acts on F without fixed points. □

We take $N = d + 1$ so that, by Construction-Proposition 4.3(2), every point of D_α is outside at least two of the divisors $B_{\alpha,1}, \dots, B_{\alpha,N}$ that Construction-Proposition 4.3 attaches to D_α.

Construction-Proposition 4.3. Notation and conditions as in Set-up 4.1. There exist line bundles $\mathcal{H}_\alpha$ on X and non-empty smooth divisors $B_{\alpha,j} \in |\mathcal{H}_\alpha^{m_\alpha}(-D_\alpha)|$, for $1 \leq \alpha \leq r$ and $1 \leq j \leq N$, such that $D + \sum_{\alpha,j} B_{\alpha,j}$ is a reduced simple normal crossing divisor and no two of the $B_{\alpha,j}$ have a common irreducible component. Fix such divisors, defining sections s_{D_α} and $s_{B_{\alpha,j}}$, and primitive m_α-th roots of unity ζ_α. For each (α, j) let $S_{\alpha,j}$ be the cyclic cover of X defined by $y_{\alpha,j}^{m_\alpha} = s_{D_\alpha} s_{B_{\alpha,j}}$ in the total space of $\mathcal{H}_\alpha$, where $y_{\alpha,j}$ is the fiber coordinate of the total space of $\mathcal{H}_\alpha$ and $s_{D_\alpha} s_{B_{\alpha,j}} \in H^0(X, \mathcal{H}_\alpha^{m_\alpha})$. Let $\pi : S \to X$ be the normalization of the fiber product over X of the rN covers $S_{\alpha,j}$, let $A := \prod_{\alpha=1}^r (\mathbb{Z}/m_\alpha\mathbb{Z})^N$ act on S so that the basis element $e_{\alpha,j}$ multiplies $y_{\alpha,j}$ by ζ_α, and let $\tau_\alpha := \sum_{j=1}^N e_{\alpha,j}$. We call $\pi : S \to X$ the *Kummer cover*. For $x \in X$ let $I(x) := \{\alpha \mid x \in D_\alpha\}$ and $J(x) := \{(\alpha, j) \mid x \in B_{\alpha,j}\}$, and let z_α ($\alpha \in I(x)$), $z_{\alpha,j}$ ($(\alpha, j) \in J(x)$), and $t_1, \dots, t_c$, where $c := d - |I(x)| - |J(x)|$, be holomorphic local coordinates at x such that $D_\alpha = \{z_\alpha = 0\}$ and $B_{\alpha,j} = \{z_{\alpha,j} = 0\}$ near x. Then the following hold.

(1) The variety S is connected, smooth, and projective, and π is a finite Galois morphism with Galois group A.

(2) For every $x \in X$ and every $\alpha \in I(x)$, the number of j with $(\alpha, j) \in J(x)$ is at most $d - 1$.

(3) Near any point $s \in \pi^{-1}(x)$, the variety S has holomorphic local coordinates u_α ($\alpha \in I(x)$), $v_{\alpha,j}$ ($(\alpha,j) \in J(x)$), and $t_1, \dots, t_c$ with $\pi^* z_\alpha = u_\alpha^{m_\alpha}$, $\pi^* z_{\alpha,j} = v_{\alpha,j}^{m_\alpha}$, and $\pi^* t_l = t_l$.
(4) The stabilizer of such a point s in A is the internal direct sum of the subgroups $\langle \tau_\alpha \rangle$ with $\alpha \in I(x)$ and $\langle e_{\alpha,j} \rangle$ with $(\alpha,j) \in J(x)$, and an element

$$g = \sum_{\alpha \in I(x)} a_\alpha \tau_\alpha + \sum_{(\alpha,j) \in J(x)} b_{\alpha,j} e_{\alpha,j}$$

of the stabilizer multiplies u_α by $\zeta_\alpha^{a_\alpha}$ and $v_{\alpha,j}$ by $\zeta_\alpha^{b_{\alpha,j}}$ and fixes the coordinates t_l.
(5) The fixed locus of $e_{\alpha,j}$ on S is the reduced inverse image of $B_{\alpha,j}$, and at a general point of each irreducible component of the fixed locus the stabilizer is $\langle e_{\alpha,j} \rangle$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct $\mathcal{H}_\alpha$ and $B_{\alpha,j}$.

Let $\mathcal{L}$ be an ample line bundle on X and $\mathcal{M}$ a line bundle on X. By the definition of ampleness [Har77, Chapter II, §7], $\mathcal{M} \otimes \mathcal{L}^{k_0}$ is generated by global sections for $k_0 \gg 0$. By [Har77, Chapter II, Exercise 7.5(e)], there is n_0 such that $\mathcal{L}^n$ is very ample for all $n \geq n_0$, so by [Har77, Chapter II, Exercise 7.5(d)], $\mathcal{M} \otimes \mathcal{L}^k$ is very ample for all $k \geq k_0 + n_0$. Taking $\mathcal{M} = \mathcal{O}_X(-D_\alpha)$ and an integer k_α with $m_\alpha k_\alpha \geq k_0 + n_0$, the line bundle $\mathcal{H}_\alpha := \mathcal{L}^{k_\alpha}$ is such that $\mathcal{H}_\alpha^{m_\alpha}(-D_\alpha) = \mathcal{O}_X(-D_\alpha) \otimes \mathcal{L}^{m_\alpha k_\alpha}$ is very ample.

We fix the divisors $B_{\alpha,j}$ one at a time, in any order of the pairs (α,j). Assume that some of them are fixed and that the sum D' of D and the fixed divisors is a reduced simple normal crossing divisor. Let $\mathcal{Z}$ be the finite set consisting of X and the connected components of intersections of irreducible components of D'. Any $Z \in \mathcal{Z}$ is smooth, and the very ample linear system $|\mathcal{H}_\alpha^{m_\alpha}(-D_\alpha)|$ induces an immersion of Z into a projective space. By Bertini's theorem [Har77, Chapter III, Corollary 10.9], for a general member B of $|\mathcal{H}_\alpha^{m_\alpha}(-D_\alpha)|$, $B \cap Z$ is smooth for any $Z \in \mathcal{Z}$. Moreover, for a general member B, B contains no $Z \in \mathcal{Z}$ with $\dim Z > 0$ and is disjoint from any $Z \in \mathcal{Z}$ with $\dim Z = 0$, since $\mathcal{Z}$ is finite and each of the conditions on B is open and non-empty. For such a member B, $B \cap Z$ is a smooth divisor on Z for any $Z \in \mathcal{Z}$ with $\dim Z > 0$. Hence $D' + B$ is a reduced simple normal crossing divisor, B is smooth, and B has no common irreducible component with D'. Since $\dim X \geq 1$ and $\mathcal{H}_\alpha^{m_\alpha}(-D_\alpha)$ is very ample, $B \neq \emptyset$. We let $B_{\alpha,j} := B$.

Step 2. In this step we prove that S is connected and projective, and that π is a finite Galois morphism with Galois group A.

Let $\mathbb{C}(X)$ be the function field of X. We fix a non-zero rational section of each $\mathcal{H}_\alpha$, so that $s_{D_\alpha} s_{B_{\alpha,j}}$ corresponds to a function $f_{\alpha,j} \in \mathbb{C}(X)^*$ and $y_{\alpha,j}$ corresponds to a root of $Y^{m_\alpha} - f_{\alpha,j}$. Another rational section of $\mathcal{H}_\alpha$ multiplies $f_{\alpha,j}$ by an m_α-th power in $\mathbb{C}(X)^*$. Let L be the field generated over $\mathbb{C}(X)$ by roots $y_{\alpha,j}$ of the polynomials $Y^{m_\alpha} - f_{\alpha,j}$ in an algebraic closure of $\mathbb{C}(X)$, which does not depend on the rational sections. Since $\mathbb{C}(X)$ contains all roots of unity, L is the splitting field over $\mathbb{C}(X)$ of the polynomials $Y^{m_\alpha} - f_{\alpha,j}$, and any $\sigma \in \mathrm{Gal}(L/\mathbb{C}(X))$ satisfies $\sigma(y_{\alpha,j}) = \zeta_\alpha^{c_{\alpha,j}} y_{\alpha,j}$ for a unique $c_{\alpha,j} \in \mathbb{Z}/m_\alpha\mathbb{Z}$. The map $\sigma \to \sum_{\alpha,j} c_{\alpha,j} e_{\alpha,j}$ is an injective homomorphism $\mathrm{Gal}(L/\mathbb{C}(X)) \to A$.

We show that $[L : \mathbb{C}(X)] = |A|$. Order the pairs (α,j) as $q_1, \dots, q_{rN}$, and let $L_k \subset L$ be the field generated over $\mathbb{C}(X)$ by $y_{q_1}, \dots, y_{q_k}$. Fix k, write $q_k = (\alpha,j)$, and let P be an irreducible component of $B_{\alpha,j}$ and v_P the valuation of $\mathbb{C}(X)$ along P. Taking the rational sections non-vanishing at the generic point of P, we have $v_P(f_{q_k}) = 1$ and $v_P(f_q) = 0$ for $q \neq q_k$, because $D + \sum_{\beta,l} B_{\beta,l}$ is reduced and no two of the $B_{\beta,l}$ have a common irreducible component. For a unit f of $\mathcal{O}_{X,P}$ and an integer $n \geq 1$, the algebra $\mathcal{O}_{X,P}[Y]/(Y^n - f)$ is finite étale over $\mathcal{O}_{X,P}$, as Y and nY^{n-1} are units in it. The integral closure of $\mathcal{O}_{X,P}$ in L_{k-1} is a direct factor of the tensor product over $\mathcal{O}_{X,P}$ of the algebras $\mathcal{O}_{X,P}[Y]/(Y^{m_\beta} - f_{q_i})$, $i < k$, where $q_i = (\beta,l)$, so it is étale over $\mathcal{O}_{X,P}$. Thus $w(f_{q_k}) = 1$ for any discrete valuation w of L_{k-1} extending v_P, and $Y^{m_\alpha} - f_{q_k}$ is an Eisenstein polynomial for the valuation ring of w. By the Eisenstein criterion

[Lan02, Chapter IV, Theorem 3.1], $Y^{m_\alpha} - f_{q_k}$ is irreducible over L_{k-1}, so $[L_k : L_{k-1}] = m_\alpha$. By induction on k,

$$[L : \mathbb{C}(X)] = \prod_{\alpha=1}^{r} m_\alpha^N = |A|.$$

Therefore $\mathrm{Gal}(L/\mathbb{C}(X)) \to A$ is an isomorphism.

The cover $S_{\alpha,j}$ is the relative spectrum over X of the locally free algebra $\bigoplus_{k=0}^{m_\alpha-1} \mathcal{H}_\alpha^{-k}$, with multiplication given by $s_{D_\alpha} s_{B_{\alpha,j}}$. Let S' be the fiber product over X of the covers $S_{\alpha,j}$ and $\pi' : S' \to X$ the projection. Then π' is finite and locally free of rank $|A|$, and the generic fiber of π' is the spectrum of an algebra of dimension $|A|$ over $\mathbb{C}(X)$ which maps onto L. Hence the generic fiber of π' is $\operatorname{Spec} L$. Since $\pi'_* \mathcal{O}_{S'}$ is locally free, it injects into the generic fiber, so S' is integral. Since S' is of finite type over $\mathbb{C}$, the normalization $S \to S'$ is finite [Har77, Chapter I, Theorem 3.9A]. Thus π is finite, S is projective because X is projective, and S is connected because S is integral. The group A acts on S' over X by letting $e_{\alpha,j}$ multiply $y_{\alpha,j}$ by ζ_α, and the action of A on S' extends to the normalization S by the universal property of normalization. On $\mathbb{C}(S) = L$, the element $e_{\alpha,j}$ acts by the automorphism multiplying $y_{\alpha,j}$ by ζ_α, so A acts on L as $\mathrm{Gal}(L/\mathbb{C}(X))$. For an affine open subset $U \subset X$, any A-invariant regular function on $\pi^{-1}(U)$ is in $\mathbb{C}(X)$ and is integral over $\mathcal{O}_X(U)$, so it is in $\mathcal{O}_X(U)$ because X is normal. Thus $X = S/A$, and π is a finite Galois morphism with Galois group A.

Step 3. In this step we compute S near $\pi^{-1}(x)$ for $x \in X$, and we prove (2)–(5) and the smoothness of S.

Fix $x \in X$. At most d irreducible components of the simple normal crossing divisor $D + \sum_{\beta,l} B_{\beta,l}$ contain x. Distinct divisors $B_{\beta,l}$ have no common irreducible component, so for $\alpha \in I(x)$ the divisor D_α and the divisors $B_{\alpha,j}$ with $(\alpha, j) \in J(x)$ contribute at least $1 + |\{j \mid (\alpha, j) \in J(x)\}|$ distinct components through x. This proves (2). Since $N = d + 1$, for $\alpha \in I(x)$ there exist at least two indices j with $(\alpha, j) \notin J(x)$, and we fix one of them, j_α.

Let W be a polydisc centered at x in the coordinates z_α, $z_{\alpha,j}$, t_l, small enough that W intersects only the components of $D + \sum_{\beta,l} B_{\beta,l}$ through x and that each $\mathcal{H}_\alpha$ is trivial on W. In trivializations of $\mathcal{H}_\alpha$ on W, $s_{D_\alpha} s_{B_{\alpha,j}} = \lambda_{\alpha,j}^{m_\alpha} \phi_{\alpha,j}$, where $\lambda_{\alpha,j}$ is a nowhere vanishing holomorphic function on W, which exists because W is simply connected, and

$$\phi_{\alpha,j} = \begin{cases} z_\alpha z_{\alpha,j} & \text{if } \alpha \in I(x),\ (\alpha,j) \in J(x), \\ z_\alpha & \text{if } \alpha \in I(x),\ (\alpha,j) \notin J(x), \\ z_{\alpha,j} & \text{if } \alpha \notin I(x),\ (\alpha,j) \in J(x), \\ 1 & \text{if } \alpha \notin I(x),\ (\alpha,j) \notin J(x). \end{cases}$$

Replacing $y_{\alpha,j}$ with $y_{\alpha,j}/\lambda_{\alpha,j}$, which does not change the action of A because $\lambda_{\alpha,j}$ is a function on W, we may assume that $\pi'^{-1}(W)$ is defined by the equations $y_{\alpha,j}^{m_\alpha} = \phi_{\alpha,j}$.

Let $\mu_m \subset \mathbb{C}^*$ be the group of m-th roots of unity. Let $\mathcal{E}$ be the set of families $\epsilon = (\epsilon_{\alpha,j})_{(\alpha,j)\notin J(x)}$ with $\epsilon_{\alpha,j} \in \mu_{m_\alpha}$ and $\epsilon_{\alpha,j_\alpha} = 1$ for $\alpha \in I(x)$. Let $\widetilde{W}$ be the polydisc of points $(u_\alpha, v_{\alpha,j}, t_l)$ with $(u_\alpha^{m_\alpha}, v_{\alpha,j}^{m_\alpha}, t_l) \in W$, and for $\epsilon \in \mathcal{E}$ let $\psi_\epsilon : \widetilde{W} \to \pi'^{-1}(W)$ be the map given by $z_\alpha = u_\alpha^{m_\alpha}$, $z_{\alpha,j} = v_{\alpha,j}^{m_\alpha}$, $t_l = t_l$, and

$$y_{\alpha,j} = \begin{cases} u_\alpha v_{\alpha,j} & \text{if } \alpha \in I(x),\ (\alpha,j) \in J(x), \\ \epsilon_{\alpha,j} u_\alpha & \text{if } \alpha \in I(x),\ (\alpha,j) \notin J(x), \\ v_{\alpha,j} & \text{if } \alpha \notin I(x),\ (\alpha,j) \in J(x), \\ \epsilon_{\alpha,j} & \text{if } \alpha \notin I(x),\ (\alpha,j) \notin J(x). \end{cases}$$

The values $y_{\alpha,j}$ satisfy $y_{\alpha,j}^{m_\alpha} = \phi_{\alpha,j}$, so ψ_ϵ is well defined, and ψ_ϵ is finite. Let $W^\circ \subset W$ be the complement of the coordinate hyperplanes $z_\alpha = 0$ and $z_{\alpha,j} = 0$. Over W°, the map

$$\psi := \bigsqcup_{\epsilon\in\mathcal{E}} \psi_\epsilon : \mathcal{E} \times \widetilde{W} \to \pi'^{-1}(W)$$

is bijective, with inverse given by $u_\alpha = y_{\alpha,j_\alpha}$ for $\alpha \in I(x)$; $\epsilon_{\alpha,j} = y_{\alpha,j}/u_\alpha$ for $\alpha \in I(x)$ and $(\alpha,j) \notin J(x)$; $v_{\alpha,j} = y_{\alpha,j}/u_\alpha$ for $\alpha \in I(x)$ and $(\alpha,j) \in J(x)$; $\epsilon_{\alpha,j} = y_{\alpha,j}$ for $\alpha \notin I(x)$ and $(\alpha,j) \notin J(x)$; $v_{\alpha,j} = y_{\alpha,j}$ for $\alpha \notin I(x)$ and $(\alpha,j) \in J(x)$. Both $\psi^{-1}(\pi'^{-1}(W^\circ)) \subset \mathcal{E} \times \widetilde{W}$ and $\pi'^{-1}(W^\circ)$ are unramified covers of W°, so ψ is an isomorphism over W°. Since $\mathcal{E} \times \widetilde{W}$ is smooth and ψ is finite, ψ is the normalization of the complex space $\pi'^{-1}(W)$ [GR84, Chapter 8, §4]. The analytification of the normalization $S \to S'$ is the normalization of the complex space $(S')^{\mathrm{an}}$ [Gro71, Exp. XII, Proposition 2.1], [GR84, Chapter 8, §4]. Hence there exists an isomorphism $\pi^{-1}(W) \cong \mathcal{E} \times \widetilde{W}$ over $\pi'^{-1}(W)$, and A acts on $\mathcal{E} \times \widetilde{W}$ by the unique lift of the action on $\pi'^{-1}(W)$.

Each $W_\epsilon := \{\epsilon\} \times \widetilde{W}$ contains exactly one point s_ϵ over x, namely the origin, and $\pi^{-1}(x) = \{s_\epsilon \mid \epsilon \in \mathcal{E}\}$. On W_ϵ, $\pi^* z_\alpha = u_\alpha^{m_\alpha}$, $\pi^* z_{\alpha,j} = v_{\alpha,j}^{m_\alpha}$, and $\pi^* t_l = t_l$. This proves (3), and S is smooth because x is arbitrary. By **Step 2** and the smoothness of S, (1) holds.

We prove (4). Let $g = \sum_{\alpha,j} c_{\alpha,j} e_{\alpha,j} \in A$ with $c_{\alpha,j} \in \mathbb{Z}/m_\alpha\mathbb{Z}$, so that g multiplies $y_{\alpha,j}$ by $\zeta_\alpha^{c_{\alpha,j}}$. Comparing with the inverse of ψ over W° and taking closures, g maps W_ϵ onto $W_{\epsilon'}$, where

$$\epsilon'_{\alpha,j} = \zeta_\alpha^{c_{\alpha,j} - c_{\alpha,j_\alpha}} \epsilon_{\alpha,j} \text{ for } \alpha \in I(x), \qquad \epsilon'_{\alpha,j} = \zeta_\alpha^{c_{\alpha,j}} \epsilon_{\alpha,j} \text{ for } \alpha \notin I(x).$$

Moreover, g multiplies u_α by $\zeta_\alpha^{c_{\alpha,j_\alpha}}$, multiplies $v_{\alpha,j}$ by $\zeta_\alpha^{c_{\alpha,j} - c_{\alpha,j_\alpha}}$ if $\alpha \in I(x)$ and by $\zeta_\alpha^{c_{\alpha,j}}$ if $\alpha \notin I(x)$, and fixes the coordinates t_l. Since the sets W_ϵ are pairwise disjoint, g fixes s_ϵ if and only if $\epsilon' = \epsilon$, that is, $c_{\alpha,j} = c_{\alpha,j_\alpha}$ for $\alpha \in I(x)$ and $(\alpha,j) \notin J(x)$, and $c_{\alpha,j} = 0$ for $\alpha \notin I(x)$ and $(\alpha,j) \notin J(x)$. Setting $a_\alpha := c_{\alpha,j_\alpha}$ for $\alpha \in I(x)$, $b_{\alpha,j} := c_{\alpha,j} - a_\alpha$ for $\alpha \in I(x)$ and $(\alpha,j) \in J(x)$, and $b_{\alpha,j} := c_{\alpha,j}$ for $\alpha \notin I(x)$ and $(\alpha,j) \in J(x)$, the conditions $\epsilon' = \epsilon$ say exactly that

$$g = \sum_{\alpha\in I(x)} a_\alpha \tau_\alpha + \sum_{(\alpha,j)\in J(x)} b_{\alpha,j} e_{\alpha,j},$$

and then g multiplies u_α by $\zeta_\alpha^{a_\alpha}$ and $v_{\alpha,j}$ by $\zeta_\alpha^{b_{\alpha,j}}$. If such an element is 0, then for $\alpha \in I(x)$ the coefficient of e_{α,j_α} is $a_\alpha = 0$, and then all $b_{\alpha,j} = 0$. Thus the sum is direct, which proves (4).

We prove (5). Let $s \in S$ be a point fixed by $e_{\alpha,j}$, and let $x := \pi(s)$. If $(\alpha,j) \notin J(x)$ and $\alpha \notin I(x)$, then by (4) the coefficient of $e_{\alpha,j}$ in any element of the stabilizer of s is 0, a contradiction because $m_\alpha \geq 2$. If $(\alpha,j) \notin J(x)$ and $\alpha \in I(x)$, then by (2) there exists $j' \neq j$ with $(\alpha,j') \notin J(x)$, and by (4) the coefficients of $e_{\alpha,j}$ and of $e_{\alpha,j'}$ in any element of the stabilizer of s are both equal to a_α, whereas in $e_{\alpha,j}$ they are 1 and 0, a contradiction. Thus $x \in B_{\alpha,j}$. Conversely, if $x \in B_{\alpha,j}$, then by (4) the element $e_{\alpha,j}$ fixes any $s \in \pi^{-1}(x)$, multiplies $v_{\alpha,j}$ by ζ_α, and fixes the other coordinates. So near s the fixed locus of $e_{\alpha,j}$ is the smooth reduced hypersurface $v_{\alpha,j} = 0$, which is the reduced inverse image of $B_{\alpha,j}$ because $\pi^* z_{\alpha,j} = v_{\alpha,j}^{m_\alpha}$. Hence the fixed locus of $e_{\alpha,j}$ is the reduced inverse image of $B_{\alpha,j}$. Let G be an irreducible component of the fixed locus. Then $\dim G = d-1$ and π is finite, so $\pi(G)$ is an irreducible component P of $B_{\alpha,j}$. Let $P^\circ \subset P$ be the dense open subset of points not contained in any other irreducible component of $D + \sum_{\beta,l} B_{\beta,l}$. Then $G \cap \pi^{-1}(P^\circ)$ is a dense open subset of G, and for $s \in G \cap \pi^{-1}(P^\circ)$ we have $I(\pi(s)) = \emptyset$ and $J(\pi(s)) = \{(\alpha,j)\}$. By (4), the stabilizer of s is $\langle e_{\alpha,j} \rangle$. This proves (5). □

We use the following lemma in this section and for Proposition 5.2 in Section 5.

Lemma 4.4. *Let X be a normal projective variety, and let A be a finite group acting on X by automorphisms. Then there exist a normal projective variety X/A and a finite surjective morphism $\sigma : X \to X/A$ such that the fibers of σ are the A-orbits and $\mathcal{O}_{X/A} = (\sigma_* \mathcal{O}_X)^A$.*

Proof. Since X is projective, any A-orbit in X is contained in an affine open subset. Thus by [Mum70, §7, Theorem (p. 66)], the quotient X/A exists, the quotient morphism $\sigma : X \to X/A$ is

finite and surjective, the fibers of σ are the A-orbits, and

$$\mathcal{O}_{X/A} = (\sigma_* \mathcal{O}_X)^A.$$

We show that X/A is projective. Let $\mathcal{L}_0$ be an ample line bundle on X, and let $\mathcal{L} := \bigotimes_{g\in A} g^*\mathcal{L}_0$. Then $\mathcal{L}$ is ample, and the permutation of the factors is an A-linearization of $\mathcal{L}$. Let

$$R := \bigoplus_{k\geq 0} H^0(X, \mathcal{L}^k).$$

The ring R is a finitely generated $\mathbb{C}$-algebra, and any $t \in R$ is a root of the monic polynomial $\prod_{g\in A}(T - g \cdot t)$, whose coefficients are in R^A. Hence R is integral over R^A. Since R is a finitely generated $\mathbb{C}$-algebra, it is a finite R^A-module, so R^A is a finitely generated $\mathbb{C}$-algebra [Sta, Tag 00IS]. Let $z \in X$. Since $\mathcal{L}$ is ample and the orbit Az is finite, there exist $k \geq 1$ and $t \in H^0(X, \mathcal{L}^k)$ that does not vanish at any point of Az. Then $\prod_{g\in A} g \cdot t \in R^A$ does not vanish at z. Thus X is covered by the A-invariant affine open subsets X_s, where $s \in R^A$ is homogeneous of positive degree. The quotient of X_s by A is the spectrum of the degree-zero part of $(R^A)_s$, and the spectra of the degree-zero parts of the rings $(R^A)_s$ glue to $\operatorname{Proj} R^A$. Hence $X/A \cong \operatorname{Proj} R^A$ is projective.

Let $\operatorname{Spec} B \subset X$ be an A-invariant affine open subset. Since X is normal and irreducible, B is a normal domain. An element of the fraction field of B^A that is integral over B^A is integral over B, hence it belongs to B, and hence to B^A since it is A-invariant. Thus X/A is a normal variety. □

The quotient Q in Construction-Proposition 4.5 below satisfies most of the hypotheses of Proposition 5.5. For a connected smooth projective variety Y with a birational morphism $\rho : Y \to Q$ which is an isomorphism over $h^{-1}(V)$, condition (1) of Proposition 5.5 follows from (3) below, condition (2) from (6), condition (4) from (5), and condition (3) for the prime divisors on X other than the D_α and the irreducible components of the $B_{\alpha,j}$ from (6), since such a divisor intersects U. Here (3) holds for any proper birational ρ from a smooth variety, since the roots of the local equations of the D_α exist at any point of Q, also over the intersections of the D_α. The remaining part of condition (3), for the irreducible components of the $B_{\alpha,j}$, is treated in Proposition 5.2, whose morphism ρ is an isomorphism over $h^{-1}(V)$, so that (5) and (6) persist on Y.

Construction-Proposition 4.5. Notation and conditions as in Set-up 4.1. Let $\pi : S \to X$ be the Kummer cover of Construction-Proposition 4.3, with Galois group A, basis elements $e_{\alpha,j}$, and divisors $B_{\alpha,j}$. For $1 \leq \alpha \leq r$ let E_α be a complex elliptic curve and $p_\alpha \in E_\alpha$ a point of order exactly m_α, and let F be the abelian variety with the action of A of Construction-Proposition 4.2 for the integers r, N, and m_α of Set-up 4.1, the curves E_α, and the points p_α, where the basis elements $e_{\alpha,j}$ of Construction-Proposition 4.2 are those of Construction-Proposition 4.3. Let A act diagonally on $S \times F$, let $Q := (S \times F)/A$, and let $h : Q \to X$ be the morphism induced by π. Let $V := X \setminus \bigcup_{\alpha,j} B_{\alpha,j}$ and $U := V \setminus D$. Then the following hold.

(1) The variety Q is connected, normal, and projective.
(2) For every $y \in Q$ and every α with $h(y) \in D_\alpha$, if z_α is a local equation of D_α near $h(y)$, then $h^* z_\alpha = w_\alpha^{m_\alpha}$ for a holomorphic function w_α near y.
(3) If Y is a smooth variety and $\rho : Y \to Q$ is a proper birational morphism, then m_α divides $\operatorname{mult}_P (h \circ \rho)^* D_\alpha$ for every prime divisor P on Y and every α.
(4) The variety $h^{-1}(V)$ is smooth.
(5) For every $y \in h^{-1}(V)$, every system of holomorphic local coordinates z_α, for the α with $h(y) \in D_\alpha$, and $t_1, \ldots, t_c$ at $h(y)$ such that $D_\alpha = \{z_\alpha = 0\}$ near $h(y)$, where $c := d - |\{\alpha \mid h(y) \in D_\alpha\}|$, and every choice of functions w_α as in (2), the map $((w_\alpha)_\alpha, h^* t_1, \ldots, h^* t_c)$ is a submersion at y.
(6) The restriction $h^{-1}(U) \to U$ is an analytically locally trivial fiber bundle with fiber F.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct the quotient morphism $\sigma : S \times F \to Q$ and prove (1).

By Construction-Proposition 4.3(1), S is a connected smooth projective variety, so $S \times F$ is a connected smooth projective variety on which A acts diagonally by automorphisms. By Lemma 4.4, the quotient Q is a normal projective variety, the quotient morphism $\sigma : S \times F \to Q$ is finite and surjective, the fibers of σ are the A-orbits, and $\mathcal{O}_Q = (\sigma_* \mathcal{O}_{S\times F})^A$. Since σ is surjective and $S \times F$ is connected, Q is connected. This proves (1). The morphism $\pi \circ \mathrm{pr}_S : S \times F \to X$ is A-invariant, so it factors through σ, and

$$h \circ \sigma = \pi \circ \mathrm{pr}_S .$$

Step 2. In this step we show that the stabilizer of any point of $S \times F$ fixes the root coordinates u_α of Construction-Proposition 4.3(3).

Let $(s, f) \in S \times F$, let $x := \pi(s)$, and let $H \subset A$ be the stabilizer of (s, f). Let z_α $(\alpha \in I(x))$, $z_{\alpha,j}$ $((\alpha, j) \in J(x))$, and $t_1, \dots, t_c$ be holomorphic local coordinates at x such that $D_\alpha = \{z_\alpha = 0\}$ and $B_{\alpha,j} = \{z_{\alpha,j} = 0\}$ near x, and let u_α, $v_{\alpha,j}$, and $t_1, \dots, t_c$ be the local coordinates of S at s of Construction-Proposition 4.3(3). Since any $g \in H$ fixes s, by Construction-Proposition 4.3(4) we may write

$$g = \sum_{\alpha \in I(x)} a_\alpha \tau_\alpha + \sum_{(\alpha,j)\in J(x)} b_{\alpha,j} e_{\alpha,j}.$$

We show that $a_\alpha = 0$ for any $\alpha \in I(x)$. Suppose that $a_\alpha \neq 0$ for some $\alpha \in I(x)$. By Construction-Proposition 4.3(2), at most $d - 1 = N - 2$ indices j satisfy $(\alpha, j) \in J(x)$. Thus there exist indices $i \neq j$ such that $(\alpha, i) \notin J(x)$ and $(\alpha, j) \notin J(x)$. The coefficients of $e_{\alpha,i}$ and of $e_{\alpha,j}$ in g are both equal to a_α, so g acts on F without fixed points by Construction-Proposition 4.2(2). This contradicts $g(f) = f$. Therefore $a_\alpha = 0$ for any $\alpha \in I(x)$, and by Construction-Proposition 4.3(4), any $g \in H$ satisfies

$$u_\alpha \circ g = u_\alpha \quad (\alpha \in I(x)) \qquad \text{and} \qquad t_l \circ g = t_l \quad (1 \leq l \leq c)$$

near s.

Step 3. In this step we show that holomorphic functions on Q near a point are H-invariant holomorphic functions on $S \times F$, and we prove (2).

Let $y \in Q$, let $(s, f) \in \sigma^{-1}(y)$, and let x, H, and the coordinates be as in **Step 2**. Then $x = h(y)$. Let W be an H-invariant open neighborhood of (s, f) in $(S \times F)^{\mathrm{an}}$ such that the coordinates u_α, $v_{\alpha,j}$, and t_l are defined on $\mathrm{pr}_S(W)$ and $g(W) \cap W = \emptyset$ for any $g \in A \backslash H$. Since the fibers of σ are the A-orbits, and since A acts freely on $\pi^{-1}(U)$ by Construction-Proposition 4.3(4), the morphism $W/H \to Q^{\mathrm{an}}$ induced by σ is a finite, bijective, and bimeromorphic map onto an open neighborhood of y. Here $\sigma(W)$ is open, since $Q^{\mathrm{an}} \backslash \sigma(W)$ is the image of the closed A-invariant subset $(S \times F)^{\mathrm{an}} \backslash \bigcup_{g\in A} g(W)$ under the finite, hence closed, map σ, and the fibers of σ are the A-orbits. The complex space W/H is normal [Car57, Théorème 4], and Q^{an} is normal since Q is normal [Gro71, Exp. XII, Proposition 2.1]. Hence $W/H \to Q^{\mathrm{an}}$ is an isomorphism onto an open neighborhood of y [GR84, Chapter 8, §4]. Thus the holomorphic functions near y are exactly the H-invariant holomorphic functions on W.

Let α be such that $h(y) \in D_\alpha$, so $\alpha \in I(x)$. By **Step 2**, $u_\alpha \circ \mathrm{pr}_S$ is H-invariant on W, so there exists a holomorphic function w_α near y such that $w_\alpha \circ \sigma = u_\alpha \circ \mathrm{pr}_S$ on W. Since $\pi^* z_\alpha = u_\alpha^{m_\alpha}$ and $h \circ \sigma = \pi \circ \mathrm{pr}_S$, we obtain $h^* z_\alpha = w_\alpha^{m_\alpha}$ near y. Let z'_α be any local equation of D_α near x. Then $z'_\alpha = \epsilon z_\alpha$ for a holomorphic function ϵ without zeros near x. Possibly shrinking the neighborhood of x, we may assume that $\epsilon = \eta^{m_\alpha}$ for a holomorphic function η. Then

$$h^* z'_\alpha = \big((h^*\eta) w_\alpha\big)^{m_\alpha}$$

near y. This proves (2).

Step 4. In this step we prove (3).

Let Y be a smooth variety, $\rho : Y \to Q$ a proper birational morphism, P a prime divisor on Y, and $1 \le \alpha \le r$. If $(h \circ \rho)(P) \not\subset D_\alpha$, then $\mathrm{mult}_P (h \circ \rho)^* D_\alpha = 0$. Assume that $(h \circ \rho)(P) \subset D_\alpha$. Let y' be a general point of P, let $y := \rho(y')$, and let z_α be a local equation of D_α near $h(y)$. By (2), $h^* z_\alpha = w_\alpha^{m_\alpha}$ for a holomorphic function w_α near y. Since Y is smooth, $\mathrm{mult}_P (h \circ \rho)^* D_\alpha$ is the order of vanishing along P of the local equation $(h \circ \rho)^* z_\alpha$ of $(h \circ \rho)^* D_\alpha$ near y'. Thus

$$\mathrm{mult}_P (h \circ \rho)^* D_\alpha = \mathrm{ord}_P \left((\rho^* w_\alpha)^{m_\alpha} \right) = m_\alpha \, \mathrm{ord}_P (\rho^* w_\alpha),$$

which is divisible by m_α. This proves (3).

Step 5. In this step we show that A acts freely over V, and we prove (4) and (5).

Let $y \in h^{-1}(V)$, let $(s, f) \in \sigma^{-1}(y)$, and let $x := h(y)$. Since $x \in V$, the set $J(x)$ is empty, so the stabilizer H of (s, f) is trivial by **Step 2**. Thus A acts freely on $\pi^{-1}(V) \times F$, and by **Step 3**, σ maps a neighborhood W of (s, f) biholomorphically onto a neighborhood of y. Since $S \times F$ is smooth, $h^{-1}(V)$ is smooth. This proves (4).

Let z_α ($\alpha \in I(x)$) and $t_1, \dots, t_c$, where $c = d - |I(x)|$, be holomorphic local coordinates at x such that $D_\alpha = \{z_\alpha = 0\}$ near x, and let w_α ($\alpha \in I(x)$) be holomorphic functions near y with $h^* z_\alpha = w_\alpha^{m_\alpha}$. Since $J(x)$ is empty, Construction-Proposition 4.3(3) implies that S has holomorphic local coordinates u_α ($\alpha \in I(x)$) and $\pi^* t_1, \dots, \pi^* t_c$ at s with $\pi^* z_\alpha = u_\alpha^{m_\alpha}$. Together with holomorphic local coordinates of F at f, the pullbacks to $S \times F$ of u_α ($\alpha \in I(x)$) and $\pi^* t_1, \dots, \pi^* t_c$ form holomorphic local coordinates at (s, f). Since

$$\prod_{\epsilon^{m_\alpha} = 1} \left(w_\alpha \circ \sigma - \epsilon u_\alpha \circ \mathrm{pr}_S \right) = (w_\alpha \circ \sigma)^{m_\alpha} - (u_\alpha \circ \mathrm{pr}_S)^{m_\alpha} = 0$$

and the ring of germs of holomorphic functions at (s, f) is a domain, $w_\alpha \circ \sigma = \epsilon_\alpha u_\alpha \circ \mathrm{pr}_S$ near (s, f) for an m_α-th root of unity ϵ_α. Therefore, in the local coordinates at (s, f) above,

$$\left((w_\alpha)_\alpha, h^* t_1, \dots, h^* t_c \right) \circ \sigma = \left((\epsilon_\alpha u_\alpha)_\alpha, t_1, \dots, t_c \right)$$

is a submersion at (s, f). Since $\sigma|_W$ is biholomorphic, $((w_\alpha)_\alpha, h^* t_1, \dots, h^* t_c)$ is a submersion at y. This proves (5).

Step 6. In this step we prove (6).

Let $x \in U$. Then $I(x)$ and $J(x)$ are empty, so by Construction-Proposition 4.3(3) and (4), π is a local biholomorphism near any point of $\pi^{-1}(x)$, and A acts freely on $\pi^{-1}(U)$. Since π is a finite Galois morphism with Galois group A, the fiber $\pi^{-1}(x)$ is one A-orbit. Let $U_x \subset U$ be an open ball around x in a holomorphic chart. Then $\pi^{-1}(U_x)$ is a disjoint union of $|A|$ open subsets, each of which is mapped biholomorphically onto U_x by π, and A permutes them simply transitively. Let U_x' be one of them.

By **Step 5**, σ restricted to $\pi^{-1}(V) \times F$ is a local biholomorphism whose fibers are the A-orbits. Moreover, $h^{-1}(U_x) = \sigma(\pi^{-1}(U_x) \times F)$ since $h \circ \sigma = \pi \circ \mathrm{pr}_S$ and σ is surjective. Any A-orbit in $\pi^{-1}(U_x) \times F$ contains exactly one point of $U_x' \times F$, so σ maps $U_x' \times F$ bijectively, and hence biholomorphically, onto $h^{-1}(U_x)$. Composing the inverse of $\sigma|_{U_x' \times F}$ with $\pi|_{U_x'} \times \mathrm{id}_F$, we obtain a biholomorphic map $h^{-1}(U_x) \to U_x \times F$ over U_x. Thus $h^{-1}(U) \to U$ is an analytically locally trivial fiber bundle with fiber F. This proves (6). □

5. The projective case

In this section, we prove Theorem 5.1, the projective case of the main theorem, without using the induction of Section 7. Section 7 uses Theorem 5.1 for compact Kähler orbifold pairs (X, Δ) with $a(X) = \dim X$, and, when $0 < a(X) < \dim X$, for the gcd base $(Z, \Delta_{\mathrm{gcd}}(f))$ (Definition 7.1) of a holomorphic model $f : X' \to Z$ of the algebraic reduction of X, defined on a modification X' of X (Proposition 7.5), where Z is a smooth projective variety. In Subsection 5.3, we construct a special smooth projective variety Y with a surjective homomorphism from $\pi_1(Y)$ onto $\pi_1^{\mathrm{orb}}(X, \Delta)$, and Theorem 2.13 shows that $\pi_1(Y)$ is virtually abelian.

Theorem 5.1. *Let (X, Δ) be an orbifold pair such that X is a smooth projective variety. Assume that (X, Δ) is special. Then $\pi_1^{\mathrm{orb}}(X, \Delta)$ is virtually abelian.*

5.1. **Extraction of auxiliary divisors.** In this subsection, we prove Proposition 5.2, which, applied in Subsection 5.3 to the irreducible components of the divisors $B_{\alpha,j}$ of Construction-Proposition 4.3, gives a prime divisor of multiplicity one over each of them. Remark 5.3 explains why blowing up the fixed loci once, as in [OAI26b, Proposition A.3], does not suffice for this purpose.

In the proof, for each prime divisor in the statement, we define a weighted monomial valuation v of $\mathbb{C}(S \times F)$ at the generic point of a component of a fixed locus, and we extract the restriction of v to $\mathbb{C}(Q)$, divided by the order of the stabilizer, by [KM98, Lemma 2.45]. The local parameters used to define v are parameters only of the local ring at that generic point, and the stabilizers of other points of that component can be larger. Thus we use v only at the generic point, and we do not construct a global weighted blow-up or a toric resolution of Q.

Condition (3) of Proposition 5.5 asks that, over a dense open subset of any prime divisor of X not contained in $\operatorname{Supp}\Delta$, each fiber of g contain a point at which g is a submersion. As explained before Construction-Proposition 4.5, for the quotient Q condition (3) remains to be shown only over the irreducible components of the $B_{\alpha,j}$, and the last conclusion of Proposition 5.2 implies it there.

Proposition 5.2. *Let $\pi : S \to X$ be a finite Galois morphism between connected smooth projective varieties with Galois group A, let F be a connected smooth projective variety with an action of A, let $Q := (S \times F)/A$ for the diagonal action, and let $h : Q \to X$ be the morphism induced by π. Let $P_1, \dots, P_b$ be distinct prime divisors on X. Assume that for each i there exists a prime divisor T_i on S with $\pi(T_i) = P_i$ such that the subgroup $H_i \leq A$ of elements acting trivially on the function field of T_i is cyclic of order $m_i \geq 2$ and acts faithfully on the conormal line of T_i, and that the fixed locus F^{H_i} is non-empty. Then there exist a smooth projective variety Y, a projective birational morphism $\rho : Y \to Q$ which is an isomorphism over $Q_{\mathrm{reg}} \setminus h^{-1}(P_1 \cup \dots \cup P_b)$, and prime divisors $E_1, \dots, E_b$ on Y such that, for $g := h \circ \rho$ and any i, we have $g(E_i) = P_i$ and $\operatorname{mult}_{E_i} g^* P_i = 1$, and g is a submersion at every point of a dense open subset of E_i whose image contains a dense open subset of P_i.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix i and construct eigenfunctions of a generator of H_i that form a regular system of parameters at the generic point of $T_i \times V_i$, where V_i is a component of F^{H_i}. We keep i fixed until the end of **Step 5**.

Let $n := \dim Q = \dim(S \times F)$. By Lemma 4.4, applied to the action of A on the connected smooth projective variety $S \times F$, the variety Q is normal and projective, the quotient morphism $S \times F \to Q$ is finite and surjective, and $\mathcal{O}_Q$ is the sheaf of A-invariant functions on $S \times F$. In particular, $\mathbb{C}(Q) = \mathbb{C}(S \times F)^A$. Let z_i be a uniformizer of the DVR $\mathcal{O}_{X,\eta_{P_i}}$, where η_{P_i} is the generic point of P_i.

Let σ_i be a generator of H_i. Since H_i acts trivially on $\mathbb{C}(T_i)$, the group H_i fixes T_i pointwise. Since H_i acts faithfully on the conormal line of T_i, there exists a primitive m_i-th root of unity ζ such that $\sigma_i(t) - \zeta t \in \mathfrak{m}^2_{S,\eta_{T_i}}$ for any uniformizer t of $\mathcal{O}_{S,\eta_{T_i}}$. Since π is Galois with inertia group H_i along T_i and the characteristic is zero, the ramification index of π along T_i is $|H_i| = m_i$ [Sta, Tag 09EE]. Fix a uniformizer t of $\mathcal{O}_{S,\eta_{T_i}}$ and let

$$u := \frac{1}{m_i} \sum_{k=0}^{m_i - 1} \zeta^{-k} \sigma_i^k(t).$$

Then $\sigma_i(u) = \zeta u$ and $u - t \in \mathfrak{m}^2_{S,\eta_{T_i}}$, so u is a uniformizer of $\mathcal{O}_{S,\eta_{T_i}}$, and

$$\pi^* z_i = u^{m_i} \epsilon \tag{5.1}$$

for a unit ϵ of $\mathcal{O}_{S,\eta_{T_i}}$.

Let V_i be an irreducible component of F^{H_i}, and let $r := \operatorname{codim}_F V_i$. By [Car57, Lemme 1], at any closed point x of V_i there exist local holomorphic coordinates in which σ_i acts linearly. Hence F^{H_i} is the fixed linear subspace near x, so V_i is smooth, and σ_i has no eigenvalue 1 on the normal space of V_i at x. The conormal sheaf of V_i in F is locally free and splits into eigensheaves of σ_i. Thus there exist integers $w_1, \dots, w_r \in \{1, \dots, m_i - 1\}$ and a basis of $\mathfrak{m}_{F,\eta_{V_i}}/\mathfrak{m}^2_{F,\eta_{V_i}}$ over $\mathbb{C}(V_i)$ on which σ_i acts by $\zeta^{w_1}, \dots, \zeta^{w_r}$. Lift the basis vectors to $a'_1, \dots, a'_r \in \mathfrak{m}_{F,\eta_{V_i}}$ and let

$$a_j := \frac{1}{m_i} \sum_{k=0}^{m_i-1} \zeta^{-kw_j} \sigma_i^k(a'_j) \quad \text{for } 1 \le j \le r.$$

Then $\sigma_i(a_j) = \zeta^{w_j} a_j$ and $a_j - a'_j \in \mathfrak{m}^2_{F,\eta_{V_i}}$, so $a_1, \dots, a_r$ form a regular system of parameters of $\mathcal{O}_{F,\eta_{V_i}}$.

Let $C_0 := T_i \times V_i$, $R := \mathcal{O}_{S\times F,\eta_{C_0}}$, and $k_0 := \mathbb{C}(C_0)$. Then $\dim C_0 = n - r - 1$, and R is a regular local ring of dimension $r + 1$ with residue field k_0. Since $C_0 = T_i \times V_i$, the pullbacks of $u, a_1, \dots, a_r$ to $S \times F$, which we again denote by $u, a_1, \dots, a_r$, form a regular system of parameters of R. The group H_i fixes C_0 pointwise, so H_i acts on R and acts trivially on k_0. Since ϵ is a unit at η_{T_i} and C_0 dominates T_i, the pullback of ϵ is a unit of R.

Step 2. In this step, we show that the completion $\hat{R}$ of R is a power series ring in $u, a_1, \dots, a_r$ over a coefficient field fixed by H_i.

The action of H_i on R extends continuously to $\hat{R}$. Let $\bar{y}_1, \dots, \bar{y}_e$ be a transcendence basis of k_0 over $\mathbb{C}$. Choose $y'_l \in R$ with residue $\bar{y}_l$, and let

$$y_l := \frac{1}{m_i} \sum_{k=0}^{m_i-1} \sigma_i^k(y'_l) \in R^{H_i}.$$

Since H_i acts trivially on k_0, the residue of y_l is $\bar{y}_l$. For any non-zero polynomial $\Phi \in \mathbb{C}[Y_1, \dots, Y_e]$, the residue of $\Phi(y_1, \dots, y_e)$ is $\Phi(\bar{y}_1, \dots, \bar{y}_e) \neq 0$, so $\Phi(y_1, \dots, y_e)$ is a unit of R. Thus R^{H_i} contains the field $E_0 := \mathbb{C}(y_1, \dots, y_e)$, and the residue map sends E_0 isomorphically onto $\mathbb{C}(\bar{y}_1, \dots, \bar{y}_e)$.

Since the characteristic is zero, k_0 is a finite separable extension of $\mathbb{C}(\bar{y}_1, \dots, \bar{y}_e)$. Let $\bar{\theta}$ be a primitive element, and let $p \in E_0[X]$ be the polynomial whose image in $\mathbb{C}(\bar{y}_1, \dots, \bar{y}_e)[X]$ is the minimal polynomial of $\bar{\theta}$. Since $\bar{\theta}$ is a simple root of the minimal polynomial of $\bar{\theta}$, since $\hat{R}$ is henselian [Sta, Tags 04GM and 06RR], there exists a unique $\theta \in \hat{R}$ with $p(\theta) = 0$ and residue $\bar{\theta}$. The coefficients of p are H_i-invariant and H_i acts trivially on k_0, so $\sigma_i(\theta)$ is a root of p with residue $\bar{\theta}$. Hence $\sigma_i(\theta) = \theta$ by uniqueness. Let $k'_0 := E_0(\theta) \subset \hat{R}^{H_i}$. Then the residue map sends k'_0 isomorphically onto k_0, so k'_0 is a coefficient field of $\hat{R}$ fixed by H_i.

The ring $\hat{R}$ is a complete regular local ring of dimension $r + 1$. By [Sta, Tag 0C0S], $\hat{R}$ is a formal power series ring in $r + 1$ variables over k'_0. Since $u, a_1, \dots, a_r$ generate the maximal ideal of $\hat{R}$, it follows that

$$\hat{R} = k'_0[[u, a_1, \dots, a_r]].$$

For $\lambda = (\lambda_0, \dots, \lambda_r) \in \mathbb{Z}^{r+1}_{\ge 0}$, let $\mu_\lambda := u^{\lambda_0} a_1^{\lambda_1} \cdots a_r^{\lambda_r}$ and $|\lambda|_w := \lambda_0 + w_1\lambda_1 + \cdots + w_r\lambda_r$. Since σ_i fixes k'_0, we have

$$\sigma_i\Big(\sum_\lambda c_\lambda \mu_\lambda\Big) = \sum_\lambda \zeta^{|\lambda|_w} c_\lambda \mu_\lambda \quad \text{for all } c_\lambda \in k'_0. \tag{5.2}$$

Step 3. In this step, we define a weighted monomial valuation v of $\mathbb{C}(S \times F)$.

For a non-zero $f = \sum_\lambda c_\lambda \mu_\lambda \in \hat{R}$ with $c_\lambda \in k'_0$, let

$$v(f) := \min\{|\lambda|_w \mid c_\lambda \neq 0\}, \qquad f_v := \sum_{|\lambda|_w = v(f)} c_\lambda \mu_\lambda.$$

Since $w_1, \dots, w_r \geq 1$, only finitely many λ satisfy $|\lambda|_w = v(f)$, so f_v is a non-zero element of the polynomial ring $k_0'[u, a_1, \dots, a_r]$. For non-zero $f, g \in \hat{R}$, the term of fg of weighted degree $v(f) + v(g)$ is $f_v g_v \neq 0$. Thus $v(fg) = v(f) + v(g)$. By definition, $v(f+g) \geq \min\{v(f), v(g)\}$ when $f + g \neq 0$. Hence v extends to a valuation of $\mathrm{Frac}(\hat{R})$ by $v(f/g) := v(f) - v(g)$. Since $R \subset \hat{R}$, the restriction of v to $\mathbb{C}(S \times F) = \mathrm{Frac}(R)$ is a valuation, which we also denote by v. Since $v(u) = 1$, we have $v(\mathbb{C}(S \times F)^*) = \mathbb{Z}$. A unit of $\hat{R}$ has a non-zero constant term, so v vanishes on the units of $\hat{R}$, on $\mathbb{C}^*$, and on ϵ. By (5.1),

$$v(\pi^* z_i) = m_i. \tag{5.3}$$

Step 4. In this step, we compute the residue field $\kappa(v)$ of v on $\mathbb{C}(S \times F)$ and show that $\mathrm{trdeg}_{\mathbb{C}}\, \kappa(v) = n - 1$.

Since $v(a_j) = w_j = v(u^{w_j})$, the residue $\bar{a}_j := \mathrm{res}(a_j/u^{w_j}) \in \kappa(v)$ is defined for $1 \leq j \leq r$. We show that

$$\kappa(v) = k_0(\bar{a}_1, \dots, \bar{a}_r)$$

with $\bar{a}_1, \dots, \bar{a}_r$ algebraically independent over k_0.

First, k_0 embeds in $\kappa(v)$. Any element of k_0 is the residue of an element of R. If $x, x' \in R$ have equal residues in k_0, then $x - x' \in \mathfrak{m}_R$ and $v(x - x') > 0$, so x and x' have equal residues in $\kappa(v)$. Moreover, for $c \in k_0'$ and $x \in R$ whose residues in k_0 are equal, we have $x - c \in \mathfrak{m}_{\hat{R}}$. Thus the embedding of k_0 via R coincides with the embedding via k_0' in the residue field of v on $\mathrm{Frac}(\hat{R})$.

Second, $\bar{a}_1, \dots, \bar{a}_r$ are algebraically independent over k_0. Let $\Phi = \sum_\beta c_\beta X^\beta$ be a non-zero polynomial in r variables with coefficients $c_\beta \in k_0'$, and let $M := \max\{w_1\beta_1 + \cdots + w_r\beta_r \mid c_\beta \neq 0\}$. Then

$$u^M \Phi\Big(\frac{a_1}{u^{w_1}}, \dots, \frac{a_r}{u^{w_r}}\Big) = \sum_\beta c_\beta u^{M - w_1\beta_1 - \cdots - w_r\beta_r} a_1^{\beta_1} \cdots a_r^{\beta_r}$$

is a non-zero element of $\hat{R}$ all of whose terms have weighted degree M, since distinct exponents β yield distinct monomials. Hence $v(\Phi(a_1/u^{w_1}, \dots, a_r/u^{w_r})) = 0$, and $\Phi(\bar{a}_1, \dots, \bar{a}_r) \neq 0$.

Third, the residue field of v on $\mathrm{Frac}(\hat{R})$ is contained in $k_0(\bar{a}_1, \dots, \bar{a}_r)$. Let $f, g \in \hat{R}$ with $v(f) = v(g) =: \ell$. Since $v(f - f_v) > \ell$ and $v(g - g_v) > \ell$,

$$v\Big(\frac{f}{g} - \frac{f_v}{g_v}\Big) = v\big((f - f_v)g_v - (g - g_v)f_v\big) - v(gg_v) > 2\ell - 2\ell = 0.$$

Hence the residue of f/g equals the residue of $(u^{-\ell} f_v)/(u^{-\ell} g_v)$. Writing $f_v = \sum_\lambda c_\lambda \mu_\lambda$ with $\lambda_0 = \ell - w_1\lambda_1 - \cdots - w_r\lambda_r$, we obtain

$$u^{-\ell} f_v = \sum_\lambda c_\lambda \Big(\frac{a_1}{u^{w_1}}\Big)^{\lambda_1} \cdots \Big(\frac{a_r}{u^{w_r}}\Big)^{\lambda_r},$$

and similarly for g_v. Thus the residue of f/g belongs to $k_0(\bar{a}_1, \dots, \bar{a}_r)$.

The three statements imply $\kappa(v) = k_0(\bar{a}_1, \dots, \bar{a}_r)$. Since $\mathrm{trdeg}_{\mathbb{C}}\, k_0 = \dim C_0 = n - r - 1$, we obtain $\mathrm{trdeg}_{\mathbb{C}}\, \kappa(v) = n - 1$.

Step 5. In this step, we show that $v_i := m_i^{-1} v|_{\mathbb{C}(Q)}$ is a valuation of $\mathbb{C}(Q)$ with $v_i(\mathbb{C}(Q)^*) = \mathbb{Z}$ and $\mathrm{trdeg}_{\mathbb{C}}\, \kappa(v_i) = n - 1$, and that $v_i|_{\mathbb{C}(X)} = \mathrm{ord}_{P_i}$.

Let $f = \sum_\lambda c_\lambda \mu_\lambda \in \hat{R}$ be non-zero and H_i-invariant. By (5.2), $c_\lambda \neq 0$ implies $\zeta^{|\lambda|_w} = 1$, that is, $m_i \mid |\lambda|_w$. Hence $v(f) \in m_i\mathbb{Z}$. Now let $f \in \mathbb{C}(S \times F)^{H_i}$ be non-zero and write $f = f_1/f_2$ with $f_1, f_2 \in R$. Let $f_3 := \prod_{k=0}^{m_i - 1} \sigma_i^k(f_2)$. Then f_3 and $ff_3 = f_1 \prod_{k=1}^{m_i - 1} \sigma_i^k(f_2)$ are non-zero H_i-invariant elements of R. Thus

$$v(f) = v(ff_3) - v(f_3) \in m_i\mathbb{Z}.$$

Since $\mathbb{C}(Q) = \mathbb{C}(S \times F)^A \subset \mathbb{C}(S \times F)^{H_i}$, the function v_i is an integer-valued valuation of $\mathbb{C}(Q)$. By (5.3), $v_i(h^* z_i) = 1$, so $v_i(\mathbb{C}(Q)^*) = \mathbb{Z}$.

The valuation rings of v_i and $v|_{\mathbb{C}(Q)}$ coincide, so $\kappa(v_i) \subset \kappa(v)$. Since $\mathbb{C}(S \times F)$ is a finite extension of $\mathbb{C}(Q)$, the extension $\kappa(v)/\kappa(v_i)$ is finite [Bou89, VI, §8, no. 1, Lemma 2]. By **Step 4**, $\operatorname{trdeg}_{\mathbb{C}} \kappa(v_i) = n - 1$.

Since $\pi(T_i) = P_i$, the subvariety C_0 of $S \times F$ dominates P_i. Hence any unit of $\mathcal{O}_{X,\eta_{P_i}}$ pulls back to a unit of R and has value 0 under v_i. Since moreover $v_i(z_i) = 1$, we obtain

$$v_i|_{\mathbb{C}(X)} = \operatorname{ord}_{P_i}. \tag{5.4}$$

Step 6. In this step, we construct Y, ρ, and $E_1, \dots, E_b$ with $g(E_i) = P_i$ and $\operatorname{mult}_{E_i} g^* P_i = 1$.

Steps 1–5 apply to any index i and construct valuations $v_1, \dots, v_b$ of $\mathbb{C}(Q)$. Let $Q_0 := Q$. For $k = 1, \dots, b$ in turn, we proceed as follows. By **Step 5**, the valuation ring of v_k is a DVR with residue field of transcendence degree $n - 1 = \dim Q_{k-1} - 1$. By [KM98, Lemma 2.45], blowing up the closure of the center of v_k on Q_{k-1} repeatedly, we obtain a projective birational morphism $Q_k \to Q_{k-1}$ [Har77, Chapter II, Proposition 7.16] and a point $q_k \in Q_k$ whose local ring is the valuation ring of v_k. Let E'_k be the closure of q_k. Since the local ring is a DVR, E'_k is a prime divisor on Q_k.

We use two facts. First, by (5.4), the center of v_k on X is η_{P_k}. Hence, for any proper birational morphism $Q' \to Q$, the center of v_k on Q' maps into P_k. In particular, the generic point of E'_k maps to η_{P_k}, and all centers blown up in Q_{k-1} are contained in the preimage of P_k. Thus $Q_b \to Q$ is an isomorphism over $Q \backslash h^{-1}(P_1 \cup \cdots \cup P_b)$ [Har77, Chapter II, Proposition 7.13(b)]. Second, let $i < k$. The generic point of the strict transform of E'_i on Q_{k-1} maps to η_{P_i}, which is not contained in P_k because $P_i \neq P_k$. Hence the generic point of the strict transform of E'_i is not contained in any center blown up in Q_{k-1}, and $Q_k \to Q_{k-1}$ is an isomorphism near the generic point of the strict transform of E'_i [Har77, Chapter II, Proposition 7.13(b)]. Thus, for any i, the local ring of Q_b at the generic point of the strict transform of E'_i is the valuation ring of v_i.

The variety Q_b may be non-normal. However, the generic point of the strict transform of E'_i is a regular point of Q_b, since the local ring of Q_b at that point is a DVR. By [Kol05, Theorem 27] (or [BM97, Theorem 1.6]), there exists a projective birational morphism $Y \to Q_b$ from a smooth variety Y which is an isomorphism over the smooth locus of Q_b. The smooth locus of Q_b contains the generic points of the strict transforms of $E'_1, \dots, E'_b$ and the preimage of $Q_{\mathrm{reg}} \backslash h^{-1}(P_1 \cup \cdots \cup P_b)$. Let $\rho : Y \to Q$ be the composition, and let E_i be the strict transform of E'_i on Y. Then Y is a smooth projective variety, ρ is a projective birational morphism which is an isomorphism over $Q_{\mathrm{reg}} \backslash h^{-1}(P_1 \cup \cdots \cup P_b)$, and $\operatorname{ord}_{E_i} = v_i$ on $\mathbb{C}(Y) = \mathbb{C}(Q)$.

Let $g := h \circ \rho$. By (5.4), the generic point of E_i maps to η_{P_i}. Since g is proper, $g(E_i) = P_i$. Moreover,

$$\operatorname{mult}_{E_i} g^* P_i = \operatorname{ord}_{E_i}(g^* z_i) = v_i(h^* z_i) = 1.$$

Step 7. In this step, we show that g is a submersion at any point of a dense open subset of E_i whose image contains a dense open subset of P_i.

Fix i and let $d := \dim X$. Recall that z_i is a uniformizer of $\mathcal{O}_{X,\eta_{P_i}}$. There exists an open subset $X^\circ \subset X$ such that z_i is regular on X°, the divisor of z_i on X° is $P_i \cap X^\circ$, and $P_i \cap X^\circ$ is non-empty and smooth. Let Y° be the complement in $g^{-1}(X^\circ)$ of the prime divisors other than E_i in $\operatorname{Supp} g^* P_i$. Since $\operatorname{mult}_{E_i} g^* P_i = 1$, the divisor of $g^* z_i$ on Y° is $E_i \cap Y^\circ$. Since Y is smooth, near any point of $E_i \cap Y^\circ$ we have

$$g^* z_i = s\epsilon', \tag{5.5}$$

where s is a local equation of E_i and ϵ' is a nowhere vanishing regular function.

Since $g(E_i) = P_i$, by generic smoothness [Har77, Chapter III, Corollary 10.7] applied to $(E_i)_{\mathrm{reg}} \to P_i$, there exists a non-empty open subset $P_i^1 \subset P_i$ such that $(E_i)_{\mathrm{reg}} \cap g^{-1}(P_i^1) \to P_i^1$ is smooth. Let

$$E_i^\circ := (E_i)_{\mathrm{reg}} \cap Y^\circ \cap g^{-1}(P_i^1 \cap X^\circ).$$

Then E_i° is an open subset of E_i containing the generic point of E_i, hence a dense open subset of E_i.

Let $y \in E_i^\circ$ and $x := g(y)$. Since $P_i \cap X^\circ$ is smooth and z_i is a local equation of P_i near x, there exist local holomorphic coordinates $(z_i, x_2, \dots, x_d)$ of X at x. Since P_i and E_i are smooth at x and y and $E_i \to P_i$ is smooth at y, the differential $T_y E_i \to T_x P_i$ is surjective. Thus the restrictions of $d(g^* x_2)_y, \dots, d(g^* x_d)_y$ to $T_y E_i$ are linearly independent. Since $s(y) = 0$, (5.5) implies

$$d(g^* z_i)_y = \epsilon'(y) ds_y,$$

which is non-zero and vanishes on $T_y E_i$. Hence $d(g^* z_i)_y, d(g^* x_2)_y, \dots, d(g^* x_d)_y$ are linearly independent on $T_y Y$, so g is a submersion at y. When $d = 1$, the covector $d(g^* z_i)_y$ is non-zero on $T_y Y$ and $\dim T_x X = 1$, so g is a submersion at y.

Finally, the morphism $E_i^\circ \to P_i^1$ is smooth, hence flat, and therefore open [Har77, Chapter III, Exercise 9.1]. Thus $g(E_i^\circ)$ is a non-empty open subset of P_i^1, hence a dense open subset of P_i. □

Remark 5.3. We use the notation of Proposition 5.2. Without an extraction, the image of $T_i \times F$ in Q need not give a prime divisor of multiplicity one over P_i: if H_i acts faithfully on F, then $S \times F \to Q$ is unramified along $T_i \times F$, and the image of $T_i \times F$ in Q has multiplicity m_i in $h^* P_i$. When $m_i = 2$, [OAI26b, Proposition A.3] blows up the fixed loci of the stabilizers in $S \times F$. A stabilizer of order two acts by -1 on all normal directions of the fixed locus. Thus the quotient of the blow-up is smooth, and the image of the exceptional divisor has multiplicity one in the pullback of P_i.

When $m_i \geq 3$, blowing up the fixed loci once does not suffice. For instance, let σ_i be a generator of H_i and V_i a component of F^{H_i}, and assume that σ_i acts on the normal directions of $T_i \times V_i$ with weight 1 along S and weights $1, \dots, m_i - 1$ along F. For $0 < k < m_i$, the weights of σ_i^k on the normal directions of $T_i \times V_i$ include k and $2k$, which are distinct modulo m_i, so σ_i^k does not fix a general point of the projectivized normal space. Hence the exceptional divisor of the blow-up of $T_i \times V_i$ has trivial generic stabilizer, and the image of the exceptional divisor in the quotient still has multiplicity m_i in the pullback of P_i.

5.2. **Specialness of the total space.** In this subsection, we show that if (X, Δ) is special, then so is Y, for a morphism $g : Y \to X$ satisfying the conditions of Proposition 5.5. Proposition 5.5 is used in **Step 3** of the proof of Theorem 5.1. We argue by contradiction with Theorem 3.1: a line bundle on Y of Iitaka dimension p in Ω_Y^p gives a line bundle of Iitaka dimension p in $\Omega_{\mathcal{X}}^p$ on the root orbifold $\mathcal{X}$ of (X, Δ).

The inequality in the following lemma is [Bog79, Lemma 12.4]. We deduce the equality case from [Cam11a, Théorème 3.12].

Lemma 5.4. *Let X be a connected smooth projective variety, let $1 \leq p \leq \dim X$, and let $i : \mathcal{L} \to \Omega_X^p$ be a non-zero map from a line bundle $\mathcal{L}$. Then the following hold.*

(1) We have $\kappa(X, \mathcal{L}) \leq p$.

(2) If $\kappa(X, \mathcal{L}) = p$, then there exist $q \geq 1$ and sections $s_0, \dots, s_p \in H^0(X, \mathcal{L}^q)$ with $s_0 \neq 0$ such that the rational functions $u_a := s_a / s_0$, $1 \leq a \leq p$, are algebraically independent and, at a general point of X, the image of i is spanned by $du_1 \wedge \cdots \wedge du_p$.

Proof. Let $\mathcal{L}' \subset \Omega_X^p$ be the saturation of $i(\mathcal{L})$. By Lemma 3.2, $\mathcal{L}'$ is a line bundle, and $\mathcal{L}'^q$ is saturated in $\operatorname{Sym}^q \Omega_X^p$ for any $q \geq 1$. Since X is connected and $i \neq 0$, the induced map $\mathcal{L} \to \mathcal{L}'$ is injective, and it is an isomorphism over a non-empty Zariski open subset of X. Thus $H^0(X, \mathcal{L}^q) \subset H^0(X, \mathcal{L}'^q)$ for any $q \geq 1$, and the map defined by $H^0(X, \mathcal{L}^q)$ is the composite of the map defined by $H^0(X, \mathcal{L}'^q)$ with a linear projection. Hence

$$\kappa(X, \mathcal{L}) \leq \kappa(X, \mathcal{L}').$$

Since $\mathcal{L}'^q$ is saturated in $\operatorname{Sym}^q \Omega_X^p$ for any $q \geq 1$, $\kappa(X, \mathcal{L}')$ is the Iitaka dimension of $\mathcal{L}'$ in the sense of [Cam11a, §3.1 and Lemme 3.1]. By [Bog79, Lemma 12.4], $\kappa(X, \mathcal{L}') \leq p$. This proves (1).

Now assume that $\kappa(X, \mathcal{L}) = p$. By (1), $\kappa(X, \mathcal{L}') = p$. Choose $q \geq 1$ such that the image of the map $X \dashrightarrow \mathbb{P}(H^0(X, \mathcal{L}^q)^*)$ defined by $H^0(X, \mathcal{L}^q)$ is of dimension p, and let $s_0, \dots, s_r$ be a basis of $H^0(X, \mathcal{L}^q)$. The function field of the image is generated by the ratios s_a / s_0, $1 \leq a \leq r$,

and it is of transcendence degree p. Renumbering $s_1, \dots, s_r$, we may assume that $u_a := s_a/s_0$, $1 \le a \le p$, are algebraically independent.

By [Cam11a, Théorème 3.12(2)] with $D = 0$, there exists a fibration $f : X \dashrightarrow W$ with $\dim W = p$ (a meromorphic fibration in the sense of [Cam11a, Définition 4.1], which is a rational map to a projective variety W since it is defined by sections of powers of $\mathcal{L}'$) such that f is the Iitaka fibration defined by the sections of the saturated powers of $\mathcal{L}'$, and $\mathcal{L}'$ is the saturation of the image of f^*K_W in Ω^p_X. For $N \ge 1$ sufficiently divisible, u_a^N is the ratio of the sections s_a^N and s_0^N of $\mathcal{L}'^{qN}$, so $u_a^N \in f^*\mathbb{C}(W)$. Since f is a fibration, its general fibers are connected [Cam11a, Définition 4.1]. By [Har77, Chapter III, Corollary 11.5] for a resolution of the indeterminacy of f, $f^*\mathbb{C}(W)$ is algebraically closed in $\mathbb{C}(X)$. Hence $u_1, \dots, u_p$ are algebraically independent elements of $f^*\mathbb{C}(W)$. Since $\dim W = p$, the differentials $du_1, \dots, du_p$ form a basis of $\Omega_{f^*\mathbb{C}(W)/\mathbb{C}}$ [Har77, Chapter II, Proposition 8.3A and Theorem 8.6A]. Thus $du_1 \wedge \cdots \wedge du_p$ spans the image of f^*K_W at a general point of X. Since $\mathcal{L}'$ is the saturation of the image of f^*K_W and $\mathcal{L} \to \mathcal{L}'$ is an isomorphism at a general point, $du_1 \wedge \cdots \wedge du_p$ spans the image of i at a general point of X. This proves (2). □

The four conditions in the following proposition play separate roles. Condition (1) yields local lifts of g to the charts of $\mathcal{X}$, along which orbifold line bundles and their maps to $\Omega^p_{\mathcal{X}}$ pull back to Y. Condition (2) implies that the ratios of sections and the coefficients of tensors on Y are constant on the general fibers of g, so they descend to X. Conditions (3) and (4) yield, over a general point of any prime divisor of X, a point at which g or a local lift of g to a chart of $\mathcal{X}$ is a submersion, and local sections through such points exclude poles of the descended tensors. Proposition 5.5 generalizes [OAI26b, Proposition 3.1], which treats the case where Δ has a single component, with coefficient $\frac{1}{2}$.

Proposition 5.5. *Let $g : Y \to X$ be a surjective morphism between connected smooth projective varieties, let $d := \dim X$, and let (X, Δ) be an orbifold pair with $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$. Assume that (X, Δ) is special and that the following hold.*

(1) For every i and every prime divisor E on Y, the integer m_i divides $\operatorname{mult}_E g^* D_i$.
(2) There exists a non-empty Zariski open subset $U \subset X \setminus \operatorname{Supp} \Delta$ such that $g^{-1}(U) \to U$ is smooth with connected fibers, and every fiber Y_x with $x \in U$ satisfies $\Omega^1_{Y_x} \cong \mathcal{O}_{Y_x}^{\oplus \dim Y_x}$.
(3) For every prime divisor P on X different from the D_i, there exists a dense open subset $P^\circ \subset P$ such that every fiber of g over P° contains a point at which g is a submersion.
(4) For every i, there exists a dense open subset $D_i^\circ \subset D_i \setminus \bigcup_{j \ne i} D_j$ such that, for any $x \in D_i^\circ$ and local coordinates $z_1, \dots, z_d$ at x with $D_i = \{z_1 = 0\}$ near x, the fiber of g over x contains a point y for which there exist holomorphic functions $\tilde{g}_1, \dots, \tilde{g}_d$ near y with $g^* z_1 = \tilde{g}_1^{m_i}$ and $g^* z_k = \tilde{g}_k$ for $2 \le k \le d$ such that $(\tilde{g}_1, \dots, \tilde{g}_d)$ is a submersion at y. Such a map $(\tilde{g}_1, \dots, \tilde{g}_d)$ is a local lift of g to the chart $(w_1, z_2, \dots, z_d) \to (w_1^{m_i}, z_2, \dots, z_d)$ of $\mathcal{X}$ (Definition 2.8).

Then Y is special.

Proof of Proposition 5.5. The proof is long, so we divide it into several steps.

Let $\varpi : \mathcal{X} \to X$ be the coarse map of the root orbifold $\mathcal{X}$ of (X, Δ). Definition 2.8 writes π for the coarse map, but in this section π denotes a finite Galois cover $S \to X$, which is the Kummer cover in Subsection 5.3. Let $D := \sum_{i=1}^r D_i$, so that $\operatorname{Supp} D = \operatorname{Supp} \Delta$. Suppose that Y is not special. By Theorem 3.1, there exist an integer $1 \le p \le \dim Y$, a line bundle $\mathcal{L}$ on Y, and a non-zero map $\phi : \mathcal{L} \to \Omega^p_Y$ such that $\kappa(Y, \mathcal{L}) = p$. By GAGA [Ser56, n° 12, Théorèmes 2–3], $\mathcal{L}$, ϕ, and the sections of the powers of $\mathcal{L}$ are algebraic. By Lemma 5.4 with $i = \phi$, there exist $q \ge 1$ and $s_0, \dots, s_p \in H^0(Y, \mathcal{L}^q)$ with $s_0 \ne 0$ such that the rational functions $u_a := s_a/s_0$, $1 \le a \le p$, are algebraically independent and, at a general point of Y, the image of ϕ is spanned by $du_1 \wedge \cdots \wedge du_p$.

Step 1. In this step we construct local lifts of g to the charts of $\mathcal{X}$, and we pull back orbifold line bundles along them.

Let $y \in Y$ and $x := g(y)$. Let $W \subset X$ be a coordinate polydisc at x with coordinates $z_1, \dots, z_d$ such that $D_{i_j} \cap W = \{z_j = 0\}$ for $1 \le j \le k$ and $D_i \cap W = \emptyset$ for $i \notin \{i_1, \dots, i_k\}$, and let $\widetilde{W} \to W$ be the chart of $\mathcal{X}$ over W (Definition 2.8). Let $N \subset g^{-1}(W)$ be a coordinate polydisc at y. Since g is surjective, $g^* z_j \not\equiv 0$ on N. The local ring $\mathcal{O}_{Y,y}$ is factorial [GR65, Chapter II, §B]. Let $\eta \in \mathcal{O}_{Y,y}$ be a prime factor of $g^* z_j$. Then $\{\eta = 0\}$ is a local branch of a prime divisor E of Y, and the exponent of η in $g^* z_j$ is $\operatorname{mult}_E g^* D_{i_j}$, which is divisible by m_{i_j} by condition (1). Thus, after shrinking N,

$$g^* z_j = \epsilon_j \xi_j^{m_{i_j}}, \qquad 1 \le j \le k,$$

where ϵ_j is a nowhere vanishing holomorphic function and ξ_j is a holomorphic function on N. Since N is simply connected, there exists a holomorphic function δ_j on N with $\delta_j^{m_{i_j}} = \epsilon_j$. Replacing ξ_j with $\delta_j \xi_j$, we may assume that $\epsilon_j = 1$. Then

$$\tilde{g} := (\xi_1, \dots, \xi_k, g^* z_{k+1}, \dots, g^* z_d) : N \to \widetilde{W}$$

is a lift of $g|_N$. If $\tilde{g}'$ is another lift of $g|_N$ to $\widetilde{W}$, with coordinates ξ_j', then ξ_j'/ξ_j is a meromorphic function on the connected set N whose m_{i_j}-th power is 1, hence a constant in $\mu_{m_{i_j}}$. Thus any two lifts of $g|_N$ to $\widetilde{W}$ differ by an element of $\prod_{j=1}^k \mu_{m_{i_j}}$.

Let $\mathcal{G}$ be an orbifold line bundle with a map $\mathcal{G} \to \Omega^p_{\mathcal{X}}$. Since line bundles, maps, and sections on $\mathcal{X}$ are equivariant on the charts and compatible on overlaps, the line bundles $\tilde{g}^* \mathcal{G}$ on the neighborhoods N, the maps $\tilde{g}^* \mathcal{G} \to \tilde{g}^* \Omega^p_{\widetilde{W}} \to \Omega^p_N$, and the pullbacks of invariant holomorphic or meromorphic sections of $\mathcal{G}^n$, $n \ge 1$, and of $\operatorname{Sym}^q \Omega^p_{\mathcal{X}}$ do not depend on the choice of the lift, and they glue. We write $f : Y \to \mathcal{X}$ for the family of local lifts, as notation only. We obtain a line bundle $f^* \mathcal{G}$ on Y, a map $f^* \mathcal{G} \to \Omega^p_Y$, and pullback maps $H^0(\mathcal{X}, \mathcal{G}^n) \to H^0(Y, (f^* \mathcal{G})^n)$ for $n \ge 1$. Since $\varpi \circ \tilde{g} = g$ for any lift $\tilde{g}$, we have $\tilde{g}^* \varpi^* \omega = g^* \omega$ for any meromorphic function or meromorphic symmetric tensor ω on X.

Step 2. In this step we show that $u_a = g^* v_a$ for algebraically independent $v_1, \dots, v_p \in \mathbb{C}(X)$.

We first show that any $\psi \in \mathbb{C}(Y)$ which is constant on Y_x for general $x \in U$ is of the form $g^* v$ with $v \in \mathbb{C}(X)$. Let $Z_\psi \subset X \times \mathbb{P}^1$ be the closure of the image of $(g, \psi) : Y \dashrightarrow X \times \mathbb{P}^1$. Since ψ is constant on Y_x for general x, the image of (g, ψ) intersects the fiber over a general point x of X in one point. Thus $Z_\psi \to X$ is generically finite, and ψ is algebraic over $g^* \mathbb{C}(X)$. By condition (2), the general fiber of g is connected. Let $Y \to Y' \to X$ be the Stein factorization of g [Har77, Chapter III, Corollary 11.5]. Since the general fiber of g is connected, $Y' \to X$ is finite and birational, hence an isomorphism because X is normal. Thus $g_* \mathcal{O}_Y = \mathcal{O}_X$, and $\mathbb{C}(X)$ is algebraically closed in $\mathbb{C}(Y)$. Hence $\psi = g^* v$ for some $v \in \mathbb{C}(X)$.

Let $Z \subsetneq Y$ be the union of the zero locus of s_0 and the set of points at which ϕ vanishes as a map of vector bundles. Since g is smooth over U, the set $U_1 := g(g^{-1}(U) \backslash Z)$ is open, and $U_1 \ne \emptyset$ because $g^{-1}(U)$ is a non-empty open subset of the irreducible variety Y. Let $x \in U_1$. Then Y_x is smooth and connected, hence irreducible, and $\phi|_{Y_x} \ne 0$ and $s_0|_{Y_x} \ne 0$. We have the exact sequence

$$0 \to g^* \Omega^1_X|_{Y_x} \to \Omega^1_Y|_{Y_x} \to \Omega^1_{Y_x} \to 0$$

[Har77, Chapter II, Proposition 8.12 and Theorem 8.17, Chapter III, Proposition 10.4], where $g^* \Omega^1_X|_{Y_x} \cong \mathcal{O}_{Y_x}^{\oplus d}$. For $0 \le l \le p$, let $G_l \subset \Omega^p_Y|_{Y_x}$ be the image of $\bigwedge^{p-l} (g^* \Omega^1_X|_{Y_x}) \otimes \Omega^l_Y|_{Y_x}$, and let $G_{-1} := 0$. Then

$$G_l / G_{l-1} \cong \bigwedge^{p-l} \mathcal{O}_{Y_x}^{\oplus d} \otimes \Omega^l_{Y_x}, \qquad 0 \le l \le p,$$

which is a trivial vector bundle by condition (2). Let l be the smallest index such that $\phi|_{Y_x}$ factors through G_l. Then the composite $\mathcal{L}|_{Y_x} \to G_l / G_{l-1}$ is non-zero, and a coordinate projection of the trivial bundle G_l / G_{l-1} yields a non-zero map $\mathcal{L}|_{Y_x} \to \mathcal{O}_{Y_x}$, that is, a non-zero section

$t \in H^0(Y_x, \mathcal{L}^{-1}|_{Y_x})$. The section $t^q s_0|_{Y_x}$ of $\mathcal{O}_{Y_x}$ is non-zero since Y_x is irreducible, and it is constant since Y_x is compact and connected. Similarly, $t^q s_a|_{Y_x}$ is constant for $1 \leq a \leq p$. Thus

$$u_a|_{Y_x} = \frac{t^q s_a|_{Y_x}}{t^q s_0|_{Y_x}}$$

is constant. Since u_a is constant on Y_x for any $x \in U_1$, the descent shown above implies that $u_a = g^* v_a$ for some $v_a \in \mathbb{C}(X)$, $1 \leq a \leq p$. Since $g^* : \mathbb{C}(X) \to \mathbb{C}(Y)$ is injective, $v_1, \ldots, v_p$ are algebraically independent. Thus $p \leq d$, and

$$\sigma := dv_1 \wedge \cdots \wedge dv_p$$

is a non-zero rational p-form on X [Har77, Chapter II, Proposition 8.3A and Theorem 8.6A]. Since $g^*\sigma = du_1 \wedge \cdots \wedge du_p$, the image of ϕ at a general point of Y is spanned by $g^*\sigma$.

Step 3. In this step we show that the coefficients of the tensors $\phi^q(s_a)$ with respect to $(g^*\sigma)^q$ descend to X.

Let $\phi^q : \mathcal{L}^q \to \operatorname{Sym}^q \Omega^p_Y$ be the map induced by ϕ, and let $\theta_a := \phi^q(s_a) \in H^0(Y, \operatorname{Sym}^q \Omega^p_Y)$ for $0 \leq a \leq p$. By **Step 2**, there exist $c_a \in \mathbb{C}(Y)$ such that

$$\theta_a = c_a (g^*\sigma)^q, \qquad 0 \leq a \leq p.$$

Choose a non-empty Zariski open subset $U' \subset U$ on which σ is a nowhere vanishing section of Ω^p_X. Since g is smooth over U', the map $g^*\Omega^p_X \to \Omega^p_Y$ is injective on fibers at the points of $g^{-1}(U')$, and $(g^*\sigma)^q$ is a nowhere vanishing section of $\operatorname{Sym}^q \Omega^p_Y$ on $g^{-1}(U')$. Dividing a coefficient of θ_a by a non-vanishing coefficient of $(g^*\sigma)^q$ locally, we see that c_a is holomorphic on $g^{-1}(U')$. For $x \in U'$, Y_x is compact and connected, so $c_a|_{Y_x}$ is constant. By **Step 2**, $c_a = g^* b_a$ for some $b_a \in \mathbb{C}(X)$. Since Y is connected and $\phi \neq 0$, the map ϕ^q is injective. Thus $\theta_0 \neq 0$ and $b_0 \neq 0$. Since $s_a = u_a s_0$, we have $\theta_a = u_a \theta_0$ and $c_a = u_a c_0$. Hence

$$b_a = v_a b_0, \qquad 1 \leq a \leq p.$$

Step 4. In this step we construct a saturated orbifold line bundle $\mathcal{M} \subset \Omega^p_{\mathcal{X}}$ which contains the direction of $\varpi^*\sigma$.

Let $\sigma' := \varpi^*\sigma$, an invariant meromorphic p-form on the charts of $\mathcal{X}$. Let $\widetilde{W}$ be a chart. Near any point of $\widetilde{W}$, choose a holomorphic function $\lambda \not\equiv 0$ such that $\lambda\sigma'$ is holomorphic, and let $\mathcal{F}_0 := \mathcal{O} \cdot (\lambda\sigma') \subset \Omega^p_{\widetilde{W}}$. The saturation of $\mathcal{F}_0$ is the sheaf of local holomorphic sections of $\Omega^p_{\widetilde{W}}$ which are meromorphic multiples of σ', so it does not depend on the choice of λ. Thus the saturations of the sheaves $\mathcal{F}_0$ glue on $\widetilde{W}$. Since σ' is invariant under the chart groups and chart changes, they define a saturated subsheaf $\mathcal{M} \subset \Omega^p_{\mathcal{X}}$. By Lemma 3.2, $\mathcal{M}$ is an orbifold line bundle, and $\mathcal{M}^q$ is saturated in $\operatorname{Sym}^q \Omega^p_{\mathcal{X}}$. In particular, if a holomorphic section of $\operatorname{Sym}^q \Omega^p_{\widetilde{W}}$ is a meromorphic multiple of σ'^q, then it is a section of $\mathcal{M}^q$.

Step 5. In this step we show that the invariant meromorphic sections

$$T_a := (\varpi^* b_a)\sigma'^q, \qquad 0 \leq a \leq p,$$

of $\operatorname{Sym}^q \Omega^p_{\mathcal{X}}$ are global sections of $\mathcal{M}^q$.

By **Step 1** and **Step 3**, for any local lift $\tilde{g}$ of g to a chart of $\mathcal{X}$,

$$\tilde{g}^* T_a = (g^* b_a)(g^*\sigma)^q = c_a (g^*\sigma)^q = \theta_a,$$

which is holomorphic. Suppose that the coefficients of T_a with respect to the coordinate basis of $\operatorname{Sym}^q \Omega^p_{\widetilde{W}}$ have a pole along a prime divisor P' of a chart $\widetilde{W} \to W$. Over $W \backslash D$, the map $\widetilde{W} \to W$ is étale and T_a is the pullback of $b_a \sigma^q$. Since $\widetilde{W} \to W$ is finite, the image of P' is an irreducible analytic hypersurface contained in D or in the polar locus of $b_a\sigma^q$. Thus the image of P' contains a non-empty open subset of a prime divisor P of X. It is not contained in $D_i \cap D_j$ for $i \neq j$, which is of codimension at least two. Since P° and D_i° are dense open subsets, the points of P' over $P^\circ \backslash D$ if P is not a component of D, and over D_i° if $P = D_i$, form a non-empty

open subset of P'. We show that T_a is holomorphic at any point of P' over $P^\circ \backslash D$ or over D_i°, a contradiction.

First assume that P is not a component of D, and let $x \in P^\circ \backslash D$. By condition (3), there exists $y \in g^{-1}(x)$ at which g is a submersion. By the holomorphic submersion theorem, there exist a neighborhood W_x of x with $W_x \cap D = \emptyset$ and a holomorphic section $\ell : W_x \to Y$ of g with $\ell(x) = y$. Over W_x, the root orbifold $\mathcal{X}$ is W_x and $\tilde{g} = g$. Thus

$$b_a \sigma^q = \ell^* g^* (b_a \sigma^q) = \ell^* \theta_a$$

is holomorphic on W_x, and T_a is holomorphic at the points over W_x.

Next assume that $P = D_i$, and let $x \in D_i^\circ$. Choose local coordinates $z_1, \dots, z_d$ at x with $D_i = \{z_1 = 0\}$, and let $\widetilde{W}_x \to W_x$ be the chart $(w_1, z_2, \dots, z_d) \to (w_1^{m_i}, z_2, \dots, z_d)$. By condition (4), there exist $y \in g^{-1}(x)$ and a local lift $\tilde{g}$ of g to $\widetilde{W}_x$ which is a submersion at y. By the holomorphic submersion theorem, there exists a holomorphic section ℓ of $\tilde{g}$ on a neighborhood of $\tilde{g}(y)$ with $\ell(\tilde{g}(y)) = y$. Thus $T_a = \ell^* \tilde{g}^* T_a = \ell^* \theta_a$ is holomorphic near $\tilde{g}(y)$, which is the unique point of $\widetilde{W}_x$ over x. Since chart changes of $\mathcal{X}$ are equivariant biholomorphisms and T_a is invariant, T_a is holomorphic at the points of P' over x.

Thus the coefficients of T_a have no pole along any prime divisor of a chart. Let $\widetilde{W}$ be a chart, let $\tilde{w} \in \widetilde{W}$, and write a coefficient of T_a near $\tilde{w}$ as α/β with coprime $\alpha, \beta \in \mathcal{O}_{\widetilde{W},\tilde{w}}$, using that $\mathcal{O}_{\widetilde{W},\tilde{w}}$ is factorial [GR65, Chapter II, §B]. If β is not a unit, then α/β has a pole along the zero locus of a prime factor of β, a contradiction. Hence T_a is holomorphic on $\widetilde{W}$. Since T_a is a meromorphic multiple of σ'^q, it is a section of $\mathcal{M}^q$ by **Step 4**. Since T_a is invariant, $T_a \in H^0(\mathcal{X}, \mathcal{M}^q)$.

Step 6. In this step we show that $\kappa(\mathcal{X}, \mathcal{M}) = p$, and we conclude the proof.

Since $\tilde{g}^* T_0 = \theta_0 \neq 0$, we have $T_0 \neq 0$. By **Step 3**,

$$\frac{T_a}{T_0} = \varpi^* \frac{b_a}{b_0} = \varpi^* v_a, \qquad 1 \leq a \leq p.$$

Thus the map $X \dashrightarrow \mathbb{P}(H^0(\mathcal{X}, \mathcal{M}^q)^*)$ defined by $H^0(\mathcal{X}, \mathcal{M}^q)$, composed with the linear projection to the coordinates corresponding to $T_0, \dots, T_p$, is the map $(1 : v_1 : \cdots : v_p) : X \dashrightarrow \mathbb{P}^p$. Since $v_1, \dots, v_p$ are algebraically independent, $(1 : v_1 : \cdots : v_p)$ is dominant. Hence

$$\kappa(\mathcal{X}, \mathcal{M}) \geq p.$$

By **Step 1** applied to $\mathcal{G} = \mathcal{M}$, we have a line bundle $f^*\mathcal{M}$ on Y and a map $f^*\mathcal{M} \to \Omega_Y^p$. Over $U \subset X \backslash \operatorname{Supp} \Delta$, the root orbifold $\mathcal{X}$ is U, and the map $f^*\mathcal{M} \to \Omega_Y^p$ is the composite $g^*\mathcal{M} \to g^*\Omega_X^p \to \Omega_Y^p$. Since $\mathcal{M} \neq 0$ and g is smooth over U, the map $f^*\mathcal{M} \to \Omega_Y^p$ is non-zero. Let $n \geq 1$ and $T, T' \in H^0(\mathcal{X}, \mathcal{M}^n)$ with $T' \neq 0$. The ratio T/T' is an invariant meromorphic function on the charts of $\mathcal{X}$, hence a meromorphic function on X, and

$$\frac{f^*T}{f^*T'} = g^* \frac{T}{T'}.$$

Thus the map $Y \dashrightarrow \mathbb{P}(H^0(\mathcal{X}, \mathcal{M}^n)^*)$ defined by the pullbacks of the sections of $\mathcal{M}^n$ is the composite of g with the map defined by $H^0(\mathcal{X}, \mathcal{M}^n)$. Since g is surjective, the image of the map defined by the pullbacks is equal to the image of the map defined by $H^0(\mathcal{X}, \mathcal{M}^n)$. Moreover, the map defined by the pullbacks is the composite of the map defined by $H^0(Y, (f^*\mathcal{M})^n)$ with a linear projection. Hence $\kappa(\mathcal{X}, \mathcal{M}) \leq \kappa(Y, f^*\mathcal{M})$. By GAGA [Ser56, n° 12, Théorèmes 2–3], $f^*\mathcal{M}$ and the map $f^*\mathcal{M} \to \Omega_Y^p$ are algebraic. By Lemma 5.4(1), $\kappa(Y, f^*\mathcal{M}) \leq p$. Therefore

$$\kappa(\mathcal{X}, \mathcal{M}) = p.$$

Since $1 \leq p \leq d$ and the inclusion $\mathcal{M} \to \Omega_{\mathcal{X}}^p$ is non-zero, Theorem 3.1 implies that (X, Δ) is not special, a contradiction. Hence Y is special. □

5.3. **Proof of the projective case.** In this subsection, we prove Theorem 5.1.

Proof of Theorem 5.1. The proof is long, so we divide it into several steps.

Step 1. In this step we treat the cases $\dim X = 0$ and $\Delta = 0$, and we fix the data of Set-up 4.1.

If $\dim X = 0$, then X is a point, $\Delta = 0$, and $\pi_1^{\mathrm{orb}}(X, \Delta)$ is trivial. Suppose that $\dim X \geq 1$ and $\Delta = 0$. Then $\pi_1^{\mathrm{orb}}(X, \Delta) = \pi_1(X)$. Since X is a smooth projective variety, X is a compact Kähler manifold [GH78, Chapter 0, Section 7]. Since X is special, the second sentence of Theorem 3.1 implies that, for any $1 \leq p \leq \dim X$, no line bundle $\mathcal{L}$ on X admits a non-zero map $\mathcal{L} \to \Omega_X^p$ with $\kappa(X, \mathcal{L}) = p$. Thus $\pi_1(X)$ is virtually abelian by Theorem 2.13.

From now on we assume that $d := \dim X \geq 1$ and $\Delta \neq 0$, and we write

$$\Delta = \sum_{\alpha=1}^{r} \Big(1 - \frac{1}{m_\alpha}\Big) D_\alpha$$

with $r \geq 1$. By Definition 2.2 and since X is projective by assumption, X, $D := \sum_{\alpha=1}^r D_\alpha$, the integers $m_1, \ldots, m_r$, and $N := d + 1$ satisfy the conditions of Set-up 4.1.

Step 2. In this step we construct a connected smooth projective variety Y and a surjective morphism $g : Y \to X$ by Proposition 5.2.

Let $\pi : S \to X$ be the Kummer cover of Construction-Proposition 4.3, with Galois group A, basis elements $e_{\alpha,j}$, divisors $B_{\alpha,j}$, and primitive m_α-th roots of unity ζ_α. For each α, let E_α be a complex elliptic curve and let $p_\alpha \in E_\alpha$ be a point of order exactly m_α. Let F be the abelian variety with the action of A of Construction-Proposition 4.2 for the curves E_α and the points p_α, let $Q := (S \times F)/A$, and let $h : Q \to X$ be the morphism of Construction-Proposition 4.5. Recall that

$$V = X \backslash \bigcup_{\alpha,j} B_{\alpha,j} \quad \text{and} \quad U = V \backslash D.$$

Let $P_1, \ldots, P_b$ be the irreducible components of all the $B_{\alpha,j}$. Fix l. Since no two of the $B_{\alpha,j}$ have a common irreducible component, there exists exactly one pair (α, j) such that $P_l \subset B_{\alpha,j}$. Let T_l be an irreducible component of $\pi^{-1}(P_l)$. Since A acts transitively on the fibers of π, A permutes the irreducible components of $\pi^{-1}(P_l)$ transitively, so $\pi(T_l) = P_l$. Let $H_l \leq A$ be the subgroup of elements acting trivially on the function field of T_l, that is, the elements fixing T_l pointwise. We show that

$$H_l = \langle e_{\alpha,j} \rangle.$$

Since $D + \sum_{\alpha,j} B_{\alpha,j}$ is a reduced simple normal crossing divisor and P_l is an irreducible component of $B_{\alpha,j}$ only, a general point $x \in P_l$ satisfies $I(x) = \emptyset$ and $J(x) = \{(\alpha, j)\}$. By Construction-Proposition 4.3(4), the stabilizer in A of any point of $\pi^{-1}(x)$ is $\langle e_{\alpha,j} \rangle$. The points of T_l over general points of P_l form a dense subset of T_l, so $e_{\alpha,j}$ fixes T_l pointwise, and $\langle e_{\alpha,j} \rangle \subset H_l$. Conversely, any element of H_l fixes a general point of T_l, so it belongs to $\langle e_{\alpha,j} \rangle$. Thus $H_l = \langle e_{\alpha,j} \rangle$.

Let $s \in T_l$ be a point over such a point x. By Construction-Proposition 4.3(3), S has holomorphic local coordinates $v_{\alpha,j}, t_1, \ldots, t_{d-1}$ near s with $\pi^* z_{\alpha,j} = v_{\alpha,j}^{m_\alpha}$, so $T_l = \{v_{\alpha,j} = 0\}$ near s. In particular, T_l is a prime divisor on S. By Construction-Proposition 4.3(4), $e_{\alpha,j}$ multiplies $v_{\alpha,j}$ by ζ_α and fixes the t_k. Since ζ_α is a primitive m_α-th root of unity, H_l is cyclic of order $m_\alpha \geq 2$ and acts faithfully on the conormal line of T_l, which is spanned by $dv_{\alpha,j}$ at s. Moreover, $F^{H_l} = F^{e_{\alpha,j}}$ is non-empty by Construction-Proposition 4.2(1).

By Construction-Proposition 4.3(1), S is a connected smooth projective variety and π is a finite Galois morphism with Galois group A, and F is a connected smooth projective variety. By Proposition 5.2, applied to the prime divisors $P_1, \ldots, P_b$ with T_l and H_l, there exist a smooth projective variety Y and a projective birational morphism $\rho : Y \to Q$ which is an isomorphism over the open subset

$$Q_{\mathrm{reg}} \backslash h^{-1}(P_1 \cup \cdots \cup P_b) = Q_{\mathrm{reg}} \cap h^{-1}(V) = h^{-1}(V),$$

where the last equality holds because $h^{-1}(V)$ is smooth by Construction-Proposition 4.5(4). Since ρ is birational and Q is connected and normal by Construction-Proposition 4.5(1), Y is connected. Let $g := h \circ \rho$. Since π and ρ are surjective, $g : Y \to X$ is a surjective morphism between connected smooth projective varieties. Proposition 5.2 also implies that, for any l, there exists a dense open subset $P_l^\circ \subset P_l$ such that, for any $x \in P_l^\circ$, the fiber $g^{-1}(x)$ contains a point at which g is a submersion.

Step 3. In this step we show that Y is special.

We show that the conditions (1)–(4) of Proposition 5.5 hold for g and (X, Δ), with D_α and m_α in place of D_i and m_i.

(1) By Construction-Proposition 4.5(3), for any α and any prime divisor E on Y, we have

$$m_\alpha \mid \mathrm{mult}_E\, g^* D_\alpha.$$

(2) Since $d \geq 1$ and $X \backslash U$ is the support of $D + \sum_{\alpha,j} B_{\alpha,j}$, U is a non-empty Zariski open subset of $X \backslash \operatorname{Supp} \Delta$. Since ρ is an isomorphism over $h^{-1}(V) \supset h^{-1}(U)$, the morphism $g^{-1}(U) \to U$ is isomorphic over U to $h^{-1}(U) \to U$, which is an analytically locally trivial fiber bundle with fiber F by Construction-Proposition 4.5(6). Thus $g^{-1}(U) \to U$ is smooth with connected fibers, and any fiber Y_x with $x \in U$ is isomorphic to the abelian variety F. Hence $\Omega^1_{Y_x} \cong \mathcal{O}_{Y_x}^{\oplus \dim Y_x}$, since the cotangent bundle of an abelian variety is trivial [Mum70, §4].

(3) Consider a prime divisor of X other than the D_α. If it is one of the P_l, then the dense open subset P_l° of **Step 2** satisfies condition (3). Otherwise, it is not an irreducible component of $D + \sum_{\alpha,j} B_{\alpha,j}$, so it is not contained in $X \backslash U$, and U contains a dense open subset of it. By (2), g is a submersion at any point of $g^{-1}(U)$, so condition (3) holds.

(4) Fix α and let

$$D_\alpha^\circ := (D_\alpha \cap V) \backslash \bigcup_{\beta \neq \alpha} D_\beta.$$

Since $D + \sum_{\alpha,j} B_{\alpha,j}$ is a reduced simple normal crossing divisor, D_α° is a dense open subset of $D_\alpha \backslash \bigcup_{\beta \neq \alpha} D_\beta$. Let $x \in D_\alpha^\circ$, let $z_1, \dots, z_d$ be local coordinates at x with $D_\alpha = \{z_1 = 0\}$ near x, and let $y \in g^{-1}(x)$, which exists because g is surjective. Since $x \in V$, ρ is an isomorphism near y. By Construction-Proposition 4.5(2), $h^* z_1 = w^{m_\alpha}$ for a holomorphic function w near $\rho(y)$. Since α is the only index with $x \in D_\alpha$, Construction-Proposition 4.5(5), with $c = d - 1$ and $t_k = z_{k+1}$, implies that $(w, h^* z_2, \dots, h^* z_d)$ is a submersion at $\rho(y)$. Since $(\rho^* w)^{m_\alpha} = g^* z_1$, the map $(\rho^* w, g^* z_2, \dots, g^* z_d)$ is a local lift of g to the chart $(w_1, z_2, \dots, z_d) \to (w_1^{m_\alpha}, z_2, \dots, z_d)$ of $\mathcal{X}$, and it is a submersion at y. Thus condition (4) holds.

Since (X, Δ) is special, Proposition 5.5 implies that Y is special.

Step 4. In this step we conclude the proof.

Since Y is a smooth projective variety, Y is a compact Kähler manifold [GH78, Chapter 0, Section 7]. Since Y is special, the second sentence of Theorem 3.1 and Theorem 2.13 imply that $\pi_1(Y)$ is virtually abelian.

Since Y is a connected compact complex manifold, $(Y, 0)$ is an orbifold pair as in Definition 2.2 with $r = 0$, and $\pi_1^{\mathrm{orb}}(Y, 0) = \pi_1(Y)$. By Definition 2.5, since the boundary of $(Y, 0)$ is zero, $g : (Y, 0) \to (X, \Delta)$ is a divisible orbifold morphism if and only if $m_\alpha \mid \mathrm{mult}_E\, g^* D_\alpha$ for any α and any prime divisor E on Y, also when $g(E) \subset D_\alpha \cap D_\beta$ for some $\beta \neq \alpha$. This is condition (1) of **Step 3**, so $g : (Y, 0) \to (X, \Delta)$ is a divisible orbifold morphism.

Let $Y \to Y' \to X$ be the Stein factorization of g. Since the fibers of g over U are connected, $Y' \to X$ is finite and birational, hence an isomorphism because X is normal. Thus $g_* \mathcal{O}_Y = \mathcal{O}_X$, and all fibers of g are connected [Har77, Chapter III, Corollaries 11.3 and 11.5]. By Theorem 2.7, the homomorphism

$$g_* : \pi_1(Y) \to \pi_1^{\mathrm{orb}}(X, \Delta)$$

is surjective. By Lemma 2.17(1), the image under g_* of an abelian subgroup of finite index in $\pi_1(Y)$ is an abelian subgroup of finite index in $\pi_1^{\mathrm{orb}}(X,\Delta)$. Thus $\pi_1^{\mathrm{orb}}(X,\Delta)$ is virtually abelian. □

6. Finite-index subgroups and the Albanese map

In this section, we construct finite-index models of subgroups of finite index in orbifold fundamental groups (Construction-Proposition 6.1), and we prove Proposition 6.2 and Proposition 6.5. Proposition 6.5 shows that the orbifold fundamental group of a special compact Kähler orbifold pair has a subgroup of finite index whose quotient by its finite residual is isomorphic to the fundamental group of a compact complex torus. Subsection 7.3 uses Proposition 6.5 for pairs (X,Δ) with $a(X)=0$ (Propositions 7.7 and 7.8).

6.1. **Finite-index subgroups.** In this subsection, we realize any subgroup of finite index in the orbifold fundamental group of a compact Kähler orbifold pair as the orbifold fundamental group of another compact Kähler orbifold pair. The finite cover of $X\setminus \operatorname{Supp}\Delta$ corresponding to a subgroup H extends to a finite normal analytic cover of X, which may be singular over the intersections of the components of $\operatorname{Supp}\Delta$, and the exceptional divisors of a resolution, if given multiplicity one, can give relations which do not hold in H. We thus take on any prime divisor over $\operatorname{Supp}\Delta$ the minimal multiplicity for which the morphism is divisible, so that the relations coming from exceptional divisors already hold in $\pi_1^{\mathrm{orb}}(X,\Delta)$. On the strict transform E of a branch of the cover over D_i with ramification index e, the minimal multiplicity is m_i/e, so any conjugate of $\gamma_i^{m_i}$ lifts to the (m_i/e)-th power of a meridian of E. Thus Construction-Proposition 6.1(3), the main point of the proof (**Step 4**), follows.

Construction-Proposition 6.1. Let (X,Δ) be a compact Kähler orbifold pair with $\Delta=\sum_{i=1}^r(1-\frac{1}{m_i})D_i$, let $D:=\operatorname{Supp}\Delta$, let $G:=\pi_1^{\mathrm{orb}}(X,\Delta)$, and let $H\le G$ be a subgroup of finite index. Then there exist a connected compact Kähler manifold Y and a surjective generically finite holomorphic map $f:Y\to X$ of degree $[G:H]$ with the following properties, where for any prime divisor $E\subset (f^{-1}D)_{\mathrm{red}}$ we let

$$M_E:=\operatorname{lcm}_{1\le i\le r}\frac{m_i}{\gcd(m_i,\operatorname{mult}_E f^*D_i)},\qquad \text{with } \gcd(m_i,0):=m_i,$$

and where $\Delta_Y:=\sum_E(1-\frac{1}{M_E})E$.

(1) The reduced inverse image $(f^{-1}D)_{\mathrm{red}}$ has simple normal crossing support, and f is finite over the complement of a closed analytic subset of X of codimension at least two.
(2) The restriction $f^{-1}(X\setminus D)\to X\setminus D$ is the connected finite cover corresponding to the inverse image of H in $\pi_1(X\setminus D)$.
(3) The map $f:(Y,\Delta_Y)\to(X,\Delta)$ is a divisible orbifold morphism, and f_* is an isomorphism from $\pi_1^{\mathrm{orb}}(Y,\Delta_Y)$ onto H.
(4) If (X,Δ) is special, then (Y,Δ_Y) is special.

We call any surjective generically finite holomorphic map $f:Y\to X$ of degree $[G:H]$ from a connected compact Kähler manifold Y satisfying (1)–(4) a *finite-index model* of H, and we write it as $f:(Y,\Delta_Y)\to(X,\Delta)$. Moreover, there exists a normal subgroup $H_0\trianglelefteq G$ of finite index such that, for any finite-index model $f_0:(Y_0,\Delta_{Y_0})\to(X,\Delta)$ of H_0, the kernel of the surjection $\pi_1^{\mathrm{orb}}(Y_0,\Delta_{Y_0})\cong H_0\to\pi_1(Y_0)$ is contained in $R_{\mathrm{fin}}(H_0)=R_{\mathrm{fin}}(G)$, and it equals $R_{\mathrm{fin}}(H_0)$ if $\pi_1(Y_0)$ is residually finite.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we extend the finite cover of $X\setminus D$ corresponding to H to a finite surjective holomorphic map $h:Z\to X$ from a normal compact complex space Z.

Let $\Lambda:=\pi_1(X\setminus D,x_0)$, let $N\subset\Lambda$ be the normal subgroup generated by $\gamma_1^{m_1},\dots,\gamma_r^{m_r}$, and let $\Lambda_H\subset\Lambda$ be the inverse image of H. Then $G=\Lambda/N$, $N\subset\Lambda_H$, $\Lambda_H/N=H$, and $[\Lambda:\Lambda_H]=[G:H]$.

Let $h^\circ : Z^\circ \to X\backslash D$ be the connected finite cover corresponding to Λ_H, where Z° carries the complex structure for which h° is locally biholomorphic, and fix $\zeta_0 \in (h^\circ)^{-1}(x_0)$ such that $h^\circ_*\pi_1(Z^\circ, \zeta_0) = \Lambda_H$. Then $\deg h^\circ = [G : H]$. Let

$$V := \bigcup_{i<j} (D_i \cap D_j).$$

Then V is a closed analytic subset of X with $\operatorname{codim}_X V \geq 2$.

Let $x \in X$, and let $W \subset X$ be a coordinate polydisc around x with coordinates $z_1, \dots, z_d$ such that, after renumbering the D_i, $D \cap W = \{z_1 \cdots z_k = 0\}$ and $D_j \cap W = \{z_j = 0\}$ for $1 \leq j \leq k$. Let $W^\circ := W\backslash D$, and let $\lambda_1, \dots, \lambda_k \in \pi_1(W^\circ) \cong \mathbb{Z}^k$ be the meridians of $\{z_1 = 0\}, \dots, \{z_k = 0\}$. Fix $x_1 \in W^\circ$ and a path from x_0 to x_1 in $X\backslash D$. Transport along the path maps λ_j to a meridian of D_j in Λ, which is conjugate to γ_j by Lemma 2.3(5). Thus $\lambda_j^{m_j}$ maps into N. Since N is normal in Λ and $N \subset \Lambda_H$, N is contained in any conjugate of Λ_H, so N acts trivially on Λ/Λ_H. Hence the subgroup $L := \bigoplus_{j=1}^k m_j\mathbb{Z}$ of $\pi_1(W^\circ) = \mathbb{Z}^k$ acts trivially on $(h^\circ)^{-1}(x_1)$, and any connected component of $(h^\circ)^{-1}(W^\circ)$ is the cover of W° corresponding to a subgroup $L' \subset \mathbb{Z}^k$ with $L \subset L'$.

Let W' be a polydisc with coordinates $w_1, \dots, w_d$ together with the finite surjective map $W' \to W$ given by $z_j = w_j^{m_j}$ for $1 \leq j \leq k$ and $z_j = w_j$ for $k < j \leq d$, and let W'° be the inverse image of W°. Then $W'^\circ \to W^\circ$ is the Galois cover corresponding to L, with group $\mathbb{Z}^k/L$ acting on W' by roots of unity. Thus the component corresponding to L' is $W'^\circ/(L'/L)$, and it extends to the finite map $W'/(L'/L) \to W$. By [Car57, Théorème 4], $W'/(L'/L)$ is a normal complex space. Taking the disjoint union over the components of $(h^\circ)^{-1}(W^\circ)$, we obtain a finite map from a normal complex space to W extending $(h^\circ)^{-1}(W^\circ) \to W^\circ$.

By [DG94, Theorem 3.4], originally due to [GR58], a finite extension by a normal complex space of the restriction of h° over an open subset of X exists and is unique up to a unique isomorphism. Thus the local extensions above glue to a finite surjective holomorphic map $h : Z \to X$ from a normal complex space Z with $h^{-1}(X\backslash D) = Z^\circ$, and for any coordinate polydisc $W \subset X$ with $D \cap W = \{z_1 \cdots z_k = 0\}$ and $D_j \cap W = \{z_j = 0\}$ for $1 \leq j \leq k$, $h^{-1}(W)$ is isomorphic over W to the disjoint union of the $W'/(L'/L)$. Since X is compact and h is finite, Z is compact. In any $W'/(L'/L)$ the inverse image of D is nowhere dense, so Z° is dense in Z. Since Z° is connected and Z is normal, Z is irreducible. Moreover, $\deg h = [G : H]$.

Now assume that $x \in D_i\backslash V$ for some i, and that $D \cap W = D_i \cap W = \{z_1 = 0\}$. Then $k = 1$, and $L' = e\mathbb{Z}$ for a positive integer e dividing m_i. Hence $W'/(L'/L)$ is a polydisc with coordinates $t_1 := w_1^{m_i/e}$ and $t_j := w_j$ for $2 \leq j \leq d$. We conclude that, for any $z \in h^{-1}(D_i\backslash V)$, there exist a coordinate polydisc W around $h(z)$ with $D \cap W = D_i \cap W = \{z_1 = 0\}$ and local coordinates $t_1, \dots, t_d$ at z such that

$$z_1 \circ h = t_1^e \quad \text{and} \quad z_j \circ h = t_j \quad \text{for } 2 \leq j \leq d, \tag{6.1}$$

where e is a positive integer dividing m_i. In particular, Z is smooth along $h^{-1}(X\backslash V)$, and $h^{-1}(D) \cap h^{-1}(X\backslash V)$ is a smooth divisor of $h^{-1}(X\backslash V)$.

Step 2. In this step we show that Z is a compact Kähler space in the sense of [Xia19, Definition 2.13], and we construct Y and f.

Let ω be a Kähler form on X. Since h is finite, $h_*\mathcal{O}_Z$ is a coherent $\mathcal{O}_X$-algebra. Thus there exist finitely many open subsets $U_1, \dots, U_s$ covering X and holomorphic functions $\psi_{j,1}, \dots, \psi_{j,n_j}$ on $h^{-1}(U_j)$ generating $h_*\mathcal{O}_Z|_{U_j}$ as an $\mathcal{O}_{U_j}$-algebra, so that

$$(h, \psi_{j,1}, \dots, \psi_{j,n_j}) : h^{-1}(U_j) \to U_j \times \mathbb{C}^{n_j}$$

is a closed embedding for any j. Let $\chi_1, \dots, \chi_s \geq 0$ be smooth functions on X with $\operatorname{Supp} \chi_j \subset U_j$ compact and $\sum_j \chi_j = 1$, and let

$$\varphi := \sum_{j=1}^s (\chi_j \circ h) \sum_{a=1}^{n_j} |\psi_{j,a}|^2,$$

where each summand is extended by zero outside $h^{-1}(U_j)$. Then φ is a smooth function on Z. Let $z \in Z$, and let v be a non-zero vector in the Zariski tangent space T_zZ with $dh(v) = 0$. Since the terms containing a derivative of $\chi_j \circ h$ vanish on v, we have

$$dd^c\varphi(v, \bar{v}) = c \sum_{j=1}^{s} \chi_j(h(z)) \sum_{a=1}^{n_j} |d\psi_{j,a}(v)|^2 > 0,$$

where $c > 0$ is the normalization constant of d^c. The sum is positive since $\chi_j(h(z)) > 0$ for some j, and for such j the differential of $(h, \psi_{j,1}, \ldots, \psi_{j,n_j})$ is injective on T_zZ, so $d\psi_{j,a}(v) \neq 0$ for some a.

We claim that there exists $C > 0$ such that $Ch^*\omega + dd^c\varphi$ is positive on any non-zero Zariski tangent vector of Z. Suppose that there exist $C_l \to +\infty$, points $z_l \in Z$, and vectors $v_l \in T_{z_l}Z$ of norm one with respect to finitely many local embeddings of Z in polydiscs, such that $C_l h^*\omega(v_l, \bar{v}_l) + dd^c\varphi(v_l, \bar{v}_l) \leq 0$. After passing to a subsequence, $z_l \to z$ and $v_l \to v$ with $v \in T_zZ$ and $v \neq 0$. Since $h^*\omega(v_l, \bar{v}_l) \geq 0$ and $h^*\omega(v_l, \bar{v}_l) \leq -C_l^{-1} dd^c\varphi(v_l, \bar{v}_l) \to 0$, we have $h^*\omega(v, \bar{v}) = 0$, hence $dh(v) = 0$, and $dd^c\varphi(v, \bar{v}) \leq 0$. This contradicts $dd^c\varphi(v, \bar{v}) > 0$ for non-zero $v \in T_zZ$ with $dh(v) = 0$, and the claim follows. The form $Ch^*\omega + dd^c\varphi$ has local potentials. Locally on Z, a smooth potential whose Levi form is positive on Zariski tangent vectors extends to a strictly plurisubharmonic function on a neighborhood in the ambient polydisc: extend the potential smoothly and add a large multiple of $\sum_l |u_l|^2$, where the u_l generate the ideal of Z, after shrinking the neighborhood. Thus $Ch^*\omega + dd^c\varphi$ is a Kähler form on Z in the sense of [Xia19, Definition 2.13], and Z is a compact Kähler space.

By [BM97, Theorems 1.6, 1.10, 12.2 and 13.2], there exists a finite sequence of blow-ups along smooth centers $g : Y \to Z$ such that Y is smooth, $(g^{-1}h^{-1}D)_{\mathrm{red}}$ is a simple normal crossing divisor, and g is an isomorphism over $h^{-1}(X\backslash V)$. Here g can be taken to be an isomorphism over $h^{-1}(X\backslash V)$ since Z is smooth and $h^{-1}(D)$ is a smooth divisor there by **Step 1**, and the normal crossings are simple since the exceptional set of an admissible sequence of blow-ups is a collection of smooth hypersurfaces with only normal crossings [BM97, (1.2)(1) and the paragraph following it]. By Lemma 2.14(1), applied to each blow-up, Y is a compact Kähler manifold. Since g is bimeromorphic and Z is irreducible, Y is connected. Let $f := h \circ g$. Then f is surjective and generically finite with $\deg f = \deg h = [G : H]$, and f is finite over $X\backslash V$. Since $V \subset D$, g is an isomorphism from $Y^\circ := f^{-1}(X\backslash D)$ onto Z°. Since $\operatorname{codim}_X V \geq 2$, (1) and (2) hold.

Step 3. In this step we show that $f : (Y, \Delta_Y) \to (X, \Delta)$ is a divisible orbifold morphism which is étale in codimension one, and we prove (4).

For any prime divisor $E \subset (f^{-1}D)_{\mathrm{red}}$, let $a_{E,i} := \operatorname{mult}_E f^*D_i$ for $1 \leq i \leq r$. For a positive integer M, the condition $m_i \mid Ma_{E,i}$ is equivalent to $\frac{m_i}{\gcd(m_i, a_{E,i})} \mid M$, also when $a_{E,i} = 0$. Taking the least common multiple, M_E is the smallest positive integer M such that $m_i \mid Ma_{E,i}$ for $1 \leq i \leq r$. In particular,

$$m_i \mid M_E \operatorname{mult}_E f^*D_i$$

for any i and any prime divisor $E \subset (f^{-1}D)_{\mathrm{red}}$. If E is a prime divisor on Y with $E \not\subset (f^{-1}D)_{\mathrm{red}}$, then $\operatorname{mult}_E f^*D_i = 0$ for any i. Moreover, $\operatorname{Supp} \Delta_Y \subset (f^{-1}D)_{\mathrm{red}}$ is a simple normal crossing divisor, so (Y, Δ_Y) is a compact Kähler orbifold pair. Thus $f : (Y, \Delta_Y) \to (X, \Delta)$ is a divisible orbifold morphism.

Let $E \subset (f^{-1}D)_{\mathrm{red}}$ be a prime divisor with $f(E) \not\subset V$. Then $f(E) \subset D_i$ for a unique i, and g is an isomorphism near a general point of E. By (6.1), $a_{E,i} = e$ for a positive integer e dividing m_i, and $a_{E,j} = 0$ for $j \neq i$. Hence $M_E = m_i/e$, and e is the ramification index of f along E. Since f is unramified over $X\backslash D$, by the Hurwitz formula the coefficient of E in $K_Y + \Delta_Y - f^*K_X$ is

$$(e-1) + \Big(1 - \frac{1}{M_E}\Big) = e - \frac{e}{m_i} = e\Big(1 - \frac{1}{m_i}\Big) = \operatorname{mult}_E f^*\Delta.$$

Thus $K_Y + \Delta_Y = f^*(K_X + \Delta)$ over $f^{-1}(X\backslash V)$. Since $\operatorname{codim}_X V \geq 2$, f is étale in codimension one in the sense of [Cam11a, Définition 3.10], that is, $K_Y + \Delta_Y = f^*(K_X + \Delta)$ holds outside

the inverse image of a closed analytic subset of X of codimension at least two. Equivalently, $eM_E = m_i$ for any such E [Cam11a, Proposition 3.11(1)].

Now assume that (X, Δ) is special. Since X and Y are connected compact Kähler manifolds, hence in Fujiki's class $\mathcal{C}$, with $\dim Y = \dim X$, and f is étale in codimension one, by Remark 2.12 and [Cam11a, Proposition 10.11], (Y, Δ_Y) is special. This proves (4).

Step 4. In this step we show that f_* is an isomorphism from $\pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ onto H, which completes the proof of (3).

Let $y_0 := g^{-1}(\zeta_0) \in Y^\circ$. Since $g : Y^\circ \to Z^\circ$ is an isomorphism, $f_* : \pi_1(Y^\circ, y_0) \to \Lambda$ is injective with image Λ_H. Since $\operatorname{Supp} \Delta_Y \subset (f^{-1}D)_{\mathrm{red}}$, by Lemma 2.3(1)(2), applied to $Y \setminus \operatorname{Supp} \Delta_Y$, the homomorphism $\pi_1(Y^\circ, y_0) \to \pi_1(Y \setminus \operatorname{Supp} \Delta_Y, y_0)$ is surjective with kernel the normal subgroup generated by the meridians of the prime divisors $E \subset (f^{-1}D)_{\mathrm{red}}$ with $M_E = 1$. For any prime divisor $E \subset (f^{-1}D)_{\mathrm{red}}$, let $\delta_E \in \pi_1(Y^\circ, y_0)$ be a meridian of E, and let $R \subset \pi_1(Y^\circ, y_0)$ be the normal subgroup generated by the $\delta_E^{M_E}$. By Lemma 2.3(5), R does not depend on the choice of the δ_E. By Definition 2.2,

$$\pi_1^{\mathrm{orb}}(Y, \Delta_Y) = \pi_1(Y^\circ, y_0)/R.$$

We show that $f_*(R) = N$.

First we show that $f_*(R) \subset N$. Let $E \subset (f^{-1}D)_{\mathrm{red}}$ be a prime divisor, and let $y \in E$ be a point not contained in any other component of $(f^{-1}D)_{\mathrm{red}}$. Let W be a polydisc around $f(y)$ as in **Step 1**, with coordinates $z_1, \dots, z_d$, $D \cap W = \{z_1 \cdots z_k = 0\}$, and $D_j \cap W = \{z_j = 0\}$ for $1 \le j \le k$. Let $Y_E \subset f^{-1}(W)$ be a coordinate polydisc around y such that $Y_E \cap (f^{-1}D)_{\mathrm{red}} = Y_E \cap E$ is a smooth prime divisor of Y_E. Then $f(Y_E \setminus E) \subset W^\circ$, and $\operatorname{mult}_{E \cap Y_E} f^*\{z_j = 0\} = a_{E,j}$ for $1 \le j \le k$. By Lemma 2.3(3), f_* maps a meridian of $E \cap Y_E$ to $\prod_{j=1}^k \lambda_j^{a_{E,j}} \in \pi_1(W^\circ)$. By Lemma 2.3(5), $f_*(\delta_E)$ is conjugate in Λ to the transport of $\prod_{j=1}^k \lambda_j^{a_{E,j}}$ along a path from x_0 to W°. The transports of $\lambda_1, \dots, \lambda_k$ commute, and they are meridians of $D_1, \dots, D_k$, so their images in G have orders dividing $m_1, \dots, m_k$ by Lemma 2.3(5). Since $m_j \mid M_E a_{E,j}$ for $1 \le j \le k$ by **Step 3**, the image of $f_*(\delta_E^{M_E})$ in G is trivial, that is, $f_*(\delta_E^{M_E}) \in N$. Since N is normal in Λ, $f_*(R) \subset N$.

Next we show that $N \subset f_*(R)$. The group N is generated by the elements $\beta \gamma_i^{m_i} \beta^{-1}$ with $\beta \in \Lambda$ and $1 \le i \le r$. Fix $\beta \in \Lambda$ and i. Let $x \in D_i \setminus V$ be a general point, and let W be a polydisc around x with $D \cap W = D_i \cap W = \{z_1 = 0\}$. By Lemma 2.3(5), γ_i is represented by a path σ from x_0 to a point $x_1 \in W^\circ$, followed by a loop μ in W° turning once positively around $\{z_1 = 0\}$, followed by σ^{-1}. Let τ be a loop representing β followed by σ. Then $\beta \gamma_i^{m_i} \beta^{-1}$ is represented by $\tau \mu^{m_i} \tau^{-1}$. Lift τ to a path in Z° starting at ζ_0, and let $\zeta_1 \in (h^\circ)^{-1}(x_1)$ be the last point of the lift. By **Step 1**, the component of $h^{-1}(W)$ containing ζ_1 is a polydisc with coordinates $t_1, \dots, t_d$ satisfying (6.1) for a positive integer e dividing m_i. Thus μ^e lifts to a meridian of $\{t_1 = 0\}$ based at ζ_1, and $\mu^{m_i} = (\mu^e)^{m_i/e}$ lifts to the (m_i/e)-th power of the meridian. Let E_Z be the prime divisor of Z containing $\{t_1 = 0\}$, and let $E \subset Y$ be the strict transform of E_Z. Then $f(E) \not\subset V$, so $M_E = m_i/e$ by **Step 3**. Since g is an isomorphism over $h^{-1}(X \setminus V)$, the lift of $\tau \mu^{m_i} \tau^{-1}$ starting at ζ_0 is a closed loop representing a conjugate of $\delta_E^{M_E}$ in $\pi_1(Y^\circ, y_0) = \pi_1(Z^\circ, \zeta_0)$ by Lemma 2.3(5). Hence $\beta \gamma_i^{m_i} \beta^{-1} \in f_*(R)$. Note that β is an arbitrary element of Λ, since the lift of τ may end on any sheet over x_1. Thus $N \subset f_*(R)$.

Therefore $f_*(R) = N$, and f_* induces an isomorphism

$$\pi_1^{\mathrm{orb}}(Y, \Delta_Y) = \pi_1(Y^\circ, y_0)/R \to \Lambda_H/N = H.$$

By Theorem 2.7, the homomorphism $f_* : \pi_1^{\mathrm{orb}}(Y, \Delta_Y) \to G$ is compatible with $f_* : \pi_1(Y^\circ, y_0) \to \Lambda$. Since $\pi_1(Y^\circ, y_0) \to \pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ is surjective, f_* is the isomorphism $\pi_1^{\mathrm{orb}}(Y, \Delta_Y) \to H$ induced by f_*, followed by the inclusion $H \subset G$. This proves (3).

Step 5. In this step we construct H_0 and prove the statement on the kernel of $\pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0}) \to \pi_1(Y_0)$.

Since X is compact, there exist finitely many polydiscs $W_1, \dots, W_s$ as in **Step 1** such that any point $x \in X$ has a polydisc W around x as in **Step 1** with $W \subset W_j$ for some j. For any j, fix $x_j \in W_j \backslash D$ and a path from x_0 to x_j in $X \backslash D$, and let $I_j \subset G$ be the image of $\pi_1(W_j \backslash D, x_j)$ under transport along the path. The group $\pi_1(W_j \backslash D, x_j)$ is free abelian with a basis of meridians of components of D, whose images in G have orders dividing the corresponding m_i by Lemma 2.3(5). Thus I_j is a finite abelian group. If $W \subset W_j$ is a polydisc as in **Step 1**, then the image in G of $\pi_1(W \backslash D)$ under transport along any path from x_0 is contained in a conjugate of I_j.

The set $\bigcup_{j=1}^{s} (I_j \backslash R_{\mathrm{fin}}(G))$ is finite. By Definition 2.16, for any element η of the set there exists a normal subgroup $G_\eta \trianglelefteq G$ of finite index with $\eta \notin G_\eta$. Let $H_0 := \bigcap_\eta G_\eta$, and let $H_0 := G$ if the set is empty. Then $H_0 \trianglelefteq G$ is of finite index, $R_{\mathrm{fin}}(G) \subset H_0$, and

$$I_j \cap H_0 = I_j \cap R_{\mathrm{fin}}(G)$$

for $1 \leq j \leq s$. Since H_0 and $R_{\mathrm{fin}}(G)$ are normal in G, $\xi I_j \xi^{-1} \cap H_0 = \xi I_j \xi^{-1} \cap R_{\mathrm{fin}}(G)$ for any $\xi \in G$. By Lemma 2.17(2), $R_{\mathrm{fin}}(H_0) = R_{\mathrm{fin}}(G)$.

Let $f_0 : (Y_0, \Delta_{Y_0}) \to (X, \Delta)$ be a finite-index model of H_0, let $Y_0^\circ := f_0^{-1}(X \backslash D)$, and fix $y_0' \in Y_0^\circ$ over x_0. Let $\Lambda_{H_0} \subset \Lambda$ be the inverse image of H_0. By (2), f_{0*} maps $\pi_1(Y_0^\circ, y_0')$ isomorphically onto Λ_{H_0}, since Λ_{H_0} is normal in Λ. By (1), $(f_0^{-1}D)_{\mathrm{red}}$ is a simple normal crossing divisor. By Lemma 2.3(1)(2), the homomorphism $\pi_1(Y_0^\circ, y_0') \to \pi_1(Y_0, y_0')$ is surjective, and the kernel is the normal subgroup generated by the meridians δ_E of the prime divisors $E \subset (f_0^{-1}D)_{\mathrm{red}}$. Since $\pi_1(Y_0^\circ, y_0') \to \pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0})$ is surjective, the kernel of $\pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0}) \to \pi_1(Y_0)$ is the normal subgroup generated by the images of the δ_E. By (3) and Theorem 2.7, the image of δ_E in $\pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0}) \cong H_0$ is the image of $f_{0*}(\delta_E)$ in G.

Let $E \subset (f_0^{-1}D)_{\mathrm{red}}$ be a prime divisor, and let $y \in E$ be a point not contained in any other component of $(f_0^{-1}D)_{\mathrm{red}}$. Let $W \subset W_j$ be a polydisc around $f_0(y)$ as in **Step 1**, and let $Y_E \subset f_0^{-1}(W)$ be a coordinate polydisc around y such that $Y_E \cap (f_0^{-1}D)_{\mathrm{red}} = Y_E \cap E$ is a smooth prime divisor of Y_E. Then $f_0(Y_E \backslash E) \subset W \backslash D$. By Lemma 2.3(3) and (5), $f_{0*}(\delta_E)$ is conjugate in Λ to the transport of an element of $\pi_1(W \backslash D)$. Thus the image of $f_{0*}(\delta_E)$ in G is contained in a conjugate of I_j. Since $f_{0*}(\delta_E) \in \Lambda_{H_0}$, the image is also contained in H_0. Hence the image of $f_{0*}(\delta_E)$ in G belongs to $R_{\mathrm{fin}}(G) = R_{\mathrm{fin}}(H_0)$. Since $R_{\mathrm{fin}}(H_0)$ is normal in H_0, the kernel of $\pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0}) \cong H_0 \to \pi_1(Y_0)$ is contained in $R_{\mathrm{fin}}(H_0)$.

Finally, assume that $\pi_1(Y_0)$ is residually finite, and let $\eta \in R_{\mathrm{fin}}(H_0)$. If the image of η in $\pi_1(Y_0)$ is non-trivial, then there exists a normal subgroup of finite index in $\pi_1(Y_0)$ not containing the image of η, and the inverse image of the subgroup in H_0 is a normal subgroup of finite index not containing η, a contradiction. Thus $R_{\mathrm{fin}}(H_0)$ is contained in the kernel of $H_0 \to \pi_1(Y_0)$, and the kernel equals $R_{\mathrm{fin}}(H_0)$. □

6.2. **The orbifold Albanese map.** In this subsection, we prove that the Albanese map of a special compact Kähler orbifold pair induces a surjection from the orbifold fundamental group onto the fundamental group of the Albanese torus, with kernel equal to the image of the orbifold fundamental group of a general fiber (Proposition 6.2).

Proposition 6.2 generalizes [OAI26c, Lemma 6.1], which treats compact Kähler manifolds, and we follow its method. A meridian of a prime divisor J contained in $A \backslash A^\circ$ may map to an element of finite order $m_J > 1$ in G/N (**Step 2**). For compact Kähler manifolds, $m_J > 1$ only if the fibers of the Albanese map over the general points of J are multiple, while for orbifold pairs, components of $\operatorname{Supp} \Delta$ mapping onto J also allow $m_J > 1$. Multiple fibers in codimension one are excluded in [Cam23, Theorem 1.2(3)] only for maps with connected fibers to abelian varieties. For maps to compact complex tori that are not abelian varieties, this case is not treated there in general [Cam23, Remarks 1.3(1)]. Here the Albanese torus need not be algebraic. **Step 3** shows that $m_J = 1$ for any such divisor by Theorem 3.1.

Proposition 6.2. *Let (X, Δ) be a special compact Kähler orbifold pair, let $G := \pi_1^{\mathrm{orb}}(X, \Delta)$, let $\alpha : X \to A := \mathrm{Alb}(X)$ be the Albanese map, and let $\Lambda := \pi_1(A)$. Then α is surjective with*

connected fibers, and there exists a non-empty Zariski open subset $A^\circ \subset A$ such that α is smooth over A° and the restriction of α to each intersection of components of $\operatorname{Supp}\Delta$ *is a submersion over A°. For $a \in A^\circ$ let $X_a := \alpha^{-1}(a)$ and $\Delta_{X_a} := \Delta|_{X_a}$, and let $N \leq G$ be the image of $\pi_1^{\mathrm{orb}}(X_a, \Delta_{X_a}) \to G$. Then N is a finitely generated normal subgroup of G which does not depend on the choice of $a \in A^\circ$, and the sequence*

$$1 \to N \to G \xrightarrow{\alpha_*} \Lambda \to 1$$

is exact.

Proof. The proof is long, so we divide it into several steps. Write $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i})D_i$, $D := \operatorname{Supp}\Delta$, and $n := \dim X$.

Step 1. In this step we construct A°, we show that N is a finitely generated normal subgroup of G which does not depend on a, and we construct a surjection $\psi : \pi_1(A^\circ) \to G/N$.

By Corollary 3.6(1), X is special. Thus α is surjective with connected fibers by [Cam23, Theorem 1.2(1)(2)]. By Lemma 2.4, applied to α and D, there exists a proper closed analytic subset $T' \subsetneq A$ with the following properties, where $A^\circ := A\backslash T'$ and $X^\circ := \alpha^{-1}(A^\circ)\backslash D$: α is smooth over A°, the restriction of α to each intersection of components of D is a submersion over A°, and $\alpha|_{X^\circ} : X^\circ \to A^\circ$ is a locally trivial fiber bundle. Since A is a connected complex manifold, A° is a non-empty connected Zariski open subset of A.

Fix $a \in A^\circ$ and $x_0 \in X_a\backslash D$. We use x_0 as the base point of the fundamental groups of $X_a\backslash D$, X°, and $X\backslash D$, and of G, and we use $a = \alpha(x_0)$ as the base point of the fundamental groups of A° and A. Since α and the restriction of α to each intersection of components of D are submersions at the points of X_a, the fiber X_a is a connected compact Kähler manifold, X_a is transverse to each D_i, and $D \cap X_a$ is a simple normal crossing divisor on X_a. Thus (X_a, Δ_{X_a}) is an orbifold pair, and any prime component E of $D \cap X_a$ is contained in D_i for exactly one i, with $\operatorname{mult}_E \Delta_{X_a} = 1 - \frac{1}{m_i}$.

We construct the homomorphism $\pi_1^{\mathrm{orb}}(X_a, \Delta_{X_a}) \to G$. Let E be a prime component of $D \cap X_a$ with $E \subset D_i$, and let $y \in E$ be a point which is not contained in D_j for $j \neq i$. Since X_a is transverse to D_i, there exists a polydisc $W \subset X$ around y with coordinates $z_1, \dots, z_n$ such that $D \cap W = \{z_1 = 0\}$ and $X_a \cap W = \{z_{c+1} = \cdots = z_n = 0\}$, where $c := \dim X_a$. In particular, $\operatorname{mult}_E(z_1|_{X_a \cap W}) = 1$, and by Lemma 2.3(3), applied with $k = 1$ to the inclusion $X_a \cap W \to W$, the inclusion maps a meridian of $E \cap W$ to the meridian of $\{z_1 = 0\}$ in $\pi_1(W\backslash D) \cong \mathbb{Z}$. Hence the map $\pi_1(X_a\backslash D) \to \pi_1(X\backslash D)$ maps any meridian of E to a meridian of D_i, which is conjugate to γ_i by Lemma 2.3(5). Thus the m_i-th power of the image of any meridian of E in G is 1. Therefore the composite $\pi_1(X_a\backslash D) \to \pi_1(X\backslash D) \to G$ factors through $\pi_1^{\mathrm{orb}}(X_a, \Delta_{X_a})$. Since $\pi_1(X_a\backslash D) \to \pi_1^{\mathrm{orb}}(X_a, \Delta_{X_a})$ is surjective, N is equal to the image of $\pi_1(X_a\backslash D)$ in G. By Lemma 2.3(4), $\pi_1(X_a\backslash D)$ is finitely generated, hence N is finitely generated.

By Lemma 2.4 (see also [Hat02, Theorem 4.41 and Proposition 4.48]), the sequence

$$\pi_1(X_a\backslash D) \to \pi_1(X^\circ) \xrightarrow{\rho} \pi_1(A^\circ) \to 1$$

is exact, where $\rho := (\alpha|_{X^\circ})_*$. Let $N^\circ \leq \pi_1(X^\circ)$ be the image of $\pi_1(X_a\backslash D)$, so that $N^\circ = \ker\rho$. By Lemma 2.3(1), applied to the manifold $X\backslash D$ and the closed analytic subset $\alpha^{-1}(T')\backslash D \subsetneq X\backslash D$, the map $\pi_1(X^\circ) \to \pi_1(X\backslash D)$ is surjective. Thus the composite $\chi : \pi_1(X^\circ) \to \pi_1(X\backslash D) \to G$ is surjective, and $\chi(N^\circ) = N$. Since N° is normal in $\pi_1(X^\circ)$, N is normal in G. For another point $a' \in A^\circ$, a path in X° from x_0 to a point of $X_{a'}\backslash D$ induces isomorphisms of fundamental groups which commute with ρ and carry the image of $\pi_1(X_{a'}\backslash D)$ onto $\ker\rho = N^\circ$. Since N is normal in G, N does not depend on the choice of $a \in A^\circ$.

Let $\bar\chi : \pi_1(X^\circ) \to G/N$ be the composite of χ with the quotient map. Since $\bar\chi(\ker\rho) = \bar\chi(N^\circ) = 1$ and ρ is surjective, there exists a unique homomorphism $\psi : \pi_1(A^\circ) \to G/N$ such that

$$\psi \circ \rho = \bar\chi. \tag{6.2}$$

Since $\bar\chi$ is surjective, ψ is surjective.

Step 2. In this step we show that ψ maps the meridians of the prime divisors contained in $A\backslash A^\circ$ to elements of finite order.

Let $V \subset A$ be the union of the irreducible components of $T' = A\backslash A^\circ$ of codimension one. By Lemma 2.3(1), the map $\pi_1(A^\circ) \to \pi_1(A\backslash V)$ is an isomorphism, and we use it to regard meridians of the irreducible components of V (Definition 2.1) as elements of $\pi_1(A^\circ)$. For an irreducible component J of V, let $\mu_J \in \pi_1(A^\circ)$ be a meridian of J. By Lemma 2.3(5), any two meridians of J are conjugate, so the order $m_J \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$ of $\psi(\mu_J)$ in G/N does not depend on μ_J.

For a prime divisor E on X, let $r_E := m_i$ if $E = D_i$, and $r_E := 1$ if $E \not\subset D$. We construct loops around the prime divisors of X over T'. Let E be a prime divisor on X with $E \subset \alpha^{-1}(T') \cup D$. Since $\alpha^{-1}(T') \neq X$, E is an irreducible component of $\alpha^{-1}(T') \cup D$. Let $x \in E$ be a general point. Then x is not contained in any irreducible component of $\alpha^{-1}(T') \cup D$ other than E, and E is smooth at x. Let $\mathbb{D} \subset \mathbb{C}$ be a small disc around 0 and let $\phi : \mathbb{D} \to X$ be a holomorphic embedding with $\phi(0) = x$ and $\phi^{-1}(E) = \{0\}$ which is transverse to E at x. Then $\phi(\mathbb{D}\backslash\{0\}) \subset X^\circ$. Let $\delta \in \pi_1(X^\circ)$ be the class of a path in X° from x_0 to a point of $\phi(\partial\mathbb{D}')$, followed by the loop $\phi(\partial\mathbb{D}')$ turning once positively, followed by the inverse path, where $\mathbb{D}' \subset \mathbb{D}$ is a smaller concentric disc. If $E \not\subset D$, then $x \notin D$, the loop $\phi(\partial\mathbb{D}')$ bounds the disc $\phi(\mathbb{D}') \subset X\backslash D$, and the image of δ in $\pi_1(X\backslash D)$ is 1. If $E = D_i$, then the image of δ in $\pi_1(X\backslash D)$ is a meridian of D_i, which is conjugate to γ_i by Lemma 2.3(5). In both cases,

$$\chi(\delta)^{r_E} = 1. \tag{6.3}$$

Now let J be an irreducible component of V. Since α is proper and surjective, $\alpha(\alpha^{-1}(J)) = J$, and $\alpha^{-1}(J) = \operatorname{Supp} \alpha^* J$. Since J is irreducible, there exists a prime divisor $E \subset \alpha^{-1}(J)$ with $\alpha(E) = J$. Let $e := \operatorname{mult}_E(\alpha^* J) \geq 1$, and let $x \in E$, ϕ, and δ be the general point, the disc, and the loop defined for E before (6.3). Since $\alpha(E) = J$ and x is general, $\alpha(x)$ is a general point of J. Thus there exists a polydisc $W \subset A$ around $\alpha(x)$ and a coordinate y_1 on W such that $T' \cap W = J \cap W = \{y_1 = 0\}$, and, after shrinking $\mathbb{D}$, $\alpha(\phi(\mathbb{D})) \subset W$. Since ϕ is transverse to E at x and $\alpha^* J = eE$ near x,

$$\operatorname{mult}_0 \left((\alpha \circ \phi)^* \{y_1 = 0\}\right) = e.$$

By Lemma 2.3(3), applied with $k = 1$ to $\alpha \circ \phi : \mathbb{D} \to W$, the loop $\alpha(\phi(\partial\mathbb{D}'))$ represents the e-th power of the meridian of $\{y_1 = 0\}$ in $\pi_1(W\backslash J) \cong \mathbb{Z}$. Since $W \cap A^\circ = W\backslash J$, Lemma 2.3(5) shows that $\rho(\delta)$ is conjugate to μ_J^e in $\pi_1(A^\circ)$. By (6.2) and (6.3), $\psi(\rho(\delta))^{r_E} = \bar{\chi}(\delta)^{r_E} = 1$. Thus $\psi(\mu_J)^{r_E e} = 1$, and

$$m_J \mid r_E \operatorname{mult}_E(\alpha^* J).$$

In particular, m_J is finite. We define the $\mathbb{Q}$-divisor $\Delta_A \geq 0$ by

$$\Delta_A := \sum_J \left(1 - \frac{1}{m_J}\right) J$$

on A, where J runs over the irreducible components of V.

Step 3. In this step we show that $\Delta_A = 0$.

Suppose that $\Delta_A \neq 0$. By [OAI26c, Lemma 2.2], there exist a quotient homomorphism $p : A \to B$ with connected fibers onto an abelian variety B with $b := \dim B > 0$ and an ample $\mathbb{Q}$-divisor $\Delta_B \geq 0$ on B such that $\Delta_A = p^* \Delta_B$. Since p is smooth with connected fibers, for any prime divisor J_B on B, $p^{-1} J_B$ is a prime divisor on A and $\operatorname{mult}_{p^{-1}J_B}(p^* J_B) = 1$. Thus the coefficient of J_B in Δ_B is equal to the coefficient of $p^{-1} J_B$ in Δ_A. For a prime component J_B of $\operatorname{Supp} \Delta_B$, $p^{-1} J_B$ is an irreducible component of V, and we let $m_{J_B} := m_{p^{-1}J_B}$. Then

$$\Delta_B = \sum_{J_B} \left(1 - \frac{1}{m_{J_B}}\right) J_B,$$

where J_B runs over the prime components of $\operatorname{Supp} \Delta_B$. Let $u := p \circ \alpha : X \to B$. Since u is surjective, $1 \leq b \leq n$.

We show the following divisibility. Let J_B be a prime component of $\operatorname{Supp}\Delta_B$ and let E be a prime divisor on X with $u(E) = J_B$. Then

$$m_{J_B} \mid r_E \operatorname{mult}_E(u^*J_B). \tag{6.4}$$

Let $J := p^{-1}J_B$ and $e := \operatorname{mult}_E(u^*J_B)$. Since $\alpha(E) \subset J \subset T'$, $E \subset \alpha^{-1}(T')$. Let $x \in E$ be a general point, and let ϕ and δ be as in **Step 2** for E and x. Since $u(E) = J_B$ and x is general, J_B is smooth at $u(x)$ and $u(x)$ is not contained in any prime component of $\operatorname{Supp}\Delta_B$ other than J_B. Let $y_1, \dots, y_b$ be coordinates on B around $u(x)$ with $J_B = \{y_1 = 0\}$. Since p is smooth, $p^*y_1, \dots, p^*y_b$ are part of a coordinate system on a polydisc $W \subset A$ around $\alpha(x)$, and, after shrinking W, $J \cap W = \{p^*y_1 = 0\}$ and $W \cap p^{-1}J'_B = \emptyset$ for any prime component $J'_B \neq J_B$ of $\operatorname{Supp}\Delta_B$. Let $R_W \subset W$ be the union of the irreducible components of $V \cap W$ other than $J \cap W$. By Lemma 2.3(1), the map $\pi_1(W \cap A^\circ) \to \pi_1(W \backslash (J \cup R_W))$ is an isomorphism, since $(T' \cap W)\backslash(J \cup R_W)$ has codimension at least two in $W\backslash(J \cup R_W)$. By Lemma 2.3(1)(2), applied to the manifold $W\backslash J$ and $R_W\backslash J$, the map $\pi_1(W\backslash(J \cup R_W)) \to \pi_1(W\backslash J)$ is surjective, and the kernel is normally generated by meridians of the irreducible components of $R_W\backslash J$. Fix a path in A° from a to a point of $W \cap A^\circ$, and let $\psi_W : \pi_1(W \cap A^\circ) \to G/N$ be the composite of the induced map $\pi_1(W \cap A^\circ) \to \pi_1(A^\circ)$ with ψ. A meridian of an irreducible component of $R_W\backslash J$ maps to a meridian of an irreducible component $J' \neq J$ of V which intersects W. Since $\operatorname{Supp}\Delta_A \cap W = p^{-1}(\operatorname{Supp}\Delta_B) \cap W = J \cap W$, $J' \not\subset \operatorname{Supp}\Delta_A$, that is, $m_{J'} = 1$. Thus ψ_W factors through $\pi_1(W\backslash J) \cong \mathbb{Z}$. The meridian λ of $\{p^*y_1 = 0\}$ is the image of a meridian of $J \cap W$ in $\pi_1(W \cap A^\circ)$, which maps to a meridian of J in $\pi_1(A^\circ)$. Hence ψ_W maps an element with image λ^k in $\pi_1(W\backslash J)$ to a conjugate of $\psi(\mu_J)^k$.

After shrinking the disc $\mathbb{D}$ of **Step 2**, $\alpha(\phi(\mathbb{D})) \subset W$ and $\alpha(\phi(\mathbb{D}\backslash\{0\})) \subset W \cap A^\circ$. Since $J = p^*J_B$ as divisors, $\alpha^*J = u^*J_B$, and since ϕ is transverse to E at x,

$$\operatorname{mult}_0\left((u \circ \phi)^*\{y_1 = 0\}\right) = \operatorname{mult}_E(u^*J_B) = e.$$

By Lemma 2.3(3), applied with $k = 1$ to $\alpha \circ \phi : \mathbb{D} \to W$ and the coordinate p^*y_1, the loop $\alpha(\phi(\partial\mathbb{D}'))$ maps to λ^e in $\pi_1(W\backslash J)$. Thus $\psi(\rho(\delta))$ is conjugate to $\psi(\mu_J)^e$. By (6.2) and (6.3), $\psi(\rho(\delta))^{r_E} = 1$. Thus $\psi(\mu_J)^{r_E e} = 1$, and $m_{J_B} = m_J$ divides $r_E e$. This proves (6.4).

We construct a root line. Let $\pi : \mathcal{X} \to X$ be the root orbifold of (X, Δ). Let η be a non-zero translation-invariant holomorphic b-form on B, so that η vanishes nowhere. Since u is surjective, u is a submersion at a general point of X, so $u^*\eta$ is a non-zero section of Ω^b_X. Let $\mathcal{F} \subset \Omega^b_X$ be the saturation of $\mathcal{O}_X \cdot u^*\eta$, and let $\mathcal{M} := \mathcal{M}_{\mathcal{F}} \subset \Omega^b_{\mathcal{X}}$ be the root line of $\mathcal{F}$ (Definition-Theorem 3.5(1)). Let $\sigma := \pi^*(u^*\eta)$, a section of $\Omega^b_{\mathcal{X}}$. Since the restriction of $\mathcal{M}$ over $X\backslash D$ is $\pi^*\mathcal{F}$, σ is a section of $\mathcal{M}$ over $\pi^{-1}(X\backslash D)$. Since $\mathcal{M}$ is saturated in $\Omega^b_{\mathcal{X}}$ (in the sense of Lemma 3.2), $\sigma \in H^0(\mathcal{X}, \mathcal{M})$, and $\sigma \neq 0$.

For a prime divisor E on X we define two numbers a_E and v_E. Let $x \in E$ be a general point. There exist a neighborhood U of x with coordinates $t, z_2, \dots, z_n$ such that $E \cap U = \{t = 0\}$ and $D \cap U \subset E$, and a root chart $U' \to U$ of $\mathcal{X}$ over U with coordinates $w, z_2, \dots, z_n$ on U' such that $t = w^{r_E}$, with r_E as in **Step 2**. Let $a_E \in \mathbb{Z}_{\geq 0}$ be the vanishing order of $\sigma|_{U'}$ along $\{w = 0\}$ as a section of the line bundle $\mathcal{M}|_{U'}$, and let

$$v_E := r_E \operatorname{mult}_E(u^*\Delta_B) \in \mathbb{Q}_{\geq 0},$$

which is the coefficient of $\{w = 0\}$ in the pullback of $u^*\Delta_B$ to U'. Neither number depends on the choice of the general point x. We show that there exists a rational number $0 < \epsilon \leq 1$ such that

$$\epsilon v_E \leq a_E \quad \text{for any prime divisor } E \text{ on } X. \tag{6.5}$$

There are three cases.

(1) If $u(E) \not\subset \operatorname{Supp}\Delta_B$, then $v_E = 0 \leq a_E$.
(2) Assume that $u(E) = J_B$ for a prime component J_B of $\operatorname{Supp}\Delta_B$, and let $e := \operatorname{mult}_E(u^*J_B)$. Let $y_1, \dots, y_b$ be coordinates on B around $u(x)$ with $J_B = \{y_1 = 0\}$, and write $\eta =$

$c_1 dy_1 \wedge \cdots \wedge dy_b$, where c_1 is a nowhere vanishing holomorphic function. Since $u^* J_B = eE$ near x, $u^* y_1 = c_2 t^e$ near x for a nowhere vanishing holomorphic function c_2. On U' we have $u^* y_1 = c_2 w^{r_E e}$, so

$$\sigma|_{U'} = w^{r_E e - 1}(c_1 \circ u)\big(r_E e c_2 dw + w dc_2\big) \wedge d(u^* y_2) \wedge \cdots \wedge d(u^* y_b).$$

Since $\mathcal{M}$ is saturated in $\Omega^b_{\mathcal{X}}$, $w^{1-r_E e}\sigma|_{U'}$ is a section of $\mathcal{M}|_{U'}$. Thus $a_E \geq r_E e - 1$. By (6.4), $m_{J_B} \leq r_E e$. Since J_B is the only prime component of $\operatorname{Supp} \Delta_B$ containing $u(x)$, $v_E = r_E e(1 - \frac{1}{m_{J_B}})$. Therefore

$$a_E \geq r_E e - 1 \geq r_E e - \frac{r_E e}{m_{J_B}} = v_E.$$

(3) Assume that $u(E) \subset \operatorname{Supp} \Delta_B$ and $\operatorname{codim}_B u(E) \geq 2$. The differential of $u|_E$ at x has rank at most $\dim u(E) \leq b - 2$, so the differential of u at x has rank at most $b - 1$. Thus $u^*\eta$ vanishes at x. Since x is general, $u^*\eta$ vanishes along E, so $\sigma|_{U'}$ vanishes along $\{w = 0\}$ and $a_E \geq 1$.

Only finitely many prime divisors E satisfy $v_E > 0$, since they are components of $\operatorname{Supp} u^* \Delta_B$. Thus

$$\epsilon := \min\Big(\{1\} \cup \Big\{ \frac{1}{v_E} \;\Big|\; E \text{ as in (3) with } v_E > 0 \Big\}\Big)$$

satisfies (6.5).

We compute the Iitaka dimension of $\mathcal{M}$. Since $\epsilon \Delta_B$ is an ample $\mathbb{Q}$-divisor on the abelian variety B, there exists an integer $j \geq 1$ such that $j\epsilon\Delta_B$ is a very ample divisor with integer coefficients. We regard a section $s \in H^0(B, \mathcal{O}_B(j\epsilon\Delta_B))$ as a meromorphic function on B with $\operatorname{div}(s) + j\epsilon\Delta_B \geq 0$. Since u is surjective, $u^* s$ is a meromorphic function on X, and $\sigma^j \cdot \pi^* u^* s$ is a meromorphic section of $\mathcal{M}^j$. For a prime divisor E on X and a root chart U' with coordinates $w, z_2, \ldots, z_n$ as in the definition of a_E and v_E, the pole order of $\pi^* u^* s$ along $\{w = 0\}$ is at most $r_E \operatorname{mult}_E(u^*(j\epsilon\Delta_B)) = j\epsilon v_E$. By (6.5), the vanishing order of $\sigma^j \cdot \pi^* u^* s$ along $\{w = 0\}$ is at least $j a_E - j\epsilon v_E \geq 0$. Thus $\sigma^j \cdot \pi^* u^* s$ is holomorphic over the complement of a closed analytic subset of X of codimension at least two. By the Hartogs extension theorem on the root charts, it extends to a section $\tau_s \in H^0(\mathcal{X}, \mathcal{M}^j)$, and the extensions on different charts are compatible by uniqueness. The linear map $s \to \tau_s$ is injective because u is surjective and $\sigma \neq 0$. Let $s_0, \ldots, s_\ell$ be a basis of $H^0(B, \mathcal{O}_B(j\epsilon\Delta_B))$, and let $\iota : B \to \mathbb{P}^\ell$ be the embedding defined by $s_0, \ldots, s_\ell$. Since $\tau_{s_l}/\tau_{s_0} = \pi^* u^*(s_l/s_0)$ for $0 \leq l \leq \ell$, the meromorphic map $\Psi : X \dashrightarrow \mathbb{P}^\ell$ defined by $\tau_{s_0}, \ldots, \tau_{s_\ell}$ is $\iota \circ u$, whose image $\iota(B)$ is of dimension b. The map Ψ is the composite of the meromorphic map $\Phi : X \dashrightarrow \mathbb{P}(H^0(\mathcal{X}, \mathcal{M}^j)^*)$ defined by $H^0(\mathcal{X}, \mathcal{M}^j)$ with a linear projection, so $\dim \Phi(X) \geq b$. Thus

$$\kappa(\mathcal{X}, \mathcal{M}) \geq b$$

by Definition 2.9. On the other hand, Definition-Theorem 3.5(4), applied to the inclusion $\mathcal{M} \to \Omega^b_{\mathcal{X}}$, implies that $\kappa(\mathcal{X}, \mathcal{M}) \leq b$. Thus $\kappa(\mathcal{X}, \mathcal{M}) = b$ with $1 \leq b \leq \dim X$, and the inclusion $\mathcal{M} \to \Omega^b_{\mathcal{X}}$ is non-zero. This contradicts Theorem 3.1, since (X, Δ) is special. Therefore $\Delta_A = 0$.

Step 4. In this step we prove that the sequence $1 \to N \to G \to \Lambda \to 1$ is exact.

By **Step 3**, $m_J = 1$ for any irreducible component J of V, that is, ψ maps any meridian of an irreducible component of V to 1. By Lemma 2.3(1)(2), applied to A and V, the map $\varpi : \pi_1(A^\circ) \cong \pi_1(A \backslash V) \to \Lambda$ is surjective, and the kernel is normally generated by the meridians of the irreducible components of V. Thus there exists a homomorphism $\bar{\psi} : \Lambda \to G/N$ such that $\bar{\psi} \circ \varpi = \psi$. On the other hand, $\alpha_* : G \to \Lambda$ is the composite of $G \to \pi_1(X)$ with the map $\pi_1(X) \to \Lambda$ induced by α. Since α maps X_a to the point a, $\alpha_*(N) = 1$. Thus α_* induces a homomorphism $\beta : G/N \to \Lambda$, and $\beta \circ \bar{\chi} = \varpi \circ \rho$, since both are induced by $\alpha|_{X^\circ}$. By (6.2),

$$\beta \circ \bar{\psi} \circ (\varpi \circ \rho) = \beta \circ \psi \circ \rho = \beta \circ \bar{\chi} = \varpi \circ \rho \quad \text{and} \quad \bar{\psi} \circ \beta \circ \bar{\chi} = \bar{\psi} \circ \varpi \circ \rho = \psi \circ \rho = \bar{\chi}.$$

Since $\varpi \circ \rho$ and $\bar{\chi}$ are surjective, $\beta \circ \bar{\psi} = \mathrm{id}_\Lambda$ and $\bar{\psi} \circ \beta = \mathrm{id}_{G/N}$. Thus β is an isomorphism, that is, α_* is surjective with kernel N. □

6.3. **Maximal irregularity.** In this subsection, we show that the orbifold fundamental group of a special compact Kähler orbifold pair contains a subgroup of finite index which is an extension of a free abelian group by a finitely generated group without proper subgroups of finite index (Proposition 6.5).

Definition 6.3. For a finitely generated group P, we call $q(P) := \frac{1}{2}\dim_{\mathbb{Q}} \mathrm{Hom}(P, \mathbb{Q})$ the *irregularity* of P. The *virtual irregularity* $\tilde{q}(P)$ of P is the supremum of $q(P')$ over all subgroups $P' \leq P$ of finite index.

Example 6.4. Let $p_1, \ldots, p_4 \in \mathbb{P}^1$ be distinct points and let $\Delta := \sum_{i=1}^4 \frac{1}{2} p_i$. The group $\pi_1(\mathbb{P}^1 \backslash \{p_1, \ldots, p_4\})$ is generated by meridians $\gamma_1, \ldots, \gamma_4$ of $p_1, \ldots, p_4$ with the single relation $\gamma_1\gamma_2\gamma_3\gamma_4 = 1$, for meridians joined to the base point by disjoint paths in the cyclic order of the points. Thus

$$G := \pi_1^{\mathrm{orb}}(\mathbb{P}^1, \Delta) = \langle \gamma_1, \ldots, \gamma_4 \mid \gamma_1^2, \ldots, \gamma_4^2, \gamma_1\gamma_2\gamma_3\gamma_4 \rangle,$$

the abelianization of G is $(\mathbb{Z}/2\mathbb{Z})^3$, and $q(G) = 0$. Let H be the kernel of the homomorphism $G \to \mathbb{Z}/2\mathbb{Z}$ sending each γ_i to 1, and let $C \to \mathbb{P}^1$ be the double cover branched over $p_1, \ldots, p_4$, which is an elliptic curve by the Hurwitz formula. The restriction of $C \to \mathbb{P}^1$ over $\mathbb{P}^1 \backslash \{p_1, \ldots, p_4\}$ is the connected double cover corresponding to the kernel Q of the homomorphism $\pi_1(\mathbb{P}^1 \backslash \{p_1, \ldots, p_4\}) \to \mathbb{Z}/2\mathbb{Z}$ sending each γ_i to 1. For $\xi \in Q$, the element $\xi\gamma_i^2\xi^{-1}$ lifts to a meridian of the ramification point over p_i, and

$$\xi\gamma_i \cdot \gamma_i^2 \cdot (\xi\gamma_i)^{-1} = \xi\gamma_i^2\xi^{-1}.$$

Thus the normal subgroup of $\pi_1(\mathbb{P}^1 \backslash \{p_1, \ldots, p_4\})$ generated by $\gamma_1^2, \ldots, \gamma_4^2$ is the normal subgroup of Q generated by the meridians of the four ramification points, and $H \cong \pi_1(C) \cong \mathbb{Z}^2$ by Lemma 2.3(2). So $q(H) = 1$. Any subgroup P' of finite index in G contains $P' \cap H$ as a subgroup of finite index, and $P' \cap H \cong \mathbb{Z}^2$. In general, if $P'' \leq P'$ is a subgroup of finite index, then for any $x \in P'$ some power x^k with $k \geq 1$ belongs to P'', as there are only finitely many cosets $x^l P''$. Since $\mathbb{Q}$ is torsion free, the restriction map $\mathrm{Hom}(P', \mathbb{Q}) \to \mathrm{Hom}(P'', \mathbb{Q})$ is injective, and so $q(P'') \geq q(P')$. Applied to the subgroup $P' \cap H \leq P'$ of finite index, the inequality $q(P'') \geq q(P')$ shows that $q(P') \leq q(P' \cap H) = 1$. Hence $\tilde{q}(G) = 1 > q(G)$.

The irregularity of a group can increase on passing to a subgroup of finite index, as Example 6.4 shows, and so we take the supremum over all subgroups of finite index.

Theorem 2.13 concerns compact Kähler manifolds, not orbifold pairs. To use it, we pass to a finite-index model given by the last sentence of Construction-Proposition 6.1: by that sentence, if the underlying manifold of the model has residually finite fundamental group, then the kernel of the natural map from the orbifold fundamental group of the model to the fundamental group of the underlying manifold is the finite residual. Theorem 2.13 shows that the fundamental group of the underlying manifold is virtually abelian, hence residually finite. After passing to a finite-index model of maximal irregularity and then to a finite unramified cover whose fundamental group is free abelian, the fundamental group of the manifold maps isomorphically onto the fundamental group of the Albanese torus, and the only unknown is the finite residual, a finitely generated group without proper subgroups of finite index.

Proposition 6.5. *Let (X, Δ) be a special compact Kähler orbifold pair and $G := \pi_1^{\mathrm{orb}}(X, \Delta)$. Then $q(H)$ is an integer and $q(H) \leq \dim X$ for any subgroup $H \leq G$ of finite index. In particular, $\tilde{q}(G)$ is an integer and $\tilde{q}(G) = q(H)$ for some subgroup $H \leq G$ of finite index. Moreover, there exist a subgroup $G_Y \leq G$ of finite index, finite-index models $(X_1, \Delta_{X_1}) \to (X, \Delta)$ and $(Y_1, \Delta_{Y_1}) \to (X_1, \Delta_{X_1})$ (Construction-Proposition 6.1) of a subgroup of finite index in G and of a subgroup of finite index in $\pi_1^{\mathrm{orb}}(X_1, \Delta_{X_1})$, respectively, and a finite unramified cover $Y \to Y_1$ with the following properties, where Δ_Y is the pullback of Δ_{Y_1} to Y, the map $f : Y \to X$ is the composite $Y \to Y_1 \to X_1 \to X$, $\alpha_Y : Y \to A_Y := \mathrm{Alb}(Y)$ is the Albanese map, and $\Lambda := \pi_1(A_Y)$.*

(1) *The pair* (Y, Δ_Y) *is a special compact Kähler orbifold pair, the map* f *is generically finite and surjective,* f *induces an isomorphism* f_* *from* $\pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ *onto* G_Y*, and* $\dim A_Y = q(G_Y) = \tilde{q}(G)$.
(2) *The homomorphism* $(\alpha_Y)_* : \pi_1(Y) \to \Lambda$ *is an isomorphism.*
(3) *Let* $G_Y \to \pi_1(Y)$ *be the composite of the inverse of the isomorphism* f_* *in (1) with the natural map* $\pi_1^{\mathrm{orb}}(Y, \Delta_Y) \to \pi_1(Y)$*, and let* $G_Y \to \Lambda$ *be the composite of* $G_Y \to \pi_1(Y)$ *with* $(\alpha_Y)_*$*. Then the subgroups* $\ker(G_Y \to \Lambda)$, $\ker(G_Y \to \pi_1(Y))$*, and* $R_{\mathrm{fin}}(G_Y)$ *of* G_Y *are equal.*
(4) *The group* $R_{\mathrm{fin}}(G_Y)$ *is finitely generated and has no subgroup of finite index other than* $R_{\mathrm{fin}}(G_Y)$*. In particular, either* $R_{\mathrm{fin}}(G_Y) = \{1\}$*, or* $R_{\mathrm{fin}}(G_Y)$ *is infinite and* $R_{\mathrm{fin}}(G_Y) = [R_{\mathrm{fin}}(G_Y), R_{\mathrm{fin}}(G_Y)]$.

Proof. The proof is long, so we divide it into several steps. Write $\Delta = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$, $D := \operatorname{Supp} \Delta$, and $n := \dim X$. By Lemma 2.3(4), $\pi_1(X \backslash D)$ is finitely generated, so G and any subgroup of finite index in G are finitely generated.

Step 1. In this step we prove the first two sentences of the proposition.

Let $(Y', \Delta_{Y'})$ be an orbifold pair such that Y' is a connected compact Kähler manifold. We show that

$$q\big(\pi_1^{\mathrm{orb}}(Y', \Delta_{Y'})\big) = \dim \operatorname{Alb}(Y'). \tag{6.6}$$

By Lemma 2.3(1)(2), the map $\pi_1(Y' \setminus \operatorname{Supp} \Delta_{Y'}) \to \pi_1(Y')$ is surjective, and the kernel is normally generated by meridians of the components of $\operatorname{Supp} \Delta_{Y'}$. Thus the natural map $\pi_1^{\mathrm{orb}}(Y', \Delta_{Y'}) \to \pi_1(Y')$ is surjective, and the kernel is normally generated by the images of the meridians of the components of $\operatorname{Supp} \Delta_{Y'}$, which are of finite order. Any homomorphism $\pi_1^{\mathrm{orb}}(Y', \Delta_{Y'}) \to \mathbb{Q}$ maps elements of finite order to 0, hence

$$\operatorname{Hom}\big(\pi_1^{\mathrm{orb}}(Y', \Delta_{Y'}), \mathbb{Q}\big) = \operatorname{Hom}\big(\pi_1(Y'), \mathbb{Q}\big).$$

By Hodge theory and the definition of the Albanese torus, $\dim_{\mathbb{Q}} \operatorname{Hom}(\pi_1(Y'), \mathbb{Q}) = b_1(Y') = 2 \dim \operatorname{Alb}(Y')$ [Voi02, Chapter 6 and Section 12.1.3]. This proves (6.6).

Let $H \le G$ be a subgroup of finite index and let $f_H : (Y_H, \Delta_{Y_H}) \to (X, \Delta)$ be a finite-index model of H (Construction-Proposition 6.1). By Construction-Proposition 6.1(3), $\pi_1^{\mathrm{orb}}(Y_H, \Delta_{Y_H}) \cong H$, so $q(H) = \dim \operatorname{Alb}(Y_H)$ by (6.6). In particular, $q(H)$ is an integer. By Construction-Proposition 6.1(4), (Y_H, Δ_{Y_H}) is special, so Y_H is special by Corollary 3.6(1). Thus the Albanese map of Y_H is surjective by [Cam23, Theorem 1.2(1)], and

$$0 \le q(H) = \dim \operatorname{Alb}(Y_H) \le \dim Y_H = n.$$

Hence $\tilde{q}(G)$ is the maximum of a non-empty set of integers in $[0, n]$, so $\tilde{q}(G)$ is an integer and $\tilde{q}(G) = q(H)$ for some subgroup $H \le G$ of finite index. For special $\mathcal{C}$-pairs [KR24a] on compact Kähler spaces, the analogous bound for the Albanese irregularities of covers is [KR24b, Corollary 8.2 and Remark 8.4].

Step 2. In this step we construct X_1, Y_1, Y, f, and G_Y, and we prove (1) except the equalities $\dim A_Y = q(G_Y) = \tilde{q}(G)$.

By **Step 1**, there exists a subgroup $H \le G$ of finite index with $q(H) = \tilde{q}(G)$. Let $f_1 : (X_1, \Delta_{X_1}) \to (X, \Delta)$ be a finite-index model of H, and let $G_1 := \pi_1^{\mathrm{orb}}(X_1, \Delta_{X_1})$. By Construction-Proposition 6.1(1)(3)(4), (X_1, Δ_{X_1}) is a special compact Kähler orbifold pair, f_1 is a divisible orbifold morphism, and $(f_1)_* : G_1 \to H$ is an isomorphism. By the last sentence of Construction-Proposition 6.1, applied to (X_1, Δ_{X_1}), there exists a normal subgroup $H_0 \le G_1$ of finite index such that, for any finite-index model $f_0 : (Y_0, \Delta_{Y_0}) \to (X_1, \Delta_{X_1})$ of H_0, the kernel of $\pi_1^{\mathrm{orb}}(Y_0, \Delta_{Y_0}) \cong H_0 \to \pi_1(Y_0)$ is $R_{\mathrm{fin}}(H_0)$ if $\pi_1(Y_0)$ is residually finite. Let $f_0 : (Y_1, \Delta_{Y_1}) \to (X_1, \Delta_{X_1})$ be a finite-index model of H_0, let $G_2 := \pi_1^{\mathrm{orb}}(Y_1, \Delta_{Y_1})$, and let $P_1 := \pi_1(Y_1)$. By Construction-Proposition 6.1(3)(4), f_0 is a divisible orbifold morphism, $(f_0)_* : G_2 \to H_0$ is an isomorphism, and (Y_1, Δ_{Y_1}) is special. By Corollary 3.6(1), Y_1 is special.

By the second sentence of Theorem 3.1 and by Theorem 2.13, P_1 is virtually abelian. Since Y_1 is a compact manifold, P_1 is finitely generated (Lemma 2.3(4)), so P_1 is residually finite by Lemma 2.17(4). Since $(f_0)_*$ is an isomorphism onto H_0, it follows that

$$\ker(G_2 \to P_1) = R_{\mathrm{fin}}(G_2). \tag{6.7}$$

A finitely generated abelian subgroup of finite index in P_1 contains a free abelian subgroup of finite index. Let $P_2 \leq P_1$ be a free abelian subgroup of finite index, and let $v : Y \to Y_1$ be the connected covering space corresponding to P_2. Then v is a finite unramified cover, Y has a unique complex structure for which v is holomorphic, and the pullback of a Kähler form on Y_1 is a Kähler form on Y. Thus Y is a connected compact Kähler manifold and $\pi_1(Y) \cong P_2$. Let $\Delta_Y := v^*\Delta_{Y_1}$. Since v is a local biholomorphism, $\operatorname{Supp}\Delta_Y = v^{-1}\operatorname{Supp}\Delta_{Y_1}$ is a simple normal crossing divisor, and for any prime component E of $\operatorname{Supp}\Delta_Y$, the coefficient of E in Δ_Y is equal to the coefficient of $v(E)$ in Δ_{Y_1}. Thus (Y, Δ_Y) is an orbifold pair. Since $v^*(K_{Y_1} + \Delta_{Y_1}) = K_Y + \Delta_Y$, the map $v : (Y, \Delta_Y) \to (Y_1, \Delta_{Y_1})$ is étale in codimension one in the sense of [Cam11a, Définition 3.10]. Since Y and Y_1 are connected compact Kähler manifolds, hence in Fujiki's class $\mathcal{C}$, [Cam11a, Proposition 10.11] implies that (Y, Δ_Y) is special in the sense of [Cam11a, Définition 8.1], hence special in the sense of Definition 2.11 by Remark 2.12.

The map $v : (Y, \Delta_Y) \to (Y_1, \Delta_{Y_1})$ is a divisible orbifold morphism: for a prime divisor E on Y and a prime component D' of $\operatorname{Supp}\Delta_{Y_1}$, $\operatorname{mult}_E(v^*D') \in \{0, 1\}$, and if $\operatorname{mult}_E(v^*D') = 1$, then the coefficient of E in Δ_Y is equal to the coefficient of D' in Δ_{Y_1}. A composite of divisible orbifold morphisms is a divisible orbifold morphism [Cam11a, Remarque 2.5(2)]. Thus the composite $f := f_1 \circ f_0 \circ v : (Y, \Delta_Y) \to (X, \Delta)$ is a divisible orbifold morphism, and f is generically finite and surjective. By Theorem 2.7, f, f_1, f_0, and v induce homomorphisms f_*, $(f_1)_*$, $(f_0)_*$, and v_* of orbifold fundamental groups.

We show that $f_* = (f_1)_* \circ (f_0)_* \circ v_*$. Let $Y^\circ := Y \backslash f^{-1}(D)$. By Construction-Proposition 6.1, $\operatorname{Supp}\Delta_{X_1} \subset f_1^{-1}(D)$ and $\operatorname{Supp}\Delta_{Y_1} \subset f_0^{-1}(\operatorname{Supp}\Delta_{X_1})$, so $\operatorname{Supp}\Delta_Y \subset f^{-1}(D)$. Thus $Y^\circ \subset Y \backslash \operatorname{Supp}\Delta_Y$, $v(Y^\circ) \subset Y_1 \backslash f_0^{-1}(\operatorname{Supp}\Delta_{X_1})$, and $f_0(v(Y^\circ)) \subset X_1 \backslash f_1^{-1}(D)$, and the right-hand sides are the complements on which the compatibility in Theorem 2.7 holds for v, f_0, and f_1, respectively. Hence f_* and $(f_1)_* \circ (f_0)_* \circ v_*$ are both compatible with the map $\pi_1(Y^\circ) \to \pi_1(X \backslash D)$ induced by f. By Lemma 2.3(1) and the definition of π_1^{orb}, the map $\pi_1(Y^\circ) \to \pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ is surjective, so $f_* = (f_1)_* \circ (f_0)_* \circ v_*$.

We show that v_* is injective and that the image of v_* is the inverse image Q_2 of P_2 under $G_2 \to P_1$. Let $S_1 := \operatorname{Supp}\Delta_{Y_1}$ and $S := \operatorname{Supp}\Delta_Y = v^{-1}(S_1)$. The restriction $Y \backslash S \to Y_1 \backslash S_1$ of v is the connected covering space corresponding to the inverse image Q of P_2 under $\pi_1(Y_1 \backslash S_1) \to P_1$. By Lemma 2.3(2), any meridian of a component of S_1 is trivial in P_1, so all meridians of components of S_1 belong to Q. Since v is a local biholomorphism, a meridian of a component S_1' of S_1 lifts to a meridian of a component S' of S with $v(S') = S_1'$, the coefficient of S' in Δ_Y is equal to the coefficient of S_1' in Δ_{Y_1}, and a meridian of a component of S maps to a meridian of a component of S_1. Since the conjugate of a meridian of a component S_1' of S_1 by an element of $\pi_1(Y_1 \backslash S_1)$ is again a meridian of S_1', the normal subgroup of $\pi_1(Y_1 \backslash S_1)$ generated by the m-th powers of the meridians of the components of S_1 with coefficient $1 - \frac{1}{m}$ is equal to the normal subgroup of $Q \cong \pi_1(Y \backslash S)$ generated by the m-th powers of the meridians of the components of S with coefficient $1 - \frac{1}{m}$. Thus v_* is injective, and the image of v_* is Q_2.

Let $G_Y := f_*(\pi_1^{\mathrm{orb}}(Y, \Delta_Y)) = (f_1)_*((f_0)_*(Q_2)) \leq H$. Since $(f_1)_*$, $(f_0)_*$, and v_* are injective, f_* is an isomorphism from $\pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ onto G_Y. Since $[G : H]$, $[G_1 : H_0]$, and $[G_2 : Q_2] = [P_1 : P_2]$ are finite, G_Y is a subgroup of finite index in H and in G. This proves (1) except the equalities $\dim A_Y = q(G_Y) = \tilde{q}(G)$.

Step 3. In this step we prove the equalities $\dim A_Y = q(G_Y) = \tilde{q}(G)$, (2), and (3).

By (6.6) and **Step 2**, $\dim A_Y = q(\pi_1^{\mathrm{orb}}(Y, \Delta_Y)) = q(G_Y)$. Since $G_Y \leq H$ is of finite index, $q(G_Y) \geq q(H) = \tilde{q}(G)$ by the inequality $q(P'') \geq q(P')$ of Example 6.4. By Definition 6.3, $q(G_Y) \leq \tilde{q}(G)$. Thus $\dim A_Y = q(G_Y) = \tilde{q}(G)$.

The group $\pi_1(Y) \cong P_2$ is free abelian. The Albanese torus satisfies

$$\pi_1(A_Y) = H_1(Y,\mathbb{Z})/H_1(Y,\mathbb{Z})_{\mathrm{tors}},$$

and $(\alpha_Y)_* : \pi_1(Y) \to \Lambda$ is the composite of the abelianization $\pi_1(Y) \to H_1(Y,\mathbb{Z})$ with the quotient by torsion [Voi02, Section 12.1.3]. Since $\pi_1(Y)$ is free abelian, both maps are isomorphisms. This proves (2).

By (2), $\ker(G_Y \to \Lambda) = \ker(G_Y \to \pi_1(Y))$. The natural maps $\pi_1^{\mathrm{orb}}(Y,\Delta_Y) \to \pi_1(Y)$ and $G_2 \to P_1$ are compatible with v_* and with the isomorphism $v_* : \pi_1(Y) \to P_2$, since both are induced by v on $Y\backslash S$. Thus v_* maps $\ker(\pi_1^{\mathrm{orb}}(Y,\Delta_Y) \to \pi_1(Y))$ onto

$$\ker(Q_2 \to P_2) = \ker(G_2 \to P_1) = R_{\mathrm{fin}}(G_2) = R_{\mathrm{fin}}(Q_2),$$

where the first equality holds because $\ker(G_2 \to P_1) \leq Q_2$, the second is (6.7), and the third is Lemma 2.17(2). Since f_* and v_* are isomorphisms onto G_Y and Q_2, respectively, $\ker(G_Y \to \pi_1(Y)) = R_{\mathrm{fin}}(G_Y)$. This proves (3).

Step 4. In this step we prove (4).

Let $R := R_{\mathrm{fin}}(G_Y)$. By Proposition 6.2, applied to (Y,Δ_Y), the kernel N_Y of the homomorphism $\pi_1^{\mathrm{orb}}(Y,\Delta_Y) \to \Lambda$ induced by α_Y is finitely generated. Since f_* is an isomorphism onto G_Y compatible with the homomorphisms to Λ, we have $f_*(N_Y) = \ker(G_Y \to \Lambda)$, which is R by (3). Thus R is finitely generated. By (2) and (3), $G_Y/R \cong \Lambda \cong \mathbb{Z}^{2\dim A_Y}$.

Let $R_0 \leq R$ be a subgroup of finite index. By Lemma 2.17(3), R has only finitely many subgroups of index $[R : R_0]$. Let R_1 be their intersection. Then $R_1 \leq R_0$, R_1 is of finite index in R, and R_1 is characteristic in R, since automorphisms of R permute the subgroups of index $[R : R_0]$. Since R is normal in G_Y, R_1 is normal in G_Y, and we obtain an exact sequence

$$1 \to R/R_1 \to G_Y/R_1 \to \mathbb{Z}^{2\dim A_Y} \to 1$$

with R/R_1 finite. By Lemma 2.17(5), G_Y/R_1 has a free abelian subgroup P_3 of finite index with $P_3 \cap (R/R_1) = \{1\}$. Let $G_0 \leq G_Y$ be the inverse image of P_3. Then G_0 is of finite index in G_Y and $G_0 \cap R = R_1$. By Lemma 2.17(2), $R \leq G_0$. Thus $R = R_1 \leq R_0$, that is, $R_0 = R$.

If R is finite, then $\{1\}$ is a subgroup of finite index in R, so $R = \{1\}$. If $R \neq [R,R]$, then $R/[R,R]$ is a non-zero finitely generated abelian group, which has a proper subgroup of finite index, and the inverse image in R of a proper subgroup of finite index in $R/[R,R]$ is a proper subgroup of finite index in R, a contradiction. Thus either $R = \{1\}$, or R is infinite and $R = [R,R]$. This proves (4). □

7. Reduction to algebraic dimension zero

In this section, we reduce Theorem 1.2, by induction on the dimension, to the case of Proposition 7.8, in which $a(X) = 0$, $\pi_1(X) = \{1\}$, and the group $G := \pi_1^{\mathrm{orb}}(X,\Delta)$ is infinite, satisfies $G = [G,G]$, and has no proper subgroup of finite index. When $a(X) > 0$ we use the algebraic reduction of X, and when $a(X) = 0$ we use the Albanese maps of finite-index models (Construction-Proposition 6.1) of subgroups of finite index of $\pi_1^{\mathrm{orb}}(X,\Delta)$.

7.1. Orbifold bases of fibrations.

In this subsection, we define the orbifold base and the gcd base of a fibration, and neat fibrations.

Definition 7.1. Let (X,Δ) be an orbifold pair and $f : X \to Z$ a surjective holomorphic map with connected fibers onto a compact complex manifold Z. For a prime divisor E on X let m_E be the integer with $\mathrm{mult}_E \Delta = 1 - \frac{1}{m_E}$. For a prime divisor P on Z, let

$$m_{\inf}(P) := \min_E \left(m_E \,\mathrm{mult}_E\, f^*P\right) \qquad \text{and} \qquad m_{\gcd}(P) := \gcd_E \left(m_E \,\mathrm{mult}_E\, f^*P\right),$$

where E runs through the prime divisors on X with $f(E) = P$. The $\mathbb{Q}$-divisors $\Delta_{\inf}(f) := \sum_P (1 - \frac{1}{m_{\inf}(P)})P$ and $\Delta_{\gcd}(f) := \sum_P (1 - \frac{1}{m_{\gcd}(P)})P$ on Z are called the *orbifold base* and the *gcd base* of f.

The multiplicities $m_{\mathrm{inf}}(P)$ are those of [Cam11a, Définition 4.2]. The orbifold base is the base used in the definition of special pairs by fibrations and in [Cam11a, Corollaire 9.7], while the gcd base is the base which appears in the exact sequence of [Cam11a, Proposition 12.9]. In this paper, specialness is used on the orbifold base: in Proposition 7.5 the orbifold base is special by [Cam11a, Corollaire 9.7], and in Lemma 7.6 the orbifold base is the boundary on the base of the relative core. The fundamental groups are computed on the gcd base: in Proposition 7.5 the exact sequence of [Cam11a, Proposition 12.9], on which [CC11, Theorem 4.1] is based, is taken over the gcd base, and Theorem 5.1 is applied to the gcd base.

Definition 7.2. Let (X,Δ) be an orbifold pair and $f : X \to Z$ a surjective holomorphic map with connected fibers onto a compact complex manifold Z. We say that f is *neat* if there exist an orbifold pair (X_0,Δ_0), a divisible elementary modification $\mu : (X,\Delta) \to (X_0,\Delta_0)$, a surjective holomorphic map $f_0 : X_0 \to Z_0$ with connected fibers onto a compact complex manifold Z_0, and a bimeromorphic holomorphic map $v : Z \to Z_0$ with $f_0 \circ \mu = v \circ f$, such that any prime divisor E on X with $\operatorname{codim}_Z f(E) \geq 2$ is μ-exceptional.

This is the divisible version of [Cam11a, Définition 4.8]. If f is neat, then the prime divisors on X whose images in Z have codimension at least two are exceptional for the divisible elementary modification μ, and μ_* is an isomorphism of orbifold fundamental groups (Theorem 2.7), so the orbifold data along such divisors are determined by (X_0,Δ_0). Neatness is the hypothesis of [Cam11a, Proposition 12.9], and neat fibrations exist after modifications of the base and of the total space [Cam11a, Proposition 4.10].

Remark 7.3. We compare the notion of specialness of [Lu02, Definition 6.1] with Definition 2.11. Let (X,Δ) be a compact Kähler orbifold pair, let $g : X \to Y$ be a surjective holomorphic map with connected fibers onto a compact complex manifold Y with $p := \dim Y > 0$, and let $\mathcal{L}_g \subset \Omega^p_X$ be the saturation of the image of g^*K_Y. Let E be a component of $\operatorname{Supp}\Delta$, and let ω be a local generator of K_Y. We show that the composite $\mathcal{L}_g|_E \to \Omega^p_X|_E \to \Omega^p_E$ vanishes at a general point of E if and only if $g(E) \neq Y$. If $g(E) = Y$, then g is a submersion at a general point of E, $g^*\omega$ generates $\mathcal{L}_g$ there, and the composite is $(g|_E)^*\omega \neq 0$. Suppose that $g(E) \neq Y$, let e be a general point of E, and let σ be a local generator of $\mathcal{L}_g$ near e. Since $\mathcal{L}_g$ is saturated, $\sigma(e) \neq 0$. Where g is a submersion, σ is a non-zero multiple of $g^*\omega$, so σ is decomposable with integrable kernel $\ker dg$. Both conditions are closed, so by the Frobenius theorem, near e we have $\sigma = c \cdot (dy'_1 \wedge \cdots \wedge dy'_p)$ for holomorphic functions $y'_1,\dots,y'_p$ and a unit c, and g is constant on the leaves, so $g = \varphi \circ y'$ for $y' := (y'_1,\dots,y'_p)$ and a holomorphic map φ. If $\sigma|_E(e) \neq 0$, then $y'|_E$ is a submersion at e, so $g(E)$ contains a non-empty open subset of Y, and $g(E) = Y$ since $g(E)$ is an irreducible closed analytic subset of Y, a contradiction. Hence the composite vanishes at a general point of E.

Let Δ^g be the sum of the terms of Δ whose components E satisfy $g(E) \neq Y$. By Definition-Theorem 3.5(2), for any positive integer q divisible by all the multiplicities of Δ,

$$(\mathcal{L}_g)_{q,\Delta} = \mathcal{L}_g^q \otimes \mathcal{O}_X(q\Delta^g).$$

The right-hand side is the q-th power of the $\mathbb{Q}$-line sheaf L_g of [Lu02, Definition 3.4], which Lu uses to define fibrations of general type [Lu02, Definition 5.5], while [Cam11a, Définitions 5.1 and 5.16] uses the algebra $\bigoplus_{q\geq 0} H^0(X,(\mathcal{L}_g)_{q,\Delta})$. Lu writes L_{g^∂} for L_g [Lu02, Definition 3.4] and L_g for $\mathcal{L}_g$ [Lu02, Definition 2.1]. The algebra $\bigoplus_{q\geq 0} H^0(X,(\mathcal{L}_g)_{q,\Delta})$ and the section algebra of L_g have equal Veronese subalgebras in degrees divisible by all the multiplicities of Δ, hence equal Iitaka dimensions. For a meromorphic fibration, both definitions pass to a modification of X on which it is holomorphic: Lu's by [Lu02, Lemma 3.11(1) and Lemma 3.20], and Campana's by [Cam11a, Théorème 3.5]. Thus (X,Δ) is special in the sense of [Lu02, Definition 6.1] if and only if it is special in the sense of [Cam11a, Définition 8.1], that is, in the sense of Definition 2.11 by Remark 2.12.

7.2. **Positive algebraic dimension.** In this subsection, we prove the induction step when the algebraic dimension is positive.

Set-up 7.4. Let d be as follows.

(1) $d \geq 1$ is an integer.
(2) Theorem 1.2 holds for any compact Kähler orbifold pair of dimension less than d.

We use the algebraic reduction of X for two reasons. The base of the algebraic reduction is projective, so that Theorem 5.1 can be used for the gcd base, and the very general fibers are special by the relative core of [Lu02, Proposition 8.7]: if a very general fiber were not special, there would be a Moishezon space dominated by X of dimension larger than $a(X)$. For an arbitrary fibration of a special pair, the general fibers need not be special: a pencil of plane quartics, after blowing up the base points, is a rational surface fibered over $\mathbb{P}^1$ whose general fiber is not special. To pass from the fiber and the base to the total space we use [CC11, Theorem 4.1], which depends on Hodge theory through [Cam11b, Corollaire 7.6]. Group theory alone is not enough: the integral Heisenberg group is an extension of a free abelian group by a free abelian group, but it is not virtually abelian. Compared with the case of manifolds, the new part of the proof is **Step 3**, where the multiplicities of the prime divisors whose images in Z have codimension at least two are raised, so that f becomes a divisible orbifold morphism to the gcd base and neat, without changing $\Delta_{\mathrm{inf}}(f)$ and $\Delta_{\mathrm{gcd}}(f)$.

Proposition 7.5. *Notation and conditions as in Set-up 7.4. Let (X, Δ) be a special compact Kähler orbifold pair of dimension d with $a(X) > 0$. Then $\pi_1^{\mathrm{orb}}(X, \Delta)$ is virtually abelian.*

Proof. The proof is long, so we divide it into several steps. Write $\Delta = \sum_{i=1}^{r}(1 - \frac{1}{m_i})D_i$ and $D := \operatorname{Supp}\Delta$, and let $N := \operatorname{lcm}(m_1, \ldots, m_r)$, with $N := 1$ if $r = 0$.

Step 1. In this step we treat the case $a(X) = d$.

Since X is Kähler and $a(X) = \dim X$, X is projective (Lemma 2.15(3)). Thus $\pi_1^{\mathrm{orb}}(X, \Delta)$ is virtually abelian by Theorem 5.1.

From now on we assume that $0 < a(X) < d$.

Step 2. In this step we construct a fibration $f : X' \to Z$ onto a smooth projective variety Z with $\dim Z = a(X)$, together with bimeromorphic holomorphic maps $\mu_0 : X' \to X_0$ and $\mu : X' \to X$, a reduced divisor S on X', and a reduced divisor T on Z.

By [Uen75, Chapter I, Theorem 3.8], there exists a dominant meromorphic map from X to a projective variety of dimension $a(X)$ which induces an isomorphism from the field of rational functions on the image to the field of meromorphic functions on X. We replace the projective variety by a resolution [BM97, Theorem 1.6], resolve the indeterminacy of the meromorphic map by blow-ups with smooth centers, and make the inverse image of D a simple normal crossing divisor by further blow-ups with smooth centers [BM97, Theorem 1.10 and (1.2)(1)]. We obtain a composite of blow-ups with smooth centers $h : X_0 \to X$ and a surjective holomorphic map $f_0 : X_0 \to Z_0$ onto a smooth projective variety Z_0 with $\dim Z_0 = a(X)$, such that the reduced divisor $S_0 := h^{-1}(D) \cup \operatorname{Exc}(h)$ is a simple normal crossing divisor and f_0^* is an isomorphism from the field of rational functions on Z_0 to the field of meromorphic functions on X_0. The manifold X_0 is compact Kähler by Lemma 2.14(1). Since f_0^* is an isomorphism of function fields, the finite part of the Stein factorization of f_0 is bimeromorphic, hence an isomorphism by Zariski's main theorem because Z_0 is normal. Thus f_0 has connected fibers.

By generic smoothness, there exists a closed analytic subset $T_0 \subsetneq Z_0$ such that f_0 and the restrictions of f_0 to all intersections of components of S_0 are smooth over $Z_0 \backslash T_0$, and such that any intersection of components of S_0 which does not dominate Z_0 is mapped into T_0. By Chow's theorem, T_0 is algebraic.

By the flattening theorem [Hir75], there exists a composite of finitely many blow-ups along coherent ideal sheaves $v_1 : Z_1 \to Z_0$ such that the irreducible component W_1 of $X_0 \times_{Z_0} Z_1$ which is bimeromorphic to X_0 is flat over Z_1. Since Z_0 is projective, the ideal sheaves of the blow-ups

are algebraic [Ser56, n° 12, Théorèmes 1–3], so Z_1 is projective. Let $V \subsetneq Z_0$ be the closed subset over which v_1 is not an isomorphism. By [BM97, Theorems 1.6 and 1.10 and (1.2)(1)], there exists a composite of blow-ups with smooth algebraic centers $v_2 : Z \to Z_1$, which is an isomorphism over $Z_1 \backslash v_1^{-1}(T_0 \cup V)$, such that Z is smooth and, for $v := v_1 \circ v_2$, the reduced subset

$$T := v^{-1}(T_0 \cup V)$$

is a simple normal crossing divisor on Z. Then Z is a smooth projective variety with $\dim Z = a(X)$, and v is an isomorphism over $Z_0 \backslash (T_0 \cup V)$.

Let W be the irreducible component of $X_0 \times_{Z_0} Z$ which is bimeromorphic to X_0, with the reduced structure, and let $e := d - a(X)$. We show that any fiber of the projection $W \to Z$ has pure dimension e. Since W is contained in $W_1 \times_{Z_1} Z$, the fiber W_z of W over a point $z \in Z$ is contained in the fiber $(W_1)_{v_2(z)}$, and $\dim (W_1)_{v_2(z)} = \dim W_1 - \dim Z_1 = e$ because W_1 is flat over Z_1. Conversely, the dimension of any irreducible component of a fiber of the surjective map $W \to Z$ is at least $\dim W - \dim Z = e$.

The space W is a closed subspace of the compact Kähler manifold $X_0 \times Z$, so W is a compact Kähler space [Xia19, Example 2.15]. By Lemma 2.14(2), there exists a resolution $X' \to W$ by a compact Kähler manifold X' which is an isomorphism over the smooth locus of W. Let $\mu_0 : X' \to X_0$ and $f : X' \to Z$ be the composites of $X' \to W$ with the projections $W \to X_0$ and $W \to Z$, and let $\mu := h \circ \mu_0$. Let S be the reduced divisor on X' whose components are the strict transforms of the D_i and the μ-exceptional prime divisors. By [BM97, Theorems 1.10 and 12.2], there exists a composite of blow-ups of X' with smooth centers contained in the locus where S is not a simple normal crossing divisor, after which S is a simple normal crossing divisor, and the blown-up manifold is compact Kähler by Lemma 2.14(1). Possibly replacing X' with such a blow-up, we may assume that S is a simple normal crossing divisor. By construction,

$$f_0 \circ \mu_0 = v \circ f.$$

The fibration f has the following properties.

(a) Over $Z \backslash T$, the map v is an isomorphism, and the part of W over $Z \backslash T$ is isomorphic to the open subset $f_0^{-1}(Z_0 \backslash (T_0 \cup V))$ of X_0, on which S_0 is a simple normal crossing divisor. Over the open subset $f^{-1}(Z \backslash T)$ of X', the strict transforms of the D_i and the μ-exceptional prime divisors are the inverse images of the components of S_0, so the blow-ups above are isomorphisms there. Thus μ_0 is an isomorphism over $f^{-1}(Z \backslash T)$, and $S \cap f^{-1}(Z \backslash T) = \mu_0^{-1}(S_0) \cap f^{-1}(Z \backslash T)$. By the choice of T_0, f and the restrictions of f to all intersections of components of S are smooth over $Z \backslash T$, and any component E of S with $E \cap f^{-1}(Z \backslash T) \neq \emptyset$ dominates Z.

(b) The map f has connected fibers. The general fibers of f are fibers of f_0 by (a), so the finite part of the Stein factorization of f is bimeromorphic over the normal variety Z, hence an isomorphism.

(c) Any prime divisor E on X' with $\operatorname{codim}_Z f(E) \geq 2$ is μ_0-exceptional. The image of E in W is contained in the inverse image W_E of $f(E)$ under $W \to Z$, and

$$\dim W_E \leq \dim f(E) + e \leq a(X) - 2 + e = d - 2$$

because the fibers of $W \to Z$ are of dimension e. The projection $W \to X_0$ does not increase dimensions, so $\dim \mu_0(E) \leq d - 2$.

(d) $a(X') = a(X) = \dim Z$ by Lemma 2.15(1).

Step 3. In this step we define a boundary Δ' on X' and show that f is a divisible orbifold morphism to $(Z, \Delta_{\mathrm{gcd}}(f))$, an orbifold morphism to $(Z, \Delta_{\mathrm{inf}}(f))$, and neat.

Let

$$\Delta_0 := \sum_{i=1}^{r} \Big(1 - \frac{1}{m_i}\Big) h_*^{-1} D_i + \sum_{E_0} \Big(1 - \frac{1}{N}\Big) E_0,$$

where E_0 runs through the h-exceptional prime divisors on X_0. Since $\operatorname{Supp}\Delta_0 \subset S_0$, (X_0, Δ_0) is an orbifold pair. For a prime divisor E on X' with $\operatorname{codim}_Z f(E) \leq 1$, let

$$m_E := \begin{cases} m_i & \text{if } E \text{ is the strict transform of } D_i, \\ N & \text{if } E \text{ is } \mu\text{-exceptional}, \\ 1 & \text{otherwise.} \end{cases}$$

For a prime divisor P on Z, the integers $m_{\inf}(P)$ and $m_{\gcd}(P)$ of Definition 7.1 only involve the prime divisors E with $f(E) = P$, so they are determined by the integers m_E above. Since f is surjective, $f^{-1}(P)$ has a component E with $f(E) = P$, so $m_{\inf}(P)$ and $m_{\gcd}(P)$ are positive integers. If $P \not\subset T$ and $f(E) = P$, then $E \cap f^{-1}(Z \backslash T) \neq \emptyset$, so $E \not\subset S$ and $\operatorname{mult}_E f^*P = 1$ by (a). Thus

$$m_{\inf}(P) = m_{\gcd}(P) = 1 \quad \text{for any prime divisor } P \not\subset T.$$

Let N' be the least common multiple of N and of the integers $m_{\inf}(P)$, where P runs through the components of T. For a prime divisor E on X' with $\operatorname{codim}_Z f(E) \geq 2$, let $m_E := N'$, and let

$$\Delta' := \sum_E \Big(1 - \frac{1}{m_E}\Big) E.$$

By (c), any prime divisor E with $m_E \geq 2$ is a component of S, so (X', Δ') is a compact Kähler orbifold pair. The integer m_E is the integer m'_E of Definition 2.5 for μ and μ_0, and $m_{\inf}(P)$, $m_{\gcd}(P)$ are the integers of Definition 7.1 for (X', Δ') and f. Moreover, $\operatorname{Supp}\Delta_{\inf}(f) \cup \operatorname{Supp}\Delta_{\gcd}(f) \subset T$, so $(Z, \Delta_{\inf}(f))$ and $(Z, \Delta_{\gcd}(f))$ are orbifold pairs.

We check the divisibility conditions. The multiplicities of the components of Δ_0 are m_i and N, and they all divide N, which divides N'. If E is the strict transform of a component E_1 of Δ or of Δ_0, then m_E is the multiplicity of E_1. For a μ-exceptional prime divisor E, $m_E \in \{N, N'\}$. For any other prime divisor E, $\operatorname{mult}_E \mu^* D_i = 0$ and $\operatorname{mult}_E \mu_0^* \Delta_0 = 0$. Thus $\mu : (X', \Delta') \to (X, \Delta)$ and $\mu_0 : (X', \Delta') \to (X_0, \Delta_0)$ are divisible orbifold morphisms. Since $\mu_* \Delta' = \Delta$ and $\mu_{0*} \Delta' = \Delta_0$, both are divisible elementary modifications.

Let E be a prime divisor on X' and P a prime divisor on Z with $\operatorname{mult}_E f^*P > 0$. If $f(E) = P$, then by Definition 7.1,

$$m_E \operatorname{mult}_E f^*P \geq m_{\inf}(P) \qquad \text{and} \qquad m_{\gcd}(P) \mid m_E \operatorname{mult}_E f^*P.$$

If $\operatorname{codim}_Z f(E) \geq 2$, then the relations $m_E \operatorname{mult}_E f^*P \geq m_{\inf}(P)$ and $m_{\gcd}(P) \mid m_E \operatorname{mult}_E f^*P$ hold because $m_{\gcd}(P) \mid m_{\inf}(P) \mid N' = m_E$. If $f(E)$ is not contained in P, then $\operatorname{mult}_E f^*P = 0$. Thus $f : (X', \Delta') \to (Z, \Delta_{\gcd}(f))$ is a divisible orbifold morphism. Besides $f(X') \not\subset \operatorname{Supp}\Delta_{\inf}(f)$ and the $\mathbb{Q}$-factoriality of Z, which hold here since f is surjective and Z is smooth, an orbifold morphism in the sense of [Cam11a, Définition 2.3] only requires the inequality $m_E \operatorname{mult}_E f^*P \geq m_{\inf}(P)$ whenever $\operatorname{mult}_E f^*P > 0$ (see the paragraph after Definition 2.5), so $f : (X', \Delta') \to (Z, \Delta_{\inf}(f))$ is an orbifold morphism in the sense of [Cam11a, Définition 2.3].

Finally, by (b), (c), and $f_0 \circ \mu_0 = v \circ f$, the fibration f is neat in the sense of Definition 7.2, with the orbifold pair (X_0, Δ_0), the divisible elementary modification μ_0 in place of μ, the fibration f_0, and the bimeromorphic map v.

Step 4. In this step we show that (X', Δ') is special and that $\mu_* : \pi_1^{\mathrm{orb}}(X', \Delta') \to \pi_1^{\mathrm{orb}}(X, \Delta)$ is an isomorphism.

The statement on μ_* holds by Theorem 2.7, since μ is a divisible elementary modification. Suppose that $\mathcal{F}' \subset \Omega^p_{X'}$ is a Δ'-Bogomolov sheaf for some $1 \leq p \leq d$. Let $\mathcal{F} \subset \Omega^p_X$ be the saturated subsheaf of rank one which coincides with $\mathcal{F}'$ over the open subset of X over which μ is an isomorphism. Then $\mu^*\mathcal{F}$ and $\mathcal{F}'$ coincide at a general point of X', so the images of $(\mu^*\mathcal{F})^{\otimes q}$ and of $\mathcal{F}'^{\otimes q}$ in $S^q\Omega^p(X', \Delta')$ both have saturation $\mathcal{F}'_{q,\Delta'}$ for any $q \geq 1$. Hence $\kappa(X', \Delta'; \mu^*\mathcal{F}) = \kappa(X', \Delta'; \mathcal{F}')$. Since μ is divisible, $m_E \operatorname{mult}_E \mu^* D_i \geq m_i$ whenever

$\mathrm{mult}_E \mu^* D_i > 0$, and $\mu_* \Delta' = \Delta$, so μ is an elementary bimeromorphic orbifold morphism in the sense of [Cam11a, Définition 2.8]. By [Cam11a, Théorème 3.5], applied to $\mathcal{F}$ and $\mu^* \mathcal{F}$,

$$\kappa(X, \Delta; \mathcal{F}) = \kappa(X', \Delta'; \mu^* \mathcal{F}) = \kappa(X', \Delta'; \mathcal{F}') = p.$$

Thus $\mathcal{F}$ is a Δ-Bogomolov sheaf, which contradicts the assumption that (X, Δ) is special. Therefore (X', Δ') is special.

Step 5. In this step we show that for a very general point $z \in Z \backslash T$ the fiber $f^{-1}(z)$, with the restriction of Δ', is a special compact Kähler orbifold pair of dimension $d - a(X)$.

Recall that S is the reduced divisor formed by the strict transforms of the D_i and the μ-exceptional prime divisors, and that $T \subset Z$ is a simple normal crossing divisor over whose complement f and the restrictions of f to all intersections of components of S are smooth. Let $z \in Z \backslash T$ be a very general point, in particular outside a countable union of proper closed analytic subsets of Z off which the conclusion of [Lu02, Proposition 8.7] holds, and let $F := f^{-1}(z)$. By (a) and (b), F is a connected compact complex manifold with $\dim F = d - a(X)$, and the components E of Δ' with $E \cap F \neq \emptyset$ dominate Z and restrict to a simple normal crossing divisor on F. Let Δ_F be the $\mathbb{Q}$-divisor on F whose components are the connected components of the restrictions $E|_F$ of the components E of Δ', each with coefficient $1 - \frac{1}{m_E}$. The restriction of a Kähler form on X' is a Kähler form on F, so (F, Δ_F) is a compact Kähler orbifold pair.

Suppose that (F, Δ_F) is not special. Since Z is projective, [Lu02, Proposition 8.7] can be applied to (X', Δ') and f. We obtain the relative core of f: a compact complex space Y over Z and a dominant meromorphic map $X' \dashrightarrow Y$ over Z whose restriction to F is the basic fibration of (F, Δ_F) in the sense of [Lu02, Remark 7.6], onto the fiber of Y over z. Moreover, by [Lu02, Proposition 8.7], Y is dominated by a projective fibration over Z. The base of the basic fibration of a pair which is not special in the sense of [Lu02, Definition 6.1] is positive-dimensional [Lu02, Definition 7.1 and Remark 7.6], and (F, Δ_F) is not special in the sense of [Lu02, Definition 6.1] by Remark 7.3. Thus $\dim Y > \dim Z$. Since Y is dominated by a projective fibration over the projective variety Z, Y is Moishezon [Uen75, Chapter I, Corollary 3.9], so $a(Y) = \dim Y > \dim Z = a(X')$ by (d). On the other hand, $a(Y) \leq a(X')$ by Lemma 2.15(1), because $X' \dashrightarrow Y$ is dominant. This is a contradiction. Therefore (F, Δ_F) is special.

Step 6. In this step we show that $(Z, \Delta_{\inf}(f))$ and $(Z, \Delta_{\gcd}(f))$ are special.

By **Step 4** and Remark 2.12, (X', Δ') is special in the sense of [Cam11a, Définition 8.1]. By **Step 3**, $f : (X', \Delta') \to (Z, \Delta_{\inf}(f))$ is a surjective orbifold morphism, so $(Z, \Delta_{\inf}(f))$ is special in the sense of [Cam11a, Définition 8.1] by [Cam11a, Corollaire 9.7], hence special by Remark 2.12.

Suppose that $\mathcal{F} \subset \Omega^p_Z$ is a $\Delta_{\gcd}(f)$-Bogomolov sheaf for some $1 \leq p \leq \dim Z$. For any prime divisor P on Z, $m_{\gcd}(P) \mid m_{\inf}(P)$, so for any integer $t \geq 0$,

$$\left\lfloor t\Big(1 - \frac{1}{m_{\gcd}(P)}\Big) \right\rfloor \leq \left\lfloor t\Big(1 - \frac{1}{m_{\inf}(P)}\Big) \right\rfloor.$$

Thus the local generators of Definition 2.10 show that

$$S^q \Omega^p(Z, \Delta_{\gcd}(f)) \subset S^q \Omega^p(Z, \Delta_{\inf}(f)),$$

and hence $\mathcal{F}_{q, \Delta_{\gcd}(f)} \subset \mathcal{F}_{q, \Delta_{\inf}(f)}$ for any $q \geq 0$. So $\bigoplus_q H^0(Z, \mathcal{F}_{q, \Delta_{\gcd}(f)})$ is a graded subalgebra of $\bigoplus_q H^0(Z, \mathcal{F}_{q, \Delta_{\inf}(f)})$, and

$$\kappa(Z, \Delta_{\inf}(f); \mathcal{F}) \geq \kappa(Z, \Delta_{\gcd}(f); \mathcal{F}) = p.$$

By Definition-Theorem 3.5(1)(4), $\kappa(Z, \Delta_{\inf}(f); \mathcal{F}) \leq p$. Thus $\mathcal{F}$ is a $\Delta_{\inf}(f)$-Bogomolov sheaf, which contradicts the specialness of $(Z, \Delta_{\inf}(f))$. Therefore $(Z, \Delta_{\gcd}(f))$ is special.

Step 7. In this step we conclude the proof.

We check the hypotheses of [Cam11a, Proposition 12.9]: X' and Z are compact complex manifolds, Δ' and $\Delta_{\gcd}(f)$ have finite integral multiplicities and simple normal crossing supports,

f is surjective with connected fibers, $f : (X', \Delta') \to (Z, \Delta_{\mathrm{gcd}}(f))$ is divisible, and f is neat in the sense of Definition 7.2 (**Step 3**). Thus, for the fiber F of **Step 5**, the sequence

$$\pi_1^{\mathrm{orb}}(F, \Delta_F) \to \pi_1^{\mathrm{orb}}(X', \Delta') \to \pi_1^{\mathrm{orb}}(Z, \Delta_{\mathrm{gcd}}(f)) \to 1$$

is exact [Cam11a, Proposition 12.9]. The gcd base appears for the following reason. Over a general point of a prime divisor P on Z, a meridian of a prime divisor E with $f(E) = P$ is mapped to the $(\mathrm{mult}_E f^*P)$-th power of a meridian of P (Lemma 2.3(3)), so the relations $\gamma_E^{m_E}$ are mapped to the $(m_E \,\mathrm{mult}_E f^*P)$-th powers of a meridian of P, and by Bezout's identity they generate the subgroup of $m_{\mathrm{gcd}}(P)$-th powers.

Since $\dim F = d - a(X) < d$ and (F, Δ_F) is a special compact Kähler orbifold pair (**Step 5**), $\pi_1^{\mathrm{orb}}(F, \Delta_F)$ is virtually abelian by Set-up 7.4. Since Z is a smooth projective variety and $(Z, \Delta_{\mathrm{gcd}}(f))$ is special (**Step 6**), $\pi_1^{\mathrm{orb}}(Z, \Delta_{\mathrm{gcd}}(f))$ is virtually abelian by Theorem 5.1. We use [CC11, Theorem 4.1] with $(Y, \Delta_Y) = (Z, \Delta_{\mathrm{gcd}}(f))$. In [CC11, Definitions 3.5 and 3.7] the orbifold base is defined with gcd multiplicities, so the exact sequence used in [CC11, Theorem 4.1] is the exact sequence above. By [CC11, Theorem 4.1], that is, the orbifold form of [Cam11b, Corollaire 7.6] for neat fibrations between smooth compact Kähler orbifolds, $\pi_1^{\mathrm{orb}}(X', \Delta')$ is virtually abelian. By **Step 4**, μ_* is an isomorphism, so $\pi_1^{\mathrm{orb}}(X, \Delta)$ is virtually abelian. □

7.3. **Algebraic dimension zero and counterexamples of least dimension.** In this subsection, we treat special compact Kähler orbifold pairs (X, Δ) with $a(X) = 0$, and we prove Propositions 7.7 and 7.8.

When $a(X) = 0$, the base of the algebraic reduction of X is a point, so the induction runs along the Albanese map instead, and we need the very general fibers of the Albanese map to be special. For smooth projective special varieties, this is [CC16, Théorème 2.4], which answers [Cam04b, Question 5.4] in the projective case. In our setting the Albanese torus need not be algebraic, and we use $a(X) = 0$ and [Wan21, Main Theorem A(II)] instead. Campana's orbifold version of Iitaka subadditivity is proved for rational simple normal crossing boundaries on compact manifolds in Fujiki's class $\mathcal{C}$ in [OAI26a, Theorem 1.1].

Lemma 7.6. *Let (X, Δ) be a compact Kähler orbifold pair with $a(X) = 0$, and let $h : X \to T$ be a surjective holomorphic map with connected fibers onto a compact complex torus T. Then there exists a countable union T' of proper closed analytic subsets of T such that for any $t \in T \backslash T'$ the fiber $X_t := h^{-1}(t)$ is smooth, $\mathrm{Supp}(\Delta|_{X_t})$ is a simple normal crossing divisor, and $(X_t, \Delta|_{X_t})$ is a special compact Kähler orbifold pair.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we define T', we show that $(X_t, \Delta|_{X_t})$ is an orbifold pair for $t \in T \backslash T'$, and we reduce the lemma to excluding $r > 0$ for an integer r defined below.

If $\dim T = 0$, then $X_t = X$ for the unique point $t \in T$, and (X, Δ) is special by Corollary 3.6(2). Thus $T' := \emptyset$ satisfies the lemma, and we may assume that $\dim T > 0$.

Write $\Delta = \sum_{i=1}^k (1 - \frac{1}{m_i}) D_i$, and let $\Delta_t := \Delta|_{X_t}$ for $t \in T$. By Sard's theorem and Remmert's proper mapping theorem, the critical values of the restrictions of h to X and to the non-empty intersections $D_{i_1} \cap \cdots \cap D_{i_j}$ form a proper closed analytic subset $T_1 \subset T$. For $t \in T \backslash T_1$, the fiber X_t is smooth, each $D_i \cap X_t$ is smooth, and $\mathrm{Supp}\, \Delta_t = \bigcup_{i=1}^k (D_i \cap X_t)$ is a simple normal crossing divisor on X_t. We write each non-empty $D_i \cap X_t$ as the sum of the connected components of $D_i \cap X_t$. The connected components of $D_1 \cap X_t, \ldots, D_k \cap X_t$ are pairwise distinct smooth prime divisors on X_t, and

$$\Delta_t = \sum_{i=1}^k \Big(1 - \frac{1}{m_i}\Big)(D_i \cap X_t),$$

where any connected component of $D_i \cap X_t$ has coefficient $1 - \frac{1}{m_i}$. Since h has connected fibers, X_t is a connected compact Kähler manifold. Thus (X_t, Δ_t) is an orbifold pair in the sense of Definition 2.2 for any $t \in T \backslash T_1$.

Since X is Kähler, h belongs to the relative class $\mathcal{C}/T$ of [Lu02, §8, p. 39]. Thus we may take the relative core of h [Lu02, Proposition 8.7]: there exist an irreducible compact complex space Z, an almost holomorphic meromorphic map $c : X \dashrightarrow Z$ with connected general fibers, a surjective holomorphic map $p : Z \to T$ with $h = p \circ c$, and a countable union T_2 of proper closed analytic subsets of T such that for any $t \in T \backslash T_2$, the restriction $c_t : X_t \dashrightarrow Z_t := p^{-1}(t)$ of c is the basic fibration of (X_t, Δ_t) in the sense of [Lu02, Remark 7.6]. Let $r := \dim Z - \dim T$. Enlarging T_1 by a proper closed analytic subset of T, we may assume that $\dim Z_t = r$ for any $t \in T \backslash T_1$. Let $T' := T_1 \cup T_2$.

Suppose that (X_t, Δ_t) is not special for some $t \in T \backslash T'$. By Remarks 2.12 and 7.3, (X_t, Δ_t) is not special in the sense of [Lu02, Definition 6.1], so the basic fibration c_t of [Lu02, Remark 7.6 and Theorem 10.1] has $r = \dim Z_t > 0$. Thus it suffices to show that $r > 0$ leads to a contradiction, and from now on we assume that $r > 0$.

Step 2. In this step we construct compact Kähler manifolds X' and V with holomorphic maps $\mu : X' \to X$, $c' : X' \to V$, and $p' : V \to T$ such that μ is bimeromorphic, c' and p' are surjective with connected fibers, $p' \circ c' = h \circ \mu$, and $c' = \psi^{-1} \circ c \circ \mu$ for a bimeromorphic map $\psi : V \dashrightarrow Z$ with $p \circ \psi = p'$.

By Lemma 2.14(3), applied to c, the space Z is bimeromorphic to a compact Kähler manifold V_0. Let $\Gamma \subset V_0 \times T$ be the closure of the graph of the meromorphic map $V_0 \dashrightarrow Z \to T$. Since $V_0 \times T$ is a compact Kähler manifold, Γ is a compact Kähler space. Let V_1 be a resolution of Γ by a compact Kähler manifold (Lemma 2.14(2)), let $p_1 : V_1 \to T$ be the induced holomorphic map, and let $\beta : V_1 \dashrightarrow Z$ be the induced bimeromorphic map. Then $p \circ \beta = p_1$. Resolving the closure of the graph of $\beta^{-1} \circ c : X \dashrightarrow V_1$ in $X \times V_1$ by a compact Kähler manifold (Lemma 2.14(2)), we obtain a compact Kähler manifold X_1 with holomorphic maps $\mu_1 : X_1 \to X$ and $c_1 : X_1 \to V_1$ such that μ_1 is bimeromorphic and $c_1 = \beta^{-1} \circ c \circ \mu_1$. Since $p_1 \circ c_1$ and $h \circ \mu_1$ coincide on a dense open subset of X_1, we have $p_1 \circ c_1 = h \circ \mu_1$.

Let D_{V_1} be the reduced divisor on V_1 whose components are the prime divisors $Q_1 \subset V_1$ such that either Q_1 is contained in the set of critical values of c_1, or $Q_1 = c_1(E_1)$ for a prime divisor $E_1 \subset X_1$ which is μ_1-exceptional or the strict transform of some D_i. By Sard's theorem and Remmert's proper mapping theorem, the set of critical values of c_1 is a proper closed analytic subset of V_1, so D_{V_1} has finitely many components. Taking an embedded resolution of $\operatorname{Supp} D_{V_1} \subset V_1$, we obtain a composite $g : V \to V_1$ of finitely many blow-ups along smooth centers such that g is an isomorphism over $V_1 \setminus \operatorname{Supp} D_{V_1}$ and the reduced divisor $D_V := g^{-1}(\operatorname{Supp} D_{V_1})$ is a simple normal crossing divisor. By Lemma 2.14(1), V is a compact Kähler manifold. Let $p' := p_1 \circ g$ and $\psi := \beta \circ g$. From now on, V is fixed.

Resolving the closure of the graph of $g^{-1} \circ c_1 : X_1 \dashrightarrow V$ in $X_1 \times V$ (Lemma 2.14(2)), and then blowing up finitely many times along smooth centers (Lemma 2.14(1)), we obtain a compact Kähler manifold X' and a bimeromorphic holomorphic map $\mu_2 : X' \to X_1$ with the following properties: $c' := g^{-1} \circ c_1 \circ \mu_2$ is holomorphic, and for $\mu := \mu_1 \circ \mu_2 : X' \to X$, the reduced divisor whose components are the strict transforms $\mu_*^{-1} D_i$ and the μ-exceptional prime divisors is a simple normal crossing divisor. Then μ is bimeromorphic, $c' = \psi^{-1} \circ c \circ \mu$, and $p' \circ c' = h \circ \mu$, since both sides coincide on a dense open subset of X'. As c is dominant and X' is compact, c' is surjective.

Since X is smooth, the fibers of μ are connected by Zariski's main theorem, so the fibers of $h \circ \mu$ are connected. The fibers of p' are the images under c' of the fibers of $h \circ \mu$, so they are connected. Since c has connected general fibers and μ and ψ are bimeromorphic, c' has connected general fibers. Since V is smooth, the Stein factorization of c' shows that all fibers of c' are connected.

Step 3. In this step we define the orbifold base B of c' and show that (V, B) is klt.

Let $\ell := \operatorname{lcm}(m_1, \dots, m_k)$, with $\ell := 1$ if $k = 0$, and let

$$\Delta' := \sum_{i=1}^{k} \Big(1 - \frac{1}{m_i}\Big)\mu_*^{-1} D_i + \sum_{E} \Big(1 - \frac{1}{\ell}\Big) E,$$

where E runs through the μ-exceptional prime divisors. By **Step 2**, (X', Δ') is a compact Kähler orbifold pair, and the integer m_E of Definition 7.1 for (X', Δ') is m_i if $E = \mu_*^{-1} D_i$, ℓ if E is μ-exceptional, and 1 otherwise. For any i and any μ-exceptional prime divisor E, m_i divides $m_E \operatorname{mult}_E \mu^* D_i$. Thus $\mu : (X', \Delta') \to (X, \Delta)$ is a divisible orbifold morphism [Cam11a, Définition 2.4]. Since a positive multiple of m_i is at least m_i, we have $m_E \operatorname{mult}_E \mu^* D_i \geq m_i$ whenever $\operatorname{mult}_E \mu^* D_i > 0$, so μ is an orbifold morphism [Cam11a, Définition 2.3]. Since moreover μ is bimeromorphic with $\mu_* \Delta' = \Delta$, and X and X' are smooth, μ is an elementary bimeromorphic orbifold morphism [Cam11a, Définition 2.8]. We take the weights m_E divisible in this way in order to use [Cam11a, Théorème 3.5] in **Step 4**.

Let $B := \Delta_{\inf}(c')$ be the orbifold base of $c' : (X', \Delta') \to V$ (Definition 7.1), and for a prime divisor Q on V, let b_Q be the infimum multiplicity of c' over Q. Thus

$$b_Q = \min_E \big(m_E \operatorname{mult}_E c'^* Q\big) \qquad \text{and} \qquad B = \sum_Q \Big(1 - \frac{1}{b_Q}\Big) Q,$$

where E runs through the prime divisors on X' with $c'(E) = Q$. Since c' is surjective, b_Q is a positive integer.

We show that if $b_Q > 1$, then Q is a component of D_V. Suppose that Q is not a component of D_V. Then $Q_1 := g(Q)$ is a prime divisor on V_1 which is not a component of D_{V_1}, and g is an isomorphism over a neighborhood of a general point of Q_1. Since Q_1 is not contained in the set of critical values of c_1 and c_1 is surjective, there exists a prime divisor $E_1 \subset X_1$ with $c_1(E_1) = Q_1$ and $\operatorname{mult}_{E_1} c_1^* Q_1 = 1$. By the definition of D_{V_1}, E_1 is neither μ_1-exceptional nor the strict transform of any D_i. Since X_1 is smooth, μ_2 is an isomorphism over a neighborhood of a general point of E_1. Let E be the strict transform of E_1 on X'. Then $c'(E) = Q$, $\operatorname{mult}_E c'^* Q = 1$, and $E \not\subset \operatorname{Supp} \Delta'$, so $m_E = 1$. Hence $b_Q = 1$. Since b_Q is a minimum over all prime divisors E with $c'(E) = Q$, this holds even if some μ_2-exceptional prime divisors with $m_E > 1$ map onto Q.

Thus $\operatorname{Supp} B \subset D_V$. Since V is smooth, $\operatorname{Supp} B$ is a simple normal crossing divisor, and the coefficients of B are in $[0, 1)$, [KM98, Corollary 2.31(3)] implies $\operatorname{discrep}(V, B) > -1$. The computation in [KM98, Corollary 2.31(3)] is local, so it is valid for the complex manifold V. Thus (V, B) is klt in the sense of [KM98, Definition 2.34], whose conditions are also local. Let N be the least common multiple of the integers b_Q for the components Q of $\operatorname{Supp} B$, with $N := 1$ if $B = 0$. Then NB is a divisor, and $\kappa(V, \mathcal{O}_V(N(K_V + B)))$ is the Iitaka dimension of the $\mathbb{Q}$-divisor $K_V + B$ in the sense of [Wan21].

Step 4. In this step we choose a general point $t \in T$ and show that $\kappa(V_t, \mathcal{O}_{V_t}(N(K_{V_t} + B_t))) = r$, where $V_t := p'^{-1}(t)$ and $B_t := B|_{V_t}$.

Let $\Sigma \subset X'$ be the critical locus of c'. Let $\mathcal{Q}$ be the finite set of prime divisors Q on V such that Q is a component of $c'(\Sigma)$ or $Q = c'(E)$ for a prime divisor $E \subset \operatorname{Supp} \Delta'$. For $Q \in \mathcal{Q}$, fix a prime divisor E_Q on X' with $c'(E_Q) = Q$ and $m_{E_Q} \operatorname{mult}_{E_Q} c'^* Q = b_Q$. Let R be the reduced divisor on X' with $\operatorname{Supp} R = \operatorname{Supp} \Delta' \cup \bigcup_{Q \in \mathcal{Q}} c'^{-1}(Q)$.

Let $\mathcal{W}$ be the finite set of irreducible closed analytic subsets of X' and V consisting of the components of $\operatorname{Supp} \Delta'$, D_V, Σ, and $c'(\Sigma)$, the sets $c'(E)$ for the prime divisors $E \subset \operatorname{Supp} \Delta'$, the components of the critical loci of $c'|_E$ for the prime divisors $E \subset \operatorname{Supp} \Delta'$ with $c'(E) = V$, the prime divisors Q and E_Q for $Q \in \mathcal{Q}$, the components of R, and the components of the intersections of any two distinct components of R.

By generic smoothness and the upper semicontinuity of fiber dimension, there exists a proper closed analytic subset $T'' \subset T$ such that for any $t \in T \backslash T''$ the following hold, where $X'_t := (h \circ \mu)^{-1}(t)$.

(1) The point t is a regular value of $h \circ \mu$ and p', the restrictions to X'_t and V_t of the prime divisors in $\mathcal{W}$ are reduced, the restrictions of $\operatorname{Supp}\Delta'$ and D_V to X'_t and V_t are simple normal crossing divisors, and the restriction of R to X'_t is reduced.
(2) For any W which is an element of $\mathcal{W}$ or a component of $\mu(\operatorname{Exc}(\mu)) \subset X$, the intersection of W with the fiber over t of the space containing W (X'_t, V_t, or X_t, respectively) is empty if W does not dominate T, and is of pure dimension $\dim W - \dim T$ otherwise.
(3) The bimeromorphic map $\psi : V \dashrightarrow Z$ restricts to a bimeromorphic map $V_t \dashrightarrow Z_t$. In particular, $\dim V_t = \dim Z_t = r$.

Fix $t \in T \backslash (T' \cup T'')$, and let $\mu_t := \mu|_{X'_t} : X'_t \to X_t$, $c'_t := c'|_{X'_t} : X'_t \to V_t$, and $\Delta'_t := \Delta'|_{X'_t}$. Writing the restrictions of the components of $\operatorname{Supp}\Delta'$ as sums of connected components as in **Step 1**, (X'_t, Δ'_t) is a compact Kähler orbifold pair, and c'_t is surjective with connected fibers by **Step 2**. Let $\Delta_{\inf}(c'_t)$ be the orbifold base of $c'_t : (X'_t, \Delta'_t) \to V_t$, and for a prime divisor P on V_t let $m_{\inf}(P)$ be the infimum multiplicity of c'_t over P (Definition 7.1). We bound $\kappa(V_t, \mathcal{O}_{V_t}(N(K_{V_t} + B_t)))$ from below in three parts (a), (b), and (c).

(a) Since $t \in T \backslash T'$ and $r > 0$, c_t is the basic fibration of (X_t, Δ_t) with positive-dimensional base. By [Lu02, Remark 7.6], c_t is a fibration of general type in the sense of [Lu02, Definition 5.5], that is, $\kappa(L_{c_t}) = \dim Z_t = r$, where L_{c_t} is the $\mathbb{Q}$-line sheaf of c_t in [Lu02, Definition 3.4] (Remark 7.3).

(b) The map μ is an isomorphism over $X \backslash \mu(\operatorname{Exc}(\mu))$, and $\dim \mu(\operatorname{Exc}(\mu)) \le \dim X - 2$. By (2), $\mu(\operatorname{Exc}(\mu)) \cap X_t$ has codimension at least two in X_t. Hence μ_t is bimeromorphic, and for any μ-exceptional prime divisor E, the components of $E \cap X'_t$ are μ_t-exceptional. By (1), the multiplicity of $\mu_t^*(D_i \cap X_t)$ along a component of $E \cap X'_t$ is $\operatorname{mult}_E \mu^* D_i$. Thus $(\mu_t)_* \Delta'_t = \Delta_t$, and the divisibility in **Step 3** holds for μ_t. Hence $\mu_t : (X'_t, \Delta'_t) \to (X_t, \Delta_t)$ is a divisible elementary bimeromorphic orbifold morphism. Let $\mathcal{G} \subset \Omega^r_{X_t}$ be the saturation of the image of $c_t^* K_{Z_t}$, which is defined at a general point of X_t, and let $\mathcal{G}' \subset \Omega^r_{X'_t}$ be the saturation of the image of $c_t'^* K_{V_t}$. For any $q \ge 0$, μ_t^* induces an isomorphism from $H^0(X_t, \mathcal{G}_{q,\Delta_t})$ onto $H^0(X'_t, (\mu_t^* \mathcal{G})_{q,\Delta'_t})$, with the sheaves of Definition 2.10 [Cam11a, Théorème 3.5]. In particular, $\kappa(X_t, \Delta_t; \mathcal{G}) = \kappa(X'_t, \Delta'_t; \mu_t^* \mathcal{G})$. By (3), c'_t and $c_t \circ \mu_t$ coincide up to the bimeromorphic map $V_t \dashrightarrow Z_t$. By Remark 7.3, applied to c'_t, and applied to the meromorphic map c_t through the modification μ_t, which turns c_t into c'_t up to $V_t \dashrightarrow Z_t$, we have $\kappa(L_{c'_t}) = \kappa(X'_t, \Delta'_t; \mathcal{G}')$ and $\kappa(L_{c_t}) = \kappa(X_t, \Delta_t; \mathcal{G})$. The subsheaves $\mu_t^* \mathcal{G}$ and $\mathcal{G}'$ of $\Omega^r_{X'_t}$ coincide at a general point of X'_t, so $(\mu_t^* \mathcal{G})_{q,\Delta'_t} = \mathcal{G}'_{q,\Delta'_t}$ for any $q \ge 0$ by Definition 2.10. Thus

$$\kappa(L_{c'_t}) = \kappa(X'_t, \Delta'_t; \mathcal{G}') = \kappa(X'_t, \Delta'_t; \mu_t^* \mathcal{G}) = \kappa(X_t, \Delta_t; \mathcal{G}) = \kappa(L_{c_t}) = r.$$

(c) In [Lu02, Definition 3.4], which is used in [Lu02, Lemma 3.13], the boundary on the base is defined with minimal multiplicities, so it is $\Delta_{\inf}(c'_t)$. Let N' be a positive multiple of N such that $N' \Delta_{\inf}(c'_t)$ is a divisor. By (b) and [Lu02, Lemma 3.13],

$$r = \kappa(L_{c'_t}) \le \kappa\big(V_t, \mathcal{O}_{V_t}(N'(K_{V_t} + \Delta_{\inf}(c'_t)))\big).$$

We show that $B_t \ge \Delta_{\inf}(c'_t)$. Let P be a prime divisor on V_t with $m_{\inf}(P) > 1$. Then over a general point of P, either c'_t is not smooth, or any prime divisor on X'_t mapping onto P is a component of $\operatorname{Supp}\Delta'_t$. By (1), the critical locus of c'_t is $\Sigma \cap X'_t$.

In the first case, $P \subset c'_t(\Sigma \cap X'_t) \subset c'(\Sigma) \cap V_t$. By (2), for any component Q of $c'(\Sigma)$, $Q \cap V_t$ is empty or of pure dimension $\dim Q - \dim T$. Since $\dim P = \dim V_t - 1$, P is a component of $Q \cap V_t$ for a component Q of $c'(\Sigma)$ with $\dim Q = \dim V - 1$, and $Q \in \mathcal{Q}$.

In the second case, let D' be a prime divisor on X'_t with $c'_t(D') = P$. Then D' is a component of $E \cap X'_t$ for a prime divisor $E \subset \operatorname{Supp}\Delta'$, and $c'(E) \ne V$, since otherwise $c'|_E$ is a submersion at a general point of D' by (2), and $c'_t(D') = V_t$. By (2), $c'(E) \cap V_t$ has pure dimension $\dim c'(E) - \dim T$, and it contains P, so $Q := c'(E)$ is a prime divisor on V, $Q \in \mathcal{Q}$, and P is a component of $Q \cap V_t$.

Since $c'(E_Q) = Q$, some component E' of $E_Q \cap X'_t$ satisfies $c'_t(E') = P$. By (1) and (2), $Q|_{V_t}$ and $E_Q|_{X'_t}$ are reduced, and E_Q is the only component of R containing E'. Since $\operatorname{Supp} c'^* Q \subset \operatorname{Supp} R$

and $\operatorname{Supp}\Delta' \subset \operatorname{Supp} R$, we obtain $\operatorname{mult}_{E'} c_t'^* P = \operatorname{mult}_{E_Q} c'^* Q$ and $m_{E'} = m_{E_Q}$. Thus

$$1 < m_{\inf}(P) \leq m_{E_Q} \operatorname{mult}_{E_Q} c'^* Q = b_Q.$$

In particular, Q is a component of $\operatorname{Supp} B$, and by (1) the coefficient of B_t along P is $1 - \frac{1}{b_Q} \geq 1 - \frac{1}{m_{\inf}(P)}$. Hence $B_t \geq \Delta_{\inf}(c_t')$.

Since $N'(B_t - \Delta_{\inf}(c_t')) \geq 0$ is a divisor and N' is a multiple of N,

$$\begin{aligned} r &\leq \kappa\big(V_t, \mathcal{O}_{V_t}(N'(K_{V_t} + \Delta_{\inf}(c_t')))\big) \leq \kappa\big(V_t, \mathcal{O}_{V_t}(N'(K_{V_t} + B_t))\big) \\ &= \kappa\big(V_t, \mathcal{O}_{V_t}(N(K_{V_t} + B_t))\big) \leq \dim V_t = r. \end{aligned}$$

Thus $\kappa(V_t, \mathcal{O}_{V_t}(N(K_{V_t} + B_t))) = r$ for any $t \in T \backslash (T' \cup T'')$.

Step 5. In this step we derive a contradiction from [Wan21, Main Theorem A(II)].

By **Steps 2 and 3**, $p' : V \to T$ is a surjective holomorphic map with connected fibers between compact Kähler manifolds, T is a complex torus, $B \geq 0$ is a $\mathbb{Q}$-divisor, and (V, B) is klt. Since K_T is trivial, $\kappa(T) = 0$. Hence [Wan21, Main Theorem A(II)] and **Step 4** imply

$$\kappa\big(V, \mathcal{O}_V(N(K_V + B))\big) \geq \kappa\big(V_t, \mathcal{O}_{V_t}(N(K_{V_t} + B_t))\big) + \kappa(T) = r > 0$$

for a general point $t \in T$. On the other hand, $c' \circ \mu^{-1} : X \dashrightarrow V$ is a dominant meromorphic map, so $a(V) \leq a(X) = 0$ (Lemma 2.15(1)). Thus $a(V) = 0$, and Lemma 2.15(2), applied to the line bundle $\mathcal{O}_V(N(K_V + B))$, implies $\kappa(V, \mathcal{O}_V(N(K_V + B))) \leq 0$, a contradiction. Thus $r = 0$, so (X_t, Δ_t) is special for any $t \in T \backslash T'$ by **Step 1**. □

Proposition 7.7. *Notation and conditions as in Set-up 7.4. Let (X, Δ) be a special compact Kähler orbifold pair of dimension d with $a(X) = 0$ and $\tilde{q}(G) > 0$, where $G := \pi_1^{\mathrm{orb}}(X, \Delta)$. Then G has a free abelian subgroup of finite index. In particular, G is virtually abelian.*

Proof. Let G_Y, (Y, Δ_Y), f, $\alpha_Y : Y \to A_Y$, and Λ be as in Proposition 6.5, and identify G_Y with $\pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ via f_* (Proposition 6.5(1)). Since f is generically finite and surjective (Proposition 6.5(1)), $\dim Y = d$, and $a(Y) = a(X) = 0$ by Lemma 2.15(1).

By Proposition 6.2, applied to (Y, Δ_Y), which is special by Proposition 6.5(1), the map α_Y is surjective with connected fibers, and there exists a non-empty Zariski open subset $A_Y^\circ \subset A_Y$ such that for any $t \in A_Y^\circ$ (the point a in Proposition 6.2), the fiber $Y_t := \alpha_Y^{-1}(t)$ is smooth, and the sequence

$$\pi_1^{\mathrm{orb}}(Y_t, \Delta_Y|_{Y_t}) \to G_Y \to \Lambda \to 1$$

is exact. Since $R_{\mathrm{fin}}(G_Y)$ is the kernel of $G_Y \to \Lambda$ (Proposition 6.5(3)), the image of $\pi_1^{\mathrm{orb}}(Y_t, \Delta_Y|_{Y_t}) \to G_Y$ is $R_{\mathrm{fin}}(G_Y)$ for any $t \in A_Y^\circ$.

Let $A_Y' \subset A_Y$ be the countable union of proper closed analytic subsets in Lemma 7.6 for (Y, Δ_Y), $h = \alpha_Y$, and $T = A_Y$. Since $A_Y \backslash A_Y^\circ$ is a proper closed analytic subset of A_Y, the set $A_Y^\circ \backslash A_Y'$ is non-empty by the Baire category theorem. Fix $t \in A_Y^\circ \backslash A_Y'$. By Lemma 7.6, $(Y_t, \Delta_Y|_{Y_t})$ is a special compact Kähler orbifold pair, and since $\dim A_Y = \tilde{q}(G)$ (Proposition 6.5(1)),

$$\dim Y_t = \dim Y - \dim A_Y = d - \tilde{q}(G) < d.$$

By Set-up 7.4, Theorem 1.2 holds for $(Y_t, \Delta_Y|_{Y_t})$, so $\pi_1^{\mathrm{orb}}(Y_t, \Delta_Y|_{Y_t})$ is virtually abelian. By Lemma 2.17(1), $R_{\mathrm{fin}}(G_Y)$ is virtually abelian. Since $R_{\mathrm{fin}}(G_Y)$ is finitely generated and has no subgroup of finite index other than $R_{\mathrm{fin}}(G_Y)$ (Proposition 6.5(4)), Lemma 2.17(4) implies $R_{\mathrm{fin}}(G_Y) = \{1\}$. Thus $G_Y \to \Lambda$ is an isomorphism, and $G_Y \cong \Lambda \cong \mathbb{Z}^{2 \dim A_Y}$ is a free abelian group. Since $[G : G_Y] < +\infty$, G has a free abelian subgroup of finite index. □

Propositions 7.5 and 7.7 leave the case of Proposition 7.8 below. In that case G has no non-trivial finite quotient, so passing to finite covers does not change G, and since $\pi_1(X) = \{1\}$, Theorem 2.13 gives no information on G. Sections 8 and 9 exclude this case.

Proposition 7.8. *Assume that Theorem 1.2 fails, and let d be the smallest dimension of a special compact Kähler orbifold pair whose orbifold fundamental group is not virtually abelian. Then there exists a special compact Kähler orbifold pair (X, Δ) of dimension d such that $a(X) = 0$, $\pi_1(X) = \{1\}$, and $G := \pi_1^{\mathrm{orb}}(X, \Delta)$ is infinite, finitely generated, satisfies $G = [G, G]$, and has no subgroup of finite index other than G.*

Proof. Let (X_0, Δ_0) be a special compact Kähler orbifold pair of dimension d such that $G_0 := \pi_1^{\mathrm{orb}}(X_0, \Delta_0)$ is not virtually abelian. A compact Kähler orbifold pair of dimension zero is a point with trivial orbifold fundamental group, so $d \geq 1$. By the minimality of d, Theorem 1.2 holds for any compact Kähler orbifold pair of dimension less than d, so d satisfies the conditions of Set-up 7.4. By Proposition 7.5, $a(X_0) = 0$. By Proposition 6.5, $\tilde{q}(G_0)$ is a non-negative integer, so Proposition 7.7 implies $\tilde{q}(G_0) = 0$.

Let $G_Y \leq G_0$, (Y, Δ_Y), $f : Y \to X_0$, $A_Y = \mathrm{Alb}(Y)$, and $\Lambda = \pi_1(A_Y)$ be as in Proposition 6.5, applied to (X_0, Δ_0), and identify G_Y with $\pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ via f_* (Proposition 6.5(1)).

By Proposition 6.5(1), $\dim A_Y = \tilde{q}(G_0) = 0$, so A_Y is a point and $\Lambda = \{1\}$. Thus $\pi_1(Y) \cong \Lambda = \{1\}$ by Proposition 6.5(2), and $G_Y = R_{\mathrm{fin}}(G_Y)$ by Proposition 6.5(3). If $G_Y = \{1\}$, then G_0 is finite, since $[G_0 : G_Y] < +\infty$, so G_0 is virtually abelian, a contradiction. Hence, by Proposition 6.5(4), G_Y is infinite, finitely generated, satisfies $G_Y = [G_Y, G_Y]$, and has no subgroup of finite index other than G_Y. By Proposition 6.5(1), (Y, Δ_Y) is special, and f is generically finite and surjective, so $\dim Y = d$, and $a(Y) = a(X_0) = 0$ by Lemma 2.15(1). Thus $(X, \Delta) := (Y, \Delta_Y)$ satisfies the proposition, with $G = \pi_1^{\mathrm{orb}}(Y, \Delta_Y) \cong G_Y$. □

8. Γ-reductions and compact subvarieties

In this section, we prove two results which Section 9 uses to exclude the case left by Proposition 7.8: the Γ-reduction of an arbitrary quotient of an orbifold fundamental group (Definition-Theorem 8.2), and a criterion, in terms of a positive current on an orbifold pair, for the existence of compact subvarieties with finite ρ-image through almost all points (Lemma 8.4). Definition-Theorem 8.2 and Lemma 8.4 are the orbifold versions of [OAI26c, Theorem 2.3] and [OAI26c, Lemma 4.7], respectively, which concern the fundamental group of a compact Kähler manifold.

8.1. Γ-reductions of quotients.

In this subsection, we construct the Γ-reduction of a surjective homomorphism from the orbifold fundamental group of a compact Kähler orbifold pair onto an arbitrary group. We first define the image of the fundamental group of a subvariety.

Definition-Lemma 8.1. Let (X, Δ) be an orbifold pair, let $D := \mathrm{Supp}\,\Delta$, and let $G := \pi_1^{\mathrm{orb}}(X, \Delta)$. Let $V \subset X$ be a compact irreducible analytic subvariety with $V \not\subset D$, and let $r : \widehat{V} \to V$ be a resolution of singularities. The *open image* $\pi_1(V)_{(X,\Delta)} \leq G$ of V is the image of the composite $\pi_1(\widehat{V} \backslash r^{-1}(D)) \to \pi_1(X \backslash D) \to G$, defined up to conjugation. It does not depend on r, and it does not change if a proper closed analytic subset is removed from $\widehat{V} \backslash r^{-1}(D)$. For a homomorphism $\rho : G \to Q$, we say that V has *finite ρ-image* if $\rho(\pi_1(V)_{(X,\Delta)})$ is finite.

Proof. Since V is irreducible, $\widehat{V}$ is a connected complex manifold. Since $V \not\subset D$, $r^{-1}(D)$ is a proper closed analytic subset of $\widehat{V}$, so $\widehat{V} \backslash r^{-1}(D)$ is connected. Let $A \subsetneq \widehat{V} \backslash r^{-1}(D)$ be a closed analytic subset. By Lemma 2.3(1), the map

$$\pi_1\big(\widehat{V} \backslash (r^{-1}(D) \cup A)\big) \to \pi_1\big(\widehat{V} \backslash r^{-1}(D)\big)$$

is surjective. Thus removing A does not change the image in G. A change of base point changes the image by a conjugation.

Let $r' : \widehat{V}' \to V$ be another resolution of singularities. There exists a proper closed analytic subset $V_1 \subsetneq V$ with $V \cap D \subset V_1$ such that r and r' are isomorphisms over $V \backslash V_1$. Then $r^{-1}(V \backslash V_1)$ is the complement of a proper closed analytic subset of $\widehat{V} \backslash r^{-1}(D)$, and $r'^{-1}(V \backslash V_1)$ is the complement of a proper closed analytic subset of $\widehat{V}' \backslash r'^{-1}(D)$. Via the isomorphisms

$r^{-1}(V\backslash V_1) \cong V\backslash V_1 \cong r'^{-1}(V\backslash V_1)$, the maps of $r^{-1}(V\backslash V_1)$ and $r'^{-1}(V\backslash V_1)$ to $X\backslash D$ are the inclusion of $V\backslash V_1$. By the first paragraph, the images defined by r and by r' both coincide with the image of $\pi_1(V\backslash V_1) \to G$. □

If ρ is the identity, Definition-Theorem 8.2 below is a theorem of Claudon [Cam11a, Théorème 12.23], see [Cla08]. Later we take the minimum of the Γ-dimension over all infinite quotients of $\pi_1^{\mathrm{orb}}(X,\Delta)$, so we need the version for an arbitrary quotient ρ. In the proof, the branched covering with deck group Q is ramified along D_i to the order of $\rho(\gamma_i)$ in Q, which divides m_i and may be smaller than m_i, and it may have quotient singularities over the intersections of the D_i, so Claudon's orbifold Kähler metric and [Cam94, Théorème 3.5] are applied to the branched covering with deck group Q (Remark 8.3).

Definition-Theorem 8.2. Let (X,Δ) be a compact Kähler orbifold pair, let $D := \operatorname{Supp}\Delta$, and let $\rho : \pi_1^{\mathrm{orb}}(X,\Delta) \to Q$ be a surjective homomorphism. Then there exist a compact Kähler manifold Z_ρ, a dominant meromorphic map $f_\rho : X \dashrightarrow Z_\rho$ with connected general fibers, and a countable union E_ρ of proper closed analytic subsets of X, such that for any $x \in X\backslash(D \cup E_\rho)$ the following hold: f_ρ is holomorphic near the fiber X_x of f_ρ through x, the fiber X_x is a compact irreducible analytic subvariety with finite ρ-image, and a compact irreducible analytic subvariety $V \subset X$ with $x \in V$ has finite ρ-image if and only if $V \subset X_x$. The map f_ρ is unique up to bimeromorphic equivalence of Z_ρ. We call f_ρ the Γ-*reduction* of ρ and $\operatorname{gdim}(X,\Delta;\rho) := \dim Z_\rho$ the Γ-*dimension* of ρ.

Remark 8.3. For an unramified covering $\widetilde{Y}$ of a compact Kähler manifold Y, [Cam94, Théorème 3.5 and Remarque 3.6] constructs a proper meromorphic map of $\widetilde{Y}$ with connected fibers and a countable union E of proper closed analytic subsets of $\widetilde{Y}$ such that for any $y \in \widetilde{Y}\backslash E$, the fiber through y is the largest compact connected analytic subset of $\widetilde{Y}$ containing y. In **Step 4** of the proof of Definition-Theorem 8.2 below, we use the corresponding statement for the branched covering of X with deck group Q constructed in **Step 1** of that proof, which is not an unramified covering of a manifold. The proof of [Cam94, Théorème 3.5] uses the covering $\widetilde{Y} \to Y$ only through [Cam94, Lemme 3.17], the properness of the projection to $\widetilde{Y}$ of the incidence graph of an irreducible component of the cycle space whose general member is irreducible, which is deduced from [Cam94, Proposition 3.12]. The minimality argument at the end of [Cam94, §3] and the chain quotient [Cam94, Théorème 3.18] hold for any normal complex space. For the branched covering, the properness of the incidence graph is proved in **Step 3** of the proof below, from the orbifold Kähler metric of [Cla08, Proposition 2.2] and the volume bound [Cla08, Lemma 2.1]. Hence the proof of [Cam94, Théorème 3.5] is valid for the branched covering. The version for compact irreducible analytic subsets is [Cla08, Theorem 0.2 and Remark 2.2].

Proof of Definition-Theorem 8.2. The proof is long, so we divide it into several steps. Write $\Delta = \sum_{i=1}^r (1-\frac{1}{m_i})D_i$, let $d := \dim X$, and let $\rho^\circ : \pi_1(X\backslash D) \to Q$ be the composite of the quotient map $\pi_1(X\backslash D) \to \pi_1^{\mathrm{orb}}(X,\Delta)$ with ρ. By Lemma 2.3(4), $\pi_1(X\backslash D)$ is finitely generated, so Q is countable. If Q is finite, then any V as in the statement has finite ρ-image, and the statement holds for the constant map from X to a point and $E_\rho = \emptyset$. Hence we may assume that Q is infinite.

Step 1. In this step we construct a branched covering $p : W \to X$ with deck group Q.

For $1 \le i \le r$, let e_i be the order of $\rho(\gamma_i)$ in Q, and let

$$\Delta_e := \sum_{e_i \ge 2} \Big(1 - \frac{1}{e_i}\Big) D_i.$$

Since $\gamma_i^{m_i} = 1$ in $\pi_1^{\mathrm{orb}}(X,\Delta)$, e_i divides m_i. Then (X,Δ_e) is an orbifold pair. By Lemma 2.3(2), applied to $X\backslash\operatorname{Supp}\Delta_e$ and the divisor $\big(\bigcup_{e_i=1} D_i\big)\backslash\operatorname{Supp}\Delta_e$, and by Lemma 2.3(5),

$$\pi_1^{\mathrm{orb}}(X,\Delta_e) = \pi_1(X\backslash D)/\langle\langle \gamma_1^{e_1}, \dots, \gamma_r^{e_r}\rangle\rangle.$$

Since $\rho^\circ(\gamma_i^{e_i}) = 1$ for any i, ρ° factors through a surjective homomorphism $\rho_e : \pi_1^{\mathrm{orb}}(X, \Delta_e) \to Q$. If $e_i \geq 2$, then the order of γ_i in $\pi_1^{\mathrm{orb}}(X, \Delta_e)$ is e_i: it divides e_i by the relation $\gamma_i^{e_i} = 1$, and it is at least e_i since $\rho_e(\gamma_i)$ has order e_i.

Let $W^\circ \to X \backslash D$ be the connected Galois covering with deck group Q which corresponds to $\ker \rho^\circ$. By a theorem of Kato and Namba [Kat87, Nam87], in the form of [Cla08, Theorem 1.1], applied to (X, Δ_e) and the subgroup $\ker \rho_e$, there exists a covering $p : W \to X$ branched at most along $\operatorname{Supp} \Delta_e$ such that W is a connected normal complex space and $p^{-1}(X \backslash D) = W^\circ$. We describe p locally. Let $x \in X$, and let $U_x \ni x$ be a coordinate polydisc with coordinates $z_1, \dots, z_d$ such that $D \cap U_x = \{z_1 \cdots z_n = 0\}$ and $\{z_j = 0\} = D_{i_j} \cap U_x$ for $1 \leq j \leq n$. Then $\pi_1(U_x \backslash D) \cong \mathbb{Z}^n$ is generated by the meridians λ_j of $\{z_j = 0\}$ (Lemma 2.3(3)), and the image of λ_j in $\pi_1(X \backslash D)$ is conjugate to γ_{i_j} by Lemma 2.3(5). Thus the restriction of ρ° to $\pi_1(U_x \backslash D)$ factors through

$$A_x := \prod_{j=1}^{n} \mathbb{Z}/e_{i_j}\mathbb{Z},$$

the Galois group of the covering $\widetilde{U}_x \to U_x$ given by $z_j = w_j^{e_{i_j}}$ for $j \leq n$ and $z_j = w_j$ for $j > n$. Let $A'_x \subset A_x$ be the kernel of the induced homomorphism $A_x \to Q$. Then any connected component of $p^{-1}(U_x)$ is isomorphic over U_x to $\widetilde{U}_x / A'_x$, since both are normal and they are isomorphic over $U_x \backslash D$. Here $\widetilde{U}_x / A'_x$ is a normal complex space by [Car57, Théorème 4]. The group Q permutes the connected components of $p^{-1}(U_x)$ transitively, with stabilizers isomorphic to A_x / A'_x. Thus the following hold.

(1) Q acts on W by biholomorphisms, properly discontinuously and with finite stabilizers, $p \circ q = p$ for any $q \in Q$, and $W/Q = X$.
(2) p has discrete fibers, and the ramification order of p along D_i is e_i.
(3) There exists a compact subset $\Omega \subset W$ with $Q \cdot \Omega = W$.
(4) W is second countable.

For (3), cover X by finitely many polydiscs $U'_x \Subset U_x$ and let Ω be the union of the closures of the intersections of $p^{-1}(U'_x)$ with one connected component of $p^{-1}(U_x)$. The closures are compact since $\widetilde{U}_x / A'_x \to U_x$ is finite. Then (4) follows from (3) since Q is countable.

Step 2. In this step we prove that a compact irreducible analytic subvariety $V \subset X$ with $V \not\subset D$ has finite ρ-image if and only if some irreducible component of $p^{-1}(V)$ dominating V is compact. We also prove that Q acts transitively on the set of irreducible components of $p^{-1}(V)$ dominating V.

Let $V_0 := V_{\mathrm{reg}} \backslash D$, a connected complex manifold which is dense in V, and let $\sigma : \widehat{V} \to V$ be a resolution of singularities which is an isomorphism over V_{reg} [BM97, Theorem 1.6]. Then $\sigma^{-1}(V_0) \cong V_0$ is the complement of a proper closed analytic subset of $\widehat{V} \backslash \sigma^{-1}(D)$. By Definition-Lemma 8.1, the image I_V of $\pi_1(V_0) \to \pi_1(X \backslash D) \to Q$ is conjugate to $\rho(\pi_1(V)_{(X,\Delta)})$. Since $W^\circ \to X \backslash D$ is the Galois covering with group Q corresponding to $\ker \rho^\circ$, the group Q permutes the connected components of the covering $p^{-1}(V_0) \to V_0$ transitively. For a connected component V'_0 of $p^{-1}(V_0)$, the stabilizer of V'_0 in Q is conjugate to I_V, and the number of sheets of $V'_0 \to V_0$ is the index of $\pi_1(V'_0)$ in $\pi_1(V_0)$, which is $|I_V|$ [Hat02, Proposition 1.32]. Since p has discrete fibers, $\dim p^{-1}(V) \leq \dim V$, and the irreducible components of $p^{-1}(V)$ dominating V are the closures of the connected components of $p^{-1}(V_0)$. In particular, Q acts transitively on them.

Assume that I_V is finite, and let V' be the closure of a connected component V'_0 of $p^{-1}(V_0)$. Let $v \in \widehat{V}$ and $x := \sigma(v)$. Fix a ball $\mathbb{B}_v \ni v$ in $\widehat{V}$ with $\sigma(\mathbb{B}_v) \subset U_x$. Since $\mathbb{B}_v \cap \sigma^{-1}(V_0)$ is the complement of a proper closed analytic subset of $\mathbb{B}_v$, it is connected. Thus the restriction of $V'_0 \to V_0$ to $\sigma(\mathbb{B}_v) \cap V_0$ has at most $|I_V|$ connected components, and each of them is contained in a connected component of $p^{-1}(U_x)$. Cover $\widehat{V}$ by finitely many balls $\mathbb{B}'_v \Subset \mathbb{B}_v$. Then V'_0 is contained in a finite union of sets of the form $Y \cap p^{-1}(\sigma(\overline{\mathbb{B}'_v}))$, where Y is a connected component of $p^{-1}(U_x)$. The sets $Y \cap p^{-1}(\sigma(\overline{\mathbb{B}'_v}))$ are compact since $Y \to U_x$ is finite. Hence V' is compact.

Conversely, let V' be a compact irreducible component of $p^{-1}(V)$ dominating V. Then $V_0' := V' \cap p^{-1}(V_0)$ is a connected component of $p^{-1}(V_0)$, and $V_0' \to V_0$ is a covering with $|I_V|$ sheets. Since V' is compact and p has discrete fibers, the fibers of $V' \to V$ are finite. Hence I_V is finite. When ρ is the identity, W is the universal covering of (X, Δ), and **Step 2** is [Cam11a, Proposition 12.19].

Step 3. In this step we fix a Kähler metric on W and prove that the families of connected cycles on W are proper over W.

Fix a Kähler form ω on X, Hermitian metrics on $\mathcal{O}_X(D_i)$, and sections s_i with $\operatorname{div}(s_i) = D_i$. By [Cla08, Propositions 2.1 and 2.2], for $c_0 \gg 0$ the current

$$\omega_e := c_0\omega + \sum_{e_i \geq 2} \sqrt{-1}\partial\bar{\partial}|s_i|^{2/e_i}$$

pulls back to a Kähler form on any $\widetilde{U}_x$ of **Step 1**, and $\omega_W := p^*\omega_e$ is a Q-invariant Kähler metric on W in the orbifold sense. Let dist be the length metric of ω_W on W. Since Q acts on W by isometries of dist and $Q \cdot \Omega = W$, where Ω is compact (**Step 1**(3)), dist is complete. By [Cla08, Remark 2.2], the volume bound [Cla08, Lemma 2.1] holds for W: there exist $\epsilon > 0$ and $c > 0$ such that

$$\operatorname{vol}_{\omega_W}\left(Y \cap \{w \in W : \operatorname{dist}(y, w) < \epsilon\}\right) \geq c$$

for any compact irreducible analytic subset $Y \subset W$ with $\dim Y \geq 1$ and any $y \in Y$.

Let $\mathcal{C}(W)$ be the Barlet space of compact analytic cycles of W [Bar75]. It is a reduced complex space with countably many irreducible components, the universal family $I \subset \mathcal{C}(W) \times W$ is proper over $\mathcal{C}(W)$, and the reduced irreducible cycles form an open subset of $\mathcal{C}(W)$, whose complement in any irreducible component is a closed analytic subset. The group Q acts on $\mathcal{C}(W)$ by push-forward of cycles. For an irreducible component $\mathcal{P}$ of $\mathcal{C}(W)$, let $I_{\mathcal{P}} \subset \mathcal{P} \times W$ be the universal family over $\mathcal{P}$, and let $A_{\mathcal{P}} \subset W$ be the image of $I_{\mathcal{P}}$. Let $\mathcal{S}$ be the set of irreducible components $\mathcal{P}$ of $\mathcal{C}(W)$ whose general member is reduced and irreducible. For $\mathcal{P} \in \mathcal{S}$, the support of any cycle in $\mathcal{P}$ is connected, as a limit of connected supports, and the volume of the cycles in $\mathcal{P}$ with respect to ω_W is a constant $v_{\mathcal{P}}$, since ω_W is closed. We show that for $\mathcal{P} \in \mathcal{S}$, the projection $I_{\mathcal{P}} \to W$ is proper. Hence $A_{\mathcal{P}}$ is a closed analytic subset of W by Remmert's proper mapping theorem [GR84, Chapter 10, §6].

Let $k_{\mathcal{P}} \geq 1$ be the dimension of the cycles in $\mathcal{P}$, let $Z \in \mathcal{P}$, and let $y, y' \in \operatorname{Supp} Z$ with $\operatorname{dist}(y, y') = \delta$. Since $\operatorname{Supp} Z$ is connected, for any integer $0 \leq j \leq \delta/(2\epsilon)$ there exists $y_j \in \operatorname{Supp} Z$ with $\operatorname{dist}(y, y_j) = 2j\epsilon$. Let Y_j be an irreducible component of $\operatorname{Supp} Z$ containing y_j. Then $\dim Y_j = k_{\mathcal{P}} \geq 1$, and the balls of radius ϵ about the y_j are pairwise disjoint. Applying the volume bound to Y_j and y_j, we obtain

$$v_{\mathcal{P}} \geq \left(\left\lfloor \frac{\delta}{2\epsilon} \right\rfloor + 1\right)c,$$

so the diameter of $\operatorname{Supp} Z$ is at most $\delta_{\mathcal{P}} := 2\epsilon v_{\mathcal{P}}/c$. Let $\Lambda \subset W$ be a compact subset. Then the support of any $Z \in \mathcal{P}$ with $\operatorname{Supp} Z \cap \Lambda \neq \emptyset$ is contained in the closed $\delta_{\mathcal{P}}$-neighborhood Λ' of Λ, which is compact since dist is complete and W is locally compact. By Bishop's theorem [Bis64], the cycles of volume $v_{\mathcal{P}}$ with support in Λ' form a compact subset of $\mathcal{C}(W)$. Hence the preimage of Λ in $I_{\mathcal{P}}$ is compact. For the universal covering, the properness of $I_{\mathcal{P}} \to W$ is [Cla08, Proposition 2.3].

Step 4. In this step we realize the reduction of W by maximal compact connected analytic subsets as a Q-equivariant family of cycles.

By **Steps 1 and 3** and Remark 8.3, [Cam94, Théorème 3.5] holds for W: there exist a proper almost holomorphic meromorphic map $f_1 : W \dashrightarrow T_1$ with connected fibers, in the sense of [Cam94, §3.2], and a countable union E_W of proper closed analytic subsets of W, such that for any $w \in W \backslash E_W$, the fiber F_w of f_1 through w is the largest compact connected analytic subset of W containing w. Let $\mu : \widehat{W} \to W$ be a resolution of singularities which is an isomorphism

over W_{reg} [BM97, Theorems 13.2 and 13.3]. Since W is normal, the fibers of μ are connected by Zariski's main theorem. Thus the fibers of $f_1 \circ \mu$ over general points of T_1 are connected, and they are smooth by Sard's theorem, hence irreducible. Since μ maps the fiber of $f_1 \circ \mu$ over $f_1(w)$ onto F_w, after enlarging E_W, F_w is irreducible for any $w \in W \backslash E_W$. Let k be the dimension of the general fibers of f_1. Enlarging E_W again, we may assume that $\dim F_w = k$ for any $w \in W \backslash E_W$. Replacing E_W with $\bigcup_{q \in Q} qE_W$, which is a countable union since Q is countable, we may assume that E_W is Q-invariant. Then $F_{qw} = qF_w$ for any $w \in W \backslash E_W$ and $q \in Q$.

For $w \in W \backslash E_W$, the reduced cycle $[F_w]$ is irreducible, so any irreducible component of $\mathcal{C}(W)$ containing $[F_w]$ belongs to $\mathcal{S}$. Let $\mathcal{S}_0 \subset \mathcal{S}$ be the set of the components containing $[F_w]$ for some $w \in W \backslash E_W$. Then $W \backslash E_W \subset \bigcup_{\mathcal{P} \in \mathcal{S}_0} A_{\mathcal{P}}$. Since $\mathcal{S}_0$ is countable, the $A_{\mathcal{P}}$ are closed analytic (**Step 3**), and W is not a countable union of proper closed analytic subsets by the Baire category theorem, there exists $\mathcal{T} \in \mathcal{S}_0$ with $A_{\mathcal{T}} = W$. Let $v := v_{\mathcal{T}}$. Let $w \in W \backslash E_W$ and let $Z \in \mathcal{T}$ with $w \in \operatorname{Supp} Z$. Since $\operatorname{Supp} Z$ is compact and connected, $\operatorname{Supp} Z \subset F_w$. Since $\dim \operatorname{Supp} Z = k$ and F_w is irreducible, $\operatorname{Supp} Z = F_w$. Thus $Z = a[F_w]$ for a positive integer a with $a \operatorname{vol}_{\omega_W}(F_w) = v$. Hence $I_{\mathcal{T}} \to W$ has finite fibers over $W \backslash E_W$, and

$$\dim \mathcal{T} + k = d.$$

Let $\mathcal{T}^b \subsetneq \mathcal{T}$ be the closed analytic subset of the cycles which are not reduced and irreducible or which are singular points of $\mathcal{T}$. Let $E_{\mathcal{T}}$ be the union of E_W with the image in W of $I_{\mathcal{T}} \cap (\mathcal{T}^b \times W)$, which is a proper closed analytic subset of W since $\dim \mathcal{T}^b + k < d$. Then for any $w \in W \backslash E_{\mathcal{T}}$, $[F_w]$ is the unique cycle of $\mathcal{T}$ whose support contains w, and $[F_w]$ is a smooth point of $\mathcal{T}$.

Let $q \in Q$. For $w \in W \backslash (E_{\mathcal{T}} \cup qE_{\mathcal{T}})$, we have $[F_w] \in \mathcal{T}$ and $[F_w] = q[F_{q^{-1}w}] \in q\mathcal{T}$. Write $E_{\mathcal{T}} \cup qE_{\mathcal{T}} = \bigcup_{l \geq 1} H_l$, where the H_l are proper closed analytic subsets of W. For any l, the cycles of $\mathcal{T}$ with support contained in H_l form the closed analytic subset $\mathcal{T} \cap \mathcal{C}(H_l)$ of $\mathcal{T}$, which is proper since $A_{\mathcal{T}} = W$. Since $\mathcal{T}$ is irreducible, the cycles in $\mathcal{T} \backslash \mathcal{T}^b$ whose supports are contained in no H_l form a dense subset of $\mathcal{T}$ by the Baire category theorem. The support of such a cycle is irreducible, so it contains a point $w \notin E_{\mathcal{T}} \cup qE_{\mathcal{T}}$, and the cycle is $[F_w]$ by the previous paragraph. Thus the cycles $[F_w]$ with $w \in W \backslash (E_{\mathcal{T}} \cup qE_{\mathcal{T}})$ form a dense subset of $\mathcal{T}$. Hence $\mathcal{T} \subset q\mathcal{T}$, so $q\mathcal{T} = \mathcal{T}$. Thus Q acts on $\mathcal{T}$. Let $\nu : T \to \mathcal{T}$ be the normalization, let I^0 be the irreducible component of $I_{\mathcal{T}}$ dominating $\mathcal{T}$, which is unique since the general member of $\mathcal{T}$ is irreducible, and let W' be the normalization of I^0. The projections induce proper holomorphic maps

$$g : W' \to W, \qquad h : W' \to T,$$

and Q acts on T and W' so that g and h are Q-equivariant. Enlarging $E_{\mathcal{T}}$ by the image under g of the set of points where $W' \to I^0$ is not an isomorphism, we may assume that $g^{-1}(w)$ is a single point for any $w \in W \backslash E_{\mathcal{T}}$. In particular, g is bimeromorphic. For $t \in T$, let $W_t := g(h^{-1}(t))$, a compact subset of $\operatorname{Supp} \nu(t)$. For $w \in W \backslash E_{\mathcal{T}}$, let $t(w) := h(g^{-1}(w))$. Then $\nu(t(w)) = [F_w]$ and $W_{t(w)} = F_w$.

Let $\operatorname{Exc} \subset W'$ be the closed analytic subset of the points at which g is not locally biholomorphic, and let $\Sigma_T := h(\operatorname{Exc})$, a Q-invariant closed analytic subset of T. Let $t \in T \backslash \Sigma_T$ and $y \in W_t$. Since W is normal, $g^{-1}(y)$ is connected by Zariski's main theorem, and $g^{-1}(y) \cap h^{-1}(t)$ consists of isolated points of $g^{-1}(y)$. Thus $g^{-1}(y)$ is a single point. Hence $g^{-1}(W_t) = h^{-1}(t)$, and g maps a neighborhood of $h^{-1}(t)$ biholomorphically onto a neighborhood of W_t. We show that $t(w) \notin \Sigma_T$ for $w \in W \backslash E_{\mathcal{T}}$, so $\Sigma_T \neq T$. Suppose that g is not locally biholomorphic at a point $y' \in h^{-1}(t(w))$, and let $y := g(y') \in F_w$. Since g is bimeromorphic and W is normal, $g^{-1}(y)$ is connected and of positive dimension by Zariski's main theorem. Since $h^{-1}(t(w)) \to \operatorname{Supp} \nu(t(w))$ is finite, there exists $t' \neq t(w)$ with $y \in \operatorname{Supp} \nu(t')$. Then $\operatorname{Supp} \nu(t') \cup F_w$ is compact, connected, and contains w, so $\operatorname{Supp} \nu(t') \subset F_w$. Since $\dim \operatorname{Supp} \nu(t') = k$ and F_w is irreducible, $\operatorname{Supp} \nu(t') = F_w$, so $\nu(t') = a[F_w]$ with $a \operatorname{vol}_{\omega_W}(F_w) = v = \operatorname{vol}_{\omega_W}(F_w)$, so $a = 1$ and $\nu(t') = \nu(t(w))$. Since $\nu(t(w))$ is a smooth point of $\mathcal{T}$, $t' = t(w)$, a contradiction.

Finally, Q acts properly on T and on W'. If $\Lambda \subset T$ is compact and $q\Lambda \cap \Lambda \neq \emptyset$, then $h^{-1}(\Lambda)$ is compact since h is proper, and $qg(h^{-1}(\Lambda)) \cap g(h^{-1}(\Lambda)) \neq \emptyset$, and there exist only finitely many

such q by **Step 1**. If $\Lambda' \subset W'$ is compact and $q\Lambda' \cap \Lambda' \neq \emptyset$, then $qg(\Lambda') \cap g(\Lambda') \neq \emptyset$, and there exist only finitely many such q by **Step 1**. In particular, the stabilizer Q_t of any $t \in T$ is finite.

Step 5. In this step we descend the family of **Step 4** to X.

By [Car57, Théorème 4], $B := T/Q$ and W'/Q are normal complex spaces. Since g and h are Q-equivariant, they induce holomorphic maps

$$\bar{g} : W'/Q \to W/Q = X, \qquad \bar{h} : W'/Q \to B.$$

Let Ω be as in **Step 1**. Then $g^{-1}(\Omega)$ is compact and surjects onto W'/Q. Thus W'/Q is compact, so $\bar{g}$ is proper and $B = \bar{h}(W'/Q)$ is compact. Since W is normal and connected, W is irreducible. Hence W', T, W'/Q, and B are irreducible. Since $g^{-1}(w)$ is a single point for $w \in W\backslash E_{\mathcal{T}}$, $\bar{g}$ is generically injective, hence a proper modification. Thus $\varphi := \bar{h} \circ \bar{g}^{-1} : X \dashrightarrow B$ is a dominant meromorphic map whose graph is the image of W'/Q in $X \times B$.

Let $t \in T\backslash\Sigma_T$, let $b \in B$ be the image of t, and let $X_b := p(W_t)$. If $q \in Q\backslash Q_t$, then $qW_t \cap W_t = W_{qt} \cap W_t = \emptyset$, since $g^{-1}(W_t) = h^{-1}(t)$. Since g maps a Q_t-invariant neighborhood of $h^{-1}(t)$ biholomorphically onto a neighborhood of W_t, $\bar{g}$ maps a neighborhood of $\bar{h}^{-1}(b)$ biholomorphically onto a neighborhood of X_b. Hence φ is holomorphic near X_b, and $X_b = \varphi^{-1}(b)$. For $w \in W\backslash E_{\mathcal{T}}$ and b the image of $t(w)$, we have $X_b = p(F_w)$, which is connected.

Step 6. In this step we construct the countable exceptional set and prove the characterization of the subvarieties with finite ρ-image.

Let $\mathcal{P} \in \mathcal{S}$ with $A_{\mathcal{P}} \neq W$, and let $Q_{\mathcal{P}} := \{q \in Q \mid q\mathcal{P} = \mathcal{P}\}$. We show that $p(A_{\mathcal{P}})$ is a proper closed analytic subset of X. The group $Q_{\mathcal{P}}$ acts properly on $\mathcal{P}$: if $\Lambda \subset \mathcal{P}$ is compact, then the supports of the cycles in Λ are contained in a compact subset of W since $I \to \mathcal{C}(W)$ is proper, and **Step 1**(1) allows only finitely many q with $q\Lambda \cap \Lambda \neq \emptyset$. Let $\mathcal{P}^\nu$ be the normalization of $\mathcal{P}$. Then $\mathcal{P}^\nu/Q_{\mathcal{P}}$ is a normal complex space [Car57, Théorème 4]. We show that $\mathcal{P}^\nu/Q_{\mathcal{P}}$ is compact. Let (Z_j) be a sequence in $\mathcal{P}$. Fix $q_j \in Q$ with $q_j \operatorname{Supp} Z_j \cap \Omega \neq \emptyset$. By **Step 3**, the supports of the $q_j Z_j$ are contained in the closed $\delta_{\mathcal{P}}$-neighborhood of Ω, which is compact. By Bishop's theorem [Bis64], after passing to a subsequence, $q_j Z_j$ converges in $\mathcal{C}(W)$. Since the irreducible components of $\mathcal{C}(W)$ are locally finite, after passing to a further subsequence, $q_j\mathcal{P} = \mathcal{P}_1$ for a fixed component $\mathcal{P}_1$. Fix $q_0 \in Q$ with $q_0\mathcal{P} = \mathcal{P}_1$. Then $q_0^{-1}q_j \in Q_{\mathcal{P}}$, and $q_0^{-1}q_j Z_j$ converges in $\mathcal{P}$. Hence $\mathcal{P}/Q_{\mathcal{P}}$ is compact, and so is $\mathcal{P}^\nu/Q_{\mathcal{P}}$ since $\mathcal{P}^\nu \to \mathcal{P}$ is finite.

Over any relatively compact open subset of $\mathcal{P}$, the union of the supports of the cycles is compact, so p is proper on the union. Since p has discrete fibers, the push-forward p_* defines a holomorphic map $\mathcal{P} \to \mathcal{C}(X)$, where $\mathcal{C}(X)$ is the Barlet space of compact cycles of X [Bar75]. It is $Q_{\mathcal{P}}$-invariant, so it induces a holomorphic map $\mathcal{P}^\nu/Q_{\mathcal{P}} \to \mathcal{C}(X)$. By Remmert's proper mapping theorem [GR84, Chapter 10, §6], the image $\overline{\mathcal{P}}$ is a compact analytic subset of $\mathcal{C}(X)$. For unramified coverings of compact Kähler manifolds, the closedness of the image of the push-forward map on an irreducible component of the cycle space is [Cam94, Lemme 3.15 and Remarque 3.6]. The universal family over $\overline{\mathcal{P}}$ is compact, and the image of the universal family in X is $p(A_{\mathcal{P}})$. Thus $p(A_{\mathcal{P}})$ is closed analytic by Remmert's proper mapping theorem [GR84, Chapter 10, §6]. If $p(A_{\mathcal{P}}) = X$, then $W = \bigcup_{q\in Q} qA_{\mathcal{P}}$ is a countable union of proper closed analytic subsets, which contradicts the Baire category theorem. Hence $p(A_{\mathcal{P}}) \neq X$. Let

$$E_1 := \bigcup_{\mathcal{P}\in\mathcal{S}, A_{\mathcal{P}}\neq W} p(A_{\mathcal{P}}),$$

a countable union of proper closed analytic subsets of X, since $\mathcal{C}(W)$ has countably many irreducible components.

Let $\mathcal{P} \in \mathcal{S}$ with $A_{\mathcal{P}} = W$. We show that for any $Z \in \mathcal{P}$ there exists $t \in T$ with $\operatorname{Supp} Z \subset W_t$. Let $\mathcal{P}^c \subset \mathcal{P}$ be the set of the cycles $Z \in \mathcal{P}$ such that $\operatorname{Supp} Z \subset W_t$ for some $t \in T$. If $Z \in \mathcal{P}$ and $\operatorname{Supp} Z$ contains a point $w \in W\backslash E_{\mathcal{T}}$, then $\operatorname{Supp} Z \subset F_w = W_{t(w)}$ since $\operatorname{Supp} Z$ is compact and connected, so $Z \in \mathcal{P}^c$. Since $A_{\mathcal{P}} = W$ and $E_{\mathcal{T}}$ is a countable union of proper closed analytic subsets, the cycles $Z \in \mathcal{P}$ with $\operatorname{Supp} Z \not\subset E_{\mathcal{T}}$ form a dense subset of $\mathcal{P}$ by the Baire

category theorem. Moreover, we show that $\mathcal{P}^c$ is closed. Let $Z_j \in \mathcal{P}^c$ converge to $Z \in \mathcal{P}$, with $\operatorname{Supp} Z_j \subset W_{t_j}$. The supports of the Z_j are contained in a compact subset of W. Since g and h are proper, after passing to a subsequence, t_j converges to some $t \in T$, and any limit of points of W_{t_j} is contained in W_t. Since $\operatorname{Supp} Z$ is contained in the limit of the $\operatorname{Supp} Z_j$, $\operatorname{Supp} Z \subset W_t$. Thus $\mathcal{P}^c$ is closed and contains a dense subset of $\mathcal{P}$, so $\mathcal{P}^c = \mathcal{P}$.

Let $E_2' := \bar{g}(\bar{h}^{-1}(\Sigma_T/Q))$, a proper closed analytic subset of X. Let $x \in X\backslash(D \cup E_1 \cup E_2')$, let $w \in p^{-1}(x)$, and let $t \in T\backslash\Sigma_T$ with $w \in W_t$. Then $g^{-1}(w)$ is a single point y with $h(y) = t$ by **Step 4**. Let $X_x := p(W_t)$ and let $V \subset X$ be a compact irreducible analytic subvariety with $x \in V$. We show that V has finite ρ-image if and only if $V \subset X_x$.

Assume that V has finite ρ-image. By **Step 2**, some irreducible component V' of $p^{-1}(V)$ dominating V is compact. Since $p(V')$ is closed and dense in V, $p(V') = V \ni x$. Since Q acts transitively on $p^{-1}(x)$, we may assume that $w \in V'$ by **Step 2**. The reduced irreducible cycle $[V']$ belongs to some $\mathcal{P} \in \mathcal{S}$. If $A_{\mathcal{P}} \neq W$, then $x \in p(A_{\mathcal{P}}) \subset E_1$, a contradiction. Thus $A_{\mathcal{P}} = W$, and $V' \subset W_{t'}$ for some $t' \in T$. Since $w \in W_{t'}$ and $g^{-1}(w) = \{y\}$, $t' = h(y) = t$. Hence $V \subset p(W_t) = X_x$.

Conversely, assume that $V \subset X_x$. Then $p^{-1}(V) \cap W_t$ is a compact analytic subset which surjects onto V. Since V is irreducible, some irreducible component V' of $p^{-1}(V) \cap W_t$ satisfies $p(V') = V$. Since p has discrete fibers, $\dim V' = \dim V \geq \dim p^{-1}(V)$, so V' is a compact irreducible component of $p^{-1}(V)$ dominating V. By **Step 2**, V has finite ρ-image.

Step 7. In this step we replace B with a compact Kähler manifold, define E_ρ, and prove uniqueness.

Since B is an irreducible compact complex space and $\varphi : X \dashrightarrow B$ is a dominant meromorphic map, Lemma 2.14(3) implies that there exists a compact Kähler manifold Z_ρ with a bimeromorphic map $\beta : B \dashrightarrow Z_\rho$. Fix proper closed analytic subsets $\Sigma \subsetneq B$ and $\Sigma' \subsetneq Z_\rho$ such that $\Sigma_T/Q \subset \Sigma$ and β induces an isomorphism $B\backslash\Sigma \cong Z_\rho\backslash\Sigma'$. Let $f_\rho := \beta \circ \varphi$, a dominant meromorphic map. Let $E_2 := \bar{g}(\bar{h}^{-1}(\Sigma))$, a proper closed analytic subset of X containing E_2', and let $X^\circ := X\backslash E_2$. By **Step 5**, $f_\rho|_{X^\circ} : X^\circ \to Z_\rho\backslash\Sigma'$ is a proper surjective holomorphic map, and for $x \in X^\circ$, f_ρ is holomorphic near the fiber X_x of f_ρ through x, which is $X_x = p(W_t)$ as in **Step 6**. By **Step 5**, the general fibers of f_ρ are connected. Since Z_ρ is smooth, the finite part of the Stein factorization of $f_\rho|_{X^\circ}$ [GR84, Chapter 10, §6] is an isomorphism, so any fiber of $f_\rho|_{X^\circ}$ is connected.

Let $\Gamma_\rho \subset X \times Z_\rho$ be the closure of the graph of f_ρ, and let $\widehat{\Gamma}_\rho \to \Gamma_\rho$ be a resolution of singularities which is an isomorphism over the smooth locus of Γ_ρ [BM97, Theorem 1.6]. By Remmert's proper mapping theorem and Sard's theorem, the critical values of $\widehat{\Gamma}_\rho \to Z_\rho$ form a proper closed analytic subset $C_\rho \subsetneq Z_\rho$. The part of Γ_ρ over $Z_\rho\backslash\Sigma'$ is the graph of $f_\rho|_{X^\circ}$, which is smooth. Thus for $x \in X^\circ$ with $f_\rho(x) \notin C_\rho$, the fiber X_x is smooth and connected, hence irreducible. Let E_3 be the image of $\Gamma_\rho \cap (X \times C_\rho)$ in X, a proper closed analytic subset of X. Let

$$E_\rho := E_1 \cup E_2 \cup E_3,$$

a countable union of proper closed analytic subsets of X. Let $x \in X\backslash(D \cup E_\rho)$. Then f_ρ is holomorphic near X_x, and X_x is a compact irreducible analytic subvariety with $X_x \not\subset D$. By **Step 6**, a compact irreducible analytic subvariety $V \subset X$ with $x \in V$ has finite ρ-image if and only if $V \subset X_x$. Applying the equivalence to $V = X_x$, we deduce that X_x has finite ρ-image.

Finally, let $f' : X \dashrightarrow Z'$ be another dominant meromorphic map with connected general fibers and a countable union E' of proper closed analytic subsets of X which satisfy the conclusion. Let $x \in X\backslash(D \cup E_\rho \cup E')$, and let X_x' be the fiber of f' through x. Since X_x' has finite ρ-image, $X_x' \subset X_x$. By symmetry, $X_x \subset X_x'$, so $X_x = X_x'$. Thus the general fibers of f_ρ and f' coincide. Let $\Gamma \subset Z_\rho \times Z'$ be the closure of the image of X under (f_ρ, f'). Then the projections $\Gamma \to Z_\rho$ and $\Gamma \to Z'$ are surjective and injective over the complement of a countable union of proper closed analytic subsets. Hence they are bimeromorphic, and f_ρ is unique up to bimeromorphic equivalence of Z_ρ. □

8.2. **Compact subvarieties from positive currents.** In this subsection, we prove Lemma 8.4, which bounds the Γ-dimension of a homomorphism ρ from $\pi_1^{\mathrm{orb}}(X,\Delta)$ by the dimension of a Kähler manifold W whose Kähler form, pulled back by a map to W/I, extends to a closed positive current. For a homomorphism from the fundamental group of a compact Kähler manifold, the analogous bound is [OAI26c, Lemma 4.7]. Since the meridians of $\operatorname{Supp}\Delta$ in general have non-trivial image under ρ, we take the closures of the fibers of the map to W/I in a compact Kähler manifold X' over X, and compute monodromy only on the complement of $\operatorname{Supp}\Delta$.

Lemma 8.4. *Let (X,Δ) be a compact Kähler orbifold pair of dimension d, let $D := \operatorname{Supp}\Delta$, let I be a countable group acting properly discontinuously and effectively by holomorphic isometries on a connected Kähler manifold (W,ω_W) of dimension s with $1 \le s \le d$, and let $\rho : \pi_1^{\mathrm{orb}}(X,\Delta) \to I$ be a homomorphism. Let $\mu : X' \to X$ be a proper modification with X' a compact Kähler manifold, and let $U \subset X\backslash D$ be a dense Zariski open subset of X, that is, the complement of a closed analytic subset of X, over which μ is an isomorphism. Let $\widetilde{U} \to U$ be the connected covering space corresponding to the kernel of the composite*

$$\pi_1(U) \to \pi_1(X\backslash D) \to \pi_1^{\mathrm{orb}}(X,\Delta) \xrightarrow{\rho} I.$$

Assume that there exists a holomorphic submersion $\widetilde{f} : \widetilde{U} \to W$ that is equivariant for the deck group of $\widetilde{U} \to U$, identified with the image of $\pi_1(U)$ in I, and let $f : U \to W/I$ be the map induced by $\widetilde{f}$. Since ω_W is I-invariant, the form $f^*\omega_W$ on U, computed with local lifts of f to W, does not depend on the choice of lifts.

Assume that there exist a closed positive $(1,1)$-current T on X' and a real number $c > 0$ such that $T = c\mu^* f^*\omega_W$ on $\mu^{-1}(U)$.

Then for almost every $u \in U$, with respect to any smooth positive volume form on U, there exists a compact irreducible analytic subvariety $V_u \subset X$ of dimension $d-s$ with $u \in V_u$ and finite ρ-image. In particular, if ρ is surjective, then $\operatorname{gdim}(X,\Delta;\rho) \le s$.

Proof. The proof is long, so we divide it into several steps. Let $k := d - s$. Fix a Kähler form ω on X'. Since μ is an isomorphism over U, we measure subsets of U with the volume form $\omega^d/d!$ on $\mu^{-1}(U)$, transported to U by μ.

Step 1. In this step we prove the mass bound

$$\int_{\mu^{-1}(U)} T^s \wedge \omega^k < +\infty.$$

Here $T = c\mu^* f^*\omega_W$ is a smooth semipositive $(1,1)$-form on $\mu^{-1}(U)$, so the integrand is a smooth semipositive form.

Since the cohomology of closed currents on X' is isomorphic to the cohomology of smooth closed forms, the dd^c-lemma [Dem12, Chapter VI, Lemma 8.6] implies that

$$T = \alpha + dd^c\varphi,$$

where α is a smooth closed real $(1,1)$-form on X' and φ is a quasi-plurisubharmonic function on X'. On $\mu^{-1}(U)$, $dd^c\varphi = T - \alpha$ is smooth, so φ is smooth on $\mu^{-1}(U)$ by elliptic regularity. Let $a > 0$ be a real number such that $\beta := \alpha + a\omega$ is a Kähler form. For any positive integer j, let $\varphi_j := \max(\varphi, -j)$. Since $\beta + dd^c\varphi = T + a\omega \ge 0$, we have $\beta + dd^c\varphi_j \ge 0$. Locally $\beta + dd^c\varphi_j$ is dd^c of a bounded plurisubharmonic function, so the Bedford–Taylor products of $\beta + dd^c\varphi_j$ are defined [BT82] (see [Dem12, Chapter III, §3]), and

$$\int_{X'} (\beta + dd^c\varphi_j)^s \wedge \omega^k = \int_{X'} \beta^s \wedge \omega^k.$$

In fact,

$$\int_{X'} (\beta + dd^c\varphi_j)^s \wedge \omega^k - \int_{X'} \beta^s \wedge \omega^k = \int_{X'} dd^c(\varphi_j R_j) \wedge \omega^k,$$

where $R_j := \sum_{l=0}^{s-1}(\beta + dd^c\varphi_j)^l \wedge \beta^{s-1-l}$ is closed, and $\int_{X'} dd^c(\varphi_j R_j) \wedge \omega^k = 0$ by the Stokes formula since X' is compact [Dem12, Chapter III, §3].

Let $\Omega_j := \{\varphi > -j\} \cap \mu^{-1}(U)$, an open subset of X'. On Ω_j, $\varphi_j = \varphi$, so by locality of the Bedford–Taylor products,

$$(\beta + dd^c\varphi_j)^s \wedge \omega^k = (T + a\omega)^s \wedge \omega^k \geq T^s \wedge \omega^k \geq 0 \quad \text{on } \Omega_j,$$

where the first inequality holds since T and ω are smooth semipositive $(1,1)$-forms on Ω_j. Since $(\beta + dd^c\varphi_j)^s \wedge \omega^k$ is a positive measure on X',

$$\int_{\Omega_j} T^s \wedge \omega^k \leq \int_{X'} \beta^s \wedge \omega^k.$$

Since φ is finite on $\mu^{-1}(U)$, the sets Ω_j increase to $\mu^{-1}(U)$ as $j \to +\infty$. Thus the mass bound follows from the monotone convergence theorem.

Step 2. In this step we construct a subset $U_2 \subset U$ such that $U\backslash U_2$ has measure zero and, for any $u \in U_2$, the fiber of $f \circ \mu$ through $\mu^{-1}(u)$ has finite volume.

Let $W^\circ \subset W$ be the set of points whose stabilizer in I is trivial. Since the action of I is properly discontinuous, any point of W has a relatively compact neighborhood N such that $hN \cap N \neq \emptyset$ for only finitely many $h \in I$. Since the action is effective, by the identity theorem any $h \in I\backslash\{1\}$ fixes no non-empty open subset of W, so the fixed locus $\mathrm{Fix}(h)$ is a proper closed analytic subset of W. Thus $W\backslash W^\circ = \bigcup_{h\neq 1} \mathrm{Fix}(h)$ is a locally finite union of proper closed analytic subsets of W, and W°/I is a complex manifold. Since I acts by isometries, ω_W induces a Kähler form on W°/I, which we also denote by ω_W. Let $U_1 := f^{-1}(W^\circ/I)$. Locally on U, $U\backslash U_1$ is the inverse image of $W\backslash W^\circ$ under a local lift of f, which is a submersion, so $U\backslash U_1$ is a nowhere dense closed analytic subset of U. Thus $U\backslash U_1$ has measure zero.

Let $g := f \circ \mu : \mu^{-1}(U_1) \to W^\circ/I$, a holomorphic submersion with $T = cg^*\omega_W$. Pointwise complex linear algebra, with the Riemannian metrics of ω and ω_W, implies that

$$g^*\frac{\omega_W^s}{s!} \wedge \frac{\omega^k}{k!} = J_g \frac{\omega^d}{d!},$$

where $J_g > 0$ is the normal Jacobian of g, that is, the product of the squares of the s complex singular values of dg. By **Step 1**,

$$\int_{\mu^{-1}(U_1)} J_g \frac{\omega^d}{d!} = \frac{1}{c^s s!k!} \int_{\mu^{-1}(U_1)} T^s \wedge \omega^k < +\infty.$$

The fibers of g are complex submanifolds of $\mu^{-1}(U_1)$ of dimension k, and by the Wirtinger formula the $2k$-dimensional volume of $g^{-1}(y)$ is $\int_{g^{-1}(y)} \omega^k/k!$. By the coarea formula [Fed69, 3.2.22],

$$\int_{W^\circ/I} \left(\int_{g^{-1}(y)} \frac{\omega^k}{k!} \right) d\,\mathrm{vol}_{\omega_W}(y) = \int_{\mu^{-1}(U_1)} J_g \frac{\omega^d}{d!} < +\infty.$$

Thus the set Y_0 of $y \in W^\circ/I$ with $\int_{g^{-1}(y)} \omega^k/k! = +\infty$ has measure zero. The coarea formula, applied to the indicator function of $g^{-1}(Y_0)$, implies that $\int_{g^{-1}(Y_0)} J_g\omega^d/d! = 0$. Since $J_g > 0$, $g^{-1}(Y_0)$ has measure zero. Let $U_2 := U_1\backslash\mu(g^{-1}(Y_0))$. Then $U\backslash U_2$ has measure zero, and $\int_{g^{-1}(f(u))} \omega^k/k! < +\infty$ for any $u \in U_2$. The ratio of two smooth positive volume forms on U is a continuous positive function, so a subset of U has measure zero with respect to one of them if and only if it has measure zero with respect to the other. Thus the meaning of "almost every $u \in U$" does not depend on the volume form.

Step 3. In this step we fix $u \in U_2$ and construct V_u.

Let $y := f(u) \in W^\circ/I$, and let F be the connected component of $g^{-1}(y)$ containing $\mu^{-1}(u)$. Since the action of I is properly discontinuous, W/I is Hausdorff, so $g^{-1}(y) = \mu^{-1}(f^{-1}(y))$ is closed in $\mu^{-1}(U)$. Since g is a submersion, F is a connected complex submanifold of $\mu^{-1}(U)$

of dimension k, closed in $\mu^{-1}(U)$. In particular, F is an irreducible closed analytic subset of $\mu^{-1}(U)$, and $\int_F \omega^k/k! < +\infty$ by **Step 2**. If $k = 0$, then $F = \{\mu^{-1}(u)\}$, we let $V_u := \{u\}$, and the open image of V_u is trivial. Assume that $k > 0$.

The current of integration $[F]$ on $\mu^{-1}(U)$ is a closed positive current of bidimension (k, k) with finite mass $\int_F \omega^k/k!$. The set $X'\backslash\mu^{-1}(U)$ is a closed analytic subset of X', so it is complete pluripolar in the sense of [Dem12, Chapter III, Definition 2.1], which is a local condition (locally a closed analytic subset $\{h_1 = \cdots = h_m = 0\}$ is the polar set of $\log \sum_i |h_i|^2$). By the Skoda–El Mir extension theorem [Sko82], [Mir84, Théorème II.1], in the form of [Dem12, Chapter III, Theorem 2.3] on complex manifolds, the trivial extension $\widetilde{[F]}$ of $[F]$ by zero is a closed positive current on X'. By Siu's theorem [Siu74] (see [Dem12, Chapter III, Corollary 8.5]),

$$\Sigma_F := \{x \in X' : \nu(\widetilde{[F]}, x) \geq 1\}$$

is a closed analytic subset of X' with $\dim \Sigma_F \leq k$.

We show that the closure $\overline{F}$ of F in X' is an irreducible closed analytic subset of X' with $\overline{F} \cap \mu^{-1}(U) = F$. Lelong numbers are local, and $\widetilde{[F]} = [F]$ on $\mu^{-1}(U)$. Since F is a closed complex submanifold of $\mu^{-1}(U)$, we have $\nu([F], x) = 1$ for any $x \in F$ and $\nu([F], x) = 0$ for any $x \in \mu^{-1}(U)\backslash F$ [Dem12, Chapter III, Remark 5.9]. Thus

$$\Sigma_F \cap \mu^{-1}(U) = F.$$

Since X' is compact, Σ_F has finitely many irreducible components. Let $V_1', \ldots, V_m'$ be the irreducible components of Σ_F that intersect F. For any i, $V_i' \cap \mu^{-1}(U)$ is a non-empty Zariski open subset of V_i', so it is dense in V_i', and $V_i' \cap \mu^{-1}(U) \subset F$. Thus $V_i' \subset \overline{F}$. Since $F \subset V_1' \cup \cdots \cup V_m'$ and the right-hand side is closed, $\overline{F} = V_1' \cup \cdots \cup V_m'$. Moreover, $F = \bigcup_i (V_i' \cap \mu^{-1}(U))$ is a finite union of closed analytic subsets of $\mu^{-1}(U)$, and F is irreducible, so $F = V_{i_0}' \cap \mu^{-1}(U)$ for some i_0. Thus $\overline{F} = V_{i_0}'$ is irreducible, $\overline{F} \cap \mu^{-1}(U) = F$, and $\dim \overline{F} = k$.

Let $V_u := \mu(\overline{F})$. By Remmert's proper mapping theorem [Rem57] (see also [Dem12, Chapter II, (8.8)]), V_u is a compact analytic subset of X, and V_u is irreducible since $\overline{F}$ is irreducible. Since μ is an isomorphism over U and $\overline{F} \cap \mu^{-1}(U) = F$, we have $V_u \cap U = \mu(F) \cong F$, and $\mu(F)$ is dense in V_u. Thus $\dim V_u = k = d - s$ and $u \in V_u$. Since $V_u \cap U \neq \emptyset$ and $U \cap D = \emptyset$, $V_u \not\subset D$. Let $r : \widehat{V}_u \to V_u$ be a resolution of singularities which is an isomorphism over the smooth locus of V_u [BM97, Theorem 1.6].

Step 4. In this step we show that V_u has finite ρ-image.

We keep the assumption $k > 0$. For $k = 0$, the open image of $V_u = \{u\}$ is trivial by **Step 3**. Since $\mu(F) = V_u \cap U$ is open in V_u and smooth, r is an isomorphism over $\mu(F)$, and $r^{-1}(\mu(F))$ is the complement of a proper closed analytic subset of $\widehat{V}_u\backslash r^{-1}(D)$. By Definition-Lemma 8.1, $\pi_1(V_u)_{(X,\Delta)}$ is the image of the composite $\pi_1(\mu(F)) \to \pi_1(X\backslash D) \to \pi_1^{\mathrm{orb}}(X, \Delta)$, up to conjugation. Since $\mu(F) \subset U$, the map $\pi_1(\mu(F)) \to \pi_1(X\backslash D)$ factors through $\pi_1(U)$. Thus $\rho(\pi_1(V_u)_{(X,\Delta)})$ is, up to conjugation, the image of $\pi_1(\mu(F))$ under the composite $\pi_1(U) \to I$ in the statement.

Fix $x_1 \in \mu(F)$ and a point $\tilde{x}_1 \in \widetilde{U}$ over x_1, and let $z := \widetilde{f}(\tilde{x}_1)$. Let λ be a loop in $\mu(F)$ based at x_1, and let $h \in I$ be the image of $[\lambda]$. The lift $\tilde{\lambda}$ of λ to $\widetilde{U}$ starting at $\tilde{x}_1$ ends at $h\tilde{x}_1$. Since f is constant with value y on $\mu(F)$, the path $\widetilde{f} \circ \tilde{\lambda}$ takes values in the fiber of $W \to W/I$ over y, which is the orbit Iz. Since the action of I is properly discontinuous, Iz is a discrete subset of W, so the path $\widetilde{f} \circ \tilde{\lambda}$ is constant. By equivariance of $\widetilde{f}$,

$$hz = \widetilde{f}(h\tilde{x}_1) = \widetilde{f}(\tilde{x}_1) = z.$$

Thus h belongs to the stabilizer I_z of z, which is finite since the action of I is properly discontinuous (and trivial since $z \in W^\circ$). Thus $\rho(\pi_1(V_u)_{(X,\Delta)})$ is contained in a conjugate of I_z, so V_u has finite ρ-image.

Step 5. In this step we assume that ρ is surjective and prove that $\mathrm{gdim}(X, \Delta; \rho) \leq s$. Let $f_\rho : X \dashrightarrow Z_\rho$ be the Γ-reduction of ρ, and let E_ρ be the countable union of proper closed analytic

subsets of X in Definition-Theorem 8.2. From Definition-Theorem 8.2 we use that Z_ρ is a compact Kähler manifold, that f_ρ is a dominant meromorphic map, and that, for $x \in X\backslash(D \cup E_\rho)$, the fiber X_x is a compact irreducible analytic subvariety which contains every compact irreducible analytic subvariety V with $x \in V$ and finite ρ-image.

Let $X_1 \subset X$ be the open subset on which f_ρ is holomorphic, and let $E_1 \subset X_1$ be the set of points at which the differential of f_ρ has rank less than $\dim Z_\rho$. Then $X\backslash X_1$ is a proper closed analytic subset of X, and since f_ρ is dominant, E_1 is a nowhere dense closed analytic subset of X_1. Each of E_ρ, $X\backslash X_1$, and E_1 is a countable union of sets each of which is locally contained in the zero set of a non-zero holomorphic function. Thus E_ρ, $X\backslash X_1$, and E_1 have measure zero. Recall that $U_2 \subset U$ is the set in **Step 2**, and $U\backslash U_2$ has measure zero. Thus there exists

$$u \in U_2\backslash\big(E_\rho \cup (X\backslash X_1) \cup E_1\big).$$

Since $U \cap D = \emptyset$, $u \in X\backslash(D \cup E_\rho)$. By **Steps 3 and 4**, V_u is a compact irreducible analytic subvariety of X with $u \in V_u$, $\dim V_u = d - s$, and finite ρ-image, so $V_u \subset X_u$ by Definition-Theorem 8.2. Since f_ρ is holomorphic at u and the differential of f_ρ at u has the maximal rank $\dim Z_\rho$, the rank is $\dim Z_\rho$ near u, and by the constant rank theorem X_u is a submanifold of dimension $d - \dim Z_\rho$ near u. Since X_u is irreducible, $\dim X_u = d - \dim Z_\rho$. Thus

$$d - s = \dim V_u \leq \dim X_u = d - \dim Z_\rho,$$

that is, $\operatorname{gdim}(X, \Delta; \rho) = \dim Z_\rho \leq s$. □

9. Orbifold bases of general type

In this section, we prove Proposition 9.1, which Section 10 uses to exclude the orbifold pair of Proposition 7.8. We prove Theorem 9.2 in Subsection 9.1, and we deduce Proposition 9.1 in Subsection 9.2.

Proposition 9.1. *Let (X, Δ) be a compact Kähler orbifold pair, and let $G := \pi_1^{\mathrm{orb}}(X, \Delta)$. Assume that G is infinite and that every subgroup of finite index in G equals G. Then there exist a compact Kähler orbifold pair (Z, Δ_Z) with $\dim Z \geq 1$ and a dominant meromorphic map $f : X \dashrightarrow Z$ such that $K_Z + \Delta_Z$ is big.*

9.1. Bigness from minimal Γ-dimension.

In this subsection, we prove Theorem 9.2, a version of [OAI26c, Theorem 4.1] in which the base is dominated by a compact Kähler orbifold pair instead of a compact Kähler manifold. Condition (2) of Theorem 9.2 replaces the minimality condition (M) of [OAI26c, Theorem 4.1], and Lemma 8.4 replaces [OAI26c, Lemma 4.7]. The results on the base, [OAI26c, Lemma 4.2, Proposition 4.3, Lemmas 4.4 and 4.5, Proposition 4.6], are used as they are, together with the facts collected in Remark 9.3.

Theorem 9.2. *Let (Z, Δ_Z) be a compact Kähler orbifold pair of dimension $d > 0$, let $q : \pi_1^{\mathrm{orb}}(Z, \Delta_Z) \to Q$ be a surjective homomorphism, let (X, Δ) be a compact Kähler orbifold pair, let $f : X \dashrightarrow Z$ be a dominant meromorphic map, and let $\rho : \pi_1^{\mathrm{orb}}(X, \Delta) \to Q$ be a surjective homomorphism. Assume that Q is infinite, that every subgroup of finite index in Q equals Q, that every homomorphism $Q \to \mathrm{GL}_k(\mathbb{C})$, $k \geq 1$, is trivial, and that the following hold.*

(1) There exists a closed analytic subset $A \subsetneq X$ containing $\operatorname{Supp}\Delta$ such that f is holomorphic on $X^\circ := X\backslash A$, $f(X^\circ) \subset Z\backslash\operatorname{Supp}\Delta_Z$, and the homomorphisms $\pi_1(X^\circ) \to Q$ induced by ρ and by $q \circ (f|_{X^\circ})_*$ coincide.

(2) For every surjective homomorphism $\psi : Q \to H$ onto an infinite group H, we have $\operatorname{gdim}(X, \Delta; \psi \circ \rho) \geq d$.

Then Z is projective and $K_Z + \Delta_Z$ is big.

Remark 9.3. The proof of Theorem 9.2 below follows the proof of [OAI26c, Theorem 4.1] on the root orbifold $\mathcal{Z}$ of (Z, Δ_Z), and besides the statements of [OAI26c, Lemma 4.5 and Proposition 4.6] it uses the following facts from their proofs, in the notation of **Step 1** below. In the proof of [OAI26c, Lemma 4.5], the end of the third paragraph shows that a positive power

of the saturated orbifold line $\mathcal{L}$ is the pullback of a line bundle $\mathcal{A}$ on Z and that the metric of $\mathcal{L}$ induces a singular Hermitian metric on $\mathcal{A}$ with plurisubharmonic local weights, and the last paragraph shows that the loci where the line tensor or the map Φ_τ of image lines degenerates are closed analytic subsets. In the proof of [OAI26c, Proposition 4.6], the part headed "The manifold of image germs" shows that the germs of the images of Φ_τ at the points where Φ_τ has rank s form a connected complex manifold W of dimension s, that Φ_τ is there the composite of a holomorphic submersion to W with a holomorphic immersion $\iota : W \to \mathbb{P}(\mathcal{H})$, that $\iota^*\omega_{\mathrm{FS}}$ is a Kähler form on W, and that Q acts on W by holomorphic isometries for which the submersion is equivariant. The part headed "The manifold of image germs" also shows that the isometry group of W is a Lie group acting properly on W, and the last two paragraphs of that proof show that the image of Q in the isometry group of W is infinite and discrete, which is the statement of [OAI26c, Proposition 4.6].

Proof of Theorem 9.2. The proof is long, so we divide it into several steps. We fix base points compatible under the maps below and omit them from the notation.

Step 1. In this step, by [OAI26c, §4] for (Z, Δ_Z) and q, we obtain an integer $1 \leq s \leq d$, an isometric action of a group I on a Kähler manifold W with $\dim W = s$, and a line bundle $\mathcal{A}$ on Z whose curvature current is, over a dense open subset of Z, the pullback of the Kähler form of W by a submersion.

Let $\mathcal{Z}$ be the root orbifold of (Z, Δ_Z), and write $\Delta_Z = \sum_{i=1}^r (1 - \frac{1}{m_i}) D_i$. By Definition 2.2, this is the setting of [OAI26c, Theorem 4.1]. Since any homomorphism $Q \to \mathrm{GL}_k(\mathbb{C})$ is trivial, any representation of Q on a complex vector space V with $\dim V < +\infty$ has trivial, hence finite, image. Thus Q satisfies the hypothesis on the group in [OAI26c, Theorem 4.1] (see also [OAI26c, Lemma 4.2]). The finite covers of $\mathcal{Z}$ which appear in [OAI26c, §4] correspond to the subgroups $q^{-1}(Q_0)$ with $Q_0 \leq Q$ of finite index. Since Q has no proper subgroup of finite index, all of them are equal to $\mathcal{Z}$, and the finite-index refinement in [OAI26c, Proposition 4.6] is trivial. The results of [OAI26c, §4] used below concern only $\mathcal{Z}$ and q (see the setting of [OAI26c, Theorem 4.1]).

Recall from [OAI26c, §4.2] that a *line tensor* on $\mathcal{Z}$ is a non-zero holomorphic section τ of $(\Omega^1_{\mathcal{Z}})^{\otimes m} \otimes \mathbb{V}$ for some integer $m \geq 1$, where $\mathbb{V}$ is the flat bundle on $\mathcal{Z}$ defined by q and a unitary representation of Q on a Hilbert space $\mathcal{H}$, such that the morphism from the dual of $(\Omega^1_{\mathcal{Z}})^{\otimes m}$ to $\mathbb{V}$ defined by τ has rank one at a general point. On the covering of $\mathcal{Z}$ corresponding to $\ker q$, the bundle $\mathbb{V}$ is trivial with fiber $\mathcal{H}$, and the image lines of τ define a holomorphic map Φ_τ from a dense open subset of that covering to the projective space $\mathbb{P}(\mathcal{H})$ of lines in $\mathcal{H}$. Among the line tensors τ on $\mathcal{Z}$ for which the closed span $\mathcal{H}_\tau \subset \mathcal{H}$ of the image lines satisfies $\dim \mathcal{H}_\tau = +\infty$, we choose one for which the generic rank s of Φ_τ is minimal [OAI26c, §4.3, first paragraph]. Such line tensors exist by [OAI26c, Proposition 4.3 and Lemma 4.4], and $1 \leq s \leq d$.

By [OAI26c, Lemma 4.5], τ determines a line bundle $\mathcal{L}$ on $\mathcal{Z}$ which is a saturated subsheaf of $(\Omega^1_{\mathcal{Z}})^{\otimes m}$, and a singular Hermitian metric on $\mathcal{L}$ with positive curvature current, which equals the pullback of the Fubini–Study form ω_{FS} by Φ_τ on the locus where Φ_τ is regular. Moreover, by Remark 9.3, there exists an integer $N \geq 1$ such that $\mathcal{L}^{\otimes N}$ is the pullback of a line bundle $\mathcal{A}$ on Z under the coarse map $\mathcal{Z} \to Z$, and the metric induces a singular Hermitian metric h on $\mathcal{A}$ with plurisubharmonic local weights. By [OAI26c, Proposition 4.6] and Remark 9.3, there exist a connected Kähler manifold (W, ω_W) with $\dim W = s$, an infinite countable group I acting properly discontinuously and effectively on W by holomorphic isometries, and a surjective homomorphism $\phi : Q \to I$, such that, on the locus where Φ_τ is regular, Φ_τ is the composite of a ϕ-equivariant holomorphic submersion to W and a holomorphic immersion $W \to \mathbb{P}(\mathcal{H})$, and ω_W is the pullback of ω_{FS} by the immersion.

Let T_Z be the curvature current of h, a closed positive $(1,1)$-current on Z. Combining [OAI26c, Lemma 4.5], [OAI26c, Proposition 4.6], and Remark 9.3, we obtain the following. There exist a closed analytic subset $B \subsetneq Z$ containing $\operatorname{Supp} \Delta_Z$ such that h is smooth on $Z_h := Z \backslash B$, a real number $c > 0$, and a ϕ-equivariant holomorphic submersion $\varpi : \widetilde{Z}_h \to W$, where $\widetilde{Z}_h \to Z_h$ is the

connected covering space corresponding to the kernel of the composite

$$q_h : \pi_1(Z_h) \to \pi_1(Z \setminus \operatorname{Supp} \Delta_Z) \to \pi_1^{\mathrm{orb}}(Z, \Delta_Z) \xrightarrow{q} Q,$$

such that, writing $\widetilde{T}_Z$ for the pullback of $T_Z|_{Z_h}$ to $\widetilde{Z}_h$,

$$\widetilde{T}_Z = c\varpi^* \omega_W. \tag{9.1}$$

Since Z_h is the complement of a proper closed analytic subset of the connected manifold $Z \setminus \operatorname{Supp} \Delta_Z$, the homomorphism q_h is surjective, and Q is the deck group of $\widetilde{Z}_h \to Z_h$.

Step 2. In this step we suppose that $s < d$, resolve the graph of f, and pull back T_Z to a closed positive $(1,1)$-current T on the resolution. **Steps 2–4** derive a contradiction from $s < d$.

Suppose that $s < d$. Let $\Gamma_f \subset X \times Z$ be the closure of the graph of f, which is a compact Kähler space [Xia19, Example 2.15]. Since f is holomorphic on X°, the projection $\Gamma_f \to X$ is an isomorphism over X°. In particular, Γ_f is smooth over X°. By Lemma 2.14(2), there exists a resolution $X' \to \Gamma_f$ by a compact Kähler manifold X' which is an isomorphism over the smooth locus of Γ_f. Let $\mu : X' \to X$ and $f' : X' \to Z$ be the composites of $X' \to \Gamma_f$ with the projections $\Gamma_f \to X$ and $\Gamma_f \to Z$, respectively. Then μ is a proper modification which is an isomorphism over X°, $f' = f \circ \mu$ on $\mu^{-1}(X^\circ)$, and f' is surjective, since f is dominant and X' is compact.

The pullback of a plurisubharmonic function by a holomorphic map from a connected complex manifold is plurisubharmonic or identically $-\infty$ [Dem12, Chapter I, Theorem 5.11]. Since f' is surjective, $f'^{-1}(Z \setminus Z_h)$ is a proper closed analytic subset of the connected manifold X'. Hence any non-empty open subset of X' contains a point which f' maps to Z_h, where the local weights of h are smooth. Thus the local weights of $f'^* h$ are plurisubharmonic, and the curvature current T of $f'^* h$ is a closed positive $(1,1)$-current on X' with $T = f'^* T_Z$ on $f'^{-1}(Z_h)$.

Step 3. In this step we construct a dense Zariski open subset $U \subset X^\circ$ and a holomorphic map $g : U \to W/I$ with monodromy $\phi \circ \rho$ such that $T = c\mu^* g^* \omega_W$ on $\mu^{-1}(U)$.

Let $V' \subset X'$ be the union of $f'^{-1}(Z \setminus Z_h)$ and the closed analytic subset where the differential of f' has rank less than d. Since f' is surjective, f' is a submersion at a general point of X' by Sard's theorem, so V' is a proper closed analytic subset of X'. By Remmert's proper mapping theorem [Rem57] (see also [Dem12, Chapter II, (8.8)]), $\mu(V')$ is a closed analytic subset of X, and $\mu(V') \neq X$ since $\dim \mu(V') \leq \dim V' < \dim X$. Let $U := X^\circ \setminus \mu(V')$. Then U is a dense Zariski open subset of X with

$$U \subset X^\circ \subset X \setminus \operatorname{Supp} \Delta,$$

μ is an isomorphism over U, $f(U) \subset Z_h$, and $f|_U : U \to Z_h$ is a submersion. Since X is connected, U is connected.

Let $\rho_U : \pi_1(U) \to Q$ be the composite

$$\pi_1(U) \to \pi_1(X \setminus \operatorname{Supp} \Delta) \to \pi_1^{\mathrm{orb}}(X, \Delta) \xrightarrow{\rho} Q.$$

Since $U \subset X^\circ$ and $f(U) \subset Z_h$, condition (1) implies that $\rho_U = q_h \circ (f|_U)_*$. Since U is the complement of a proper closed analytic subset of $X \setminus \operatorname{Supp} \Delta$ and ρ is surjective, ρ_U is surjective. Let $U_1 \to U$ be the connected covering space corresponding to $\ker \rho_U$, with deck group Q. The image of $\pi_1(U_1) = \ker \rho_U$ under $(f|_U)_*$ is contained in $\ker q_h$, so by the lifting criterion [Hat02, Proposition 1.33], the composite $U_1 \to U \xrightarrow{f} Z_h$ lifts to a holomorphic map $\tilde{f} : U_1 \to \widetilde{Z}_h$. For $\beta \in Q$, the maps $\tilde{f} \circ \beta$ and $\beta \circ \tilde{f}$ are lifts of the composite $U_1 \to U \xrightarrow{f} Z_h$ which coincide at the base point, since $\rho_U = q_h \circ (f|_U)_*$. By the uniqueness of lifts, $\tilde{f} \circ \beta = \beta \circ \tilde{f}$.

Thus $\varpi \circ \tilde{f} : U_1 \to W$ is a ϕ-equivariant holomorphic map. Since $f|_U$ is a submersion, the covering maps are local biholomorphisms, and ϖ is a submersion onto the s-dimensional manifold W, the map $\varpi \circ \tilde{f}$ has rank s at any point. The subgroup $\ker \phi \subset Q$ acts trivially on W, so $\varpi \circ \tilde{f}$ descends to a holomorphic submersion

$$\widetilde{U} := U_1 / \ker \phi \longrightarrow W,$$

which is equivariant for $I = Q/\ker\phi$. Here $\widetilde{U} \to U$ is the connected covering space corresponding to the kernel of $\phi \circ \rho_U$, and I is the deck group of $\widetilde{U} \to U$. Let $g : U \to W/I$ be the induced map.

Let $u \in U$ and let σ be a local section of $U_1 \to U$ near u. Then $\tilde{f} \circ \sigma$ is a local lift of $f|_U$ to $\widetilde{Z}_h$, and $\varpi \circ \tilde{f} \circ \sigma$ is a local lift of g to W. By (9.1), near u we have

$$(f|_U)^* T_Z = (\tilde{f} \circ \sigma)^* (c\varpi^* \omega_W) = c(\varpi \circ \tilde{f} \circ \sigma)^* \omega_W = c g^* \omega_W.$$

Since $f' = f \circ \mu$ on $\mu^{-1}(U) \subset f'^{-1}(Z_h)$ and $T = f'^* T_Z$ on $f'^{-1}(Z_h)$ (**Step 2**), we obtain

$$T = c\mu^* g^* \omega_W \quad \text{on } \mu^{-1}(U).$$

Step 4. In this step we bound $\operatorname{gdim}(X, \Delta; \phi \circ \rho)$ from above by Lemma 8.4 and from below by condition (2), and conclude that $s = d$.

Since f is dominant, we have

$$1 \le s \le d = \dim Z \le \dim X,$$

so $1 \le s \le \dim X$. The group I is countable, the space (W, ω_W) is a connected Kähler manifold with $\dim W = s$, and I acts on W properly discontinuously and effectively by holomorphic isometries (**Step 1**). The map $\mu : X' \to X$ is a proper modification with X' a compact Kähler manifold, and $U \subset X^\circ \subset X \setminus \operatorname{Supp}\Delta$ is a dense Zariski open subset over which μ is an isomorphism (**Step 3**). The hypotheses of Lemma 8.4 hold for (X, Δ), I, (W, ω_W), the homomorphism $\phi \circ \rho : \pi_1^{\mathrm{orb}}(X, \Delta) \to I$, μ, U, the current T, and the real number c, with g in place of f (**Step 3**). Since $\phi \circ \rho$ is surjective, Lemma 8.4 implies that

$$\operatorname{gdim}(X, \Delta; \phi \circ \rho) \le s.$$

On the other hand, $\phi : Q \to I$ is a surjective homomorphism onto the infinite group I, so condition (2) with $\psi = \phi$ implies that

$$\operatorname{gdim}(X, \Delta; \phi \circ \rho) \ge d.$$

Thus $d \le s$, which contradicts $s < d$. Hence $s = d$.

Step 5. In this step we show that $\mathcal{A}$ is big and that Z is projective.

By **Step 4**, $s = d$, so the submersion $\varpi : \widetilde{Z}_h \to W$ is a local biholomorphism and $\varpi^* \omega_W$ is a Kähler form on $\widetilde{Z}_h$. By (9.1), $T_Z|_{Z_h}$ is a Kähler form on Z_h. Let $T_{Z,\mathrm{ac}}$ be the absolutely continuous part of T_Z. It is a positive $(1,1)$-form with locally integrable coefficients, and $T_{Z,\mathrm{ac}} = T_Z$ on Z_h. Hence

$$\int_Z (T_{Z,\mathrm{ac}})^d \ge \int_{Z_h} T_Z^d > 0.$$

Here

$$\operatorname{vol}(\mathcal{A}) := \limsup_{k \to +\infty} \frac{d!}{k^d} h^0(Z, \mathcal{A}^k)$$

is the volume of $\mathcal{A}$. Since $\mathcal{A}$ carries the closed positive $(1,1)$-current T_Z, [Bou02, Theorem 1.2(i)] implies that $\operatorname{vol}(\mathcal{A}) \ge \int_Z (T_{Z,\mathrm{ac}})^d > 0$. Thus

$$h^0(Z, \mathcal{A}^k) \ge \frac{\operatorname{vol}(\mathcal{A})}{2 \cdot d!} k^d$$

for infinitely many integers $k \ge 1$. By [Uen75, Theorem 8.1], there exists a real number $C > 0$ such that $h^0(Z, \mathcal{A}^k) \le C k^{\kappa(Z, \mathcal{A})}$ for any integer $k \ge 1$. Hence $\kappa(Z, \mathcal{A}) \ge d$, so $\kappa(Z, \mathcal{A}) = \dim Z$, that is, $\mathcal{A}$ is big in the sense of Definition 2.9. By Lemma 2.15(2), $a(Z) \ge \kappa(Z, \mathcal{A}) = \dim Z$, so $a(Z) = \dim Z$. By Lemma 2.15(3), Z is projective.

Step 6. In this step we pull back the inclusion $\mathcal{L} \subset (\Omega^1_Z)^{\otimes m}$ to a Kawamata cover of Z, and conclude by [CP19, Theorem 7.11].

Since Z is projective (**Step 5**), by [CP19, Lemma 5.2] there exists a Kawamata cover $\eta : Z' \to Z$ adapted to (Z, Δ_Z). Thus Z' is a smooth projective variety, η is a finite Galois cover, and any point of Z' has local coordinates $y_1, \dots, y_d$, and the image point has local coordinates $z_1, \dots, z_d$

on a coordinate polydisc $U_Z \subset Z$ as in Definition 2.8, with $D_{i_j} \cap U_Z = \{z_j = 0\}$ for $1 \le j \le k$ and $D_i \cap U_Z = \emptyset$ for $i \notin \{i_1, \dots, i_k\}$, such that η is given by

$$z_j = y_j^{m_{i_j}} \ (1 \le j \le k), \qquad z_j = y_j^{a_j} \ (k < j \le d)$$

for some integers $a_j \ge 1$. Let $\Omega^1_\eta(Z, \Delta_Z)$ be the orbifold cotangent bundle of [CP19, Definition 5.3]. In the coordinates above, by [CP19, §5.2, second remark after Definition 5.3], it is the subsheaf of $\Omega^1_{Z'}$ generated by

$$dy_1, \dots, dy_k, \ \eta^* dz_{k+1}, \dots, \eta^* dz_d, \qquad \text{where } \eta^* dz_j = a_j y_j^{a_j - 1} dy_j.$$

Let $\widetilde{U}_Z \to U_Z$ be the chart of $\mathcal{Z}$ in Definition 2.8, with coordinates $w_1, \dots, w_k, z_{k+1}, \dots, z_d$ on $\widetilde{U}_Z$. In the coordinates above, the map

$$(y_1, \dots, y_d) \to (y_1, \dots, y_k, y_{k+1}^{a_{k+1}}, \dots, y_d^{a_d})$$

is a local lift of η to $\widetilde{U}_Z$, and two local lifts differ by an element of the product over $1 \le j \le k$ of the groups of m_{i_j}-th roots of unity, acting on w_j. Since orbifold vector bundles on $\mathcal{Z}$ and morphisms between them are equivariant on the charts, pulling back along the local lifts defines a vector bundle $\eta^*\mathcal{V}$ on Z' for any orbifold vector bundle $\mathcal{V}$ on $\mathcal{Z}$, and similarly for morphisms. On $\widetilde{U}_Z$, the bundle $\Omega^1_{\mathcal{Z}}$ has basis $dw_1, \dots, dw_k, dz_{k+1}, \dots, dz_d$, whose pullbacks by the local lift are $dy_1, \dots, dy_k, \eta^* dz_{k+1}, \dots, \eta^* dz_d$. Thus pulling back forms along the local lifts is an isomorphism

$$\eta^* \Omega^1_{\mathcal{Z}} \xrightarrow{\cong} \Omega^1_\eta(Z, \Delta_Z).$$

The inclusion $\mathcal{L} \subset (\Omega^1_{\mathcal{Z}})^{\otimes m}$ is non-zero at a general point, and the N-th tensor power of a non-zero vector is non-zero, so the morphism $\mathcal{L}^{\otimes N} \to (\Omega^1_{\mathcal{Z}})^{\otimes mN}$ is non-zero. By **Step 1**, $\mathcal{L}^{\otimes N}$ is the pullback of $\mathcal{A}$. Pulling back along the local lifts of η, we obtain a morphism

$$\eta^* \mathcal{A} \longrightarrow \Omega^1_\eta(Z, \Delta_Z)^{\otimes mN},$$

which is non-zero since η is étale over a dense open subset of $Z \setminus \operatorname{Supp} \Delta_Z$, where the local lifts are local biholomorphisms. The variety Z is projective (**Step 5**), $1 - \frac{1}{m_i} \in [0, 1)$ for $1 \le i \le r$, and $\operatorname{Supp} \Delta_Z$ is an SNC divisor, so (Z, Δ_Z) is a log canonical SNC pair. By **Step 5**, $\mathcal{A}$ is big. [CP19, Theorem 7.11] states that if Y is a smooth projective variety, (Y, Δ_Y) is a log canonical SNC pair, L is a big line bundle on Y, and there exists a non-zero morphism from the pullback of L to a tensor power of the orbifold cotangent bundle on a Kawamata cover adapted to (Y, Δ_Y), then $K_Y + \Delta_Y$ is big. For a reduced boundary, this is [CP15, Theorem 4.1]. By [CP19, Theorem 7.11] with $(Y, \Delta_Y) = (Z, \Delta_Z)$, $L = \mathcal{A}$, the cover η, and the morphism above, $K_Z + \Delta_Z$ is big. In [CP19, §7.1, item 1.5 after Definition 7.1], bigness of a $\mathbb{Q}$-line bundle L on a smooth projective variety Y means $\kappa(Y, L) = \dim Y$, which is bigness in the sense of Definition 2.9, both for the hypothesis on $\mathcal{A}$ and for the conclusion on $K_Z + \Delta_Z$ in [CP19, Theorem 7.11]. □

9.2. **The inertial base of a minimal quotient.** In this subsection, we deduce Proposition 9.1 from Theorem 9.2 for the base of the Γ-reduction of an infinite quotient of G of minimal Γ-dimension.

Theorem 9.2 needs an orbifold pair on the base whose orbifold fundamental group maps onto G/C, where C is the open image of a general fiber. The quotient map $G \to G/C$ kills C, but a meridian of a prime divisor J removed from the base may still have a non-trivial image of finite order n_J in G/C, so the homomorphism to G/C does not factor through the fundamental group of the base. Taking the orders n_J as multiplicities, we obtain a boundary for which the homomorphism factors through the orbifold fundamental group.

Construction-Proposition 9.4. Let (X, Δ) be a compact Kähler orbifold pair, let $G := \pi_1^{\mathrm{orb}}(X, \Delta)$, and let $f : X \dashrightarrow Z$ be a dominant meromorphic map with connected general fibers onto a compact Kähler manifold Z. For general $z \in Z$ let X_z be the closure of the fiber of f over z. Then there exists a normal subgroup $C \trianglelefteq G$ such that $\pi_1(X_z)_{(X,\Delta)} = C$ for any

z in a dense Zariski open subset of Z. Moreover, there exist a compact Kähler manifold Z_1 with a bimeromorphic map $Z_1 \dashrightarrow Z$, a dense Zariski open subset $Z_1^\circ \subset Z_1$ such that the divisorial part of $Z_1 \backslash Z_1^\circ$ has simple normal crossing support, and a surjective homomorphism $\psi_1 : \pi_1(Z_1^\circ) \to G/C$ with the following properties, where for a prime divisor J on Z_1 we let n_J be the order of the image under ψ_1 of a meridian of J if $J \subset Z_1 \backslash Z_1^\circ$, and $n_J := 1$ otherwise.

(1) For any prime divisor J on Z_1, the order n_J is finite.
(2) The pair (Z_1, Δ_{Z_1}) with $\Delta_{Z_1} := \sum_J (1 - \frac{1}{n_J})J$ is a compact Kähler orbifold pair, and ψ_1 factors through a surjective homomorphism $q_1 : \pi_1^{\mathrm{orb}}(Z_1, \Delta_{Z_1}) \to G/C$.
(3) There exists a dense Zariski open subset $X^\circ = X \backslash A \subset X \backslash \operatorname{Supp} \Delta$, with A a closed analytic subset of X, on which the induced map $f_1 : X \dashrightarrow Z_1$ is holomorphic, such that $f_1(X^\circ) \subset Z_1^\circ$ and the homomorphisms $\pi_1(X^\circ) \to G/C$ induced by the quotient map $G \to G/C$ and by $\psi_1 \circ (f_1|_{X^\circ})_*$ coincide.

We call (Z_1, Δ_{Z_1}) an *inertial base* of f.

Proof of Construction-Proposition 9.4. The proof is long, so we divide it into several steps. Write $\Delta = \sum_i (1 - \frac{1}{m_i}) D_i$ with prime divisors D_i, and let $D := \operatorname{Supp} \Delta$. We fix base points compatibly with the maps below. Changing base points conjugates the homomorphisms between fundamental groups, which changes neither orders of elements nor normal subgroups.

Step 1. In this step we construct C and show that $\pi_1(X_z)_{(X,\Delta)} = C$ for any z in a dense Zariski open subset of Z.

Let $\Gamma_f \subset X \times Z$ be the closure of the graph of f. Since $X \times Z$ is a compact Kähler manifold, Γ_f is a compact Kähler space [Xia19, Example 2.15]. By Lemma 2.14(2), there exists a resolution $X' \to \Gamma_f$ by a compact Kähler manifold X'. Let $\mu' : X' \to X$ and $c' : X' \to Z$ be the induced holomorphic maps, and let $B' := \mu'^{-1}(D) \cup \operatorname{Exc}(\mu')$. By [BM97, Theorem 1.10 and (1.2)(1)] applied to the ideal of B', there exists a composite $X'' \to X'$ of blow-ups with smooth centers over B' such that the inverse image of B' in X'' is a simple normal crossing divisor. The manifold X'' is compact Kähler by Lemma 2.14(1). Let $g : X'' \to X'$ be the composite of blow-ups. Since the centers map into B', g is an isomorphism over $X' \backslash B'$, so $\operatorname{Exc}(g) \subset g^{-1}(B')$, and since a composite of bimeromorphic morphisms of manifolds of equal dimension is a local isomorphism at a point if and only if both factors are local isomorphisms at the corresponding points, $\operatorname{Exc}(\mu' \circ g) = \operatorname{Exc}(g) \cup g^{-1}(\operatorname{Exc}(\mu'))$. Thus

$$g^{-1}(B') = (\mu' \circ g)^{-1}(D) \cup \operatorname{Exc}(\mu' \circ g).$$

Possibly replacing X' with X'', we may assume that B' is a simple normal crossing divisor.

The map c' is proper and surjective, and the general fibers of c' are connected. By Lemma 2.4 applied to c' and B', there exists a proper closed analytic subset $T' \subsetneq Z$ such that c' is a submersion over $Z \backslash T'$, the restriction of c' to any intersection of components of B' is a submersion over $Z \backslash T'$, and

$$U' := c'^{-1}(Z \backslash T') \backslash B' \to Z \backslash T'$$

is a locally trivial fiber bundle. The set $Z \backslash T'$ is connected, so all fibers of c' over $Z \backslash T'$ are diffeomorphic to a general fiber, hence connected. For $z \in Z \backslash T'$, let $F_z := c'^{-1}(z) \backslash B'$. Then $B' \cap c'^{-1}(z)$ is a simple normal crossing divisor or empty in the connected compact manifold $c'^{-1}(z)$, so F_z is connected. By [Hat02, Theorem 4.41 and Proposition 4.48], the sequence

$$\pi_1(F_z) \to \pi_1(U') \xrightarrow{c'_*} \pi_1(Z \backslash T') \to 1$$

is exact. Thus the image N of $\pi_1(F_z)$ in $\pi_1(U')$ is the normal subgroup $\ker(c'_*)$, and N does not depend on $z \in Z \backslash T'$.

Since $\mu'^{-1}(D) \subset B'$ and $\operatorname{Exc}(\mu') \subset B'$, the map μ' restricts to an isomorphism from U' onto

$$X^\circ := X \backslash \mu'\big(B' \cup c'^{-1}(T')\big),$$

which is a dense Zariski open subset of $X\backslash D$. Let $\eta : \pi_1(U') \to G$ be the composite of $(\mu'|_{U'})_* : \pi_1(U') \to \pi_1(X\backslash D)$ and the quotient map $\pi_1(X\backslash D) \to G$. By Lemma 2.3(1), the map $\pi_1(X^\circ) \to \pi_1(X\backslash D)$ is surjective, so η is surjective. Let $C := \eta(N)$. Since N is normal in $\pi_1(U')$ and η is surjective, C is normal in G.

Let $z \in Z\backslash T'$. Then $c'^{-1}(z)$ is a connected compact manifold not contained in B', and μ' maps the dense open subset F_z of $c'^{-1}(z)$ isomorphically into X°. Since Γ_f is the closure of the graph of f and $c'^{-1}(z)$ is irreducible, $\mu'(c'^{-1}(z)) = X_z$, so $\mu'|_{c'^{-1}(z)} : c'^{-1}(z) \to X_z$ is a resolution of singularities. We have

$$F_z = \big(c'^{-1}(z)\backslash \mu'^{-1}(D)\big)\backslash\big(c'^{-1}(z) \cap \mathrm{Exc}(\mu')\big),$$

where $c'^{-1}(z) \cap \mathrm{Exc}(\mu')$ is a proper closed analytic subset of $c'^{-1}(z)$. By Definition-Lemma 8.1, $\pi_1(X_z)_{(X,\Delta)}$ is the image of $\pi_1(F_z)$ under η, which is a conjugate of C. Since C is normal, $\pi_1(X_z)_{(X,\Delta)} = C$ for any $z \in Z\backslash T'$.

Step 2. In this step we construct Z_1, Z_1°, and ψ_1, and we prove (3).

By [BM97, Theorem 1.10 and (1.2)(1)] applied to the ideal of T', there exists a composite $\sigma : Z_1 \to Z$ of blow-ups with smooth centers such that σ is an isomorphism over $Z\backslash T'$ and $\sigma^{-1}(T')$ is a divisor with simple normal crossing support. By Lemma 2.14(1), Z_1 is a compact Kähler manifold. Let $Z_1^\circ := Z_1\backslash\sigma^{-1}(T')$. Then $Z_1\backslash Z_1^\circ$ is a divisor with simple normal crossing support, and σ restricts to an isomorphism $Z_1^\circ \to Z\backslash T'$.

The composite of η and the quotient map $G \to G/C$ maps N to 1, since $\eta(N) = C$. By the exactness in **Step 1**, the composite factors through c'_*. Thus there exists a unique homomorphism $\psi_1 : \pi_1(Z_1^\circ) \to G/C$ such that

$$\psi_1 \circ (\sigma|_{Z_1^\circ})_*^{-1} \circ c'_* = (G \to G/C) \circ \eta \quad \text{on } \pi_1(U'). \tag{9.2}$$

Since η is surjective, ψ_1 is surjective.

Let $f_1 := \sigma^{-1} \circ f : X \dashrightarrow Z_1$. On X°, we have $f = c' \circ (\mu'|_{U'})^{-1}$ and $f(X^\circ) \subset Z\backslash T'$. Hence f_1 is holomorphic on X°, $f_1(X^\circ) \subset Z_1^\circ$, and

$$f_1|_{X^\circ} = (\sigma|_{Z_1^\circ})^{-1} \circ c' \circ (\mu'|_{U'})^{-1}.$$

Identifying $\pi_1(X^\circ)$ with $\pi_1(U')$ by μ', (9.2) says that the homomorphisms $\pi_1(X^\circ) \to G/C$ induced by the quotient map $G \to G/C$ and by $\psi_1 \circ (f_1|_{X^\circ})_*$ coincide. This proves (3).

Step 3. In this step we prove (1). More precisely, for any prime divisor $J \subset Z_1\backslash Z_1^\circ$ and any prime divisor E on the resolution X_1 of the graph of f_1 constructed below that maps onto J, we define positive integers e and r_E and show that $n_J \mid e r_E$.

By Lemma 2.3(5), any two meridians of J in $\pi_1(Z_1^\circ)$ are conjugate, so n_J does not depend on the meridian. Let $\Gamma_{f_1} \subset X \times Z_1$ be the closure of the graph of f_1, which is a compact Kähler space [Xia19, Example 2.15]. By Lemma 2.14(2), there exists a resolution $X_1 \to \Gamma_{f_1}$ by a compact Kähler manifold X_1 which is an isomorphism over the smooth locus of Γ_{f_1}. Let $\mu_1 : X_1 \to X$ and $c_1 : X_1 \to Z_1$ be the induced holomorphic maps. Over X°, the space Γ_{f_1} is the graph of the holomorphic map $f_1|_{X^\circ}$, hence smooth. Thus μ_1 is an isomorphism over X°, and $c_1 = f_1 \circ \mu_1$ on $V := \mu_1^{-1}(X^\circ)$.

Let $R := c_1^{-1}(Z_1\backslash Z_1^\circ) \cup \mu_1^{-1}(D)$. Since $Z_1\backslash Z_1^\circ$ and D are divisors and c_1 and μ_1 are surjective, R is a closed analytic subset of X_1 of pure codimension one. We have $V \subset X_1\backslash R$, because $\mu_1(V) = X^\circ \subset X\backslash D$ and $c_1(V) = f_1(X^\circ) \subset Z_1^\circ$. Let $\alpha : \pi_1(X_1\backslash R) \to G/C$ be the homomorphism induced by $\mu_1 : X_1\backslash R \to X\backslash D$ and the quotient map $\pi_1(X\backslash D) \to G/C$, and let $\beta := \psi_1 \circ (c_1|_{X_1\backslash R})_*$. On $\pi_1(V)$, the homomorphism β equals $\psi_1 \circ (f_1|_{X^\circ})_* \circ (\mu_1|_V)_*$, so (3) implies that α and β coincide on the image of $\pi_1(V)$. The set V is the complement in $X_1\backslash R$ of the closed analytic subset $\mu_1^{-1}(X\backslash X^\circ)$, so $\pi_1(V) \to \pi_1(X_1\backslash R)$ is surjective by Lemma 2.3(1). Therefore $\alpha = \beta$.

Since c_1 is proper and surjective, some irreducible component of $c_1^{-1}(J)$ maps onto J. Let E be any prime divisor on X_1 with $c_1(E) = J$. Then $E \subset c_1^{-1}(J) \subset R$, so E is an irreducible

component of R. Let

$$e := \operatorname{mult}_E c_1^* J \geq 1, \qquad a_i := \operatorname{mult}_E \mu_1^* D_i,$$

and let r_E be the smallest positive integer such that $m_i \mid r_E a_i$ for all i. If $E \not\subset \operatorname{Supp} \mu_1^* D_i$, then $a_i = 0$ and the condition holds for i.

Since $c_1(E) = J$, we may choose a point $y \in E$ such that R is smooth at y and $Z_1 \backslash Z_1^\circ$ is smooth at $w := c_1(y)$. Let $x := \mu_1(y)$. Let $\mathbb{D}_w \subset Z_1$ be a coordinate polydisc centered at w with $(Z_1 \backslash Z_1^\circ) \cap \mathbb{D}_w = J \cap \mathbb{D}_w = \{z_1 = 0\}$, and let $\lambda \in \pi_1(\mathbb{D}_w \backslash J) \cong \mathbb{Z}$ be the meridian of $\{z_1 = 0\}$. Let $\mathbb{D}_x \subset X$ be a coordinate polydisc centered at x with $D \cap \mathbb{D}_x = \{z_1 \cdots z_k = 0\}$, where $k \geq 0$ and $\{z_j = 0\} = D_{i_j} \cap \mathbb{D}_x$ for distinct indices $i_1, \ldots, i_k$. Let $\mathbb{D}_y \subset X_1$ be a coordinate polydisc centered at y such that $c_1(\mathbb{D}_y) \subset \mathbb{D}_w$, $\mu_1(\mathbb{D}_y) \subset \mathbb{D}_x$, and $R \cap \mathbb{D}_y = E \cap \mathbb{D}_y$ is a smooth connected hypersurface of $\mathbb{D}_y$. Let $\epsilon \in \pi_1(X_1 \backslash R)$ be a meridian of E represented by a small loop in $\mathbb{D}_y \backslash E$ around E, joined to the base point by a path in $X_1 \backslash R$.

We have $c_1(\mathbb{D}_y \backslash E) \subset \mathbb{D}_w \backslash J$. By Lemma 2.3(3) applied to $c_1|_{\mathbb{D}_y}$, the loop maps to λ^e in $\pi_1(\mathbb{D}_w \backslash J)$. Joining $\mathbb{D}_w \backslash J$ to the base point by the image of the path under c_1, we obtain a meridian τ of J in $\pi_1(Z_1^\circ)$ with $(c_1)_* \epsilon = \tau^e$. Thus

$$\beta(\epsilon) = \psi_1(\tau)^e,$$

and $\psi_1(\tau)$ has order n_J.

If $k = 0$, then μ_1 maps the loop into the simply connected set $\mathbb{D}_x \subset X \backslash D$, so $\alpha(\epsilon) = 1$. Assume that $k \geq 1$. We have $\mu_1(\mathbb{D}_y \backslash E) \subset \mathbb{D}_x \backslash D$, since $\mu_1^{-1}(D) \cap \mathbb{D}_y \subset R \cap \mathbb{D}_y = E \cap \mathbb{D}_y$. Let $\lambda_j' \in \pi_1(\mathbb{D}_x \backslash D) \cong \mathbb{Z}^k$ be the meridian of $\{z_j = 0\}$. By Lemma 2.3(3) applied to $\mu_1|_{\mathbb{D}_y}$, the loop maps to $\prod_{j=1}^k \lambda_j'^{a_{i_j}}$, because $\operatorname{mult}_{E \cap \mathbb{D}_y} \mu_1^* \{z_j = 0\} = a_{i_j}$. Joining $\mathbb{D}_x \backslash D$ to the base point by the image of the path under μ_1, the elements λ_j' become pairwise commuting meridians δ_j of D_{i_j} in $\pi_1(X \backslash D)$, and $(\mu_1)_* \epsilon = \prod_{j=1}^k \delta_j^{a_{i_j}}$. By Lemma 2.3(5), δ_j is a conjugate of γ_{i_j}, so $\delta_j^{m_{i_j}}$ maps to 1 in G. Since $m_{i_j} \mid r_E a_{i_j}$ and the elements δ_j commute,

$$\alpha(\epsilon)^{r_E} = \prod_{j=1}^k \big(\text{image of } \delta_j^{m_{i_j}}\big)^{r_E a_{i_j}/m_{i_j}} = 1.$$

Hence $\alpha(\epsilon)^{r_E} = 1$ in both cases. Since $\alpha = \beta$,

$$\psi_1(\tau)^{e r_E} = \beta(\epsilon)^{r_E} = \alpha(\epsilon)^{r_E} = 1.$$

Thus $n_J \mid e r_E$. In particular, n_J is finite for any prime divisor $J \subset Z_1 \backslash Z_1^\circ$. Since $n_J = 1$ for the other prime divisors on Z_1, this proves (1).

Step 4. In this step we prove (2).

The divisor $Z_1 \backslash Z_1^\circ$ has finitely many irreducible components, and $n_J = 1$ for any prime divisor $J \not\subset Z_1 \backslash Z_1^\circ$. Thus Δ_{Z_1} is a finite sum, $\operatorname{Supp} \Delta_{Z_1}$ is the union of the components J of $Z_1 \backslash Z_1^\circ$ with $n_J \geq 2$, and $\operatorname{Supp} \Delta_{Z_1}$ is a simple normal crossing divisor. By (1), each coefficient of Δ_{Z_1} is $1 - \frac{1}{n_J}$ for an integer $n_J \geq 2$. Hence (Z_1, Δ_{Z_1}) is a compact Kähler orbifold pair.

Let $Z_1^\bullet := Z_1 \backslash \operatorname{Supp} \Delta_{Z_1}$. Then $Z_1^\circ \subset Z_1^\bullet$, and $Z_1^\bullet \backslash Z_1^\circ$ is the union of the components J of $Z_1 \backslash Z_1^\circ$ with $n_J = 1$, intersected with $Z_1^\bullet$. By Lemma 2.3(2), the kernel of $\pi_1(Z_1^\circ) \to \pi_1(Z_1^\bullet)$ is the normal subgroup generated by meridians of the components J of $Z_1 \backslash Z_1^\circ$ with $n_J = 1$. The homomorphism ψ_1 maps such meridians to 1. Thus ψ_1 factors through a surjective homomorphism $\psi_1^\bullet : \pi_1(Z_1^\bullet) \to G/C$.

Let J be a component of $\operatorname{Supp} \Delta_{Z_1}$. A meridian of J in $\pi_1(Z_1^\bullet)$ may be represented by a small loop around J near a point of J not contained in any other component of $Z_1 \backslash Z_1^\circ$, so it is the image of a meridian of J in $\pi_1(Z_1^\circ)$. Hence $\psi_1^\bullet$ maps the n_J-th power of any meridian of J to 1. The group $\pi_1^{\mathrm{orb}}(Z_1, \Delta_{Z_1})$ is the quotient of $\pi_1(Z_1^\bullet)$ by the normal subgroup generated by the n_J-th powers of meridians of the components J of $\operatorname{Supp} \Delta_{Z_1}$. Therefore $\psi_1^\bullet$ factors through a homomorphism $q_1 : \pi_1^{\mathrm{orb}}(Z_1, \Delta_{Z_1}) \to G/C$, which is surjective because ψ_1 is surjective. This proves (2). □

Proof of Proposition 9.1. The proof is long, so we divide it into several steps. Let $D := \operatorname{Supp}\Delta$.

Step 1. In this step we define the minimal Γ-dimension ℓ and show that $\ell \geq 1$.

Let ℓ be the minimum of $\operatorname{gdim}(X, \Delta; \rho)$ over all surjective homomorphisms ρ from G onto infinite groups. Since G is infinite, $\rho = \mathrm{id}_G$ is such a homomorphism, and each $\operatorname{gdim}(X, \Delta; \rho)$ is an integer in $[0, \dim X]$. Thus ℓ exists.

Suppose that $\ell = 0$, and let $\rho : G \to P$ be a surjective homomorphism onto an infinite group P with $\operatorname{gdim}(X, \Delta; \rho) = 0$. Then Z_ρ is a point. Since a countable union of proper closed analytic subsets of X contains no non-empty open subset, there exists $x \in X \backslash (D \cup E_\rho)$, and the fiber X_x of f_ρ through x is X. By Definition-Theorem 8.2, X has finite ρ-image. Taking the resolution id_X in Definition-Lemma 8.1, $\pi_1(X)_{(X,\Delta)}$ is the image of $\pi_1(X \backslash D) \to G$, which is G. Thus $P = \rho(G)$ is finite, a contradiction. Therefore $\ell \geq 1$.

Step 2. In this step we construct the base and show that G/C is infinite.

Let $\rho : G \to P$ be a surjective homomorphism onto an infinite group P with $\operatorname{gdim}(X, \Delta; \rho) = \ell$, and let $f_\rho : X \dashrightarrow Z_\rho$ be the Γ-reduction of ρ (Definition-Theorem 8.2). Then Z_ρ is a compact Kähler manifold with $\dim Z_\rho = \ell$, and f_ρ is a dominant meromorphic map with connected general fibers. We apply Construction-Proposition 9.4 to f_ρ. We obtain a normal subgroup $C \trianglelefteq G$ and a dense Zariski open subset $Z_\rho^\circ \subset Z_\rho$ such that $\pi_1(X_z)_{(X,\Delta)} = C$ for any $z \in Z_\rho^\circ$, together with Z_1, Z_1°, ψ_1, Δ_{Z_1}, q_1, $f_1 : X \dashrightarrow Z_1$, and X° satisfying (1), (2), and (3) of Construction-Proposition 9.4. Since Z_1 is bimeromorphic to Z_ρ, $\dim Z_1 = \ell$.

The set of points $x \in X$ at which f_ρ is holomorphic and $f_\rho(x) \in Z_\rho^\circ$ contains a dense Zariski open subset of X. Since a countable union of proper closed analytic subsets of X contains no non-empty open subset, there exists $x \notin D \cup E_\rho$ such that f_ρ is holomorphic at x and $f_\rho(x) \in Z_\rho^\circ$. Then $X_x = X_{f_\rho(x)}$, so

$$\rho(C) = \rho\big(\pi_1(X_x)_{(X,\Delta)}\big)$$

is finite by Definition-Theorem 8.2.

Since C is normal in G and ρ is surjective, $\rho(C)$ is normal in P. Since P is infinite and $\rho(C)$ is finite, $P/\rho(C)$ is infinite. The homomorphism ρ induces a surjection $G/C \to P/\rho(C)$. Thus G/C is infinite.

Step 3. In this step we verify the conditions of Theorem 9.2 applied to q_1, $Q := G/C$ and the quotient map $G \to Q$, with the pair (Z_1, Δ_{Z_1}) of dimension $\ell > 0$, the pair (X, Δ), and the map f_1.

By **Step 2**, Q is infinite. If $Q_0 \leq Q$ is a subgroup of finite index, then the inverse image of Q_0 in G has finite index, hence equals G, so $Q_0 = Q$. By Lemma 2.3(4), $\pi_1(X \backslash D)$ is finitely generated, so G and Q are finitely generated. By Lemma 2.17(6), any homomorphism $Q \to \mathrm{GL}_k(\mathbb{C})$, $k \geq 1$, is trivial. By Construction-Proposition 9.4(2), q_1 is surjective, and the quotient map $G \to Q$ is surjective.

We check condition (1) of Theorem 9.2. Since $\operatorname{Supp}\Delta_{Z_1} \subset Z_1 \backslash Z_1^\circ$, we have $f_1(X^\circ) \subset Z_1^\circ \subset Z_1 \backslash \operatorname{Supp}\Delta_{Z_1}$. The homomorphism ψ_1 is the composite of $\pi_1(Z_1^\circ) \to \pi_1^{\mathrm{orb}}(Z_1, \Delta_{Z_1})$ and q_1. Thus condition (1) is Construction-Proposition 9.4(3).

We check condition (2) of Theorem 9.2. Let $\psi : Q \to H$ be a surjective homomorphism onto an infinite group H. Then the composite of the quotient map $G \to Q$ and ψ is a surjective homomorphism from G onto the infinite group H. By the definition of ℓ in **Step 1**,

$$\operatorname{gdim}\big(X, \Delta; \psi \circ (G \to Q)\big) \geq \ell = \dim Z_1.$$

By Theorem 9.2, Z_1 is projective and $K_{Z_1} + \Delta_{Z_1}$ is big. Since $\dim Z_1 = \ell \geq 1$ and f_1 is a dominant meromorphic map, (Z_1, Δ_{Z_1}) and f_1 satisfy the conclusion of Proposition 9.1. □

10. Proofs of the main theorems

In this section, we prove Theorem 1.2 and Corollary 1.3.

Proof of Theorem 1.2. Suppose that Theorem 1.2 fails. By Proposition 7.8, there exists a special compact Kähler orbifold pair (Y, Δ_Y) such that $a(Y) = 0$ and $G := \pi_1^{\mathrm{orb}}(Y, \Delta_Y)$ is infinite and has no subgroup of finite index other than G. By Proposition 9.1, there exist a compact Kähler orbifold pair (Z, Δ_Z) with $\dim Z \geq 1$ and a dominant meromorphic map $f : Y \dashrightarrow Z$ such that $K_Z + \Delta_Z$ is big. By Definition 2.9, there exists an integer $N \geq 1$ such that $N(K_Z + \Delta_Z)$ is a divisor and the line bundle $\mathcal{L} := \mathcal{O}_Z(N(K_Z + \Delta_Z))$ satisfies $\kappa(Z, \mathcal{L}) = \dim Z$. By Lemma 2.15(1) and (2), we have

$$a(Y) \geq a(Z) \geq \kappa(Z, \mathcal{L}) = \dim Z \geq 1,$$

which contradicts $a(Y) = 0$. □

Proof of Corollary 1.3. Let $\Delta := \sum_{i=1}^{r}(1 - 1/m_i)D_i$. Since $a(X) = 0$, Corollary 3.6(2) shows that (X, Δ) is special. Thus $\pi_1^{\mathrm{orb}}(X, \Delta)$ is virtually abelian by Theorem 1.2. □

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